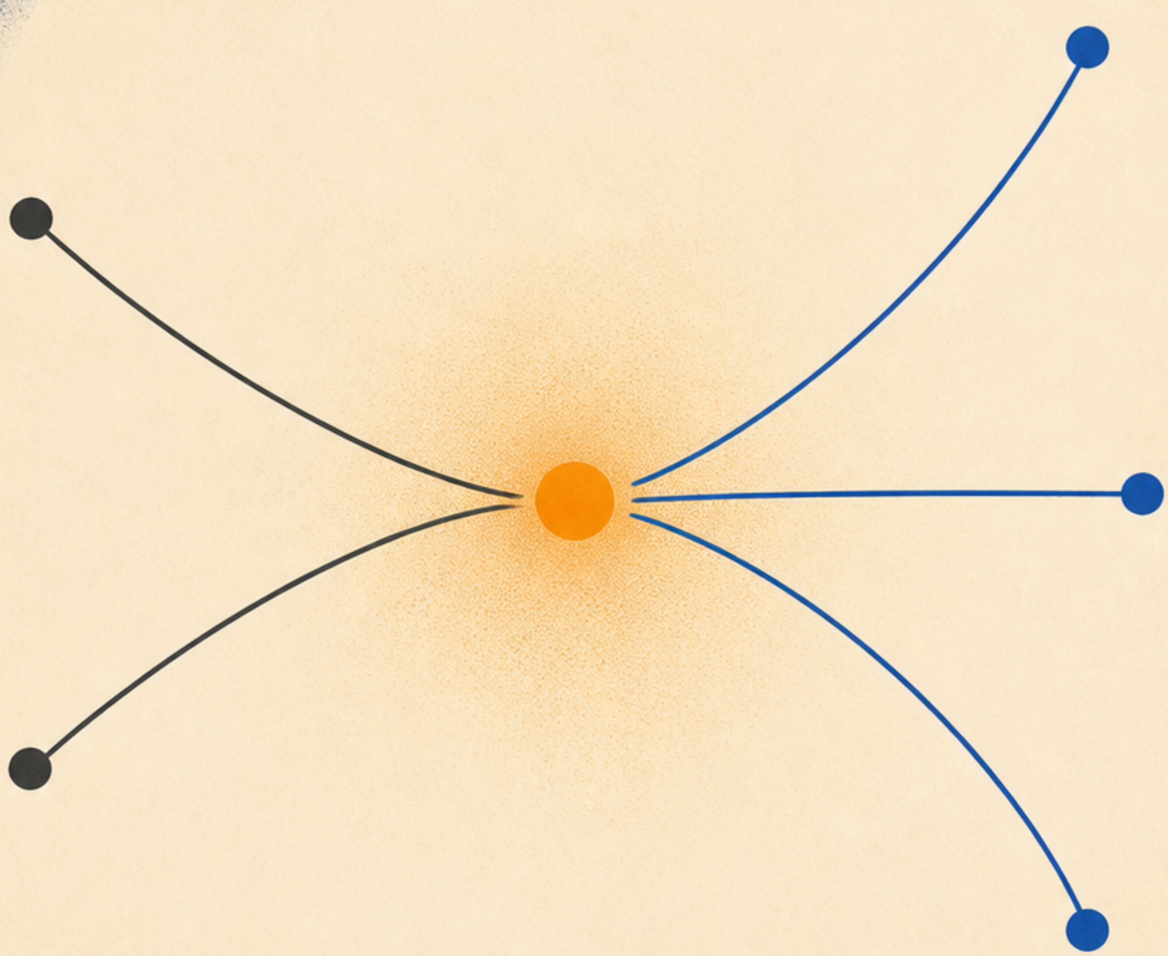
Particle Physics 1 Basic Concepts

Supplementary track, 2009 Fall



Lectures by Leonard Susskind

Companion edition by LazyingArt LLC with Video2Book

About These Notes

These independently edited companion notes follow lectures by Leonard Susskind. They were reconstructed from machine transcripts, subtitles, and selected blackboard frames, then checked against the available source material. They are not an original manuscript by Professor Susskind and have not been reviewed or endorsed by him.

The edition was prepared by LazyingArt LLC using the open-source Video2Book workflow. Machine transcripts remain available separately and may contain recognition errors. Editorial clarifications and nontrivial reconstructions are identified in footnotes.

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Chapter 1

Particles, Waves, and the Energy of Resolution

1.1 Particles or Fields?

What is particle physics about? Particles—but that answer immediately raises an ancient question. Is nature discrete? Discrete here means that matter comes in separate, indivisible units, not that it is tactful. The Greek word *atom* means indivisible, even though objects now called atoms can be broken apart. For the moment, *particle* will be the generic name for a discrete unit.

The opposing picture is a continuum. Matter is then not a collection of separate pieces but a continuous smear through space. Its natural language is the language of fields. A field is a function of position: the density of a material is a field, and the electric and magnetic fields are fields. The latter determine how charged particles move, but the fields themselves are conceived classically as continuously distributed through space.

The historical order of these ideas should not be pressed too hard. The sequence is useful because it displays the logic of the argument, not because it supplies a dependable chronology. The particle vocabulary includes atoms, molecules, collisions, and localized objects; the continuum vocabulary includes fields and continuously varying properties. Which picture is right? Quantum mechanics prevents either from being completely right, but it also prevents either from being completely wrong.

The discussion will be partly self-contained, though later arguments draw on quantum mechanics, classical mechanics, special relativity, and electromagnetism. The immediate question remains simpler: what first persuaded people that matter really is discrete?

1.2 Chemical Evidence for Discrete Matter

The first serious evidence came, at least in this logical account, from chemistry rather than physics. The distinction is artificial: chemistry is largely the physics of atoms and molecules. Dalton compared the masses of equal numbers of molecules of different substances. Establishing that the samples contained equal numbers was itself a major problem. Avogadro's name enters here, and a mole is Avogadro's number of molecules. The actual value of that number was not needed merely to compare samples containing the same number of molecules.

The striking pattern was that molar masses came approximately in integer multiples of the molar mass of hydrogen. The multipliers were not generally $3/2$, π, or arbitrary real numbers, but numbers close to integers. Carbon-12, for example, carries roughly twelve of the mass units represented by hydrogen. This strongly suggests construction from repeated building blocks of nearly the same mass.

The modern explanation removes the mystery. Ordinary atoms contain protons, neutrons, and electrons. An electron has only about one two-thousandth the mass of a proton or neutron, so electrons make a small contribution to atomic mass. Protons and neutrons carry almost all of it, and their masses differ by much less than one percent. Since the nucleus of ordinary hydrogen is one proton, the mass of an atom or molecule is approximately an integer multiple of a hydrogen-scale nuclear mass. The statement is only approximate: proton and neutron masses are not identical, and electrons contribute a little. The near-integral pattern nevertheless reveals discrete constituents.

Molecules were then understood as combinations of atoms. Nineteenth-century chemistry identified roughly a hundred kinds of atoms and organized them as elements; roughly ninety-two occur naturally in the elementary count, with further distinctions once isotopes are included. The periodic table is not periodic in the naive sense of simply repeating equal blocks, but it organized those separate species. At that stage the elements themselves looked like the elementary building blocks. Einstein's 1905 work supplied a quantitative way to establish the molecular theory of matter much more decisively, but the chemical evidence had already made the atomic picture compelling.

Near the end of the nineteenth century this apparent endpoint failed. The electron was discovered, and radioactivity revealed phenomena that the elements-as-elementary picture could not contain. Radioactivity already held, in miniature, many of the questions that became nuclear and particle physics.

1.3 Radioactivity and Separated Beams

The exact attribution and chronology are uncertain. Becquerel is tentatively associated with the observation that radioactive material happened to be near a photographic plate and affected it as though invisible radiation had illuminated it. Rather than treating the fogged plate merely as spoiled, one could ask what had reached it. Marie Curie's extensive later work belongs to the history as well, but the precise division of the early discoveries is not needed for the experimental logic.

Place the material deep inside a dense lead block and leave a narrow channel toward a photographic plate. Lead is useful because it is dense, while the channel collimates whatever escapes. Instead of a diffuse exposure, the plate develops a localized spot, suggesting a directed beam. In this logical sense it is the first particle beam, though at this stage the nature of the beam is unknown.

Now pass the collimated emission through a magnetic field perpendicular to the plane of the trajectories. The exposure separates into three branches. One continues undeflected. Two bend in opposite directions, and by unequal amounts. A magnetic force reverses direction when the sign of the charge reverses, so the oppositely bent branches behave as oppositely charged radiation; the straight branch behaves as neutral radiation.[1]

[1] The schematic reconstructs the diagram developed in the lecture video between 00:14:34 and 00:19:31.

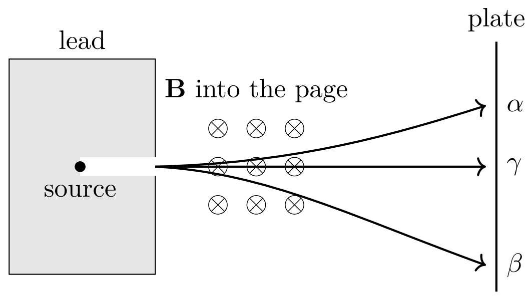

Figure 1.1: A collimated radioactive beam in a transverse magnetic field. The neutral component is undeflected, while oppositely charged components bend in opposite directions by unequal amounts.

The three components were named alpha, beta, and gamma. In modern terms,

$$\begin{aligned} &\beta\text{-radiation} = \text{electrons}, \\ &\alpha\text{-radiation} = \text{helium nuclei}, \\ &\gamma\text{-radiation} = \text{photons}. \end{aligned} \tag{1.1}$$

An alpha particle contains two protons and two neutrons, explaining its mass number four. In Dalton's approximate integer sequence, helium would therefore have occupied number four: four hydrogen-scale nuclear units. Beta rays are the negatively charged electrons already known from cathode-ray experiments. Gamma rays carry no electric charge.

With a sufficiently dilute source and a detector capable of resolving individual arrivals, the continuous-looking exposure becomes a sequence of separate events: blip, blip, blip. The same can be done for beta, alpha, and gamma radiation. A Geiger counter gives the right picture, whether or not the earliest apparatus was operated in precisely that regime. Each component arrives in discrete bundles. Gamma remains the puzzle: it is neutral, yet it is detected one event at a time. Gamma radiation is very-high-energy light, and Planck and especially Einstein connected light with quanta. Before confronting that conclusion, however, the classical evidence that light is a wave has to be taken seriously.

1.4 The Classical Wave Character of Light

Classically, light was the strongest example of continuity. It consists of electric and magnetic fields propagating as continuous waves. Newton had favored a corpuscular picture, but wave experiments displaced it. Shaking a charge produces electromagnetic radiation. A macroscopic antenna can emit radio waves, microwaves, infrared, visible light, or ultraviolet radiation; a charge moving around within an atom is also an antenna. The emitted radiation can expose a photographic plate.

A single opening gives a first indication of wave behavior. If light were a stream of bullets, a small opening would be expected to make a sharply collimated spot. Instead the illumination spreads into a fuzzy region. Spreading alone is not decisive, however, because one could imagine particles being deflected by forces at the edge of the aperture.

Two openings produce the sharper test. Two independent particle streams would give two overlapping smears, perhaps brightest where they overlap. Waves do something different. A crest from one opening can meet a crest from the other and reinforce it; a crest can also meet a trough and cancel it. The screen therefore shows alternating bright and dark bands, including dark places that either opening alone would illuminate.[2]

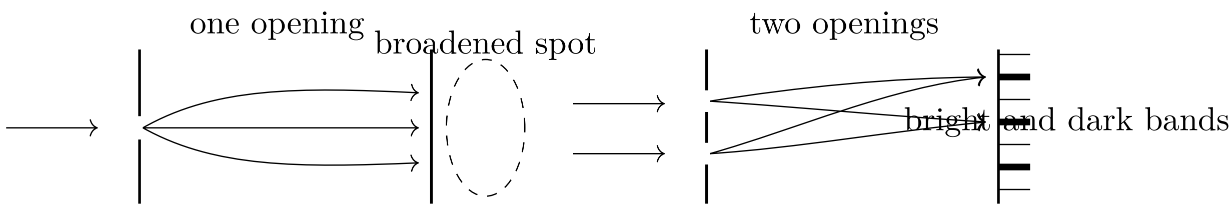

Figure 1.2: A single aperture produces a broadened image, while two apertures produce alternating constructive and destructive interference.

The same effect is easy to imagine with water. Wiggle two fingers in the surface: the circular waves spread, reinforce at some places, and cancel at others. Interference, rather than mere spreading, is the decisive wave signature.

The historical sequence is again not simple. In this schematic account Huygens is placed before Newton as an advocate of wave propagation; Thomas Young is associated with the interference demonstration; and some of Newton's own optical experiments admit a wave-interference interpretation even though he favored corpuscles. The later interference experiments established the wave character of light beyond reasonable doubt.

Maxwell then identified what oscillates. If a light wave propagates along an axis, its electric field is transverse to that axis. Its magnetic field is also transverse and is perpendicular to the electric field. The complete pattern travels at the speed of light.[3]

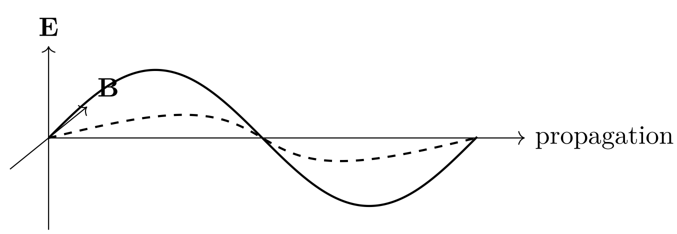

Figure 1.3: In an electromagnetic wave, the electric and magnetic fields are mutually perpendicular and both are transverse to the direction of propagation.

Ordinary visible light oscillates far too rapidly for the alternating fields to be followed directly. For an extremely long radio wave the oscillation would be correspondingly slower. In either case the same transverse-wave geometry gives interference.

1.5 Wavelength, Period, and Frequency

The wavelength λ is a spatial distance: begin at one point in the oscillation and move along the wave until the same phase recurs after one full cycle. It is measured in metres, centimetres, or any other unit of length.

[2]The schematic reconstructs the aperture and interference diagrams developed in the lecture video between 00:25:00 and 00:29:49.
[3]The schematic reconstructs the electromagnetic-wave diagram drawn in the lecture video at 00:31:08.

Now stand at one position while the wave passes. The time required for one complete cycle is the period T. Visible light has an extremely short period, roughly of order 10^{-15} s; longer wavelength corresponds to longer period, and shorter wavelength to shorter period.

During one period a wave advances by one wavelength, so a wave of speed v obeys

$$v = \frac{\lambda}{T}. \tag{1.2}$$

Water and sound waves have their own propagation speeds. For light,

$$c = \frac{\lambda}{T}. \tag{1.3}$$

The frequency f is the number of cycles per unit time:

$$f = \frac{1}{T}. \tag{1.4}$$

Hence

$$\lambda f = c. \tag{1.5}$$

Either wavelength or frequency determines the other.

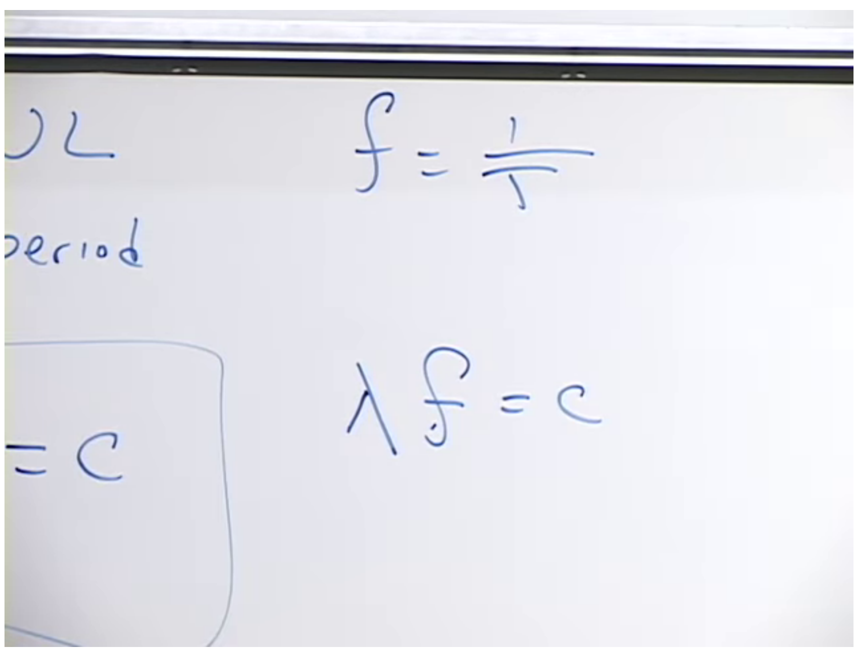

Figure 1.4: For a light wave, $f = 1/T$ and $\lambda f = c$.

Lecture frame: 00:36:24.

Physicists often measure oscillation rate in radians per second rather than complete cycles per second. Imagine uniform motion around a circle. If the point completes f circuits each second and each circuit contains 2π radians, its angular frequency is

$$\omega = 2\pi f, \qquad \frac{\omega}{2\pi} = f. \tag{1.6}$$

For now this is a definition; the practical question of whether it reduces the number of factors of 2π returns later.

Combining $f = c/\lambda$ with $\omega = 2\pi f$ gives

$$\omega = \frac{2\pi c}{\lambda}, \qquad \omega\lambda = 2\pi c. \tag{1.7}$$

For another kind of wave, c is replaced by that wave's propagation speed.

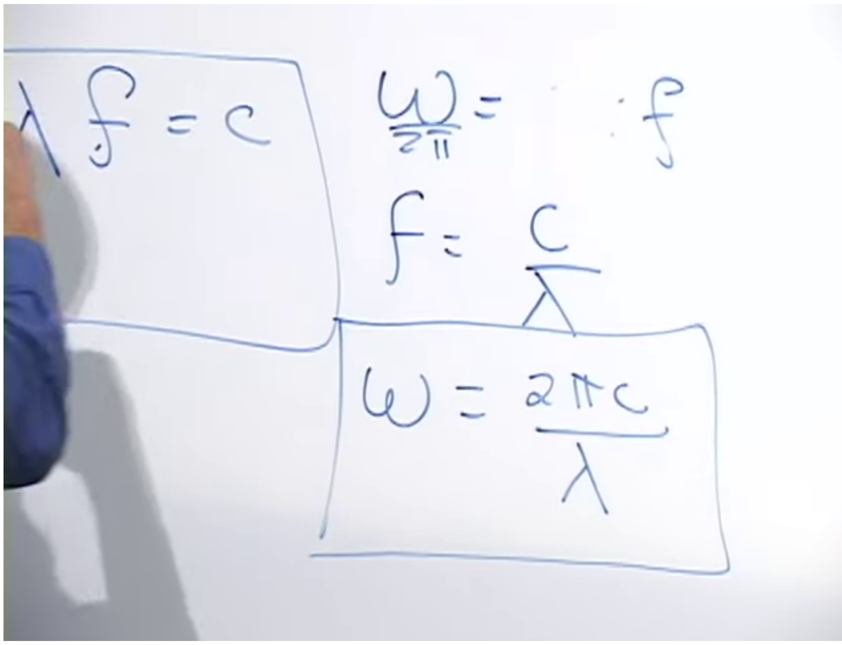

Figure 1.5: With $\omega = 2\pi f$, a light wave satisfies $\omega = 2\pi c/\lambda$.

Lecture frame: 00:39:10.

Electromagnetic radiation forms a continuous spectrum. Ordered from longer to shorter wavelength, the conventional names are

$$\begin{aligned} &\text{radio} \longrightarrow \text{microwave} \longrightarrow \text{infrared} \longrightarrow \text{visible} \\ &\longrightarrow \text{ultraviolet} \longrightarrow \text{X ray} \longrightarrow \text{gamma ray}. \end{aligned} \tag{1.8}$$

Visible light occupies only a narrow interval. The boundaries between the named regions are conventional, so no exact dividing wavelengths are needed here. Radio waves may be extremely long; gamma radiation is the very-short-wavelength end of the electromagnetic spectrum. The difficulty is now sharp: gamma radiation is classically a continuous field wave, yet it is detected in discrete blips.

1.6 Discrete Arrivals and Wave Probabilities

Thermodynamic properties of radiation led Planck and Einstein to the conclusion that light must be treated in discrete units. An imagined experiment makes the point vivid, but it is not a claim about the actual historical route to the discovery.

Shine a beam on a screen. At ordinary intensity it makes a continuous-looking illuminated blob. Put a partly opaque attenuator in the beam: the blob becomes dimmer. Increase the attenuation repeatedly and less light arrives per unit time. Eventually the appearance changes. Instead of a continuously dim blob, the detector records isolated events.[4]

[4] The schematic reconstructs the attenuation experiment developed in the lecture video between 00:42:32 and

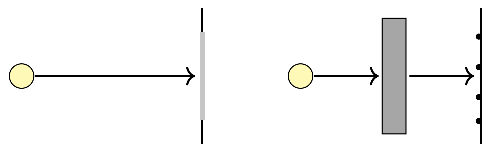

Figure 1.6: Progressive attenuation lowers the detection rate until a continuous-looking exposure resolves into isolated photon events.

Wait long enough and those events accumulate into the same spatial profile as the original blob. The bright region was therefore the unresolved sum of many individual arrivals. This low-intensity experiment can be performed, though it was not the early experiment that originally motivated the photon hypothesis. Einstein's photon proposal arose from thermodynamic and photoelectric considerations; the attenuation experiment supplies an especially direct way to picture its consequence.

Now place two apertures in the beam and reduce the intensity until photons arrive one at a time. Each detection is a localized blip. Yet after many detections the blips fill in an interference pattern, including bright and dark bands. The discrete arrivals are random, but their probability distribution is the distribution predicted by the wave pattern. The classical wave has become a wave governing probabilities.

Wave and particle are therefore not two separate kinds of entity assigned to different parts of nature. They are two manifestations of the same quantum object. Light cannot be described adequately as only a wave or only a particle.

Electrons reverse the original surprise. Electrons were accepted as particles and arrive at a detector one at a time, but with sufficiently close apertures they also build an interference pattern. Alpha particles, about eight thousand times heavier than electrons, obey the same principle, although the experiment is harder. Interference has even been observed with buckyballs: molecules of sixty carbon atoms arranged approximately like a generalized soccer ball. The same principle is expected to apply even to bowling balls, although such an experiment is fantastically impractical. The discussion can be pushed toward proposed experiments with viruses or living cells as increasingly ambitious examples. Understanding how wave and particle descriptions fit together requires quantum mechanics; the immediate task is to extract the photon properties needed for particle physics.

1.7 Photon Energy

Photons are the discrete, indivisible elements of light. Light carries energy—sunlight heats an absorbing object—so each photon must carry energy. A monochromatic wave has a definite wavelength and angular frequency, and every photon in it has

$$E_\gamma = \hbar\omega. \tag{1.9}$$

Trying to picture a continuous wave as a little mechanical collection of dots is misleading. What is definite is the experimental combination: localized detections occur with a wave-governed distribution, and the energy arrives in indivisible units.

00:45:42.

Since

$$\omega = \frac{2\pi c}{\lambda}, \tag{1.10}$$

a shorter wavelength means a larger angular frequency and therefore more energy in each photon:

$$\lambda \downarrow \implies \omega \uparrow \implies E_\gamma \uparrow . \tag{1.11}$$

A single radio photon has very little energy. A radio beam can nevertheless carry enormous total energy by containing very many photons. For a monochromatic pulse containing n photons,

$$E_{\text{pulse}} = n\hbar\omega, \qquad n = 0, 1, 2, \ldots . \tag{1.12}$$

The total may be large, but it remains an integer multiple of the one-photon energy.

This relation points toward the practical question that will govern the rest of the discussion. Why must bigger and bigger machines be built to see smaller and smaller things? Why not use a small machine for a small object? The first half of the answer is already present: resolving smaller structure requires shorter wavelength, and a shorter-wavelength quantum carries greater energy.

1.8 Rest Energy and Invariant Mass

Particle physics combines quantum mechanics with relativity. The photon relation $E_\gamma = \hbar\omega$ supplies the quantum side; the corresponding elementary fact from relativity is

$$E_{\text{rest}} = mc^2. \tag{1.13}$$

The qualification matters. Older accounts distinguished *rest mass* from a velocity-dependent *moving mass*. Modern terminology uses *mass* for the invariant mass of an object. Motion changes its energy, not its mass. Every electron has the same mass, whatever its state of motion, while an electron's energy increases as it is accelerated. Thus $E = mc^2$, in this form, is the energy of a massive object in a frame in which the object as a whole is at rest.

The object need not be internally motionless. A box of gas can stand still on a scale while its molecules move rapidly in all directions. The energy of those internal motions is part of the rest energy of the complete box. Begin with a cold box, measure its mass, and then add thermal energy while keeping the box itself at rest. Its rest mass changes by

$$\Delta m = \frac{\Delta E_{\text{internal}}}{c^2}. \tag{1.14}$$

Its weight and inertia increase with its mass. For an ordinary box, even after a large temperature increase, the effect is far too small to notice on an ordinary scale; the smallness of the effect does not change the principle.

A sharper example uses an electron and a positron. They have precisely equal masses and opposite electric charges. If they are initially at rest and annihilate, photons carry away the energy:

$$e^- + e^+ \longrightarrow \text{photons}. \tag{1.15}$$

The initial energy is

$$E_{\text{initial}} = 2m_e c^2, \tag{1.16}$$

and that energy cannot disappear. A single pair yields only a tiny amount, so enormous numbers of pairs would be needed to heat even a cup of coffee, but individual annihilation events exhibit the conversion directly.

For an object at rest, mass and energy are therefore the same physical quantity expressed in different units. The factor c^2 is the conversion factor between the customary mass and energy units, much as a numerical factor converts metres to centimetres.

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[a] **Question from the audience.** Does converting mass into released energy require a mechanism?

Response. Yes, a physical mechanism is required, but conservation laws already constrain its outcome. Even before the microscopic annihilation mechanism is described, the initial electron–positron energy must reappear somewhere. Conservation of energy can therefore determine an important part of the answer without a detailed account of the interaction.

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[a]Lecture timestamp: 01:04:39.

1.9 Dimensions and the Choice of Units

In SI units the speed of light is a very large number and Planck's constant is a very small one. The smallness of the latter is already suggested by the photon relation: the angular frequency of visible light is enormous while the energy of one photon is tiny, so $E_\gamma = \hbar\omega$ requires a small numerical value of $\hbar$ in these units. At the level of order of magnitude,

$$c \sim 10^8 \ \mathrm{m\,s^{-1}}, \qquad \hbar \sim 10^{-34} \ \mathrm{J\,s}. \tag{1.17}$$

[5] Units are essential: neither constant is merely a bare decimal. Since energy has the dimensions

$$[E] = ML^2T^{-2}, \tag{1.18}$$

and $E_\gamma = \hbar\omega$ with $[\omega] = T^{-1}$, Planck's constant has dimensions

$$[\hbar] = [E]T = ML^2T^{-1}. \tag{1.19}$$

Calling c large and $\hbar$ small already presupposes a choice of units. Measuring speed in feet per hour produces a different numerical value of c from measuring it in metres per second. In light-years per year, or light-seconds per second, the speed of light is simply one. A suitable coordinated choice of mass, length, and time units can likewise make $\hbar = 1$.

Familiar systems of units are inherited conventions, not privileged features of nature. A mile contains 5280 feet, a stone contains fourteen pounds, horses are measured in hands, and power may be measured in horsepower. Metric units are more systematic, but their scales remain human choices. Nothing in fundamental physics singles out the metre, second, or kilogram.

Length, time, and mass provide three independent basic unit choices:

$$L, \qquad T, \qquad M. \tag{1.20}$$

Other mechanical units can be built from them. The useful choices depend on the problem. A solar mass is convenient in astrophysics; a proton mass may be convenient in nuclear physics; light-years

[5]Editorial clarification: Only the orders of magnitude stated in the lecture are retained because the final board digits are indistinct.

and years make relativistic astronomical formulas compact. With three independent choices, one can assign convenient numerical values to three independent dimensional constants. Astrophysics, atomic physics, and particle physics need not make the same choices.

For particle physics, two particularly useful conventions are

$$c = 1, \qquad \hbar = 1. \tag{1.21}$$

There is no comparably distinguished particle mass for the remaining unit. One could declare the proton mass, Higgs-boson mass, or muon mass to be one, but that would merely privilege one member of a large collection.

Why choose c and $\hbar$, then? A first answer might be that they appear in many equations. Frequency of appearance is not enough: the proton mass also appears in many equations, and so does the proton's size. What distinguishes c and $\hbar$ is their universality.

The constants c and $\hbar$ are different. The speed c constrains every physical object: massive objects may approach it but cannot be accelerated through it. Planck's constant constrains every quantum object through the order-of-magnitude uncertainty relation

$$\Delta x \, \Delta p \gtrsim \hbar. \tag{1.22}$$

Neither statement singles out a particular species of particle. This universality is what makes $c = 1$ and $\hbar = 1$ natural conventions.

The proton supplies no comparable universal scale. It is only one object among many. Setting the proton mass to one would have no more fundamental standing than setting the mass of a rubidium atom to one; the electron, Higgs boson, and muon present the same difficulty.

Electric charge suggests another universal quantity, but after c and $\hbar$ have been normalized, the physically significant electromagnetic coupling is dimensionless. Changing units can move dimensional constants around an electromagnetic formula; it cannot change a genuine dimensionless coupling.

Gravity supplies a dimensional universal constant. In Newtonian form, the magnitude of the force between two masses is

$$F = G\frac{m_1 m_2}{r^2}. \tag{1.23}$$

Every mass gravitates, whereas electrically neutral objects need not participate in an electric force. Thus G has the required universality. It is nevertheless a poor practical normalization for ordinary particle physics because gravitational forces between known elementary particles are negligible in the calculations under discussion.

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[a] **Question from the audience.** Is it legitimate simply to dismiss G? At very high energies, elementary particles might not obey classical gravity in the way being assumed.

Response. Individual particles do respond to the Earth's gravitational field: a single particle falls in that field. The gravitational influence of one elementary particle on another has not been measured directly, so the universality assumption can be questioned at that level. The immediate reason for leaving G out is practical rather than absolute: gravity contributes negligibly to ordinary particle-physics experiments and equations.

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[a] Lecture timestamp: 01:20:34.

[a] **Question from the audience.** Could the electric interaction be used as the third unit-setting convention instead of gravity?

Response. Yes. One can choose electromagnetic units in which electric charge, or the constant appearing in Coulomb's law, is normalized. Such conventions are used. Electrons repel rather than attract, but that sign does not obstruct the choice. It is simply not the convention adopted here.

[a] Lecture timestamp: 01:22:01.

[a] **Question from the audience.** Once c and $\hbar$ are set equal to one, is electric charge dimensionless?

Response. Yes. A dimensionless measure associated with the square of the electric charge is approximately 1/137. The precise convention and the meaning of that number can be postponed; the point here is that it is dimensionless.

[a] Lecture timestamp: 01:23:06.

[a] **Question from the audience.** Could Coulomb's constant be chosen as the third constant set equal to one?

Response. It can be absorbed into the electromagnetic unit convention. That choice cannot turn the dimensionless coupling 1/137 into one, because changing units cannot alter a dimensionless number. No particular benefit is claimed for adopting that convention here.

[a] Lecture timestamp: 01:23:37.

1.10 Natural-Unit Equations and Reference Frames

With $c = 1$, the rest-energy relation for a massive object becomes

$$E_{\text{rest}} = m. \tag{1.24}$$

With $\hbar = 1$, the photon relation becomes

$$E_\gamma = \omega. \tag{1.25}$$

The constants can be restored whenever numerical sizes in conventional units are needed.

These formulas look similar but mean different things. In $E_{\text{rest}} = m$, E is the rest energy of a massive object. A photon has no mass and cannot be brought to rest, so its energy is not rest energy. The relation $E_\gamma = \omega$ gives the energy of a moving, massless quantum.

[a] **Question from the audience.** What does *at rest* mean when rest is always relative to something and the laboratory itself is not absolutely at rest?

Response. Rest is defined relative to a chosen frame. If a cup is stationary relative to the laboratory, then in that laboratory frame its energy is its mass when $c = 1$, with no overall kinetic contribution. Another observer moving past the cup assigns it kinetic energy as well. Different observers generally assign different energies to the same object or photon; energy is frame-dependent, and no claim of absolute rest is required.

[a] Lecture timestamp: 01:26:07.

The angular-frequency convention can now be reconsidered. Writing $E_\gamma = \hbar\omega$ is compact, but $\omega = 2\pi f$ inserted a factor of 2π earlier.

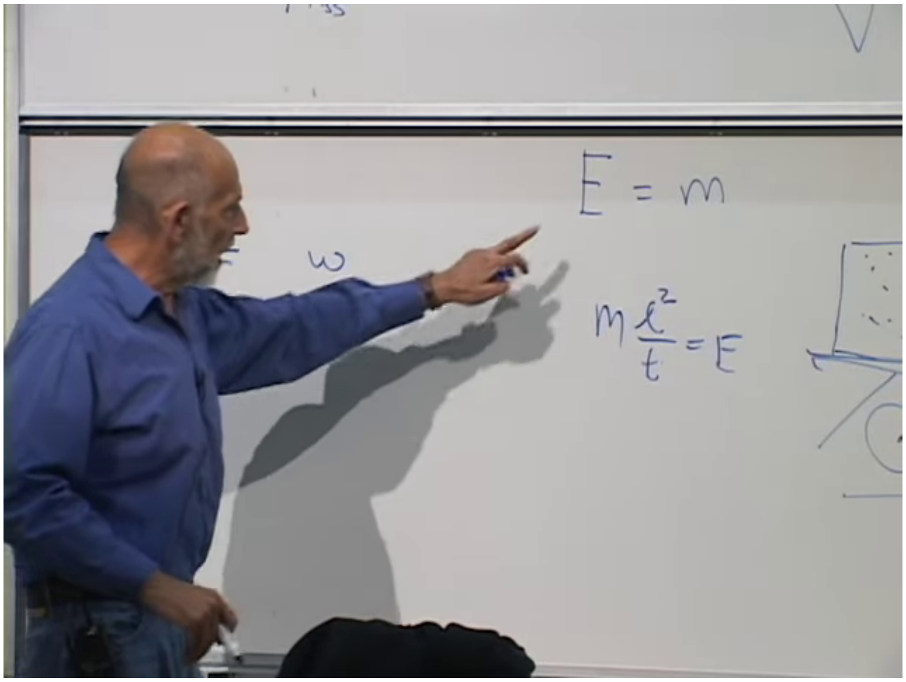

Figure 1.7: In units with $c = 1$, the rest energy of a massive object at rest is $E = m$.
Lecture frame: 01:25:40.

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[a] **Question from the audience.** Why use angular frequency ω, measured in radians per second, rather than the apparently simpler number of complete cycles per second?

Response. The choice is partly historical, and the mathematical answer is pragmatic rather than immediate. The answer offered here is tentative: write the relevant equations once with f and once with ω, then see which convention leaves fewer factors of 2π. The factors of 2π cannot be made to disappear everywhere. A possible connection with complex-exponential notation was also suggested, but it is not developed here.

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[a]Lecture timestamp: 01:28:13.

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[a] **Question from the audience.** Can two particles be accelerated until their mutual gravitational interaction becomes important?

Response. In principle, yes. In practice, present accelerators are extraordinarily far from that regime. The answer also depends on what is meant by a particle: a macroscopic gravitating body is not the elementary-particle case at issue.

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[a]Lecture timestamp: 01:29:51.

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[a] **Question from the audience.** What can serve as a source of positrons?

Response. High-energy collisions can create positrons. Some nuclear processes and decays produce them as well.

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[a]Lecture timestamp: 01:30:27.

[a] **Question from the audience.** Are Planck time and Planck distance the natural units being used here?

Response. No. Planck units arise when G, c, and $\hbar$ are all set equal to one. The resulting Planck length is far smaller than the length scales of ordinary particle physics, so it is an inconvenient measuring stick here. Such units are natural when gravity, quantum mechanics, and relativity all matter together, as in black-hole physics, cosmology, or quantum gravity. When gravity is negligible, setting only c and $\hbar$ equal to one is more useful.

[a] Lecture timestamp: 01:30:53.

1.11 Momentum, Light, and Resolution

The next basic concept is momentum. An object at rest has no momentum, even though it has the nonzero rest energy mc^2. Once it moves, it generally carries momentum. Momentum is conserved in a collision just as energy is conserved: when two objects collide, the total vector momentum before and after the collision is the same. Anything constrained by a conservation law deserves special attention.

Momentum is a vector: it points along the direction of motion. For a nonrelativistic object,

$$\mathbf{p} = m\mathbf{v}. \tag{1.26}$$

This is not the general relativistic formula, and it cannot be applied directly to light.

Light nevertheless carries momentum as well as energy. Ordinary light produces so little push that its momentum is hard to notice, but an intense laser pulse can transfer a substantial impulse. The large laser at Livermore, for example, can drive material inward with the momentum carried by its radiation.

Consider a pulse sufficiently well collimated that its rays travel nearly along one axis. A real beam may have some angular spread, with portions traveling in slightly different directions, but for an ideal collimated beam the momentum points along the beam and has magnitude

$$p = \frac{E}{c}, \qquad \text{or equivalently} \qquad E = cp. \tag{1.27}$$

This relation is taken as a relativistic fact here rather than proved. The following dimensional check only confirms that its units agree with those of ordinary momentum:

$$\left[\frac{E}{c}\right] = \frac{ML^2T^{-2}}{LT^{-1}} = MLT^{-1} = [mv] = [p]. \tag{1.28}$$

This dimensional comparison does not assign a rest mass to light. A photon has zero mass; it carries momentum because it carries energy and moves at c.

The direction of $\mathbf{p}$ is the direction of propagation. Its magnitude is usually difficult to detect because c appears in the denominator. A beam can readily reveal its energy by heating or evaporating matter, while its mechanical push is much smaller. With enough intensity, however, that push can become large.

For one photon,

$$p_\gamma = \frac{E_\gamma}{c} \tag{1.29}$$

$$= \frac{\hbar\omega}{c}. \tag{1.30}$$

To express this in terms of wavelength, return carefully to the wave relations. Since

$$f\lambda = c, \qquad \omega = 2\pi f, \tag{1.31}$$

the correct relation is

$$\omega\lambda = 2\pi c, \qquad \omega = \frac{2\pi c}{\lambda}. \tag{1.32}$$

The factor of 2π is easy to reverse if the definitions are not written out. Starting again from $f\lambda = c$ and $f = \omega/(2\pi)$ settles the issue:

$$\frac{\omega}{2\pi}\lambda = c \quad \Longrightarrow \quad \omega\lambda = 2\pi c. \tag{1.33}$$

Substitution now gives

$$p_\gamma = \frac{\hbar}{c}\frac{2\pi c}{\lambda} \tag{1.34}$$

$$= \frac{2\pi\hbar}{\lambda} \tag{1.35}$$

$$= \frac{h}{\lambda}. \tag{1.36}$$

---

[a] **Question from the audience.** Is $2\pi\hbar$ equal to h?

Response. Yes. By definition $h = 2\pi\hbar$, so $p_\gamma = 2\pi\hbar/\lambda$ is the same as $p_\gamma = h/\lambda$. Factors of 2π reappear somewhere whichever frequency convention is chosen.

---

[a]Lecture timestamp: 01:42:38.

The shorter the wavelength, the larger the photon momentum. A radio photon has a very long wavelength and a tiny momentum. A gamma-ray photon has a very short wavelength and a much larger kick. One photon still gives only a small macroscopic impulse, but its momentum is much larger than that of a radio photon. Its energy is larger as well, since $E_\gamma = cp_\gamma$.

This relation explains why smaller objects require larger machines. A photograph made with wavelengths much longer than a person's head could show only a blur. To distinguish facial features, the wavelength must be shorter than the head. To resolve one hair, it must be comparable to or smaller than the hair's thickness. To resolve the structure of an atom, the probe wavelength must be no larger than the atom:

$$\lambda_{\text{probe}} \lesssim \text{size of the structure to be resolved.} \tag{1.37}$$

Smaller structures therefore require shorter wavelengths, and shorter wavelengths require larger momentum and energy. The probe need not be a photon; high-energy particles of other kinds obey the same quantum connection between wavelength and momentum. Producing them requires accelerators that transfer more and more energy to each particle. The pair $p_\gamma = h/\lambda$ and $E_\gamma = \hbar\omega$ carries the argument: increasing the probe energy increases its momentum and shortens its wavelength.

This has been the recurring pattern of particle physics for roughly a century: higher energy, larger momentum, shorter wavelength, and finer spatial resolution.

[a] **Question from the audience.** Is there a theoretical lower limit to wavelength?

Response. Possibly. Something new is expected near the Planck length, but the answer is tentative. If spatial resolution bottoms out there, that scale is still about seventeen orders of magnitude below the distances being probed at the LHC, so present experiments remain enormously far from it.

[a] Lecture timestamp: 01:46:57.

The same chain of reasoning suggests a rough connection between the scale of an accelerator and the scale it can resolve.

[a] **Question from the audience.** Is the wavelength, or the size being resolved, roughly inversely related to the size of the accelerator?

Response. Roughly, yes, provided the maximum electric fields that accelerate the particles, and the magnetic fields used to guide them, remain fixed by the available apparatus. An accelerating gradient is an energy gain per unit length. For a linear accelerator of length L and approximately fixed gradient $\mathcal{G}$,

$$E_{\text{beam}} \sim \mathcal{G}L.$$

At ultrarelativistic energies the resolvable wavelength is inversely related to energy, so increasing the linear accelerator's length improves the resolution by roughly the inverse factor. This is only a heuristic scaling, not an exact law. Circular machines allow the same apparatus to be traversed repeatedly, but accelerated charges radiate while bending and lose energy.

[a] Lecture timestamp: 01:48:02.

[a] **Question from the audience.** How large would an accelerator have to be to reach the Planck length?

Response. On this rough scaling, it would have to be about seventeen orders of magnitude longer than CERN's accelerator. That is not the size of the universe, but it is roughly galactic in scale.

[a] Lecture timestamp: 01:49:47.

[a] **Question from the audience.** How do the energies of the highest-energy cosmic rays compare with collider energies?

Response. Individual cosmic rays can have far more laboratory energy. For the numerical comparison, take an extreme cosmic ray with laboratory energy of order 10^{21} eV $= 10^{12}$ GeV. But a cosmic ray strikes a target that is stationary in the laboratory, whereas a collider brings two energetic beams together head-on. More importantly, useful experiments require flux: many particles and many well-controlled collisions. A lone cosmic ray may miss the relevant target or produce an event from which little can be inferred. An intentionally extravagant oil-equivalent estimate put the supply needed to run a useful galactic accelerator at about one hundred trillion barrels of oil per second. Its purpose is to show that adequate collision statistics would be at least as serious an obstacle as the accelerator's length.

[a] Lecture timestamp: 01:50:09.

[6]

These are real distance scales, not merely formal ones. The practical puzzle is how to probe them without constructing a galactic accelerator and supplying its fantastical power.

[6] Editorial clarification: The oil-equivalent estimate is inconsistent in the source. It is first stated and then repeated as about 10^{14} barrels per second; a later restatement gives 10^{11} barrels per second. Both scales are retained rather than silently reconciled.

[a] **Question from the audience.** How should the LHC energy be compared with a cosmic ray of energy 10^{12} GeV?

Response. For an ultrarelativistic projectile of laboratory energy E_{lab} striking a fixed target of mass m_{target}, the center-of-mass energy behaves parametrically as

$$E_{\mathrm{cm}} \simeq \sqrt{2m_{\mathrm{target}}c^2 E_{\mathrm{lab}}}.$$

[b] With the target mass held fixed, the useful collision energy therefore grows only as the square root of the projectile energy. Thus a 10^{12} GeV cosmic ray corresponds, at order-of-magnitude level, to counter-propagating beam energies of order 10^6 GeV per beam, not 10^{12} GeV in each beam. Cosmic rays can still exceed terrestrial collider energies, but their very low flux makes systematic, controlled experiments impractical.

[a] Lecture timestamp: 01:52:55.

[b] Editorial clarification: The displayed fixed-target expression makes explicit the square-root scaling stated in the exchange.

Chapter 2

Waves, Quantized Modes, and Creation Operators

2.1 Waves, Fields, and Complex Exponentials

Quantum field theory is the basic tool of particle physics. Before giving even an elementary account of a quantum field, it is useful to collect the mathematical facts that connect waves with particles. A wave is already a field configuration: a field varies from point to point in space, and a wave is a configuration in which those values oscillate and propagate. Quantum mechanics supplies the link between such waves and particles.

Quantum mechanics and special relativity are assumed background here. The immediate purpose is to review the pieces needed for particle physics, not to replace a course in either subject.

An exponential is characterized by proportional growth. If

$$f(x) = e^{\alpha x}, \tag{2.1}$$

then

$$\frac{df}{dx} = \alpha f. \tag{2.2}$$

Over a small interval, the increase is proportional to the amount already present. A population growing in a Petri dish gives the familiar picture: the more organisms are present, the more are available to reproduce, so the rate of increase is proportional to the population itself. Multiplying $e^{\alpha x}$ by an overall constant does not change this defining property.

Sine and cosine oscillate instead of growing. They are the same curve shifted by one quarter of a cycle, and differentiation rotates them into one another:

$$\frac{d}{dx}\sin(kx) = k\cos(kx), \tag{2.3}$$

$$\frac{d}{dx}\cos(kx) = -k\sin(kx). \tag{2.4}$$

Neither function alone is an exponential, because its derivative is proportional to the other function.

The complex combination does have the exponential property. With $i^2 = -1$,

$$\frac{d}{dx}(\cos(kx) + i\sin(kx)) = -k\sin(kx) + ik\cos(kx) \tag{2.5}$$

$$= ik\big(\cos(kx) + i\sin(kx)\big). \tag{2.6}$$

Therefore

$$e^{ikx} = \cos(kx) + i\sin(kx). \tag{2.7}$$

This identity comes up repeatedly because sines and cosines describe wave-like oscillations. When x is a spatial coordinate, the function oscillates through space. When the argument is time, it oscillates in time.

2.2 Frequency, Wavelength, and Photon Momentum

The ordinary frequency f is the number of complete cycles per second. A cycle means one full oscillation. Angular frequency measures the same motion in radians per second:

$$\omega = 2\pi f. \tag{2.8}$$

The two notations describe the same frequency. One moves between them to keep unnecessary factors of 2π out of a calculation, although the factors never disappear altogether.

The wavelength λ is the distance occupied by one complete spatial cycle. If a wave propagates with speed v, then the distance per cycle times the number of cycles per second is the propagation speed:

$$\lambda f = v. \tag{2.9}$$

For light in vacuum, $v = c$, so

$$\lambda f = c. \tag{2.10}$$

For a sound wave the right-hand side is the speed of sound, and for a water wave it is the appropriate water-wave speed. The appearance of c is therefore a statement about light, not a definition common to every wave.

Quantum mechanics introduces Planck's constant in two notations,

$$\hbar = \frac{h}{2\pi}. \tag{2.11}$$

Changing from h to $\hbar$ merely moves factors of 2π from one equation to another.

The numerical values of dimensional constants depend on the units used to quote them. Metres, seconds, and kilograms are human-scale conventions, convenient for ordinary lengths and times. Traditional units comparable with the length of an arm were useful for cloth or rope; whatever the precise historical route to the metre, no fundamental principle selects this particular human scale. In SI units,

$$\begin{aligned} &h \simeq 6.6 \times 10^{-34}\ \mathrm{J\,s}, \quad \hbar \simeq 1.1 \times 10^{-34}\ \mathrm{J\,s}, \\ &c \simeq 3 \times 10^{8}\ \mathrm{m\,s^{-1}}. \end{aligned} \tag{2.12}$$

The point of these figures is their scale in conventional units: Planck's constant is extraordinarily small and the speed of light is extraordinarily large.

One quantum of a mode has energy

$$E_{\text{one quantum}} = \hbar\omega = hf. \tag{2.13}$$

This is the energy of one photon, not the total energy of an arbitrary classical electromagnetic wave. Visible light has a frequency of order 10^{15} cycles per second, so a typical optical photon has energy of order

$$E_\gamma \sim 10^{-19}\ \text{J}. \tag{2.14}$$

Even such a rapid optical oscillation gives a very small energy per quantum because h is so small. A macroscopic amount of optical energy therefore contains an enormous number of photons.

Energy and momentum deserve special attention because both are conserved. They are useful bookkeeping quantities: the total energy and momentum entering a process must be balanced by the total energy and momentum leaving it. Momentum is a vector and points in the direction of motion. For a particle moving much more slowly than light,

$$\vec{p} = m\vec{v}. \tag{2.15}$$

Now consider electromagnetic radiation collimated along one direction, rather than dispersed into many directions. Its momentum points along the beam. The nonrelativistic formula cannot be applied to a photon, because m means rest mass and a photon has zero rest mass. There is nevertheless a useful dimensional mnemonic. If one *inappropriately* assigns the photon the quantity E/c^2 as though it were a mass and then substitutes $v = c$, one finds

$$mv \longrightarrow \frac{E}{c^2}c = \frac{E}{c}. \tag{2.16}$$

This maneuver happens to give the right answer and gets the units right, but it is not a legitimate argument that a photon has rest mass E/c^2. The actual relativistic statement for collimated radiation is

$$|\vec{p}| = \frac{E}{c}. \tag{2.17}$$

The equation gives the magnitude. The direction is the direction in which the radiation travels.

For one photon,

$$|\vec{p}_\gamma| = \frac{E_\gamma}{c} = \frac{hf}{c}. \tag{2.18}$$

For light,

$$\lambda f = c, \qquad \frac{f}{c} = \frac{1}{\lambda}, \tag{2.19}$$

and therefore

$$|\vec{p}_\gamma| = \frac{hf}{c} = \frac{h}{\lambda}. \tag{2.20}$$

The shorter the wavelength, the larger the photon momentum. This is one of the most important quantum facts for particle physics. To study a small object one needs a short-wavelength probe, and the price of that short wavelength is large momentum.

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[a] **Question from the audience.** Is $E = \hbar\omega$ fundamental, or is it derived from a deeper principle?

Response. It follows from the quantum mechanics of the harmonic oscillator. That result will be used here without repeating its proof; the oscillator language developed below gives the deeper setting for it.

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[a]Lecture timestamp: 00:22:23.

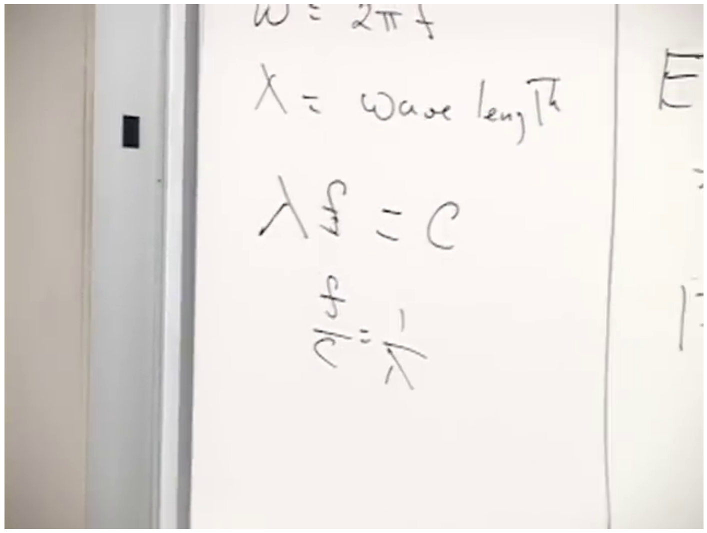

Figure 2.1: For light, $\lambda f = c$ and $f/c = 1/\lambda$, the relations used to obtain $|\vec{p}_\gamma| = h/\lambda$.
Lecture frame: 00:21:39.

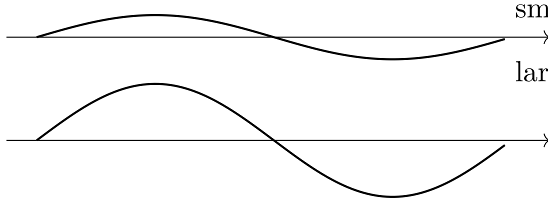

Figure 2.2: Waves with equal wavelength and frequency but different amplitudes.

2.3 Wave Amplitude and Photon Number

[a] **Question from the audience.** What does the amplitude of a wave mean in photon language?

Response. Compare two waves with the same wavelength and frequency but different amplitudes, such as $\sin(kx)$ and $A\sin(kx)$. Classically, the energy in a fixed region is proportional to the amplitude squared. Quantum mechanically, a mode containing n photons has energy $n\hbar\omega$. At fixed frequency, region, and normalization, the amplitude therefore grows in order of magnitude like $\sqrt{n}$.

[a] Lecture timestamp: 00:23:10.

The two waves are identical in their pattern of oscillation. One is simply stronger than the other.[1]

For a fixed amount of wave,

$$E_{\text{wave}} \propto A^2. \tag{2.21}$$

For an electromagnetic wave this is the familiar statement that field energy is proportional to the square of the field strength. If the same mode contains n photons, then

$$E_{\text{wave}} = n\hbar\omega. \tag{2.22}$$

[1] The schematic follows the wave comparison at 00:23:47–00:24:28.

Consequently, with the mode, frequency, normalization, and region held fixed,

$$A^2 \propto n\hbar\omega, \qquad A \propto \sqrt{n} \quad \text{at fixed } \omega. \tag{2.23}$$

No exact normalization is being asserted. The amount of wave included matters: doubling the length of an otherwise uniform wave changes its total energy without changing the local amplitude. The proportionality may therefore be understood for a fixed region, such as one wavelength.

There is also a specifically quantum qualification. Amplitude and phase cannot both be assigned exact classical values. The question "What is the amplitude of one photon?" is therefore ambiguous if it asks for a perfectly sharp classical field. Setting $n = 1$ nevertheless gives a sensible order-of-magnitude scale.

---

[a] **Question from the audience.** Does the vertical displacement in an electromagnetic-wave drawing represent the photon's position in space?

Response. No. The propagation axis is spatial, while the vertical coordinate represents the value of the electric or magnetic field. The momentum points along the propagation axis, not along the plotted field amplitude.

---

[a] Lecture timestamp: 00:28:25.

The wave may propagate in any spatial direction, but the familiar sinusoidal drawing uses only its horizontal axis as a spatial coordinate. The other plotted direction records field strength.

---

[a] **Question from the audience.** Is the stated photon number being counted within one wavelength or period of the wave?

Response. One may choose a fixed region such as one wavelength. This is why the relation is only proportional: the total energy and photon number also depend on how much of the wave is included. Two, three, or four wavelengths contain correspondingly more energy than one wavelength at the same amplitude.

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[a] Lecture timestamp: 00:29:37.

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[a] **Question from the audience.** Is the amplitude of a single photon meaningless?

Response. It is meaningful only as an order-of-magnitude statement obtained by setting $n = 1$. For an electromagnetic mode, electric- and magnetic-field information plays a role analogous to conjugate position and momentum information, so quantum uncertainty limits how sharply a classical amplitude can be assigned. The order-of-magnitude relation remains useful.

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[a] Lecture timestamp: 00:30:17.

These questions already lead into quantum field theory, where the relation between field variables and particle number can be stated precisely. For now the proportionalities are enough.

2.4 General Quantum Relations and Slow Matter Waves

Some formulas introduced for photons apply much more generally, while others do not. It is worth sorting them before turning to electrons.

The definitions of f, $\omega = 2\pi f$, and λ apply generally. The quantum relation

$$E = \hbar\omega = hf \tag{2.24}$$

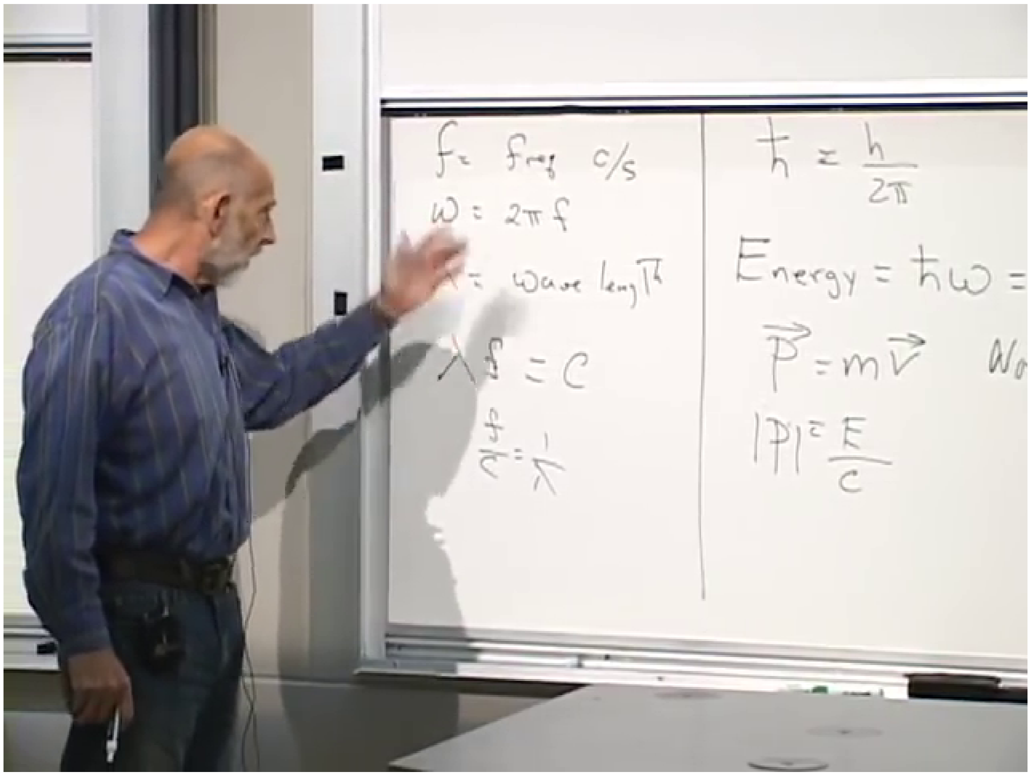

Figure 2.3: Wave definitions and the photon and nonrelativistic momentum relations compared by their domains of validity.

Lecture frame: 00:31:52.

is also general in the present discussion. By contrast, formulas containing c, such as $\lambda f = c$ and $|\vec{p}| = E/c$, apply to waves moving at the speed of light. The formula $\vec{p} = m\vec{v}$ applies to particles moving much more slowly than light. Relativistic motion between those limits requires an interpolation that is not needed yet.

The momentum–wavelength relation survives the photon example:

$$p = \frac{h}{\lambda}. \tag{2.25}$$

Here p denotes the magnitude when λ is taken as a positive wavelength; direction can be supplied separately.

An electron, a nucleus, and even a bowling ball can be associated with a wave. Electron interference is one manifestation of that fact. Write the electron wave as ψ. What replaces $\lambda f = c$ for a slowly moving electron? The missing ingredient is the nonrelativistic relation between energy and momentum.

Begin with kinetic energy and momentum:

$$E = \frac{1}{2}mv^2, \tag{2.26}$$

$$p = mv, \qquad v = \frac{p}{m}. \tag{2.27}$$

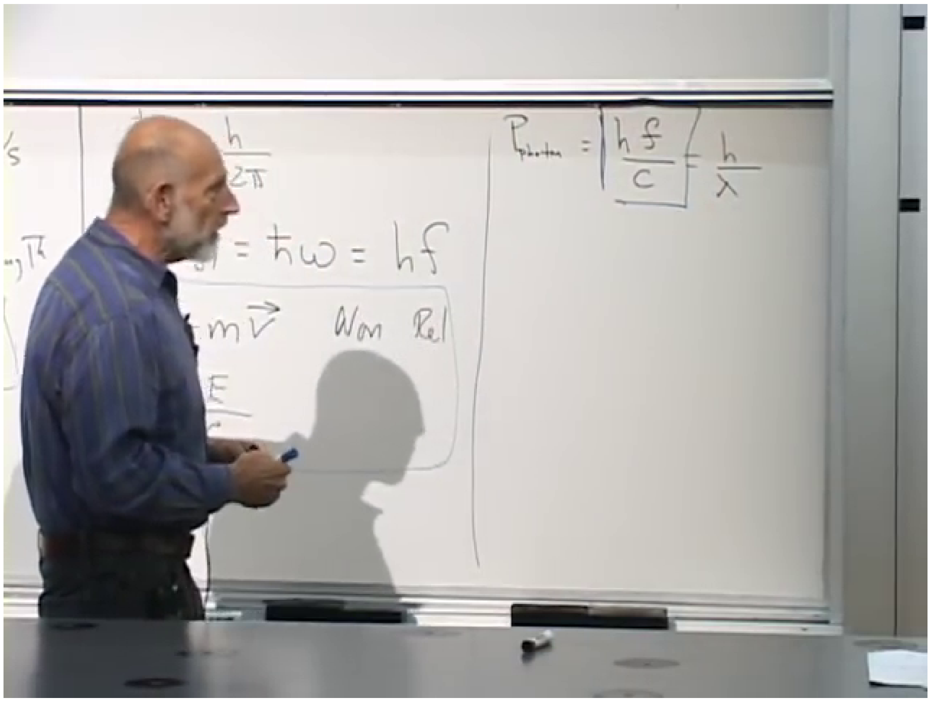

Figure 2.4: Photon momentum $|\vec{p}_\gamma| = hf/c = h/\lambda$, leading to the general de Broglie relation.
Lecture frame: 00:33:32.

Substituting the last relation into the first gives

$$E = \frac{1}{2}m\left(\frac{p}{m}\right)^2 \tag{2.28}$$

$$= \frac{p^2}{2m}. \tag{2.29}$$

The mass is upstairs in $\frac{1}{2}mv^2$ and downstairs in $p^2/(2m)$. That placement is easy to confuse and useful to rederive.

Now use the general quantum relations

$$E = hf, \qquad p = \frac{h}{\lambda}. \tag{2.30}$$

For a single slow particle,

$$hf = \frac{p^2}{2m} \tag{2.31}$$

$$= \frac{1}{2m}\left(\frac{h}{\lambda}\right)^2 \tag{2.32}$$

$$= \frac{h^2}{2m\lambda^2}. \tag{2.33}$$

Dividing by h yields

$$f = \frac{h}{2m\lambda^2}. \tag{2.34}$$

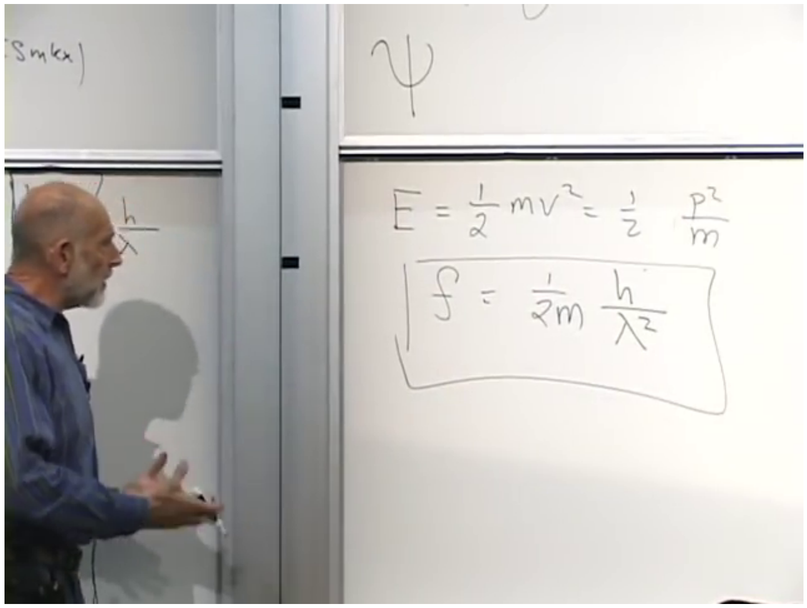

Figure 2.5: For a slow particle, $E = p^2/(2m)$ and $p = h/\lambda$ give $f = h/(2m\lambda^2)$.
Lecture frame: 00:37:47.

For light,

$$f \propto \frac{1}{\lambda}, \tag{2.35}$$

whereas for slow matter,

$$f \propto \frac{1}{\lambda^2}. \tag{2.36}$$

There is no need to memorize the second relation in isolation. It can always be reconstructed from $E = p^2/(2m)$, $E = hf$, and $p = h/\lambda$. For an electron moving slowly in an atom, it gives the frequency of the corresponding Schrödinger wave.

---

[a] **Question from the audience.** Does the nonrelativistic frequency–wavelength relation apply to particles other than electrons?

Response. Yes. It applies to any particle moving much more slowly than light. The relativistic interpolation between the slow and lightlike limits is a separate subject.

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[a] Lecture timestamp: 00:38:35.

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[a] **Question from the audience.** Which relation applies to a neutrino?

Response. It depends on the neutrino's speed. Most neutrinos encountered in nature move very close to the speed of light, so the lightlike relation is a good approximation. A neutrino moving much more slowly than light would instead obey the nonrelativistic relation.

---

[a] Lecture timestamp: 00:38:55.

[2]

[2] Editorial clarification: A brief audience exchange immediately after the neutrino question concerns amplitudes, quanta, and differing electron wavelengths, but its wording is too damaged to recover securely. The recoverable response postpones the general relation between amplitudes and quanta of different wavelengths.

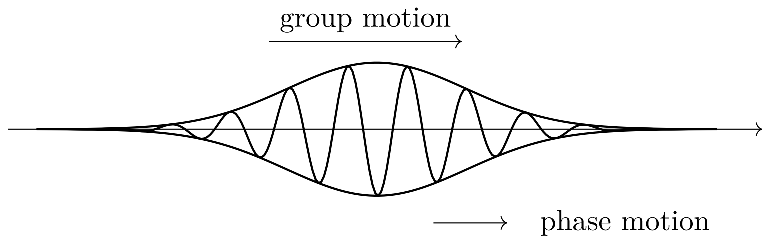

Figure 2.6: A localized wave packet whose envelope and internal phase generally move at different velocities.

[a] **Question from the audience.** Is an electron from a 20 keV accelerator slow or relativistic?

Response. It is treated as slow. In units with $c = 1$, mass, energy, and momentum can be expressed in the same units. The electron rest-energy scale is roughly 0.5 MeV, while 20 keV is much smaller. Equivalently, the nonrelativistic momentum criterion is $p \ll mc$.

[a] Lecture timestamp: 00:39:56.

The energy comparison in the example is

$$K = 20\,\mathrm{keV} \ll m_e c^2 \simeq 0.5\,\mathrm{MeV}. \tag{2.37}$$

[3]

2.5 Phase Velocity, Group Velocity, and Dispersion

[a] **Question from the audience.** Can frequency times wavelength be used to obtain the velocity of an electron wave?

Response. It gives the phase velocity, not the velocity assigned to the electron. A localized packet has both a phase velocity, describing the motion of its internal crests and troughs, and a group velocity, describing the motion of the envelope. The group velocity is identified with the particle's motion.

[a] Lecture timestamp: 00:41:46.

A wave packet looks locally like a sine or cosine but is confined by an envelope. As it moves, the envelope and the crests inside it need not move at the same speed.[4]

The phase velocity is the velocity of the peaks, troughs, and other fixed phases of a monochromatic component. For the nonrelativistic dispersion relation,

$$v_\phi = f\lambda = \frac{h}{2m\lambda}. \tag{2.38}$$

It depends on wavelength. Different wavelength components therefore have different phase velocities, so the crests can move through the packet and the packet can disperse. The group velocity is the velocity of the packet as a whole, and it is the one identified with the particle.

[3] Editorial clarification: The source passage at 00:39:56–00:41:40 introduces 20 keV as an accelerator energy and then phrases the relativistic criterion in terms of momentum compared with rest mass. Writing the kinetic-energy and momentum criteria separately avoids identifying the quoted energy with a momentum while preserving the stated nonrelativistic conclusion.

[4] The schematic follows the wave-packet discussion at 00:42:34–00:44:11.

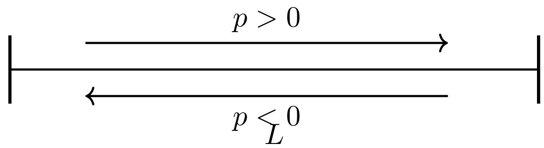

Figure 2.7: Reflection at the ends of a finite segment reverses the momentum of a traveling wave.

There is a simple special case. If all wavelength components move with the same velocity, phase and group velocities agree. Light in vacuum has this property: every wavelength propagates at c. The slow-particle relation does not; its $f\lambda$ changes with λ.

The equation

$$f = \frac{h}{2m\lambda^2} \tag{2.39}$$

is the free Schrödinger equation in a slightly disguised form. Frequency specifies the temporal dependence of a plane wave, wavelength specifies its spatial dependence, and the Schrödinger equation relates the two. Conversely, inserting a definite-frequency, definite-wavelength wave into the free Schrödinger equation reproduces this dispersion relation.

---

[a] **Question from the audience.** Can the phase velocity exceed the speed of light?

Response. Yes. Phase velocity does not carry information. Information and the localized packet propagate with the group velocity, so a superluminal phase velocity does not permit superluminal signaling.

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[a]Lecture timestamp: 00:47:02.

2.6 Finite Space and Discrete Momentum

The eventual interest is in waves in an infinite space, but a finite system is the safer starting point. Infinity is not something a computer can store or manipulate directly. One first formulates the equations for a finite system and only afterward takes the limit in which its size becomes arbitrarily large. This habit is useful throughout quantum mechanics, quantum field theory, and particle physics.

Nothing essential is lost by beginning in one spatial dimension. The wave might live on a violin string, on a rope being shaken, or in an abstract one-dimensional world. The most obvious finite system is a segment of length L with reflecting ends. A wave travels to an end, reflects, and travels back. This is a perfectly legitimate reflecting boundary condition, but it has an inconvenience: momentum is not conserved within the wave system. A right-moving wave carries rightward momentum; after reflection it carries leftward momentum. The boundary has supplied the impulse and absorbed the difference.

The momentum reversal at a reflecting end can be represented schematically.[5]

There is a different way to make the system finite while preserving momentum. Identify the two endpoints of the segment. It is possible to draw the result as a circle, but no literal circle embedded in a second dimension is required. The mathematical statement is simply that the coordinate is periodic: after traveling a total distance L, one returns to the same point.

[5]The schematic follows the reflecting-boundary argument at 00:49:24–00:51:08.

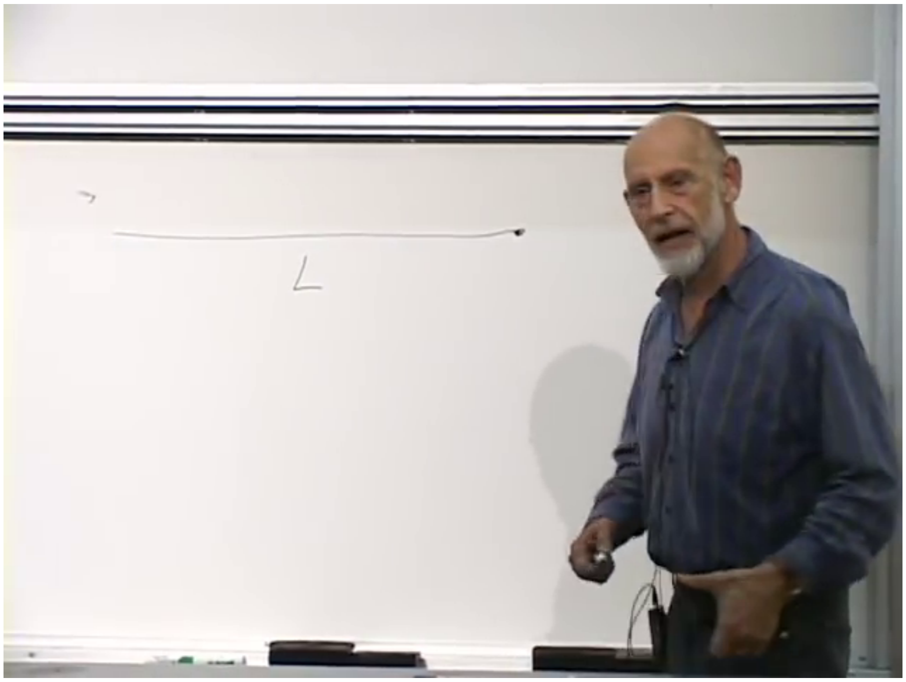

Figure 2.8: A one-dimensional path of total length L used to introduce periodic identification.
Lecture frame: 00:51:21.

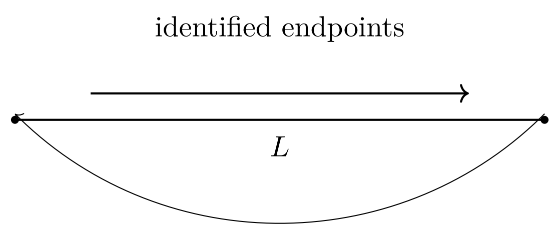

Figure 2.9: Periodic one-dimensional space formed by identifying the endpoints of a path of length L.

The endpoint identification is made explicit in the following schematic.[6]

A wave moving to the right then reaches the displayed endpoint, reappears at the beginning, and keeps moving to the right. A left-moving wave likewise keeps moving to the left. There is no distinguished endpoint at which the wave reflects, so the momentum does not reverse. Periodic space makes the volume finite and preserves momentum, but it has an important consequence: the allowed momenta are discrete.

An integer number of wavelengths must fit into the total length. For a nonconstant mode,

$$\lambda_{|N|} = \frac{L}{|N|}, \qquad |N| = 1, 2, 3, \ldots. \tag{2.40}$$

Thus the wavelength magnitude can be L, $L/2$, $L/3$, and so on. A choice such as $L/\sqrt{2}$ does not fit: after one circuit the wave would return with the wrong phase and fail to match its starting value.

[6]The schematic follows the periodic-identification argument at 00:51:09–00:52:37.

Using $p = h/\lambda$, the magnitude of the momentum for the $|N|$-th nonconstant mode is

$$|p_N| = \frac{|N|h}{L}. \tag{2.41}$$

To include both directions and the constant mode without ambiguity, it is convenient to let the mode label be signed:

$$p_N = \frac{Nh}{L}, \qquad N = 0, \pm 1, \pm 2, \dots. \tag{2.42}$$

Positive and negative N describe opposite propagation directions. The $N = 0$ mode is constant and has zero momentum; it has no finite wavelength to insert into h/λ. The basic momentum spacing is therefore h/L.

For a macroscopic L, both h and $1/L$ make this spacing extremely small, so the allowed values form a very dense set and look nearly continuous. A bead constrained to a circular wire nevertheless has exactly this discrete spectrum.

---

[a] **Question from the audience.** Is the quantized quantity the total momentum of the wave, or the momentum associated with one period?

Response. The statement applies to the momentum of each quantum. Every quantum has one of the allowed discrete momenta. The total momentum of any collection of quanta is consequently also an integer multiple of the basic unit h/L.

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[a]Lecture timestamp: 00:58:27.

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[a] **Question from the audience.** Why must the wave return to the same value after one circuit?

Response. The identified endpoints are the same physical point, so an observable field must have the same value there. A mismatch would require a sudden jump in the field at that point. Such a discontinuity is costly in energy and momentum; the admissible field configurations are continuous and periodic.

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[a]Lecture timestamp: 00:59:22.

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[a] **Question from the audience.** What happens if the wave is localized in a packet?

Response. A packet is a Fourier sum of plane waves, or equivalently of sines and cosines. In periodic space that sum may contain only the discrete allowed momentum modes. The quantization statement is therefore a statement about the Fourier components of every admissible packet.

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[a]Lecture timestamp: 01:00:09.

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[a] **Question from the audience.** Is there a problem if the periodic length L becomes extremely small?

Response. As L decreases, the basic momentum h/L becomes very large. No immediate inconsistency is implied at ordinary microscopic lengths, but attempting to push the construction to the Planck-length regime is expected to require new physics.

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[a]Lecture timestamp: 01:01:01.

Now take the circular picture literally: a bead moves on a circular wire of radius r, whose circumference is $L = 2\pi r$. Its allowed tangential momenta are

$$p_N = \frac{Nh}{2\pi r} = \frac{N\hbar}{r}. \tag{2.43}$$

For a nonrelativistic particle in a circular orbit the angular momentum is mvr. Since $p = mv$ in that regime, this is rp. The expression mvr is not relativistically general, because p is then not simply mv, but momentum times perpendicular distance remains the right relation. In vector form,[7]

$$\vec{J} = \vec{r} \times \vec{p}, \tag{2.44}$$

and for tangential circular motion its magnitude is $J = rp$. Hence

$$J_N = rp_N = N\hbar. \tag{2.45}$$

The spacing of linear momentum depends on the size of the circle, whereas the angular-momentum unit is always $\hbar$. There are exceptions to the integer rule when fermions are involved; those exceptions are postponed.

The next task is to connect fields with particles and quanta more precisely. That requires a short return to the quantum mechanics of operators and the harmonic oscillator.

2.7 Wave Modes and Quantum Operators

The quantum mechanics that follows is necessarily abstract and is meant as a review. The harmonic oscillator is the essential system because a wave is an oscillation. At a fixed point, an electric field, a magnetic field, or any other wave amplitude varies periodically in time; to say that a component has a frequency is already to say that it oscillates.

A general wave need not have only one wavelength. Light of several colors can be mixed together, and, conversely, a prism can separate mixed light into its component colors, as in Newton's decomposition of white light. Each component has its own wavelength, frequency, amplitude, energy, and momentum. Thus the mixed wave may be resolved into quanta of different energies and momenta. Each wavelength determines a frequency, and each definite-frequency component behaves as an oscillator. This is why the harmonic oscillator, rather than an arbitrary quantum system, is the natural building block for a field.

Quantum observables are represented by operators. Operators can be added and multiplied, but unlike ordinary numbers their products need not be independent of order. Let A and B represent two properties of a system, each of which might be measurable by itself. For example, A might represent an electric field and B a magnetic field. If the measured values are the ordinary numbers a and b, then

$$ab = ba. \tag{2.46}$$

Their numerical product cannot depend on the order of multiplication. In quantum mechanics, however, the corresponding operators may instead satisfy $AB - BA \neq 0$.

At first this seems nonsensical. If measuring the electric field gives one number and measuring the magnetic field gives another, multiplying those two numbers must give the same result in either order. The resolution is that when the operator products depend on order, the two quantities cannot be measured simultaneously. The experiment that measures one interferes with the measurement of the other. There is no contradiction because the two numerical values cannot be obtained together.

[7] The lecture first uses L for angular momentum, then notices its collision with L for the periodic length and changes symbols. Here J is used throughout; see 01:03:40–01:05:21.

When two quantities can be measured simultaneously, their measured values are ordinary numbers, and those numbers commute. The test for simultaneous measurability is therefore the commutator

$$[A, B] = AB - BA. \tag{2.47}$$

A vanishing commutator means that the observables are compatible; a nonzero commutator means that the corresponding measurements interfere. Reversing the entries reverses the sign:

$$[B, A] = -[A, B]. \tag{2.48}$$

For a nonrelativistic particle, the three position components commute with one another, and so do the three momentum components:

$$[x, y] = [y, z] = [z, x] = 0, \tag{2.49}$$

$$[p_x, p_y] = [p_y, p_z] = [p_z, p_x] = 0. \tag{2.50}$$

All three coordinates can be specified together, and all three momentum components can be specified together. Corresponding position and momentum components do not commute:

$$[x, p_x] = [y, p_y] = [z, p_z] = i\hbar, \tag{2.51}$$

whereas mixed components such as $[x, p_y]$ vanish. Measuring a position coordinate gives the particle a disturbance in the conjugate momentum. Equivalently, the product of the corresponding position and momentum uncertainties is controlled by $\hbar$.

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[a] **Question from the audience.** When changing from x to y, should the i in the commutator change to j?

Response. No. The symbol i is always the imaginary unit, the square root of minus one. It is neither a Cartesian unit-vector label nor one of the quaternionic symbols i, j, k.

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[a]Lecture timestamp: 01:16:30.

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[a] **Question from the audience.** Does $AB - BA$ mean measuring A first and then B, and then reversing the measurement order?

Response. No. The two terms are operator or matrix products in different algebraic orders. They do not denote a literal sequence of measurements. The connection with incompatible measurements is a consequence of the operator algebra, not the definition of the product.

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[a]Lecture timestamp: 01:17:24.

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[a] **Question from the audience.** Do these position–momentum commutators apply directly to a photon?

Response. Not directly. They apply cleanly to nonrelativistic particles. A photon's position is harder to define, so the analogous statement requires more care and is not needed here.

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[a]Lecture timestamp: 01:18:32.

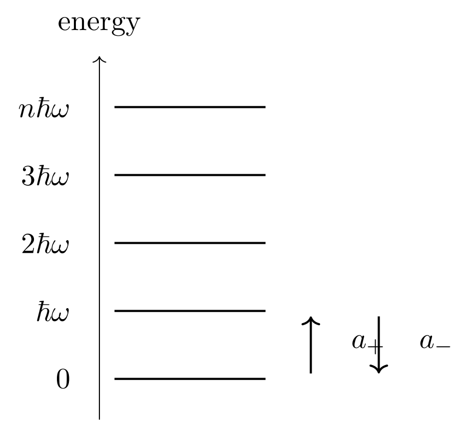

Figure 2.10: Equally spaced oscillator levels in the convention $E_n = n\hbar\omega$.

2.8 The Quantum Harmonic Oscillator

A weight on a spring, a sound wave, and a light wave all provide examples of oscillatory motion. Every ideal harmonic oscillator has a frequency. For a spring, one full period is not merely the time required for the weight to return to the same position; it must return to that position moving in the same direction. The number of complete periods per second is f, and $\omega = 2\pi f$.

Classically, an ideal oscillator can have any nonnegative energy. A gentle push gives a small energy, and a harder push gives a larger one. A real spring would eventually break or melt, but the mathematical idealization has a continuous range of positive energies. The quantum oscillator is different: its allowed energies form a discrete, equally spaced ladder. Choosing the origin of energy so that the lowest level is called zero,

$$E_n = n\hbar\omega, \qquad n = 0, 1, 2, \ldots. \tag{2.52}$$

The equally spaced levels can be pictured as follows.[8]

The spacing $\hbar\omega$ is the same energy assigned earlier to one photon of angular frequency ω. The value $2\hbar\omega$ might be the energy of one quantum belonging to an oscillator of frequency 2ω, or the energy of two quanta in an oscillator of frequency ω. The second interpretation is the one needed here. A single oscillator of fixed frequency may contain zero, one, two, three, or more quanta, with no allowed energies between the rungs.

The integer n labels the oscillator state, written as a ket $|n\rangle$. The state $|0\rangle$ is the ground state, $|1\rangle$ has one quantum of excitation, $|2\rangle$ has two, and so on. The ket is simply the notation for the state on which operators act.

Define a raising operator a_+ that moves a state up one level and a lowering operator a_- that moves it down one level:

$$a_+|n\rangle = \sqrt{n+1}\,|n+1\rangle, \tag{2.53}$$
$$a_-|n\rangle = \sqrt{n}\,|n-1\rangle. \tag{2.54}$$

The numerical square-root factors are part of the definitions. On the lowest state,

$$a_-|0\rangle = 0, \tag{2.55}$$

[8]The energy-level schematic follows the board discussion at 01:21:20–01:23:56.

because there is no state below the ground state.

It is important here to distinguish ordinary numbers from operators. The label n and coefficients such as $\sqrt{n}$ are ordinary numbers; they commute with everything. The symbols a_+ and a_- are operators, and their order must be preserved.

First lower $|n\rangle$, then raise it again. The lowering step gives

$$a_-|n\rangle = \sqrt{n}\,|n-1\rangle. \tag{2.56}$$

Acting with a_+ on the resulting state supplies another factor $\sqrt{n}$, because raising $|n-1\rangle$ returns it to $|n\rangle$:

$$\begin{aligned} a_+a_-|n\rangle &= a_+\big(\sqrt{n}\,|n-1\rangle\big) && (2.57)\\ &= \sqrt{n}\,\sqrt{n}\,|n\rangle && (2.58)\\ &= n|n\rangle. && (2.59) \end{aligned}$$

Thus a_+a_- reads the level number. For example, suppose $|\chi\rangle$ is known to be a number eigenstate but its level label is unknown. If

$$a_+a_-|\chi\rangle = q|\chi\rangle, \tag{2.60}$$

then the ordinary numerical coefficient q is the number of quanta in that state. The product is therefore the number operator,

$$\hat{n} = a_+a_-, \tag{2.61}$$

and, in the chosen energy convention, the Hamiltonian is

$$H = \hbar\omega\, a_+a_-. \tag{2.62}$$

Indeed, acting on any level gives $H|n\rangle = n\hbar\omega|n\rangle$.

Now reverse the order. Raising first gives $\sqrt{n+1}|n+1\rangle$. Lowering then returns to $|n\rangle$ and supplies a second factor $\sqrt{n+1}$:

$$\begin{aligned} a_-a_+|n\rangle &= a_-\big(\sqrt{n+1}\,|n+1\rangle\big) && (2.63)\\ &= \sqrt{n+1}\,\sqrt{n+1}\,|n\rangle && (2.64)\\ &= (n+1)|n\rangle. && (2.65) \end{aligned}$$

The two orders differ by one on every state, so

$$[a_-, a_+] = a_-a_+ - a_+a_- = 1. \tag{2.66}$$

The oscillator also has an amplitude and a phase. Its phase specifies where it is in its cycle; concretely, it may be fixed by the time at which the oscillator passes through the origin in a chosen direction. Delaying that crossing changes the phase without changing the frequency. The ladder operators combine the quantum variables associated with amplitude and phase. Their algebra is intrinsically quantum, but their action is simple: a_+ adds one quantum of oscillator energy and a_- removes one. For an electromagnetic mode, those quanta are photons, so the two operators are called creation and annihilation operators.

[a] **Question from the audience.** Do the raising and lowering operators correspond to a physical measurement?

Response. Photon number is measurable. The operators a_+ and a_- are not separately measurable observables, for additional technical reasons. Their commutator shows that the two do not commute, but that fact by itself is not the argument that either one individually is unobservable.

[a] Lecture timestamp: 01:37:34.

A collection of springs with different natural frequencies gives many independent oscillators. So do the electromagnetic modes of a cavity: one wavelength may be excited without requiring the others to have the same excitation. Quantum mechanically, creation and annihilation operators add and remove quanta of a fixed frequency in each independent oscillator. This language is central because particles will be identified with precisely those quanta.

2.9 A Field as a Collection of Oscillators

Return to periodic one-dimensional space of total length L. Its nonconstant modes have wavelength magnitudes

$$\lambda_N = \frac{L}{N}, \qquad N = 1, 2, 3, \ldots, \tag{2.67}$$

and corresponding nonnegative angular frequencies ω_N. For comparison with the earlier momentum spectrum, it is useful to relabel its signed integer by M:

$$p_M = \frac{Mh}{L}, \qquad M = 0, \pm 1, \pm 2, \ldots. \tag{2.68}$$

The sign of M records the direction of propagation. [9]

Each allowed frequency behaves as a harmonic oscillator. Energy may be added independently to one frequency and then to another. The detailed first example of a quantum field is deferred to the next lecture, but the state notation is already clear. Ordering the nonconstant modes from the longest wavelength upward, write

$$|n_1, n_2, n_3, \ldots\rangle, \tag{2.69}$$

where $n_N = 0, 1, 2, \ldots$ is the number of quanta in the oscillator of frequency ω_N. The first entry tells how many quanta occupy the longest-wavelength mode, the second tells how many occupy the next mode, and so on without end.

The harmonics of a string make this construction concrete. The progression from the fundamental through the next two harmonics is shown schematically.[10]

The fundamental moves as one broad arch and has no interior node. It is one oscillator, and it may contain $n_1 = 0, 1, 2, \ldots$ quanta; each excitation costs $\hbar\omega_1$. The second harmonic has one interior node and twice the frequency. If the fundamental sounds middle C, this mode sounds the C one octave higher. It has an independent occupation number n_2, and each of its quanta costs $\hbar\omega_2 = 2\hbar\omega_1$.

[9] Editorial clarification: At 01:42:08–01:42:24 the frequency-mode index is corrected from a signed range to one running from zero upward. The symbol M is used here only to keep signed momentum notation separate from the nonnegative frequency-mode label N; it does not introduce independent occupations for $+M$ and $-M$. The nonconstant oscillator list begins at $N = 1$.

[10] The schematic follows the string-harmonic discussion at 01:45:20–01:47:46.

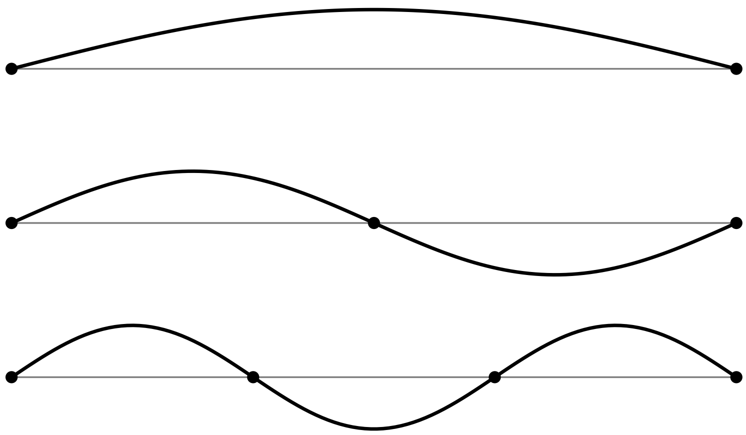

Figure 2.11: The string fundamental and its second and third harmonics, with zero, one, and two interior nodes.

The third harmonic has two interior nodes and three times the fundamental frequency. It has its own independent occupation number n_3, which may again be any nonnegative integer.

Mode by mode, in the energy convention used above, the energy is the occupation number times the quantum energy. Collecting those contributions gives

$$E = \sum_{N=1}^{\infty} n_N \hbar\omega_N. \tag{2.70}$$

[11]

The same harmonic-oscillator mathematics must be repeated for every field mode, with one creation operator and one annihilation operator for each independent oscillator. In this first elementary picture, a quantum field is that entire collection of oscillators and their creation–annihilation algebra.

This language is adapted to particle scattering. Quanta with definite energy and momentum enter an experiment, and quanta with other energies and momenta leave it. The initial particles are removed by annihilation operators; outgoing particles are inserted into the final state by creation operators for the appropriate modes.

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[a] **Question from the audience.** Is the same integer being used to label the oscillator mode and to count the number of quanta in it?

Response. There are two distinct integers. The mode label N identifies the spatial pattern; in the string example, its number of nodes determines the frequency ω_N. The occupation number n_N tells how many quanta have that frequency.

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[a]Lecture timestamp: 01:49:38.

[11]Editorial clarification: This displayed sum is the algebraic collection of the mode energies stated separately at 01:44:02–01:47:46.

Chapter 3

From Momentum Modes to Local Quantum Fields

What is a quantum field, and how is it related to particles? The basic mathematical tool is the harmonic oscillator. A field contains many possible modes of oscillation, and quantizing the field amounts to putting together one quantum harmonic oscillator for each mode. The construction is easiest to see first in a finite periodic space, where the allowed modes form a discrete list.

3.1 Periodic Space and Allowed Wave Numbers

Take one spatial coordinate x and identify points separated by a distance L. One can picture the space as a circle of circumference L. This is a useful finite-volume trick. On a finite interval with endpoints, a traveling wave reflects from the ends. On a periodic interval it can continue in either direction without reflection. Translation symmetry, and therefore momentum conservation, is retained while the complications of infinite volume are temporarily removed.

A plane wave has the form

$$e^{ikx}. \tag{3.1}$$

After one trip around the circle it must return to its original value. Since the wave equals 1 at $x = 0$, periodicity requires

$$e^{ikL} = 1. \tag{3.2}$$

Why does an integer multiple of 2π solve this condition? Euler's formula gives

$$\begin{aligned} e^{i2\pi n} &= \cos(2\pi n) + i\sin(2\pi n) && (3.3)\\ &= 1, \qquad n \in \mathbb{Z}. && (3.4) \end{aligned}$$

The cosine is 1 and the sine is 0 after any whole number of cycles. Consequently,

$$kL = 2\pi n, \qquad k = \frac{2\pi n}{L}, \qquad n \in \mathbb{Z}. \tag{3.5}$$

The quantity k is the wave number. The integer n labels an allowed wave number; it is not yet counting particles.

The same condition can be expressed in terms of wavelength. A wavelength is the distance through which the phase advances by one full cycle, so

$$k\lambda = 2\pi, \qquad k = \frac{2\pi}{\lambda}. \tag{3.6}$$

A long wavelength has a small wave number, and a short wavelength has a large wave number. The periodicity condition says that an integer number of wavelengths fits around the circle.

For a quantum-mechanical wave, the momentum is

$$p = \frac{h}{\lambda} = \frac{2\pi\hbar}{\lambda} = \hbar k. \tag{3.7}$$

Thus k and p differ only by the factor $\hbar$. Positive and negative k describe waves moving in opposite directions and particles with oppositely directed momentum. In units with $\hbar = 1$, it is common to call k the momentum without further distinction.

For every allowed k, there may be some number of quanta in that mode. This introduces a second integer:

$$n(k) = \text{number of quanta with wave number } k. \tag{3.8}$$

It is called the occupation number. The integer in $k = 2\pi n/L$ labels the mode; $n(k)$ tells how many particles occupy that mode. Occupation numbers must be given for negative as well as positive k.

---

[a] **Question from the audience.** Is k itself an integer?

Response. No. The wave number is an integer multiple of $2\pi/L$: $k = 2\pi n/L$. The integer n in that formula labels the allowed momentum, and it is distinct from the occupation number $n(k)$, which counts quanta in the mode.

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[a]Lecture timestamp: 00:11:58.

3.2 Occupation-Number States and Mode Operators

A quantum state is a vector, or linear element. State vectors can be added and multiplied by numbers, and operators act on them to produce other state vectors. Observable real quantities are represented by Hermitian operators. These elementary facts matter here because both the many-particle state and the field will be built from linear combinations and operators.

Let

$$k_j = \frac{2\pi j}{L}, \qquad j \in \mathbb{Z}. \tag{3.9}$$

A convenient label for a many-particle state is the entire string of occupation numbers. Using the shorthand $n_j \equiv n(k_j)$ for the displayed entries, write

$$\big|\{n(k_j)\}\big\rangle = \big|\ldots, n_{-2}, n_{-1}, n_0, n_1, n_2, \ldots\big\rangle. \tag{3.10}$$

The entry $n(k_1)$ is the number of particles in the first positive-momentum mode, $n(k_2)$ is the number in the second, and so on. The string extends without end in both directions.

Each momentum mode is a harmonic oscillator. It therefore has a raising operator and a lowering operator. The notation written initially as $a^+(k)$ and $a^-(k)$ will be written as $a^\dagger(k)$ and $a(k)$. Their

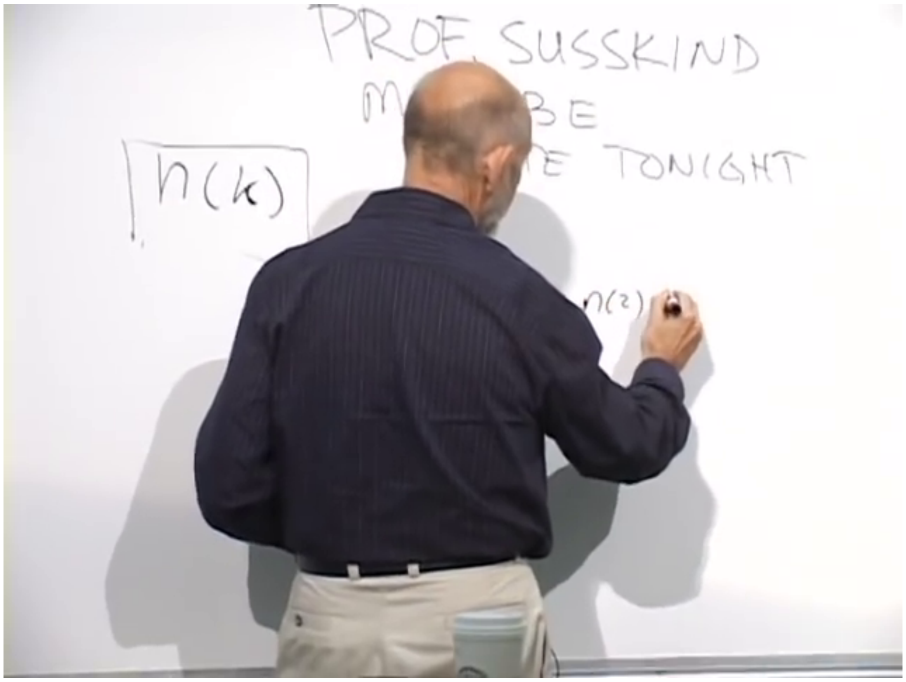

Figure 3.1: Occupation number $n(k)$ for an allowed momentum mode.
Lecture frame: 00:15:04.

actions are

$$a^\dagger(k)|\ldots,n(k),\ldots\rangle = \sqrt{n(k)+1} \\ \times |\ldots,n(k)+1,\ldots\rangle, \tag{3.11}$$

$$a(k)|\ldots,n(k),\ldots\rangle = \sqrt{n(k)}\,|\ldots,n(k)-1,\ldots\rangle. \tag{3.12}$$

The operator $a^\dagger(k)$ goes to the slot belonging to k, raises that occupation by one, and multiplies the state by $\sqrt{n(k)+1}$. The operator $a(k)$ lowers the same slot and supplies $\sqrt{n(k)}$. There is one such pair for every allowed momentum.

3.3 Fourier Coefficients Become Operators

The connection between these oscillator operators and a field begins with an ordinary classical Fourier series. A complex classical field on the periodic interval can be written

$$\psi(x) = \sum_k \alpha(k)e^{ikx}, \tag{3.13}$$

$$\psi^*(x) = \sum_k \alpha^*(k)e^{-ikx}, \tag{3.14}$$

where every sum is over the allowed values $k = 2\pi n/L$. This is equivalent to expanding the field in sines and cosines.

The complex field has a real and an imaginary part. Its complex conjugate reverses the sign of the imaginary part, and e^{ikx} becomes e^{-ikx}. Specifying all the coefficients $\alpha(k)$ specifies the classical

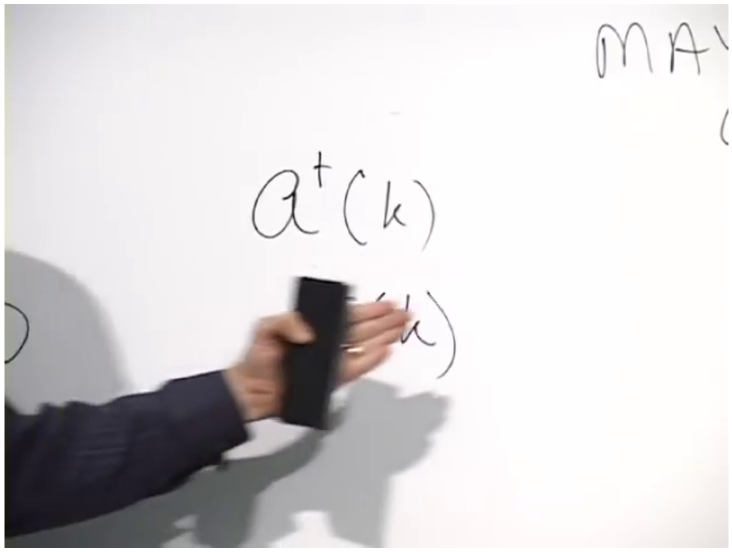

Figure 3.2: Creation and annihilation operators associated with momentum mode k.
Lecture frame: 00:17:14.

field. There is no need to specify $\alpha^*(k)$ independently once $\alpha(k)$ is known, but the real and imaginary parts of each coefficient are separate real quantities. They can be measured as the amplitudes of the corresponding Fourier components. In this sense a complex coefficient packages two classical observables.

Quantization replaces these classical Fourier coefficients by operators. An ordinary equality sign would identify objects of different kinds, so it is better to write a replacement rule:

$$\alpha(k) \rightsquigarrow a(k), \qquad \alpha^*(k) \rightsquigarrow a^\dagger(k). \tag{3.15}$$

With a consistent choice of Fourier signs, the quantum fields are defined by

$$\Psi(x) = \sum_k a(k) e^{ikx}, \tag{3.16}$$

$$\Psi^\dagger(x) = \sum_k a^\dagger(k) e^{-ikx}. \tag{3.17}$$

[1]

For the moment, this is a definition: a quantum field is a Fourier sum whose coefficients are operators. A fuller justification would have to show how these operators reproduce the classical coefficients in an appropriate limit, which requires some dynamics. That question is postponed. What is already clear is that the field itself is an operator, and the relations among its Fourier coefficients are the creation-and-annihilation relations of harmonic oscillators.

[1] Editorial clarification: The spoken discussion changes and later questions the Fourier-sign convention. The conjugate pair used here is the convention visible in the clearer field expansion and in the position-state calculation: Ψ carries e^{ikx} and $\Psi^\dagger$ carries e^{-ikx}. Reversing every sign consistently would describe the same construction.

[a] **Question from the audience.** Does introducing creation and annihilation operators mean that the dynamics necessarily creates and destroys particles?

Response. No. The operators are mathematical operations that change occupation numbers. Whether the laws of nature actually change particle number is a question about the dynamics. If the dynamics permits such changes, these operators provide the natural language for describing them.

[a]Lecture timestamp: 00:26:48.

3.4 Scattering and Variable Particle Number

First consider a process that does not change the number of particles. Suppose the only particle initially present has momentum k_7. In occupation-number notation its state has a 1 in the seventh slot and zeros everywhere else:

$$|k_7\rangle = |\ldots, 0, \underset{k_7}{1}, 0, \ldots\rangle. \tag{3.18}$$

A scattering process changes its momentum from k_7 to k_9. The operator that makes exactly this change is

$$a^\dagger(k_9)a(k_7). \tag{3.19}$$

The rightmost operator acts first: it removes the particle in the k_7 slot. The left operator then creates a particle in the k_9 slot.

[a] **Question from the audience.** Where do the square-root factors enter in this $k_7 \to k_9$ process?

Response. The annihilation factor is $\sqrt{n(k_7)} = \sqrt{1} = 1$. After that removal the system is in the all-zero state. Since the k_9 mode is empty, the creation factor is $\sqrt{n(k_9)+1} = \sqrt{1} = 1$. Thus neither factor changes the numerical coefficient in this one-particle example.

[a]Lecture timestamp: 00:31:37.

Let us carry out both steps. Annihilation gives

$$a(k_7)|k_7\rangle = \sqrt{1}\,|\ldots, 0, 0, 0, \ldots\rangle \tag{3.20}$$
$$= |0\rangle. \tag{3.21}$$

The all-zero state is the vacuum. It is the state with no particles, and in this simple collection of oscillators it is also the ground state. Creation in the empty k_9 mode then gives

$$a^\dagger(k_9)|0\rangle = \sqrt{0+1}\,|k_9\rangle = |k_9\rangle. \tag{3.22}$$

Consequently,

$$a^\dagger(k_9)a(k_7)|k_7\rangle = |k_9\rangle. \tag{3.23}$$

Slot by slot, the sole 1 has simply moved from the seventh momentum mode to the ninth. Because no occupation ever exceeded one, both square-root factors were equal to one. They will become important as soon as an already occupied mode is involved.

This one-to-one process could also be described in ordinary single-particle quantum mechanics: one particle changes its momentum. The occupation-number language becomes indispensable when particle number can change. For example,

$$a^\dagger(k_9)a^\dagger(k_{16})a(k_7)|k_7\rangle = |k_9, k_{16}\rangle \tag{3.24}$$

describes the mathematical change from one incoming particle in mode k_7 to two outgoing particles, one in k_9 and one in k_{16}. A target may absorb the incoming particle, emit the two outgoing particles, and take up whatever recoil is required. Such a process cannot be represented in a Hilbert space restricted to one particle, but it fits naturally in a state space whose occupation numbers can vary. The operator expression is not by itself a law of nature; it is the language in which a number-changing law can be written.

3.5 A Field Operator at a Point

What does the position-space operator $\Psi^\dagger(x)$ create? Quantum states form a linear space: momentum states may be added with complex coefficients to form a new state. Begin with the vacuum. Since every mode is empty,

$$\Psi(x)|0\rangle = 0. \tag{3.25}$$

The zero on the right is the zero vector: an annihilation operator cannot find any particle to remove.

The conjugate field acts differently:

$$\Psi^\dagger(x)|0\rangle = \sum_k e^{-ikx} a^\dagger(k)|0\rangle \tag{3.26}$$

$$= \sum_k e^{-ikx}|k\rangle, \tag{3.27}$$

where

$$|k\rangle = a^\dagger(k)|0\rangle \tag{3.28}$$

is the one-particle state of momentum $\hbar k$.

Formally, and up to normalization, the sum is the generalized position state of one-particle quantum mechanics.[2] Momentum eigenstates, combined with coefficients e^{-ikx}, produce localization at x. In this formal sense,

$$\Psi^\dagger(x) \text{ creates a particle at position } x, \tag{3.29}$$

while

$$\Psi(x) \text{ annihilates a particle at position } x. \tag{3.30}$$

The mode operator $a^\dagger(k)$ creates a particle with definite momentum. The field operator $\Psi^\dagger(x)$ creates the coherent superposition of allowed momentum states that is formally localized at x.

In the formal, unnormalized sum, every coefficient e^{-ikx} has magnitude one. With a finite cutoff, the normalized truncated state has equal-magnitude coefficients for the retained momenta, while their phases produce a wave packet sharply localized near x. This is the elementary connection between a quantum field and particles.

Ignoring special-relativistic complications for now, $\Psi^\dagger(x)$ creates a particle at x in this model. Relativistic fields require refinements, but the operator-valued Fourier expansion and the distinction between momentum-mode and position-space operators remain basic.

[2] With infinitely many momentum modes, an exactly localized position state is non-normalizable. Imposing a finite momentum cutoff and normalizing the truncated sum instead gives a sharply localized wave packet centered at x.

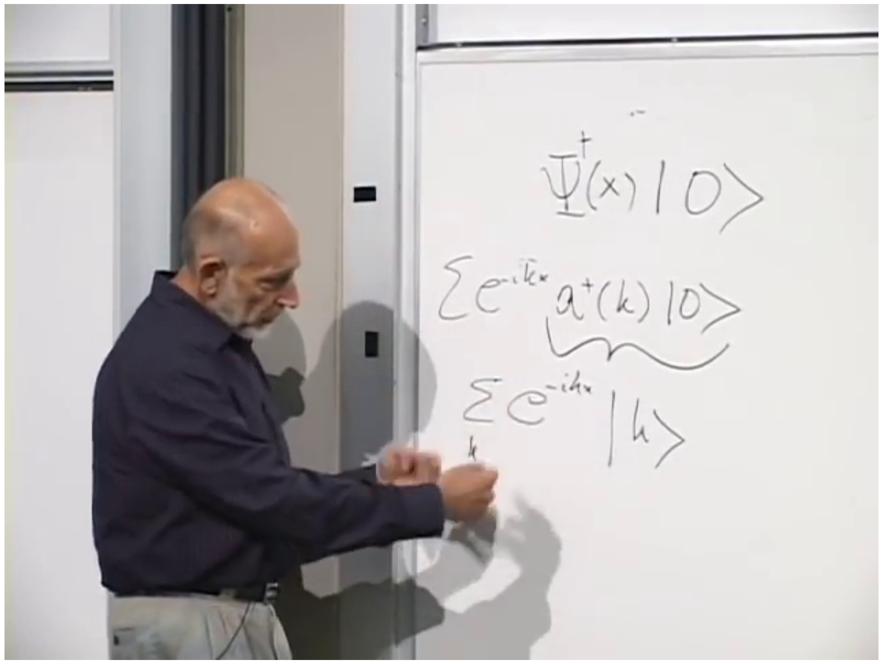

Figure 3.3: The formal position-space state $\Psi^\dagger(x)|0\rangle$ expanded in one-particle momentum states.
Lecture frame: 00:43:16.

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[a] **Question from the audience.** Is x a spot on the periodic circle, and what happens as the circle becomes large?

Response. Yes. In the finite model, x is a point on the circle. Neighboring allowed wave numbers are separated by $2\pi/L$. As L grows, that spacing becomes smaller, so for every desired momentum there is an allowed value arbitrarily close to it. In the infinite-volume limit, sums over k are replaced by integrals over k. It is useful to understand the discrete sums first.

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[a] Lecture timestamp: 00:44:59.

3.6 Local Particle Reactions

Once $\Psi(x)$ and $\Psi^\dagger(x)$ have been interpreted as annihilating and creating particles at a point, locality can be expressed directly. Imagine that a particle is absorbed by a target sharply localized at x, while one outgoing particle suddenly appears near Alpha Centauri and another near Betelgeuse. Such a reaction is nonlocal: its separated events cannot be connected by ordinary propagation.

One could first weaken the requirement and allow the outgoing particles to be emitted somewhere within a small region around the target. The precise local idealization is stronger: if the incoming particle is absorbed at x, the outgoing particles are created at that same point. Using fields evaluated at separated points would instead describe a nonlocal operator product. A local one-to-two operation is therefore

$$\Psi^\dagger(x)\Psi^\dagger(x)\Psi(x). \tag{3.31}$$

The rightmost field removes the incoming particle at x; the two fields to its left create two outgoing particles there.[3]

[3] The local one-to-two schematic follows the reaction described at 00:47:38–00:49:13.

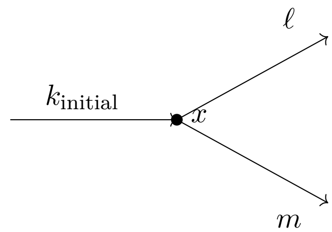

Figure 3.4: One incoming particle is absorbed at x, and two outgoing particles are created at the same point.

Let the incoming state contain one particle of definite wave number k_{initial}. To determine the outgoing momenta, substitute the momentum-space expansions of the local fields. It is important to separate a fixed momentum label from a dummy summation index. Write

$$\Psi(x) = \sum_q a(q)e^{iqx}, \tag{3.32}$$

$$\Psi^\dagger(x) = \sum_\ell a^\dagger(\ell)e^{-i\ell x}. \tag{3.33}$$

The index q is summed, whereas k_{initial} labels the particular occupied incoming mode. The two copies of $\Psi^\dagger(x)$ also require independent dummy indices, say ℓ and m. Reusing one index would incorrectly tie two independent sums together. Here ℓ and m are wave numbers; m does not denote mass.

A concrete picture is a photon striking an atom localized at x, followed by two photons leaving that point. The simple field used here is better suited to nonrelativistic particles than to a fully relativistic photon field, so the photon language is only illustrating the operator bookkeeping.

The annihilation field acts first:

$$\Psi(x)|k_{\text{initial}}\rangle = \sum_q e^{iqx}a(q)a^\dagger(k_{\text{initial}})|0\rangle. \tag{3.34}$$

Every term vanishes except the one with $q = k_{\text{initial}}$. An annihilation operator has to find a particle in its own mode. Hence

$$\Psi(x)|k_{\text{initial}}\rangle = e^{ik_{\text{initial}}x}|0\rangle. \tag{3.35}$$

The two creation fields then give

$$\Psi^\dagger(x)\Psi^\dagger(x)\Psi(x)|k_{\text{initial}}\rangle \tag{3.36}$$

$$= e^{ik_{\text{initial}}x}\sum_{\ell,m} e^{-i\ell x}e^{-imx}a^\dagger(\ell)a^\dagger(m)|0\rangle \tag{3.37}$$

$$= \sum_{\ell,m} e^{i(k_{\text{initial}}-\ell-m)x}a^\dagger(\ell)a^\dagger(m)|0\rangle. \tag{3.38}$$

For unequal ℓ and m, the operator state may be denoted $|\ell, m\rangle$: one particle occupies each of the two modes. The final result is a sum over all such two-particle momentum configurations, with a phase determined by the incoming and outgoing momenta.

[a] **Question from the audience.** Do the two creation operators commute?

Response. Yes. All creation operators commute with one another, and all annihilation operators commute with one another. A creation and an annihilation operator commute when they refer to different modes; for the same mode their commutator is nonzero.

[a]Lecture timestamp: 01:00:36.

[a] **Question from the audience.** Should the two outgoing momenta satisfy momentum conservation?

Response. The displayed particles need not conserve momentum by themselves because the atom or target can recoil. The difference $k_{\text{initial}} - \ell - m$ is the momentum transferred to the target. Total momentum, including the target, is the conserved quantity.

[a]Lecture timestamp: 01:01:34.

Since the coefficient $e^{i(k_{\text{initial}}-\ell-m)x}$ is a pure phase, its magnitude is one. At first sight this suggests equal probability weights for all pairs (ℓ, m). There is, however, an important qualification when the two labels coincide.

If $\ell = m$, the first creation operator puts one particle in mode m, and the second acts on a mode already occupied once:

$$a^\dagger(m)a^\dagger(m)|0\rangle = a^\dagger(m)|1_m\rangle \tag{3.39}$$

$$= \sqrt{2}\,|2_m\rangle. \tag{3.40}$$

Thus double creation in one mode has an extra amplitude $\sqrt{2}$. Squaring an amplitude produces a factor of 2 in probability.

[a] **Question from the audience.** Must ℓ and m differ by one because the creation coefficient contains $n+1$?

Response. No. The symbols ℓ and m label momentum modes. The n in $\sqrt{n+1}$ is the occupation number of the particular mode on which the creation operator acts. Momentum labels and particle counts are distinct integers.

[a]Lecture timestamp: 01:04:54.

The particles in this construction are identical. The $\sqrt{2}$ is therefore a bosonic occupation factor, not a label distinguishing a first particle from a second. Squaring it seems, provisionally, to make the same-momentum outcome twice as probable as a single unequal-momentum term. That comparison will need another look once the terms in the double sum are counted carefully.

[a] **Question from the audience.** In a complicated reaction, how is the order of creation and annihilation operators chosen?

Response. An expression with all creation operators to the left and annihilation operators to the right is called normal ordered. In that form the rightmost operators first remove incoming particles and the remaining operators create outgoing particles.

[a]Lecture timestamp: 01:06:43.

Reordering a mixed creation–annihilation pair is not generally a simple permutation: it must use the commutator, and for the same mode this produces an additional term.[4]

[4]This commutator qualification is an editorial correction to the informal normal-ordering discussion at 01:06:43.

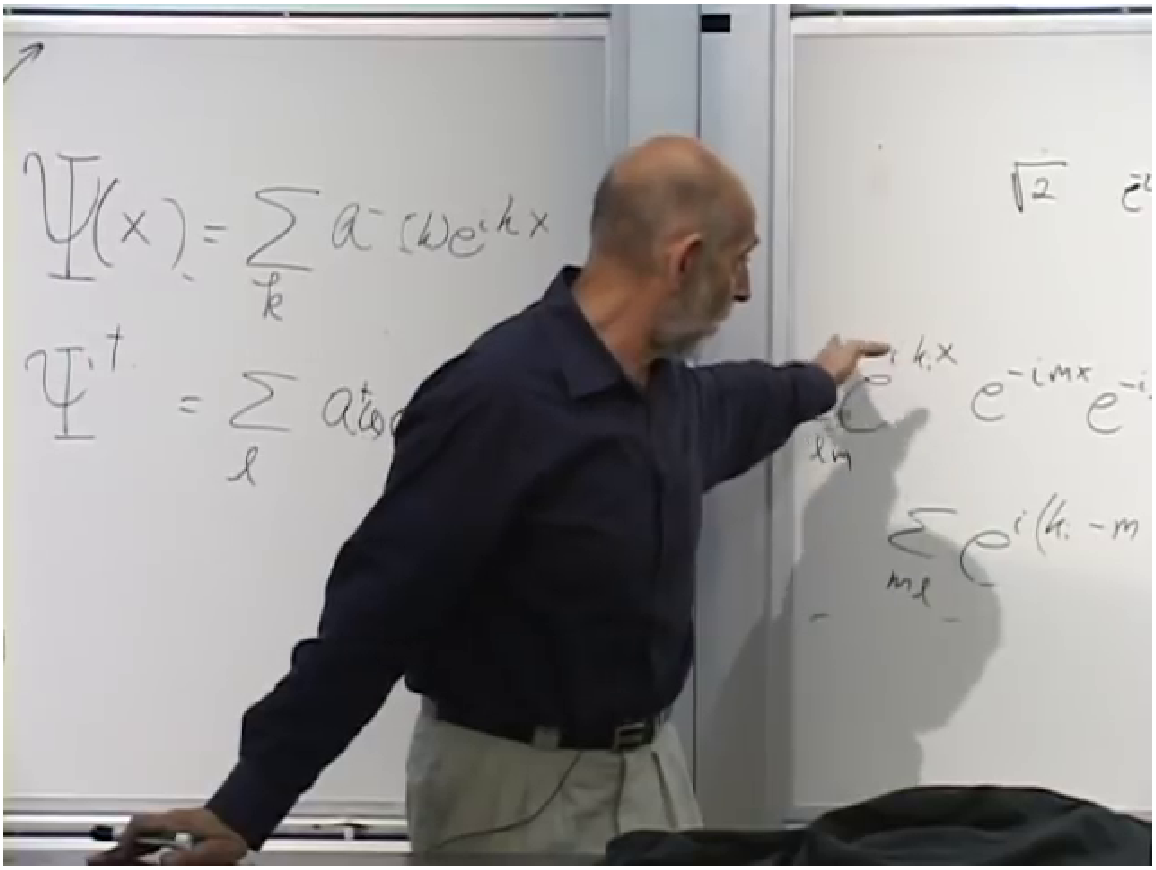

Figure 3.5: Momentum-space expansion of the local one-to-two particle amplitude.

Lecture frame: 01:05:44.

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[a] **Question from the audience.** If the momentum amplitudes all have the same magnitude, what does a statement such as “twice as probable” mean?

Response. Keep the finite cavity in mind. Its allowed momenta are discrete, so probabilities can be compared mode by mode. The comparison is not between probabilities of isolated points in a continuum.

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[a]Lecture timestamp: 01:07:30.

For $\ell \neq m$, the unrestricted double sum contains both ordered terms

$$a^\dagger(\ell)a^\dagger(m)|0\rangle \quad \text{and} \quad a^\dagger(m)a^\dagger(\ell)|0\rangle. \tag{3.41}$$

---

[a] **Question from the audience.** When $\ell \neq m$, does the same final state occur twice in the double sum, once as (ℓ, m) and once as (m, ℓ)?

Response. Yes. If $\ell \neq m$, the terms (ℓ, m) and (m, ℓ) both occur, and because the creation operators commute they describe the same physical state.

---

[a]Lecture timestamp: 01:08:24.

The two ordered terms are equal, so state normalization and this double counting must be handled before a quantitative channel comparison is made. The reliable algebraic conclusion is narrower and sufficient for what follows: creating in a mode already occupied by n identical bosons produces the factor $\sqrt{n+1}$.

3.7 Spontaneous and Stimulated Emission

A cleaner example isolates the square-root factor. Suppose an object localized at x emits one photon. If the photon field begins in the vacuum, the outgoing state is

$$\Psi^\dagger(x)|0\rangle = \sum_k e^{-ikx} a^\dagger(k)|0\rangle \tag{3.42}$$

$$= \sum_k e^{-ikx}|k\rangle. \tag{3.43}$$

Within this simplified model all momenta have equal amplitude magnitude. With no photons already present, the emission is spontaneous.

Now suppose there is already one photon in mode ℓ. Acting with the same local creation field gives two kinds of terms:

$$\Psi^\dagger(x)|\ell\rangle = \sum_{k\neq\ell} e^{-ikx}|k,\ell\rangle + e^{-i\ell x}\sqrt{2}\,|2_\ell\rangle. \tag{3.44}$$

For $k \neq \ell$, the operator creates into an empty mode, so the coefficient is $\sqrt{0+1} = 1$. For $k = \ell$, it creates into the occupied mode, so the coefficient is $\sqrt{1+1} = \sqrt{2}$. The original photon is a bystander; the new photon is preferentially emitted into its occupied mode.

[a] **Question from the audience.** Must the pre-existing photon be at the same location in order to stimulate the emission?

Response. The simplified calculation refers to occupation of the same momentum mode, not to two particles at one sharply defined position. A particle cannot have both exact position and exact momentum. Real photons are described by wave packets, while the cavity idealization provides definite discrete modes. In that setting the enhancement is into the already occupied cavity mode.

[a] Lecture timestamp: 01:16:14.

If mode ℓ initially contains n_ℓ photons, then

$$a^\dagger(\ell)|\ldots, n_\ell, \ldots\rangle = \sqrt{n_\ell + 1}\,|\ldots, n_\ell + 1, \ldots\rangle. \tag{3.45}$$

The probability is therefore enhanced by the factor $n_\ell + 1$. With one photon already present the factor is 2. With one hundred photons in the mode, the amplitude factor is $\sqrt{101}$, and the probability for emission into that mode is 101 times the empty-mode value. The more quanta already present in a bosonic mode, the more likely the next quantum is to be emitted into that mode. This is stimulated emission.

The same occupation effect appears in scattering reactions. If a collision can produce a particle in several modes, pre-existing occupation enhances production into the occupied modes. It is a general property of this oscillator algebra, not a special peculiarity of atomic decay.

[a] **Question from the audience.** In the earlier two-particle process, do two newly created particles still carry greater weight when their momenta differ?

Response. Yes. In the ordered double sum, an unequal-momentum pair occurs in two orders. The same-mode factor $\sqrt{2}$ reduces that combinatorial difference, but by itself does not erase it.

[a]Lecture timestamp: 01:18:15.

As before, a physical channel comparison still requires consistent state normalization and counting.

[a] **Question from the audience.** Does that mean bosons are attracted toward already occupied momentum states?

Response. No. There is no literal force or attraction. The enhancement is a statistical consequence of particle identity and the square-root factors in the bosonic creation operators.

[a]Lecture timestamp: 01:18:30.

It is useful to keep the two names separate. If no photons are present, the coefficient begins with $n = 0$, giving the unit factor associated with spontaneous emission. If photons are already present, the factor $n + 1$ makes emission into their mode more likely; that is stimulated emission.

There is a classical analog. A pre-existing electromagnetic wave passing an atom drives its electrons. The driven electrons radiate more strongly than they would without the wave, and the added radiation follows the direction and wave number of the driving wave. The quantum statement replaces the pre-existing wave by occupied photon modes: existing quanta stimulate the production of more quanta in the same mode.

Particles governed by this harmonic-oscillator algebra are bosons. They have a statistical tendency to accumulate in occupied states, though they do not attract one another into those states. Particles come in two statistical kinds, bosons and fermions. The present construction is bosonic; the opposite fermionic behavior is taken up later.

3.8 Bosonic Algebra and Particle Density

For every momentum mode there are creation and annihilation operators. Their algebra is

$$[a^\dagger(k), a^\dagger(\ell)] = 0, \tag{3.46}$$

$$[a(k), a(\ell)] = 0, \tag{3.47}$$

$$[a(k), a^\dagger(\ell)] = \delta_{k\ell}. \tag{3.48}$$

Here

$$[A, B] = AB - BA, \tag{3.49}$$

and the Kronecker delta is

$$\delta_{k\ell} = \begin{cases} 1, & k = \ell, \\ 0, & k \neq \ell. \end{cases} \tag{3.50}$$

Creation operators commute for equal or unequal momenta, as do annihilation operators. A mixed pair commutes for different momenta; for the same momentum it has the familiar single-oscillator commutator.

Together with the Fourier definition of $\Psi(x)$, these relations define the simplest bosonic quantum field. The field's work is to create and annihilate particles, mode by mode or at specified positions.

Consider now the operator

$$\int_0^L dx\, \Psi^\dagger(x)\Psi(x). \tag{3.51}$$

Instead of naming it in advance, work it out. Substitution of the Fourier expansions gives

$$\int_0^L dx\, \Psi^\dagger(x)\Psi(x) = \sum_{k,\ell} a^\dagger(k)a(\ell) \int_0^L dx\, e^{-ikx} e^{i\ell x} \tag{3.52}$$

$$= \sum_{k,\ell} a^\dagger(k)a(\ell) \int_0^L dx\, e^{i(\ell-k)x}. \tag{3.53}$$

If $k = \ell$, the exponential is 1 and the integral is L. If $k \neq \ell$, an integer number of sine and cosine cycles is integrated around the circle; positive and negative parts cancel. Thus

$$\int_0^L dx\, e^{i(\ell-k)x} = L\delta_{k\ell}. \tag{3.54}$$

The double sum collapses to

$$\int_0^L dx\, \Psi^\dagger(x)\Psi(x) = L\sum_k a^\dagger(k)a(k). \tag{3.55}$$

For each mode,

$$a^\dagger(k)a(k) = N_k \tag{3.56}$$

is the occupation-number operator. Summing it counts all particles:

$$N = \sum_k N_k = \sum_k a^\dagger(k)a(k). \tag{3.57}$$

With the present Fourier normalization,

$$N = \frac{1}{L}\int_0^L dx\, \Psi^\dagger(x)\Psi(x). \tag{3.58}$$

Other conventions put factors such as $1/\sqrt{L}$ into the Fourier expansion and remove the explicit $1/L$ here. In the convention being used, the local number-density operator is

$$\rho(x) = \frac{1}{L}\Psi^\dagger(x)\Psi(x), \qquad \int_0^L dx\, \rho(x) = N. \tag{3.59}$$

Thus $\Psi^\dagger(x)\Psi(x)$ is proportional to particle density. Calling it the number of particles at one mathematical point is misleading; density is the appropriate local notion.

The total number may or may not be conserved. If the dynamics creates or absorbs particles, N can change with time. If particle number is conserved by the dynamics, then N is a conserved observable.

Knowing the commutators of the mode operators also makes it possible to work out the commutators of $\Psi(x)$ and $\Psi^\dagger(x)$. That calculation is left as a useful exercise in applying the Fourier sums.

This construction describes one species of identical bosons. Identical means that all the particles are of the same kind, though they may have different positions and momenta. One would not put electrons and photons into the same field Ψ. A photon field requires relativistic refinements to this elementary model. An electron field requires a different, fermionic algebra.

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[a] **Question from the audience.** Is this construction only one-dimensional?

Response. The example so far has one spatial coordinate, but nothing essential depends on that. Higher dimensions are obtained by introducing additional coordinates and momentum components.

---

[a]Lecture timestamp: 01:35:43.

5

Examples include photons, the Higgs boson, and gravitons.[6]

3.9 Large Occupation and the Classical Field Limit

A complex field operator is not itself a single Hermitian observable. Its two Hermitian components are

$$\Psi_1(x) = \frac{\Psi(x) + \Psi^\dagger(x)}{2}, \quad \Psi_2(x) = \frac{\Psi(x) - \Psi^\dagger(x)}{2i}. \tag{3.60}$$

They are the quantum analogs of the real and imaginary parts of a classical field. They do not commute, so both cannot be measured simultaneously with arbitrary precision. This is the field version of the uncertainty between the conjugate variables of a harmonic oscillator.

When, then, can the field behave like a classical wave? Consider one highly excited momentum mode k. Its number operator is

$$a^\dagger(k)a(k) = N_k. \tag{3.61}$$

Suppose the state has a large mean occupation, $\langle N_k \rangle = n_k \gg 1$. The scale available to a macroscopic oscillator amplitude is then of order $\sqrt{n_k}$. Meanwhile, the right-hand side of

$$[a(k), a^\dagger(k)] = 1 \tag{3.62}$$

remains fixed. The commutator never vanishes, but at large occupation its effects can be small compared with the squared scale of the excitation. Large occupation supplies the scale needed for a classical limit, but it is not sufficient by itself. The state must also be appropriately phase coherent, so that the uncertainties of the two quadratures are small compared with the macroscopic field amplitude.[7]

For an electromagnetic field, large occupation means many photons. In an appropriately phase-coherent state with many photons in the relevant modes, the measurable field components behave approximately like an ordinary classical electromagnetic wave. With only a few photons, the discreteness of the quanta is visible and the classical wave description fails. A quantum field remains an operator in both regimes; what changes is the relative importance of its quantum fluctuations.

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[5]Editorial clarification: A brief exchange here appears to begin with fluorescence and stimulated emission, then turns to transparency and interference. The damaged wording does not support a reliable reconstruction of the question or its answer.

[6]Other particle names in the brief example list are indistinct and are not retained.

[7]This qualifies the large-occupation heuristic. A number eigenstate has $\langle a(k) \rangle = 0$, no definite phase, and quadrature fluctuations of order $\sqrt{n_k}$; it is not by itself a classical wave.

[a] **Question from the audience.** Is the large occupation number the initial occupation or the final occupation?

Response. If the process changes the number of photons only slightly, it does not matter for this argument. The initial and final occupations are both large, and their difference is small compared with either one.

[a]Lecture timestamp: 01:42:09.

The opposite extreme is the ground state, in which every occupation number is zero. That state is highly quantum mechanical, and its notation must be distinguished carefully from the null vector.

[a] **Question from the audience.** If an annihilation operator acts on the vacuum, does it produce the vacuum again, or no state at all?

Response. It produces the zero vector, $a(k)|0\rangle = \mathbf{0}$. The vacuum $|0\rangle$ is a genuine normalized state with no particles, satisfying $\langle 0|0\rangle = 1$. It is not the zero vector. The zero vector has zero norm and represents no physical state.

[a]Lecture timestamp: 01:42:36.

[a] **Question from the audience.** Is the field itself another kind of state?

Response. No. In quantum field theory the field is an operator. States are vectors, and the field operators act on those vectors.

[a]Lecture timestamp: 01:44:11.

3.10 Periodic Space in Practice

The periodic construction is a calculational device rather than a claim that the universe is literally a circle of some chosen circumference. Closely related discrete-mode systems do occur in practice.

[a] **Question from the audience.** How is the finite periodic model used in practice?

Response. Radiation in a cavity with reflecting walls provides a familiar discrete set of modes. A particle moving around a closed wire gives a directly periodic example. Taking L to be the size of the universe is neither necessary nor especially useful.

[a]Lecture timestamp: 01:45:03.

Fixing L gives a nonzero separation between adjacent momenta, but it does not leave only finitely many of them.

[a] **Question from the audience.** Does fixing the length L limit the total number of possible momentum states?

Response. No. The allowed values are $k_j = 2\pi j/L$, and j can be any integer, so there are still infinitely many modes. Finite L gives only finitely many modes below a chosen maximum momentum. Without such a cutoff, the momentum label remains unbounded.

[a]Lecture timestamp: 01:45:39.

The oscillator interpretation can also be read in reverse. In elementary oscillator mechanics one begins with conjugate variables and constructs a and $a^\dagger$. Here the mode operators were introduced first, so it is natural to ask where their conjugate variables appear.

[a] **Question from the audience.** For a fixed mode, what corresponds to the oscillator variables X and P when the construction begins with $a(k)$ and $a^\dagger(k)$?

Response. For each fixed momentum mode there is one oscillator. Its two Hermitian quadratures are built from $a(k)$ and $a^\dagger(k)$, just as the real and imaginary field components are built from Ψ and $\Psi^\dagger$. Up to normalization and a possible interchange of names, these are the analogs of X and P.

[a]Lecture timestamp: 01:46:43.

For the fixed mode, suppressing convention-dependent numerical factors, the quadratures are[8]

$$X_k \propto a(k) + a^\dagger(k), \qquad P_k \propto \frac{a(k) - a^\dagger(k)}{i}. \tag{3.63}$$

The position-space components $\Psi_1(x)$ and $\Psi_2(x)$ introduced above are Fourier superpositions of these mode quadratures, not the quadratures of one fixed mode. Their uncertainty relation is the same obstruction encountered when trying to regard a weakly excited quantum field as a completely classical object.

3.11 Periodic Space in Higher Dimensions

[a] **Question from the audience.** How is periodic space represented in more than one dimension?

Response. In two dimensions the analog of the circle is a torus. It can be represented by a rectangle whose left and right edges are identified and whose bottom and top edges are identified. There are two periodic directions, so both momentum components are quantized.

[a]Lecture timestamp: 01:48:21.

A torus is often pictured as the surface of a doughnut, but a rectangle is more useful for calculation. Cut the surface once and open it into a cylinder. Cut the cylinder once more and lay it flat. The resulting rectangle still carries two identifications: the left edge is the same as the right edge, and the bottom edge is the same as the top edge.[9]

A particle that crosses the right edge reappears at the corresponding point on the left. If it crosses the top, it reappears at the corresponding point on the bottom. A periodic field obeys

$$\Psi(x + L_x, y) = \Psi(x, y), \tag{3.64}$$

$$\Psi(x, y + L_y) = \Psi(x, y). \tag{3.65}$$

For a plane wave $e^{i(k_x x + k_y y)}$, these identifications require

$$k_x = \frac{2\pi n_x}{L_x}, \qquad k_y = \frac{2\pi n_y}{L_y}, \qquad n_x, n_y \in \mathbb{Z}. \tag{3.66}$$

Both components of momentum are discrete.

[8]The exchange at 01:46:43 moves loosely between one momentum mode and the full field. The displayed relation makes the fixed-mode statement explicit: $\Psi(x)$ and $\Psi^\dagger(x)$ are Fourier sums over all modes.

[9]The rectangle, its two edge identifications, the reappearing particle, and the statement that both momentum components are quantized follow 01:48:46–01:50:58. The displayed formulas for k_x and k_y are the componentwise reconstruction of the earlier condition $kL = 2\pi n$.

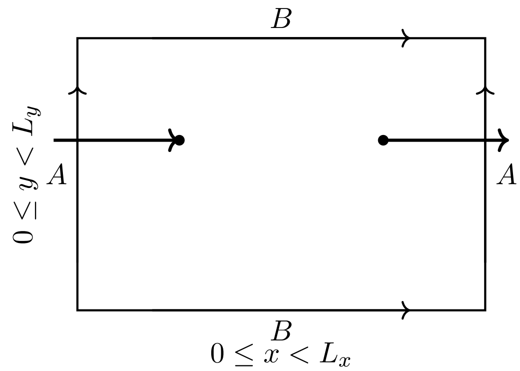

Figure 3.6: A two-dimensional periodic space: edges carrying the same label are identified, so a particle crossing one edge reappears at the corresponding point on the opposite edge.

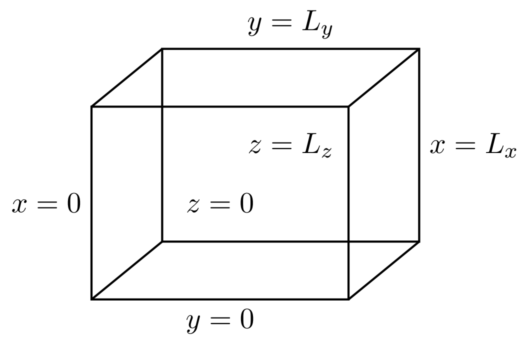

Figure 3.7: A three-dimensional periodic space as a cube with each pair of opposite faces identified.

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[a] **Question from the audience.** Does the periodic construction represent space, or the objects in space?

Response. It represents space. The objects and fields live in that space and obey its identifications. Periodicity is imposed to control the infinite-volume problem; it is removed by taking the volume large and replacing momentum sums by integrals.

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[a]Lecture timestamp: 01:50:58.

The three-dimensional analog is a cube with three pairs of opposite faces identified. The left and right faces are the same, the top and bottom faces are the same, and the front and back faces are the same. A particle crossing any face reappears through its identified partner.[10] The three momentum components obey

$$k_i = \frac{2\pi n_i}{L_i}, \qquad i = x, y, z, \tag{3.67}$$

with one integer n_i for each direction. Tori can be defined in any number of dimensions by the same rule.

[10]The periodic cube and its three pairs of identified faces follow 01:51:27–01:52:17. The displayed equation for k_i is the componentwise reconstruction of the earlier condition $kL = 2\pi n$, applied to the three periodic directions.

[a] **Question from the audience.** Does field theory choose a physical grid size for space?

Response. That is a speculative question. For practical purposes here, space is continuous. The quanta are particles, not cells of a spatial grid. In the circle construction it is momentum that becomes discrete; position remains continuous and periodic.

[a]Lecture timestamp: 01:52:20.

11

The discreteness of k came entirely from imposing a periodic wave function. It is not evidence that the coordinate x itself lies on a lattice. Questioning whether space may have a fundamental microscopic structure is legitimate, but it is a separate, speculative issue.

3.12 Fermions and External Fields

Bosonic modes allow any nonnegative occupation number. Fermions require the opposite behavior: the Pauli exclusion principle forbids two identical fermions from occupying the same state.

[a] **Question from the audience.** Does replacing commutators by anticommutators give the simplest fermion theory?

Response. Yes, in essence. The commutator is $[A, B] = AB - BA$, whereas the anticommutator is $\{A, B\} = AB + BA$. The corresponding fermionic creation-and-annihilation algebra implements Pauli exclusion: a second identical fermion cannot be put into an already occupied state. The detailed theory is deferred.

[a]Lecture timestamp: 01:55:13.

The two products differ by a sign:

$$[A, B] = AB - BA, \qquad \{A, B\} = AB + BA. \tag{3.68}$$

The commutator is antisymmetric under exchange, while the anticommutator is symmetric. If an anticommutator vanishes, then

$$\{A, B\} = 0 \quad \Longrightarrow \quad AB = -BA. \tag{3.69}$$

For such a pair, exchanging the two fermionic operators changes the sign. This is more unusual than ordinary commuting, where $[A, B] = 0$ gives $AB = BA$. Replacing the bosonic commutators by the corresponding anticommuting algebra is the mathematical change that leads to fermions; its detailed development is postponed.

Anticommutation reverses the statistical behavior seen for bosons. There is no stimulated piling-up of identical fermions in one mode; occupancy of that mode blocks another identical fermion from entering it. Since a single fermionic state can contain at most one identical fermion, no classical field limit is obtained by putting an arbitrarily large number of them into that one state.

[11]Editorial clarification: A long damaged passage during the grid-size exchange mentions Planckian scales but does not preserve a recoverable claim about fundamental spatial discreteness. No such claim is inferred here.

[a] **Question from the audience.** How can an alpha particle be a boson if it is made from fermions?

Response. A bound composite containing an even number of fermionic constituents can have bosonic statistics. Putting several such composite bosons into the same overall state is not the same as putting their individual fermionic constituents into one identical single-particle state.

[a] Lecture timestamp: 01:58:03.

Composite statistics concern the bound object as a whole. Fermionic constituents can therefore combine into an object that behaves as a boson without violating exclusion for the constituents.

[a] **Question from the audience.** How are external fields incorporated?

Response. No external background field has been included in the construction so far. Such a field can be prescribed as a classical field, or represented quantum mechanically by a state containing a large number of quanta.

[a] Lecture timestamp: 01:58:33.

For the two descriptions to agree, the large-occupation state must be appropriately prepared and phase coherent; particle number alone is not sufficient.[12] In that semiclassical regime their effects agree up to small quantum deviations. This is a basic consistency condition. The electromagnetic field must describe discrete photons when individual quanta matter, yet an appropriately phase-coherent state containing many photons must reproduce the familiar action of a classical electromagnetic wave.

[12] The exchange supplies the many-quanta scale. Phase coherence is stated here as the additional condition that prevents large occupation alone from being mistaken for a classical field.

Chapter 4

Delta Functions, Quantum Fields, and Conservation Laws

The simple quantum field introduced earlier is enough to describe scattering, creation, and annihilation in outline, but there is more to extract from it. In particular, it shows how conservation laws enter particle processes. A full course in quantum field theory takes much longer than a few lectures; here the aim is narrower. We shall use the simplest field to see, mechanically, where energy conservation comes from, why a fixed target need not conserve the momentum of the scattered particle, and how phase invariance encodes charge conservation.

Before returning to the field, two pieces of mathematics are needed: the Dirac delta function and the action of creation and annihilation operators on bra vectors.

4.1 The Dirac Delta Function

Call the coordinate y, without deciding whether it will later represent position, wave number, or something else. Begin with an ordinary lump-shaped function concentrated near $y = a$. Make the lump narrower and narrower. If its height were held fixed, its area would shrink to zero, so raise the height as the width decreases, keeping height times width equal to one. The limiting object is the Dirac delta function $\delta(y - a)$.

It vanishes away from $y = a$, is infinitely sharply concentrated at $y = a$, and has unit area:

$$\delta(y-a) = 0 \quad (y \neq a), \qquad \int_{-\infty}^{\infty} dy\, \delta(y-a) = 1. \tag{4.1}$$

The important point is the combination of localization and fixed area. Narrowing the peak does not remove it, because its height increases at the same time.

The same picture can be idealized as a vertical spike:

The delta function appears naturally in Fourier integrals. Return for a moment to a periodic spatial interval of total length L. A periodic mode is

$$e^{ikx}, \tag{4.2}$$

where k is one of the allowed wave numbers

$$k_n = \frac{2\pi n}{L}, \qquad n \in \mathbb{Z}. \tag{4.3}$$

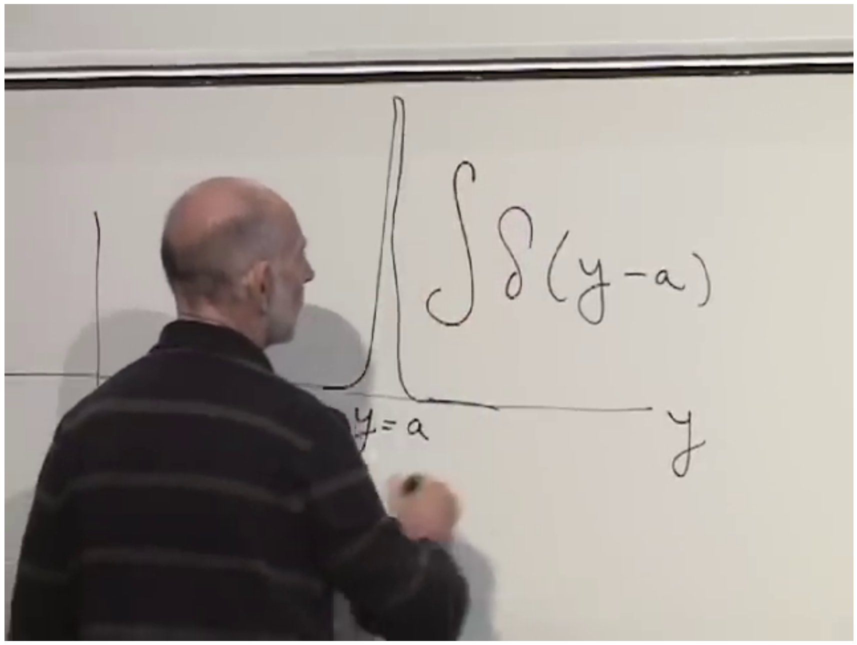

Figure 4.1: Unit-area delta-function spike centered at $y = a$.
Lecture frame: 00:05:50.

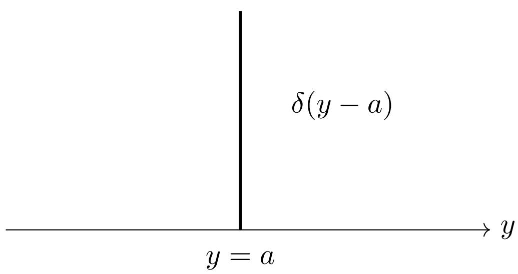

Figure 4.2: Idealized localization of $\delta(y - a)$ at the zero of its argument.

The interval may be represented as $0 \le x < L$, but it is more symmetric to put the origin in the middle and use

$$-\frac{L}{2} \le x < \frac{L}{2}. \tag{4.4}$$

The endpoints are identified: crossing $L/2$ brings one back at $-L/2$.

Integrate one allowed mode over the entire periodic interval:

$$I_L(k) = \int_{-L/2}^{L/2} dx\, e^{ikx}. \tag{4.5}$$

If $k = 0$, the integrand is identically one and the answer is L. If $k \neq 0$ is an allowed value, the positive and negative parts of the oscillation cancel over the full number of periods. Thus

$$I_L(k) = \begin{cases} L, & k = 0, \\ 0, & k \neq 0, \end{cases} \qquad k = \frac{2\pi n}{L}. \tag{4.6}$$

Now regard $I_L(k)$ as a function on the discrete set of allowed k's. The spacing between neighboring values is

$$\Delta k = \frac{2\pi}{L}. \tag{4.7}$$

At $k = 0$ the function has height L; at every other allowed value it is zero. For purposes of drawing the discrete spike, give it one k-spacing of width. Its area is then

$$(\text{height})(\text{width}) = L\frac{2\pi}{L} = 2\pi. \tag{4.8}$$

As L grows, the height grows, the spacing shrinks, and the area remains 2π. In the infinite-volume limit the spike becomes $2\pi\delta(k)$, giving the formal Fourier identity

$$\int_{-\infty}^{\infty} dx\, e^{ikx} = 2\pi\delta(k). \tag{4.9}$$

The variables x and k occur symmetrically in the exponential. Interchanging their roles gives the companion identity

$$\int_{-\infty}^{\infty} dk\, e^{ikx} = 2\pi\delta(x). \tag{4.10}$$

These two elementary rules will soon turn an integral over all interaction times into a condition of energy conservation.

4.2 Bras, Kets, and Operators Acting to the Left

A ket is symbolic notation for a quantum state. For the present purpose, an initial state will be written as a ket and a final state as a bra. Putting a bra and ket together produces an inner product, which is a number. Those numbers, or absolute squares of them, are ultimately what can be compared with experiment.

---

[a] **Question from the audience.** Had the word “ket” actually been introduced before?

Response. Yes. A ket is one half of the bra-ket notation; the other half, introduced now, is the bra.

---

[a]Lecture timestamp: 00:15:30.

For the harmonic oscillator, the occupation-number states may be written either as kets $|n\rangle$ or as bras $\langle n|$. Their inner product is

$$\langle m|n\rangle = \delta_{mn} = \begin{cases} 1, & m = n, \\ 0, & m \neq n. \end{cases} \tag{4.11}$$

The Kronecker delta δ_{mn} is the discrete analogue of the Dirac delta function.

Creation and annihilation operators act on kets in the familiar way:

$$a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle, \tag{4.12}$$
$$a|n\rangle = \sqrt{n}\,|n-1\rangle. \tag{4.13}$$

The creation and annihilation operators may also be denoted a^+ and a^-; here $a^\dagger$ and a will be used throughout.

How should the same operators act on bras? The notation puts an operator to the right of a bra when it acts to the left. The definition is chosen so that a matrix element has the same value whether the operator is first applied to the ket or first applied to the bra. Consider

$$\langle n|a^\dagger|m\rangle. \tag{4.14}$$

Acting on the ket gives

$$\langle n|a^\dagger|m\rangle = \sqrt{m+1}\,\langle n|m+1\rangle \tag{4.15}$$

$$= \sqrt{m+1}\,\delta_{n,m+1}. \tag{4.16}$$

This is nonzero only if $n = m+1$, or equivalently if $m = n-1$. To reproduce the same condition by acting on the bra, $a^\dagger$ must lower its label:

$$\langle n|a^\dagger = \sqrt{n}\,\langle n-1|. \tag{4.17}$$

The numerical factor is $\sqrt{n}$, since on the nonzero matrix element $n = m+1$. The annihilation operator obeys the complementary rule

$$\langle n|a = \sqrt{n+1}\,\langle n+1|. \tag{4.18}$$

Thus $a^\dagger$ raises a ket but lowers a bra, while a lowers a ket but raises a bra.

The advantage is visible in the number-operator matrix element

$$\langle n|a^\dagger a|n\rangle. \tag{4.19}$$

Acting to the right,

$$a|n\rangle = \sqrt{n}\,|n-1\rangle, \tag{4.20}$$

$$a^\dagger|n-1\rangle = \sqrt{n}\,|n\rangle, \tag{4.21}$$

so

$$\langle n|a^\dagger a|n\rangle = n\langle n|n\rangle = n. \tag{4.22}$$

Acting instead to the left,

$$\langle n|a^\dagger = \sqrt{n}\,\langle n-1|, \tag{4.23}$$

$$\langle n-1|a = \sqrt{n}\,\langle n|, \tag{4.24}$$

and the same answer follows. One need not decide which direction is more fundamental; the calculus is arranged so that either direction gives the same number.

4.3 A Quantum Field in Space and Time

The reason for this bookkeeping is particle physics. At microscopic scales, the main experimental handle is to collide particles and observe what emerges. An initial state with one or two particles may turn into a state with four, five, seven, nine, or some other number. Quantum fields, built from creation and annihilation operators, provide a language for such processes.

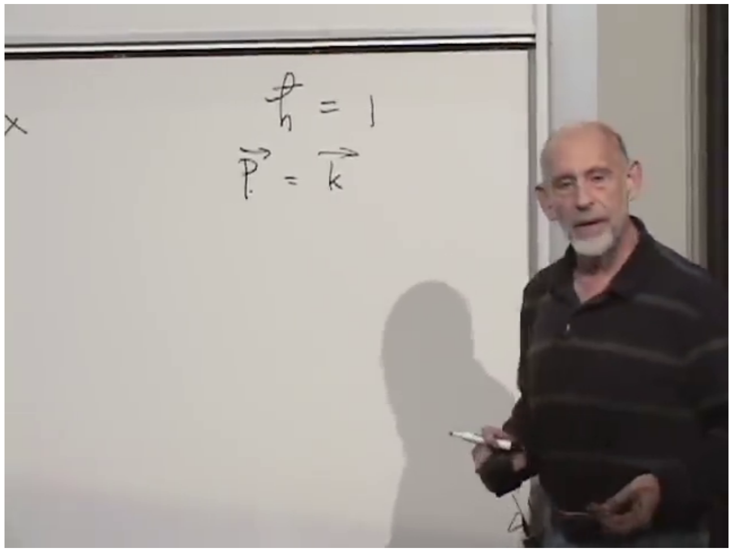

Figure 4.3: Natural units $\hbar = 1$ and the identification $\vec{p} = \vec{k}$.
Lecture frame: 00:36:56.

Work first in one spatial dimension. Nothing essential prevents the use of three dimensions: x and k simply become vectors with three components. It is also convenient to use units in which

$$\hbar = 1. \tag{4.25}$$

Then momentum and wave vector have the same numerical value,

$$\vec{p} = \vec{k}. \tag{4.26}$$

Adopt the convention that ψ annihilates and $\psi^\dagger$ creates:

$$\psi(x) = \sum_k a(k)e^{ikx}, \tag{4.27}$$

$$\psi^\dagger(x) = \sum_k a^\dagger(k)e^{-ikx}. \tag{4.28}$$

The sum is over $k = 2\pi n/L$ in periodic space and becomes continuous in infinite space. The second equation is the Hermitian conjugate of the first. It may also be thought of as the operator analogue of a complex conjugate.[1]

Why this definition? At this stage the best answer is that it is useful. It is a simple sum over momentum modes, and its coefficients are precisely the operators that remove or add particles. The detailed justification comes from the dynamics and from the calculations it makes possible.

[2]

[1] The blackboard temporarily labels the creation-field expansion $\sum_k a^\dagger(k)e^{-ikx}$ by Ψ, and later hesitates over the dagger assignment. The labels here are fixed globally so that ψ annihilates and $\psi^\dagger$ creates, exactly as in the subsequent scattering calculation.

[2] Editorial clarification: A short portion of the source immediately after the conjugate spatial field is introduced is not intelligible enough to support an additional normalization or interpretation statement. The secure content resumes with the usefulness of the definition.

So far the fields depend on position but not time. Fields oscillate, and each wave number k has an associated frequency $\omega(k)$. An object of definite frequency ω has time dependence

$$e^{i\omega t} = \cos(\omega t) + i\sin(\omega t). \tag{4.29}$$

Attach the conjugate phases to the two field expansions:

$$\psi(x,t) = \sum_k a(k) e^{ikx} e^{-i\omega(k)t}, \tag{4.30}$$

$$\psi^\dagger(x,t) = \sum_k a^\dagger(k) e^{-ikx} e^{i\omega(k)t}. \tag{4.31}$$

Now each spatial mode oscillates at the frequency appropriate to its wave number, and the field depends on both space and time.

4.4 The Nonrelativistic Field Equation

The relation between ω and k determines the differential equation obeyed by the field. For a single quantum, the wave-particle relations are

$$E = \hbar\omega, \qquad p = \hbar k. \tag{4.32}$$

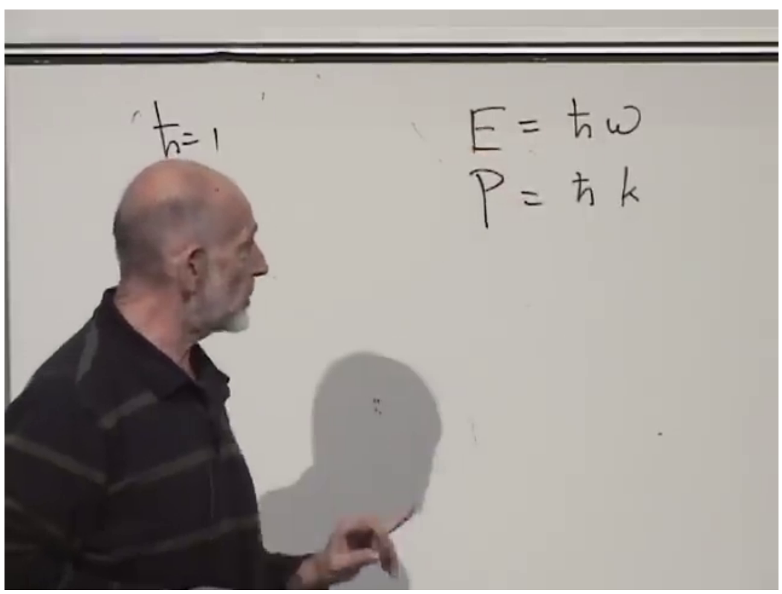

Figure 4.4: Energy-frequency and momentum-wave-number relations $E = \hbar\omega$ and $p = \hbar k$.
Lecture frame: 00:41:28.

With $\hbar = 1$, energy is frequency and momentum is wave number:

$$E = \omega, \qquad p = k. \tag{4.33}$$

For a nonrelativistic particle moving much more slowly than light, the kinetic energy and momentum obey

$$E = \frac{p^2}{2m}. \tag{4.34}$$

This is the same relation obtained from $E = \frac{1}{2}mv^2$ and $p = mv$. Combining these relations gives the nonrelativistic dispersion relation

$$\omega(k) = \frac{k^2}{2m}. \tag{4.35}$$

For the creation field, the mode expansion is

$$\psi^\dagger(x,t) = \sum_k a^\dagger(k) e^{-ikx} e^{i\omega(k)t}. \tag{4.36}$$

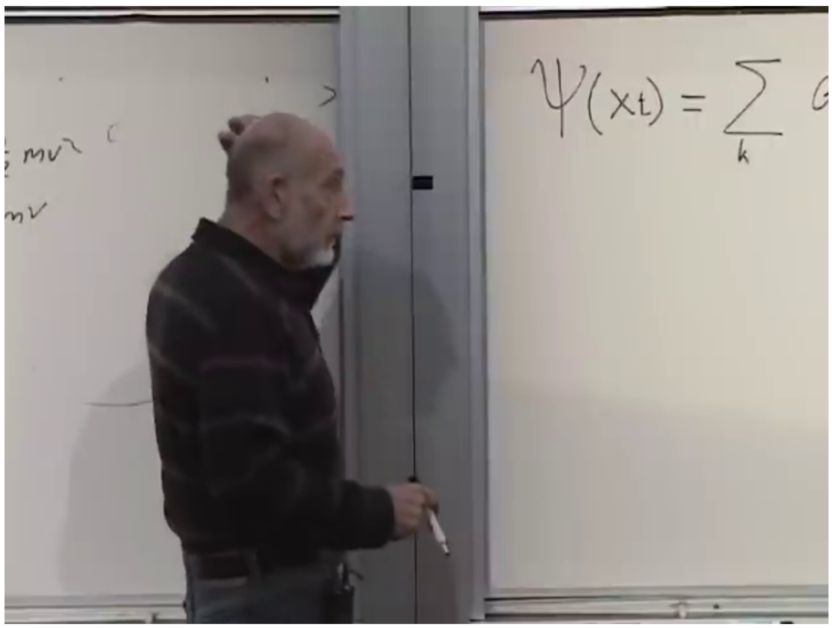

Figure 4.5: Time-dependent mode expansion used to begin the nonrelativistic field equation.
Lecture frame: 00:43:23.

Differentiate with respect to time:

$$\dot{\psi}^\dagger(x,t) = \sum_k i\omega(k) a^\dagger(k) e^{-ikx} e^{i\omega(k)t}. \tag{4.37}$$

Differentiate once with respect to position:

$$\frac{\partial \psi^\dagger}{\partial x} = \sum_k (-ik) a^\dagger(k) e^{-ikx} e^{i\omega(k)t}. \tag{4.38}$$

There is no direct first-order relation between these derivatives because the time derivative brings down $\omega(k)$, whereas one spatial derivative brings down k. To see the distinction explicitly, imagine the linear dispersion relation

$$\omega(k) = k. \tag{4.39}$$

The time derivative would then bring down $i\omega = ik$, while the first spatial derivative brings down $-ik$. Mode by mode,

$$\frac{\partial \psi^\dagger}{\partial t} = -\frac{\partial \psi^\dagger}{\partial x}. \tag{4.40}$$

That hypothetical field would satisfy a first-order equation. Here, however, $\omega = k^2/(2m)$, so a second spatial derivative is required:

$$\frac{\partial^2 \psi^\dagger}{\partial x^2} = \sum_k (-k^2) a^\dagger(k) e^{-ikx} e^{i\omega(k)t} \tag{4.41}$$

$$= -2m \sum_k \omega(k) a^\dagger(k) e^{-ikx} e^{i\omega(k)t}. \tag{4.42}$$

[a] **Question from the audience.** Is the minus sign from the spatial derivatives really supposed to be there?
Response. Yes. Each x-derivative of e^{-ikx} brings down $-ik$, and two derivatives give $(-ik)^2 = -k^2$. Substituting $k^2 = 2m\omega$ therefore gives $-2m\omega$.

[a]Lecture timestamp: 00:47:14.

Multiplying the time derivative by i gives the same mode coefficient as dividing the second spatial derivative by $2m$:

$$i\dot{\psi}^\dagger = -\sum_k \omega(k) a^\dagger(k) e^{-ikx} e^{i\omega(k)t}, \tag{4.43}$$

$$\frac{1}{2m}\frac{\partial^2 \psi^\dagger}{\partial x^2} = -\sum_k \omega(k) a^\dagger(k) e^{-ikx} e^{i\omega(k)t}. \tag{4.44}$$

Hence

$$i\frac{\partial \psi^\dagger}{\partial t} = \frac{1}{2m}\frac{\partial^2 \psi^\dagger}{\partial x^2}, \tag{4.45}$$

or, equivalently,

$$-i\frac{\partial \psi^\dagger}{\partial t} = -\frac{1}{2m}\frac{\partial^2 \psi^\dagger}{\partial x^2}. \tag{4.46}$$

[a] **Question from the audience.** Is the coefficient $1/(2m)^2$?
Response. No. It is $1/(2m)$.

[a]Lecture timestamp: 00:49:43.

The conjugate annihilation field obeys the familiar Schrödinger form

$$i\frac{\partial \psi}{\partial t} = -\frac{1}{2m}\frac{\partial^2 \psi}{\partial x^2}. \tag{4.47}$$

The two field equations are Hermitian conjugates.[3]

The equation has the same form as the Schrödinger equation of elementary quantum mechanics, but the object satisfying it is different. In elementary quantum mechanics, a wavefunction is a complex-valued function whose absolute square is interpreted probabilistically. Here ψ and $\psi^\dagger$ are operators: ψ removes a particle and $\psi^\dagger$ adds one. The conjugate pair provides a direct link between particles and fields.

[3]During the sign check, the board briefly gave the spatial term with the opposite sign. Direct differentiation of the displayed mode phases gives the conjugate pair of equations written here.

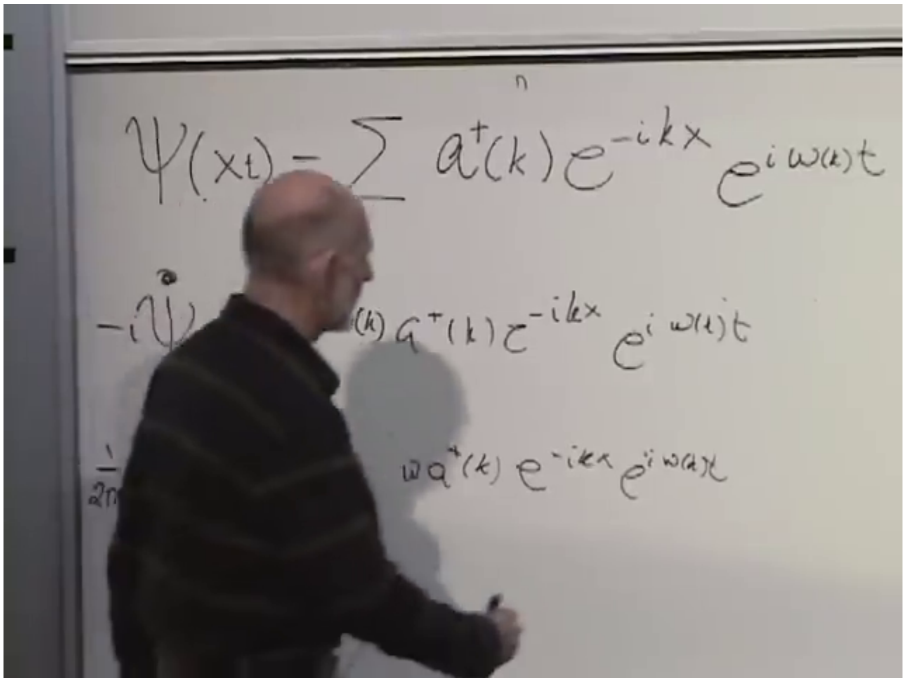

Figure 4.6: Time and spatial derivatives of the creation-field mode expansion.

Lecture frame: 00:49:06.

When many quanta occupy the relevant modes, a field can behave approximately classically, just as a strongly excited harmonic oscillator can look classical while a one-quantum oscillator does not. Large occupation makes a large classical scale possible and can make appropriate relative fluctuations small.

[4]

4.5 A Fixed Target and a Scattering Amplitude

How are these fields used? Consider the simplest possible scattering process. A particle meets a target so heavy that its recoil will be ignored. The target is fixed in space forever. A particle can be reflected, transmitted, or redirected; in three dimensions its outgoing direction can differ from its incoming direction.

The particle's momentum need not be the same before and after the collision. Momentum is conserved for the complete physical system, but the target absorbs the missing momentum. Once the target's motion is omitted from the mathematical description, the momentum of the particle alone is not conserved.

Its energy behaves differently in the idealized model. Imagine throwing a tennis ball against a wall. A real tennis ball heats and deforms, but if those effects are ignored, it returns with the same kinetic energy even though its momentum reverses. The fixed-target model is designed to reproduce that distinction: the particle's energy is conserved while its momentum may change.

[4] Editorial clarification: Large occupation by itself is not sufficient for a classical field. A phase-coherent semiclassical state is also needed; a number eigenstate has no definite phase and need not have a nonzero mean field.

Draw space horizontally and time vertically. The target is the vertical line $x = 0$. An incoming one-particle state of momentum k_i reaches the target and an outgoing one-particle state of momentum k_f leaves it.

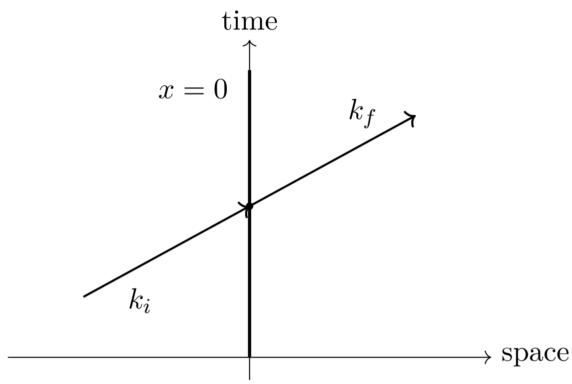

Figure 4.7: A particle of momentum k_i scatters from a target fixed at $x = 0$ and emerges with momentum k_f.

Use the language of absorption and re-emission. The target annihilates the incoming particle and immediately creates the outgoing particle, possibly with a different momentum. This is only a mathematical model of the scattering event; an actual atom, for example, can absorb a photon, remain excited for a time, and emit later.

---

[a] **Question from the audience.** Why must the particle be absorbed and re-emitted immediately?

Response. It need not be in nature. Instantaneous absorption and re-emission at one point is the simplifying model being used here.

---

[a]Lecture timestamp: 01:02:28.

The initial state contains one particle in the k_i mode and none in any other mode. In occupation-number language every entry is zero except the entry associated with k_i, which is one. It is enough to abbreviate the full list by

$$|k_i\rangle = a^\dagger(k_i)|0\rangle. \tag{4.48}$$

Similarly,

$$\langle k_f| = \langle 0|a(k_f). \tag{4.49}$$

Annihilation at the target and time t is represented by $\psi(0, t)$; immediate recreation at the same point and time is represented by $\psi^\dagger(0, t)$. There is no preferred time at which the event occurs. It can happen whenever the incoming particle reaches the target, with the same interaction strength at every time. To express that absence of a special time, integrate over all t:

$$\int_{-\infty}^{\infty} dt\, \psi^\dagger(0, t)\psi(0, t). \tag{4.50}$$

Acting on the initial ket, this operator produces a state. Its overlap with the final bra is the amplitude for the specified transition. The probability is obtained only after taking the absolute square of the amplitude.

One further number is needed. Let g measure the strength of the interaction between the particle and the target. In this bookkeeping model, write the schematic leading amplitude as[5]

$$\mathcal{A}_{i\to f} = g \left\langle k_f \left| \int_{-\infty}^{\infty} dt\, \psi^\dagger(0,t)\psi(0,t) \right| k_i \right\rangle . \tag{4.51}$$

The coefficient g is called a coupling constant. For a charged target scattering a photon, electric charge supplies the relevant measure of interaction strength: increasing the charge increases the tendency to scatter.

Different interactions have very different couplings. A meson reaching a proton undergoes a strong interaction and has a comparatively large scattering strength. A photon scattered by a charged proton has a smaller electromagnetic coupling. A neutrino interacting with a proton has a smaller coupling still.

---

[a] **Question from the audience.** Is this simple model valid for an electron?

Response. Only as a simplified stand-in for several kinds of scattering. It should not yet be read as a realistic theory of an electron or of any one particular target.

---

[a]Lecture timestamp: 01:09:09.

Gravitational scattering of a graviton from a nucleus would be extraordinarily weak. These examples do not share one universal g; particle physics contains many coupling constants.

Nor must an effective coupling always be independent of frequency.

---

[a] **Question from the audience.** Why take g to be a constant rather than a function of ω or k?

Response. It need not always be constant. If a photon is absorbed by an atom, the atom remains excited for a time, and the photon is emitted later, the effective scattering can depend on frequency. The constant g belongs to the present instantaneous model.

---

[a]Lecture timestamp: 01:10:31.

For the moment, keep g constant and calculate the consequences of this simplest choice.

---

[a] **Question from the audience.** If $g = 1$, must the scattering probability be one?

Response. No. The expression being calculated is only the leading approximation for small g. If the coupling is not small, higher-order perturbation theory is required; a probability larger than one would signal the breakdown of the one-step approximation, not a physical probability.

---

[a]Lecture timestamp: 01:12:32.

[5]State-normalization factors and the conventional overall factor of first-order perturbation theory are suppressed; the calculation uses only the operator selection rules and the energy-conserving support.

4.6 Collapsing the Mode Sums

At $x = 0$, the spatial phases equal one, so

$$\psi(0,t) = \sum_k a(k) e^{-i\omega(k)t}, \tag{4.52}$$

$$\psi^\dagger(0,t) = \sum_\ell a^\dagger(\ell) e^{i\omega(\ell)t}. \tag{4.53}$$

[a] **Question from the audience.** Since the target is at $x = 0$, should the spatial exponential be omitted? **Response.** Yes. At the target $e^{ikx} = e^{ik0} = 1$, and the conjugate spatial factor is also one.

[a]Lecture timestamp: 01:13:58.

Use different dummy labels k and ℓ in the two sums. Substitution gives

$$\mathcal{A}_{i\to f} = g \int_{-\infty}^{\infty} dt \sum_{\ell,k} e^{i\omega(\ell)t} e^{-i\omega(k)t} \langle k_f | a^\dagger(\ell) a(k) | k_i \rangle. \tag{4.54}$$

Which term in the k-sum can survive? An annihilation operator gives zero in every empty mode. The initial state contains a particle only in mode k_i, so $k = k_i$. Which term in the ℓ-sum can survive? Acting to the left, $a^\dagger(\ell)$ must remove the particle represented by the final bra, so $\ell = k_f$. Both sums collapse:

$$\mathcal{A}_{i\to f} = g \int_{-\infty}^{\infty} dt\, e^{i(\omega_f - \omega_i)t} \langle k_f | a^\dagger(k_f) a(k_i) | k_i \rangle. \tag{4.55}$$

The annihilation operator turns the initial one-particle ket into the vacuum,

$$a(k_i)|k_i\rangle = |0\rangle, \tag{4.56}$$

and the creation operator, acting to the left, turns the final one-particle bra into the vacuum bra,

$$\langle k_f | a^\dagger(k_f) = \langle 0|. \tag{4.57}$$

Their matrix element is therefore

$$\langle 0|0\rangle = 1. \tag{4.58}$$

All the operator bookkeeping reduces to one; the coupling supplies g, and the nontrivial part left over is the time integral:

$$\mathcal{A}_{i\to f} = g \int_{-\infty}^{\infty} dt\, e^{i(\omega_f - \omega_i)t}. \tag{4.59}$$

[6]

Now the earlier delta-function identity applies directly:

$$\int_{-\infty}^{\infty} dt\, e^{i(\omega_f - \omega_i)t} = 2\pi\delta(\omega_f - \omega_i). \tag{4.60}$$

[6]Editorial clarification: A brief audience question at 01:18:40 is not recoverable from the source. The response distinguishes this one-particle calculation from a relativistic or multiparticle issue; no more precise exchange can be reconstructed securely.

Hence

$$\boxed{\mathcal{A}_{i\to f} = 2\pi g\,\delta(\omega_f - \omega_i)}. \tag{4.61}$$

The amplitude vanishes unless

$$\omega_f = \omega_i. \tag{4.62}$$

Since $\hbar = 1$, this is $E_f = E_i$. Energy conservation has appeared because every interaction time was treated alike. If the time integral were restricted to a finite interval, it would produce a broadened function rather than an exact delta function. Exact invariance under translations of time is what gives the exact energy-conserving condition.

---

[a] **Question from the audience.** The amplitude must be squared to obtain a probability. What happens to the delta function when it is squared?

Response. The squared delta function requires additional care. The point needed here is secure: the result is zero unless the initial and final energies are equal. On the energy-conserving support, the schematic squared prefactor contains $4\pi^2 g^2$.

---

[a]Lecture timestamp: 01:21:30.

[7]

Nothing in this point-target interaction distinguishes one final momentum from another except the energy condition. In three dimensions, the model therefore assigns equal leading weight to all outgoing momenta with the same energy—equivalently, to all outgoing directions on the fixed-energy sphere. That isotropy is a property of this very simple point scatterer, not of every real target.

The calculation has supplied three linked ideas. The coefficient g measures the interaction strength. A bra-operator-ket expression is a scattering amplitude whose absolute square enters the probability. Most importantly, integration over all times produces the energy delta function: time-translation symmetry enforces energy conservation.

4.7 What Changes When the Target or Coupling Changes?

The origin $x = 0$ was chosen only for convenience. Put the target at $x = x_\mathrm{T}$. The spatial factors in $\psi^\dagger(x_\mathrm{T}, t)\psi(x_\mathrm{T}, t)$ give

$$e^{-ik_f x_\mathrm{T}} e^{ik_i x_\mathrm{T}} = e^{i(k_i - k_f)x_\mathrm{T}}. \tag{4.63}$$

Thus the amplitude becomes

$$\mathcal{A}_{i\to f}(x_\mathrm{T}) = e^{i(k_i - k_f)x_\mathrm{T}} \mathcal{A}_{i\to f}(0). \tag{4.64}$$

---

[a] **Question from the audience.** Can the position always be redefined so that the target is at $x = 0$?

Response. For this single fixed target, yes. If it is left at $x_\mathrm{T} \neq 0$, its position appears as the phase $e^{i(k_i - k_f)x_\mathrm{T}}$ in the amplitude. Multiplication by the complex-conjugate phase removes it from the probability.

---

[a]Lecture timestamp: 01:24:37.

[7]Editorial clarification: A normalized transition probability or rate cannot be obtained by simply discarding $\delta(\omega_f - \omega_i)^2$. It requires finite-time regularization and consistent state normalization. Thus $4\pi^2 g^2$ is retained only as the schematic squared prefactor emphasized in the source.

Indeed,

$$\left|e^{i(k_i-k_f)x_{\mathrm{T}}}\right|^2 = 1. \tag{4.65}$$

The amplitude remembers the target's position, but this one-target probability does not.

Now reverse the asymmetry. Suppose an interaction could occur at any position but only at one distinguished time. An integral over position would give

$$\int_{-\infty}^{\infty} dx\, e^{i(k_i-k_f)x} = 2\pi\delta(k_i-k_f), \tag{4.66}$$

while fixing the time would remove the energy delta function.

---

[a] **Question from the audience.** If it did not matter where the interaction happened, but it happened only at one fixed time, would a different quantity be conserved?

Response. Yes. Integration over all positions would enforce momentum conservation, while the distinguished interaction time would remove exact energy conservation. In a complete two-particle problem, the interaction may occur at any time and any place.

---

[a]Lecture timestamp: 01:26:41.

This also gives a physical interpretation of explicit time dependence. If the coupling is $g(t)$, or an outside intervention suddenly makes the scattering possible, not every time is equivalent. The reduced description may then exchange energy with whatever external dynamics were omitted.

---

[a] **Question from the audience.** Do coupling constants mostly come from experiment, and what does a time-dependent coupling mean?

Response. Yes, their values are determined experimentally. Explicit time dependence usually means that some moving or changing part of the system has been left out, much as target recoil was left out of the fixed-target model. Energy that appears missing belongs to those omitted degrees of freedom.

---

[a]Lecture timestamp: 01:28:29.

A moving target is the simplest example.

---

[a] **Question from the audience.** Is hitting a moving target an example in which the particle's energy is not conserved?

Response. Yes. If the target's prescribed motion is omitted from the energy bookkeeping, it can supply or remove energy from the particle.

---

[a]Lecture timestamp: 01:29:23.

The pointlike character of the target is another simplification.

---

[a] **Question from the audience.** When does the size of the target have to be compared with the size or wavelength of the incident wave?

Response. In the present model the target has zero size. A finite target size does produce additional effects when compared with the wavelength, but that refinement is postponed.

---

[a]Lecture timestamp: 01:29:43.

Return now to the moving target. Imagine a wall moving toward a tennis ball. The wall can give energy to the ball. If the wall's motion is prescribed and its energy is omitted, the ball's energy alone is not conserved. Mathematically this is time dependence in the target position rather than necessarily in g.

Creation and annihilation operators have therefore done exactly what was wanted: they describe a transition from one particle state to another, while the spacetime dependence of the fields determines which conservation conditions accompany that transition.

4.8 Charge Counting and Phase Invariance

Each particle species has its own field. Electrons and photons, for example, are not described by the same ψ. To expose charge counting with the algebra already available, temporarily let ψ stand for an electron-like field: ψ annihilates one such charged particle and $\psi^\dagger$ creates one. This is a bosonic stand-in for the bookkeeping; the genuine fermionic electron field requires algebra not yet introduced.

The fixed-target factor

$$\psi^\dagger\psi \tag{4.67}$$

annihilates one particle and creates one. One electron comes in and one goes out, so the charge carried by this species does not change.

Now imagine one electron entering and two leaving from the same point. The corresponding operator product would contain one annihilation field and two creation fields,

$$\psi^\dagger\psi^\dagger\psi. \tag{4.68}$$

For an electron field this would create one net electron and would violate electric charge conservation. Two electrons in and two out can instead be represented by

$$\psi^\dagger\psi^\dagger\psi\psi, \tag{4.69}$$

which has equal numbers of creation and annihilation fields. Two in and three out would require

$$\psi^\dagger\psi^\dagger\psi^\dagger\psi\psi, \tag{4.70}$$

and again changes the net charge. The immediate rule is simple: a charge-conserving interaction contains the same number of ψ's and $\psi^\dagger$'s.

---

[a] **Question from the audience.** Does the order of the field operators matter for this charge-conservation test?

Response. The immediate test concerns the number of ψ and $\psi^\dagger$ factors, not their order. Whether differently ordered products are otherwise identical is a separate question.

---

[a]Lecture timestamp: 01:34:27.

There is a mathematical diagnostic deeper than merely counting. Multiply the field by a constant phase:

$$\psi \longrightarrow e^{i\alpha}\psi. \tag{4.71}$$

Because $\psi^\dagger$ is the Hermitian conjugate, it transforms with the conjugate phase,

$$\psi^\dagger \longrightarrow e^{-i\alpha}\psi^\dagger. \tag{4.72}$$

An expression with equal numbers of the two fields is unchanged. For example,

$$\psi^\dagger\psi \longrightarrow e^{-i\alpha}e^{i\alpha}\psi^\dagger\psi = \psi^\dagger\psi, \tag{4.73}$$

$$\psi^\dagger\psi^\dagger\psi\psi \longrightarrow e^{-2i\alpha}e^{2i\alpha}\psi^\dagger\psi^\dagger\psi\psi = \psi^\dagger\psi^\dagger\psi\psi. \tag{4.74}$$

An expression with unequal numbers changes. Thus

$$\psi^\dagger\psi^\dagger\psi \longrightarrow e^{-i\alpha}\psi^\dagger\psi^\dagger\psi. \tag{4.75}$$

The allowed charge-conserving interactions are precisely the ones invariant under this global change of phase. Phase invariance and charge conservation are two descriptions of the same rule in this simple field language.

The combination

$$\psi + \psi^\dagger \tag{4.76}$$

is not invariant under an arbitrary phase rotation.

---

[a] **Question from the audience.** Isn't that the real part of ψ?

Response. Yes. It is proportional to the real part of ψ, and multiplying ψ by a phase changes that real part. It is therefore not phase invariant.

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[a]Lecture timestamp: 01:37:53.

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[a] **Question from the audience.** If an electron strikes a target and many secondary electrons are emitted, how can charge still be conserved?

Response. The target is not passive in that process. If an atom emits many electrons, the atom itself changes charge and state. Its dynamics must be included, and the target must also be described by quantum fields.

---

[a]Lecture timestamp: 01:38:05.

At this point three correspondences are visible. Time-translation invariance corresponds to energy conservation. Global phase invariance corresponds to charge conservation. Spatial-translation invariance corresponds to momentum conservation, although the full spatial calculation remains to be carried out.

4.9 Operator Order and the Limits of the Lowest-Order Model

For simplicity, the products above place all creation fields to the left and all annihilation fields to the right. An interleaved product can still have equal numbers and therefore pass the charge-counting test, but operator order can affect the expression in other ways.

---

[a] **Question from the audience.** Could the same kind of process be written with interleaved factors such as $\psi\psi^\dagger\psi\psi^\dagger$?

Response. It still has equal numbers and so satisfies the charge rule, but it is not generally the same expression as one with all daggers to the left. The rearrangement of different orderings is postponed.

---

[a]Lecture timestamp: 01:40:02.

[8]

These links between symmetries and conservation laws are instances of Noether's theorem. In classical mechanics that theorem gives the general framework; here the same connection has been displayed directly through the phases and delta functions of the quantum fields.

The leading fixed-target result also has an obvious domain of validity. A schematic probability prefactor such as $4\pi^2 g^2$ cannot be trusted once it approaches or exceeds unity.

---

[a] **Question from the audience.** Does the schematic result mean that g must be smaller than roughly $1/(2\pi)$?

Response. Either g is small enough for this first term to be useful, or the model is incomplete. A particle can scatter once, return and scatter twice, then three times, and so on. The full amplitude is an infinite series. Only for small g is the simplest term dominant; otherwise many Feynman graphs must be summed.

---

[a] Lecture timestamp: 01:43:33.

The simple spacetime pictures are the beginnings of Feynman graphs. Their vertices specify elementary possibilities: one particle in and one out corresponds to a factor such as $\psi^\dagger\psi$; two in and two out corresponds to a factor such as $\psi^\dagger\psi^\dagger\psi\psi$. Repeated vertices generate higher powers of the coupling and the higher-order terms in the series.

---

[a] **Question from the audience.** Can a coupling constant have an imaginary component?

Response. Complex coupling constants can occur. Irreducible T-odd phase combinations can signal violation of time-reversal invariance: the time-reversed process need not carry the symmetry-related amplitude.

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[a] Lecture timestamp: 01:45:39.

[9]

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[a] **Question from the audience.** Can the outgoing particle be emitted before or after the incoming particle is absorbed, rather than at the same instant?

Response. That is a fair question, but the instantaneous model does not address it. The more general time ordering is deferred.

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[a] Lecture timestamp: 01:47:02.

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[a] **Question from the audience.** How is this construction made relativistic?

Response. The brief indication is to combine the conjugate pieces and include a denominator involving a square root of $\omega(k)$. The complete construction is postponed.

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[a] Lecture timestamp: 01:47:44.

[10]

[8] Editorial clarification: The wording of an audience question at 01:41:10 is too damaged to reconstruct. The recoverable response repeats that equal numbers of ψ and $\psi^\dagger$ imply phase invariance and charge conservation, whereas unequal numbers imply neither.

[9] Editorial clarification: The QCD θ parameter was mentioned in this exchange, but it is a real coefficient multiplying a T-odd interaction, not an imaginary component of a coupling. A removable phase or an absorptive complex phase does not by itself establish T violation. The inference applies only to irreducible T-odd phase combinations.

[10] Editorial clarification: The brief spoken relativistic formula is not recoverable securely enough to typeset. Only its stated ingredients and deferral are retained.

Charge conservation must also be distinguished sharply from conservation of rest mass.

---

[a] **Question from the audience.** Is rest mass conserved in the same structural sense as electric charge?

Response. No. In Galilean nonrelativistic mechanics, mass is an additive conserved quantity. In relativistic reactions the sum of the individual particles' rest masses need not be conserved. An electron and a positron can annihilate into two photons while electric charge remains conserved.

---

[a]Lecture timestamp: 01:48:22.

11

Conservation of additive mass is therefore an effective nonrelativistic fact, not a fundamental rule on the same footing as electric charge conservation. The phase invariance found above expresses the deeper charge symmetry.

The next steps are to replace the bosonic stand-in by fermionic fields and to allow several particle species—electrons, muons, quarks, photons, and others—each with its own field and its own allowed interactions.

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[a] **Question from the audience.** Was it legitimate to call the earlier example an electron field when only bosonic fields have been developed so far?

Response. Only as a bosonic version or stand-in for an electron. The genuine fermionic theory has not yet been introduced.

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[a]Lecture timestamp: 01:50:39.

[11]Editorial clarification: For a complete closed relativistic system, total four-momentum is conserved, so the invariant mass of that system is fixed. The annihilation example concerns the sum of the individual particles' rest masses, not the invariant mass of the closed system.

Chapter 5

Group Velocity, Fermionic Exclusion, and the Dirac Sea

5.1 Phase Velocity and Group Velocity

[a] **Question from the audience.** What are group velocity and phase velocity?

Response. Phase velocity follows a point of fixed phase in a plane wave. It generally does not describe the motion of a measurable object. Group velocity follows the envelope of a wave packet and is the velocity associated with particles, signals, and probability flow.

[a]Lecture timestamp: 00:00:07.

Begin with a plane wave,

$$\psi(x,t) = e^{i(kx-\omega t)}, \tag{5.1}$$

or with its real form,

$$\psi(x,t) = \sin(kx - \omega t). \tag{5.2}$$

A crest, a trough, or any other point of fixed phase moves so that

$$kx - \omega t = \text{constant}. \tag{5.3}$$

Choosing the constant to be zero for convenience gives

$$x = \frac{\omega}{k}t, \qquad v_{\text{phase}} = \frac{\omega}{k}. \tag{5.4}$$

Since $\omega = 2\pi/T$ and $k = 2\pi/\lambda$,

$$v_{\text{phase}} = \frac{\omega}{k} = \frac{\lambda}{T}. \tag{5.5}$$

This is just the distance by which a given phase advances during one period.

For a nonrelativistic Schrödinger wave, with $\hbar = 1$, energy and momentum are ω and k, so

$$\omega(k) = \frac{k^2}{2m}. \tag{5.6}$$

Its phase velocity is therefore

$$v_{\text{phase}} = \frac{k}{2m}. \tag{5.7}$$

The ordinary velocity of a particle with momentum k, however, is k/m. The factor-of-two mismatch is the first warning that the movement of a plane-wave phase is not the movement of the particle.

There is an even sharper warning. Add the same constant C to every frequency:

$$\omega(k) \longrightarrow \omega(k) + C. \tag{5.8}$$

The constant could, for example, represent a common rest-energy contribution such as mc_{light}^2, but its particular name is unimportant. Here C is an energy offset, not the speed of light. The phase velocity changes to

$$v_{\text{phase}} \longrightarrow \frac{\omega(k) + C}{k} = v_{\text{phase}} + \frac{C}{k}. \tag{5.9}$$

Yet adding the same constant to every energy changes no energy difference and hence no nonrelativistic physics.

This can be seen directly in the wavefunction. Write a general Schrödinger field as a sum or integral of modes,

$$\psi(x,t) = \sum_k a_k e^{ikx} e^{-i\omega(k)t}, \tag{5.10}$$

with normalization factors suppressed. After the shift,

$$\psi(x,t) \longrightarrow \sum_k a_k e^{ikx} e^{-i[\omega(k)+C]t} \tag{5.11}$$

$$= e^{-iCt}\psi(x,t). \tag{5.12}$$

Every mode has acquired the same time-dependent phase. The conjugate field acquires the opposite phase, and therefore

$$\begin{aligned} \psi^*(x,t)\psi(x,t) &\longrightarrow e^{iCt}e^{-iCt}\psi^*(x,t)\psi(x,t) \\ &= \psi^*(x,t)\psi(x,t). \end{aligned} \tag{5.13}$$

The same cancellation occurs in expectation values. A quantity that changes when the zero of energy is shifted cannot be the physical propagation speed of the Schrödinger particle. Phase velocity is therefore partly conventional.

A localized particle is represented not by one infinite plane wave but by a packet. The component waves add constructively in the region of the packet and destructively outside it. The motion of this pattern can already be found from two nearby waves:

$$\psi(x,t) = \sin(kx - \omega t) + \sin(k'x - \omega' t), \tag{5.14}$$

where k' is close to k and $\omega' = \omega(k')$. A locus of constructive interference occurs where their phases agree,

$$kx - \omega t = k'x - \omega' t. \tag{5.15}$$

Thus

$$(k - k')x = (\omega - \omega')t, \tag{5.16}$$

and the reinforcing pattern moves according to

$$x = \frac{\omega - \omega'}{k - k'}t. \tag{5.17}$$

Taking the neighboring-wave limit gives

$$x = \frac{d\omega}{dk} t, \qquad v_{\text{group}} = \frac{d\omega}{dk}. \tag{5.18}$$

This is the velocity of the envelope—the localized lump produced by the interference of the component waves.

For the nonrelativistic dispersion relation, including the arbitrary offset,

$$\omega(k) = \frac{k^2}{2m} + C, \tag{5.19}$$

the group velocity is

$$v_{\text{group}} = \frac{d}{dk}\left(\frac{k^2}{2m} + C\right) = \frac{2k}{2m} = \frac{k}{m}. \tag{5.20}$$

The additive constant has disappeared, and the result is precisely the classical particle velocity.

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[a] **Question from the audience.** Does the factor of two disappear in the group velocity?

Response. Yes. Differentiating k^2 produces $2k$, which cancels the two in $2m$. The result is k/m.

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[a]Lecture timestamp: 00:16:35.

Group velocity is consequently the velocity most closely associated with classical particle motion. It is also the velocity relevant to the propagation of a packet and, in the ordinary circumstances considered here, to the propagation of signals. Phase velocity need not have either meaning.

5.1.1 Massless and Massive Relativistic Waves

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[a] **Question from the audience.** Does this have to do with claims that light can travel faster than light?

Response. No. Those claims raise a separate issue. A related calculation does show that a phase can move faster than light while the physical group velocity remains subluminal, but that does not describe a signal outrunning light.

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[a]Lecture timestamp: 00:17:47.

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[a] **Question from the audience.** Is the relevant example the familiar one in which phase velocity exceeds the speed of light while group velocity remains below it?

Response. Yes. The same pattern appears for a relativistic wave with nonzero mass, and it follows directly from the relativistic energy–momentum relation.

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[a]Lecture timestamp: 00:18:05.

Set $\hbar = c_{\text{light}} = 1$. The relativistic relation is

$$E = \sqrt{p^2 + m^2}, \qquad \omega(k) = \sqrt{k^2 + m^2}. \tag{5.21}$$

It is useful to establish the massless case first. When $m = 0$,

$$\omega = |k|. \tag{5.22}$$

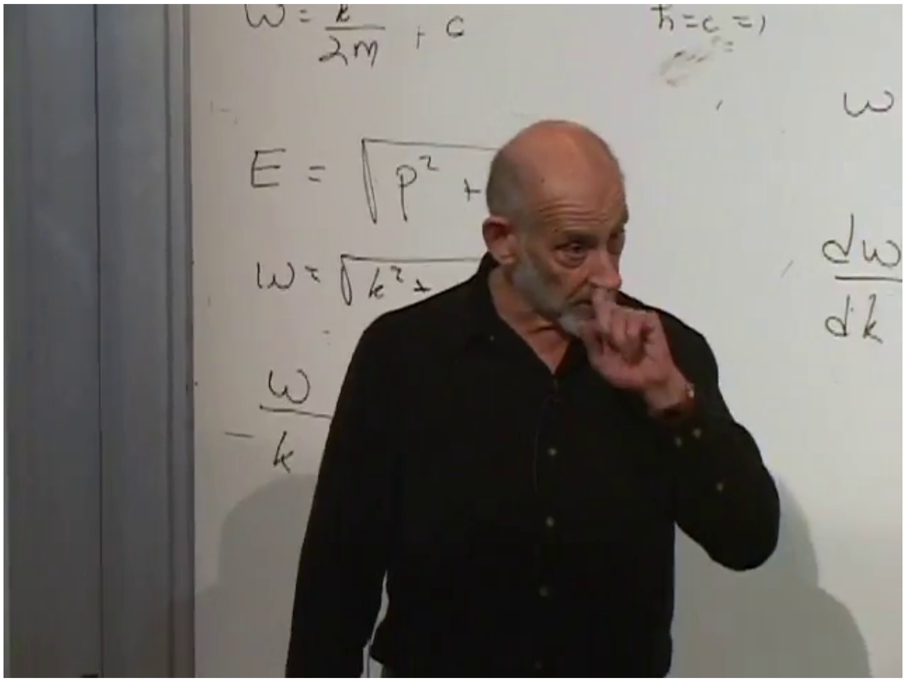

Figure 5.1: Nonrelativistic and relativistic dispersion relations, with phase velocity ω/k and group velocity $d\omega/dk$.

Lecture frame: 00:35:52.

On the right-moving branch $k > 0$, $\omega = k$, and therefore

$$v_{\text{phase}} = \frac{\omega}{k} = 1, \qquad v_{\text{group}} = \frac{d\omega}{dk} = 1. \tag{5.23}$$

On the left-moving branch both velocities have the corresponding negative sign. In either case their speed is the speed of light. Because the dispersion relation is linear on either branch, all Fourier components on that branch move with the same velocity.

For a massive particle with $k > 0$,

$$v_{\text{phase}} = \frac{\omega}{k} = \sqrt{\frac{k^2 + m^2}{k^2}} > 1, \tag{5.24}$$

whereas

$$v_{\text{group}} = \frac{d\omega}{dk} = \frac{k}{\sqrt{k^2 + m^2}} < 1. \tag{5.25}$$

For $k < 0$, both velocities reverse sign, while their magnitudes obey the same inequalities. Thus the phase can move faster than light while the packet, particle, energy, and signal move more slowly than light. For this particular dispersion relation and $k \neq 0$,

$$v_{\text{phase}} v_{\text{group}} = 1. \tag{5.26}$$

That inverse relation is an accident of the square-root dispersion relation, not a general property of waves.

[a] **Question from the audience.** Is dispersion tied to the wave speed depending on frequency?

Response. Yes. When $\omega(k)$ is linear, every Fourier component has the same group velocity and a packet can retain its shape. When $\omega(k)$ is nonlinear, different components move at different velocities, so the packet disperses and changes shape. Low-frequency sound is approximately nondispersive when its frequency is nearly proportional to wave number.

[a] Lecture timestamp: 00:23:41.

Phase velocity ordinarily carries neither signal nor energy nor probability flow. Its numerical value can even be changed by shifting the zero of energy. The group velocity is the physically relevant motion of a sufficiently narrow packet.

[a] **Question from the audience.** Can phase velocity be measured for a photon?

Response. In vacuum a photon has linear dispersion, so its phase and group velocities coincide and both equal the speed of light. In a material medium the dispersion relation changes, and phase and group velocities need not agree.

[a] Lecture timestamp: 00:25:14.

[a] **Question from the audience.** Is this discussion restricted to wave packets that do not spread with time?

Response. A packet remains nonspreading only when all of its Fourier components share one velocity, which corresponds to a linear dispersion relation over the relevant range. With nonlinear dispersion the components separate and the packet spreads.

[a] Lecture timestamp: 00:25:50.

Questions about light cones and relativistic causal structure require more than the distinction between phase and group velocity and are left separate from this calculation.

5.2 Translation Symmetry and Conservation Laws

The same plane-wave phases give a compact account of conservation laws. Begin with energy conservation. An annihilation field acting on an incoming particle of energy ω_j contributes a time factor

$$e^{-i\omega_j t}, \tag{5.27}$$

whereas a creation field producing an outgoing particle of energy ω'_r contributes

$$e^{+i\omega'_r t}. \tag{5.28}$$

For several incoming and outgoing particles, their product is

$$\exp\left[i\left(\sum_r \omega'_r - \sum_j \omega_j\right)t\right]. \tag{5.29}$$

If the interaction may occur at any time with the same amplitude, the amplitude includes an integral over the interaction time:

$$\begin{aligned}\int_{-\infty}^{\infty} dt\, \exp\left[i\left(\sum_r \omega'_r - \sum_j \omega_j\right)t\right] \\ = 2\pi\,\delta\left(\sum_r \omega'_r - \sum_j \omega_j\right).\end{aligned} \tag{5.30}$$

The delta function requires total outgoing energy to equal total incoming energy. Equal amplitude at every possible interaction time is time-translation invariance, and it produces energy conservation.

Space translation gives momentum conservation by the same mechanism. Consider a physical one-to-two decay in which an incoming particle of momentum k disappears and two particles of momenta k'_1 and k'_2 appear. A local interaction at position x contains one annihilation field and two creation fields,

$$\mathcal{O}(x) = \psi^\dagger(x)\psi^\dagger(x)\psi(x). \tag{5.31}$$

Suppressing time dependence and normalization, the relevant mode factors are

$$\psi(x) \supset a_k e^{ikx}, \tag{5.32}$$

$$\psi^\dagger(x) \supset a^\dagger_{k'_1} e^{-ik'_1 x}, \qquad \psi^\dagger(x) \supset a^\dagger_{k'_2} e^{-ik'_2 x}. \tag{5.33}$$

In the matrix element

$$\langle k'_1, k'_2|\mathcal{O}(x)|k\rangle, \tag{5.34}$$

the annihilation operator removes the incoming particle and the two creation operators supply the outgoing particles. Once those operator actions are taken, the position-dependent factor that remains is

$$e^{-ik'_1 x}e^{-ik'_2 x}e^{ikx} = e^{i(k-k'_1-k'_2)x}. \tag{5.35}$$

Because no position is privileged, one integrates over all possible decay positions:

$$\int_{-\infty}^{\infty} dx\, e^{i(k-k'_1-k'_2)x} = 2\pi\delta(k-k'_1-k'_2). \tag{5.36}$$

The surviving amplitude therefore obeys

$$k = k'_1 + k'_2. \tag{5.37}$$

In three dimensions the same calculation uses d^3x, bold momentum vectors, and a three-dimensional delta function. Nothing in the argument depends on having only one incoming particle or two outgoing particles. For arbitrary particle numbers, integration over the common interaction position produces

$$\delta^{(3)}\left(\sum_j \mathbf{k}_j - \sum_r \mathbf{k}'_r\right). \tag{5.38}$$

Equal amplitude at every position is spatial translation invariance, and it gives momentum conservation just as equal amplitude at every time gives energy conservation.

A third example is a global phase rotation,

$$\psi \longrightarrow e^{i\lambda}\psi, \qquad \psi^\dagger \longrightarrow e^{-i\lambda}\psi^\dagger. \tag{5.39}$$

An interaction amplitude is invariant when the phases contributed by its annihilation and creation fields cancel. For a field carrying electric charge, this equality of charge removed from the initial state and restored in the final state is charge conservation. Thus time translations, space translations, and global phase rotations illustrate how fields that create and annihilate particles encode energy, momentum, and charge conservation.

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[a] **Question from the audience.** Does integration over space mean integration over all three spatial dimensions?

Response. Yes. The three integrations produce a delta function for the three momentum components, so all three components of total momentum are conserved.

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[a]Lecture timestamp: 00:37:52.

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[a] **Question from the audience.** Does this model determine the relative probabilities for different ways of sharing the conserved momentum?

Response. No. This simple expression enforces the sum of the outgoing momenta but is otherwise neutral about their division. Relative probabilities require additional dynamical information.

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[a]Lecture timestamp: 00:38:03.

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[a] **Question from the audience.** Why does a sum of annihilation operators reduce to the term matching the occupied momentum?

Response. An annihilation operator acting on an empty mode gives the zero vector. If the state contains one particle only in the mode k_0, every term except the one with $k = k_0$ vanishes.

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[a]Lecture timestamp: 00:39:04.

For example,

$$\sum_k a_k e^{ikx}|1_{k_0}\rangle = e^{ik_0x}|0\rangle, \tag{5.40}$$

up to normalization: all annihilators associated with empty modes return the zero vector.

5.3 Bosonic Enhancement and Fermionic Exclusion

Bosons permit any nonnegative occupation number in a one-particle state, and their creation-operator algebra actually favors an already occupied state. For one bosonic mode, the normalized number-state convention writes the occupation enhancement as[1]

$$a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle. \tag{5.41}$$

Suppose an excited atom can emit a photon into many possible modes. In the absence of other photons, the different directions and momenta occur with their ordinary probability distribution. If a particular mode already contains many photons—all with the same momentum, energy, and frequency—emission into that mode is enhanced by the corresponding square-root occupation factor.

[1]The lecture states the square-root occupation enhancement; this indexed equation is its standard number-state form.

The atom preferentially adds its photon to the state already populated. This is stimulated emission and is a concrete expression of bosonic congregation.

Fermions behave in the opposite way. Two identical fermions cannot occupy the same quantum state. The exclusion principle first arose in atomic physics, where two electrons in an atom cannot have identical quantum numbers, but it is not confined to atoms: identical fermions cannot share a complete one-particle state anywhere.

The field expansion initially looks just like the bosonic one. In a finite box one may write

$$\psi(x) = \sum_k A_k e^{ikx}, \qquad \psi^\dagger(x) = \sum_k A_k^\dagger e^{-ikx}. \tag{5.42}$$

For continuous momentum the sums become integrals. The opposite sign in the conjugate spatial phase follows from Hermitian conjugation. This part of the construction is nonrelativistic; relativistic fermions will enter only at the end.

The state of the field is still recorded by an occupation number for every mode, but now each number can only be zero or one:

$$|n_{k_1}, n_{k_2}, \ldots\rangle, \qquad n_k \in \{0, 1\}. \tag{5.43}$$

There are no additional relativistic particle types with, say, at most three particles per state or at least seven particles per state. Establishing that bosonic and fermionic statistics are the only possibilities requires quantum mechanics together with special relativity and lies beyond the present discussion. Here the task is to describe the fermionic possibility and its consequences.

5.3.1 One Fermionic State

Focus on a single one-particle state. It is either empty, $|0\rangle$, or occupied, $|1\rangle$. Creation and annihilation act as

$$A^\dagger|0\rangle = |1\rangle, \qquad A^\dagger|1\rangle = 0, \tag{5.44}$$

$$A|0\rangle = 0, \qquad A|1\rangle = |0\rangle. \tag{5.45}$$

The unadorned zero on the right is the zero vector, not the empty state $|0\rangle$. Attempting to add a second fermion does not return the vacuum; it returns no physical state at all.

---

[a] **Question from the audience.** Are the fermionic creation and annihilation operators still linear?

Response. Yes. Quantum-mechanical operators are linear operators, including the fermionic creation and annihilation operators.

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[a]Lecture timestamp: 00:46:25.

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[a] **Question from the audience.** Does zero mean the empty state, or does it mean that the attempted operation cannot produce a state?

Response. It means the zero vector, not the empty state. A genuine vacuum or ground state has a nonzero wavefunction; the identically zero function has zero norm and is not a physical state.

---

[a]Lecture timestamp: 00:47:34.

The harmonic oscillator makes the distinction concrete. Its ground state, the state with no oscillator excitation, has a Gaussian wavefunction such as

$$\phi_0(q) \propto e^{-\alpha q^2}, \qquad \alpha > 0. \tag{5.46}$$

This is a perfectly good, normalizable state. By contrast,

$$\phi(q) \equiv 0 \tag{5.47}$$

is the zero vector: it has no probability amplitude anywhere and is not a state. Thus $A^\dagger|1\rangle = 0$ says that the attempted second creation produces the zero vector.

---

[a] **Question from the audience.** Should fermion operators use notation different from boson operators?
Response. Yes. It is clearer to reserve a and $a^\dagger$ for bosonic modes and use c and $c^\dagger$ for fermionic modes. Both conventions occur in the literature, but different letters keep the two algebras distinct.

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[a]Lecture timestamp: 00:48:42.

Adopting that notation, the single-mode rules are

$$c^\dagger|0\rangle = |1\rangle, \qquad c^\dagger|1\rangle = 0, \tag{5.48}$$
$$c|0\rangle = 0, \qquad c|1\rangle = |0\rangle. \tag{5.49}$$

Now evaluate the two possible products on both basis states. On the empty state,

$$c^\dagger c|0\rangle = 0, \qquad cc^\dagger|0\rangle = |0\rangle. \tag{5.50}$$

On the occupied state,

$$c^\dagger c|1\rangle = |1\rangle, \qquad cc^\dagger|1\rangle = 0. \tag{5.51}$$

Their sum returns whichever state it acts on:

$$(c^\dagger c + cc^\dagger)|0\rangle = |0\rangle, \tag{5.52}$$
$$(c^\dagger c + cc^\dagger)|1\rangle = |1\rangle. \tag{5.53}$$

Since $|0\rangle$ and $|1\rangle$ span the one-mode state space,

$$c^\dagger c + cc^\dagger = 1. \tag{5.54}$$

The plus-sign combination is the anticommutator,

$$\{A, B\} = AB + BA, \tag{5.55}$$

so the fermionic relation is

$$\{c, c^\dagger\} = 1. \tag{5.56}$$

It is the fermionic analog of the bosonic commutator

$$[a, a^\dagger] = aa^\dagger - a^\dagger a = 1. \tag{5.57}$$

Two creations in the same fermionic state always vanish. If the state is empty, the first creation fills it and the second fails; if it is already occupied, the first creation already fails. Similarly, two annihilations always vanish. Therefore

$$\{c^\dagger, c^\dagger\} = 2(c^\dagger)^2 = 0, \qquad \{c, c\} = 2c^2 = 0. \tag{5.58}$$

---

[a] **Question from the audience.** Is the zero in the squared-annihilation relation a zero vector?

Response. The equation states that c^2 is the zero operator. It is an operator that sends every state vector to the zero vector. The zero operator and the zero vector are different mathematical objects, although the former always produces the latter.

---

[a]Lecture timestamp: 00:56:45.

For many discrete momentum modes, the full algebra is

$$\{c_k, c_{k'}^\dagger\} = \delta_{kk'}, \tag{5.59}$$

$$\{c_k, c_{k'}\} = 0, \tag{5.60}$$

$$\{c_k^\dagger, c_{k'}^\dagger\} = 0. \tag{5.61}$$

The Kronecker delta is one when $k = k'$ and zero otherwise. For continuous momenta it is replaced, with the appropriate normalization, by a Dirac delta function. The same-type anticommutators vanish for every pair of modes, not only when their labels coincide.

5.3.2 Exclusion in Position Space

The momentum-labeled algebra first says that two identical fermions of the same species cannot occupy one momentum mode. It says more. A particle localized at position x is created by

$$\psi^\dagger(x) = \sum_k c_k^\dagger e^{-ikx}, \tag{5.62}$$

up to normalization. At the origin, trying to create two particles gives

$$\psi^\dagger(0)\psi^\dagger(0)|\Omega\rangle = \sum_{k,k'} c_k^\dagger c_{k'}^\dagger |\Omega\rangle. \tag{5.63}$$

The labels k and k' are dummy summation variables. Interchanging them leaves the double sum unchanged, so it may be averaged with its relabeled form:

$$\sum_{k,k'} c_k^\dagger c_{k'}^\dagger |\Omega\rangle = \frac{1}{2}\sum_{k,k'} \left(c_k^\dagger c_{k'}^\dagger + c_{k'}^\dagger c_k^\dagger\right) |\Omega\rangle \tag{5.64}$$

$$= \frac{1}{2}\sum_{k,k'} \{c_k^\dagger, c_{k'}^\dagger\} |\Omega\rangle = 0. \tag{5.65}$$

---

[a] **Question from the audience.** Do only terms with unequal momenta vanish in the two-particle position-state sum?

Response. No. Relabeling the dummy indices turns the whole double sum into one half of the anticommutator of two creation operators. That anticommutator vanishes for every pair of momenta, so the complete sum is zero.

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[a]Lecture timestamp: 01:02:26.

The calculation shows that exclusion is not peculiar to momentum eigenstates. Identical fermions cannot be placed in the same position state either. More generally, they cannot occupy any identical one-particle state.

5.3.3 What Constitutes the Same State?

[a] **Question from the audience.** Can two far-separated electrons have the same momentum?

Response. Exclusion concerns the complete one-particle state. Two electrons can have the same exact momentum eigenvalue when another commuting label, such as spin, differs. Spatially separated electrons are instead localized packets rather than exact momentum eigenstates; their mean momenta can be nearly equal while their spatial wavefunctions remain distinct. What is forbidden is equality of the complete states.

[a]Lecture timestamp: 01:04:30.

Spin has not yet been developed, but it supplies a familiar additional label. Two electrons can share the same momentum value when their spin states differ. Once every relevant label is included, no two identical fermions may share the same full state.

[a] **Question from the audience.** Can electrons in two separated hydrogen atoms have the same momentum or atomic state?

Response. An electron bound in an atom is not a momentum eigenstate. Its orbital wavefunction is centered on that atom. Two separated atoms may each have an electron in what is called the atomic ground state, but one orbital is centered at the first atom and the other at the second, so they are different one-particle states.

[a]Lecture timestamp: 01:05:25.

[a] **Question from the audience.** Can the more general word system replace state?

Response. No. A system is a collection of degrees of freedom; a state specifies a particular quantum configuration of that system. An electron is a system, while its orbital and spin wavefunction describes a state of that system.

[a]Lecture timestamp: 01:06:33.

[a] **Question from the audience.** Does an electron stream in a cathode-ray tube contain no two electrons in the same state?

Response. Yes. The electrons occupy distinct wave packets, separated in position or otherwise distinguished in their full states. This contrasts with photons, which may all occupy the same plane-wave mode and build a coherent beam.

[a]Lecture timestamp: 01:07:22.

An electron beam can be pictured as one localized packet followed by another. The packets need not be exact position states or exact momentum states; it is enough that their wavefunctions are distinct. A photon beam, by contrast, may have many photons in precisely the same plane-wave state.

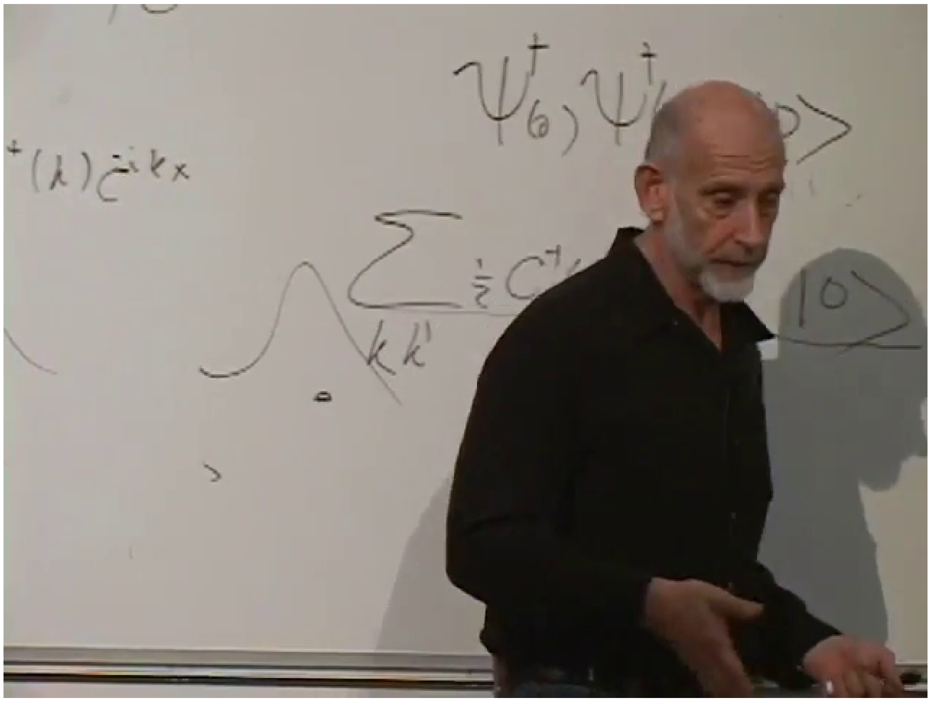

Figure 5.2: Localized one-particle wavefunctions and the two-creation-operator state used to test fermionic exclusion.

Lecture frame: 01:07:21.

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[a] **Question from the audience.** Can two electrons be at the same position with different momenta?

Response. Exact position and exact momentum are incompatible labels because the corresponding operators do not commute. One cannot specify a one-particle state by simultaneously sharp position and momentum. Spin can accompany a position or momentum label because spin commutes with those observables in this nonrelativistic setting.

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[a] Lecture timestamp: 01:08:14.

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[a] **Question from the audience.** Are separated bound electrons approximately in position states?

Response. They are not exact position eigenstates, because each orbital has finite spatial extent, but each is approximately localized near its own atom. That localization is enough to make the two orbital wavefunctions different states.

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[a] Lecture timestamp: 01:08:51.

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[a] **Question from the audience.** What is the definition of a quantum state?

Response. A state is the quantum analog of a system's configuration. It contains the information needed to assign probabilities to the possible measurements. For one electron, the state is represented by a wavefunction, together with any internal labels such as spin.

---

[a] Lecture timestamp: 01:09:31.

[a] **Question from the audience.** Can electrons localized in very distant places be in the same state or have the same momentum?

Response. They cannot be the same state because their localization differs: an electron here and one in the Andromeda Galaxy, or one in Nevada and one in Connecticut, have distinct wavefunctions. They may have nearly equal momenta. Making either momentum infinitely sharp would spread the corresponding wavefunction through space and destroy the localization that distinguished the states.

[a]Lecture timestamp: 01:10:32.

The standard-deviation form of the position–momentum uncertainty relation is[2]

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}. \tag{5.66}$$

A packet can be localized in Nevada and have a fairly narrow momentum distribution, while another packet in Connecticut has nearly the same distribution. They remain different states because their spatial wavefunctions differ. In the exact momentum-eigenstate limit, neither packet remains localized in either place.

Fermion operators are used in reactions in the same way as bosonic operators. An electron annihilation operator removes an incoming electron, an electron creation operator supplies an outgoing electron, and these factors can be multiplied by photon creation or annihilation operators. Photons are bosons, while electrons are fermions. Products of their operators provide the bookkeeping for reactions involving incoming and outgoing electrons together with emitted or absorbed photons.

[a] **Question from the audience.** Is Pauli exclusion locally communicated, or is it globally enforced when two creation processes are spatially separated?

Response. If one particle is known to be in Nevada and the other in Connecticut, then they are not in the same state.

[a]Lecture timestamp: 01:13:36.

[3]

When their locations are not definite, exclusion asks whether the full one-particle wavefunctions coincide. The Nevada–Connecticut superpositions make that distinction explicit.

Let $|N\rangle$ and $|C\rangle$ be orthonormal states localized in Nevada and Connecticut. Then

$$|+\rangle = \frac{|N\rangle + |C\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|N\rangle - |C\rangle}{\sqrt{2}}. \tag{5.67}$$

The state $|+\rangle$ gives equal probabilities of finding its one electron in Nevada or Connecticut. The state $|-\rangle$ gives the same position probabilities but differs by a relative phase. The two states are orthogonal:

$$\langle +|-\rangle = 0. \tag{5.68}$$

Two electrons cannot both occupy $|+\rangle$, and they cannot both occupy $|-\rangle$, but exclusion does not prevent one electron from occupying each state.

[2]The lecture gives the qualitative uncertainty-product statement; the factor $\hbar/2$ is supplied in the standard-deviation convention.

[3]Editorial clarification: A short portion of the reply to this question is not recoverable. The distinction between localized one-particle states and the Nevada–Connecticut superpositions is secure.

[a] **Question from the audience.** Can one electron occupy the plus Nevada–Connecticut superposition and another the minus superposition?

Response. Yes. The plus and minus combinations are orthogonal and therefore distinct one-particle states.

[a]Lecture timestamp: 01:16:25.

[a] **Question from the audience.** Does the separated-state example require a notion of simultaneity?

Response. The present model is nonrelativistic. It compares the two occupied states on one common time slice, with an absolute time shared by the spatially separated Nevada and Connecticut regions. It also ignores finite-speed operational limitations on comparing those regions. Relativistic physics introduces additional fuzziness in how particles are localized without undoing exclusion, but that refinement is postponed here.

[a]Lecture timestamp: 01:16:39.

The essential nonrelativistic statement is now fixed: fermions are described by anticommuting creation and annihilation operators, and no two identical fermions can occupy one complete one-particle state. The many-body ground state therefore differs radically from that of bosons.

5.4 Bosonic and Fermionic Ground States in a Finite Box

Consider particles confined to a finite three-dimensional box of side L, with periodic boundary conditions. In one dimension the allowed wave numbers are

$$k = \frac{2\pi n}{L}, \qquad n \in \mathbb{Z}. \tag{5.69}$$

In three dimensions this condition applies component by component:

$$\mathbf{k} = \frac{2\pi}{L}(n_x, n_y, n_z), \qquad n_x, n_y, n_z \in \mathbb{Z}. \tag{5.70}$$

The allowed momentum vectors therefore form a lattice in momentum space. This is not a lattice of positions; each point labels an allowed momentum state. Increasing L decreases the separation $2\pi/L$ between neighboring values, while the lattice continues without bound in every momentum direction.

For a nonrelativistic particle,

$$E_{\mathbf{k}} = \frac{\mathbf{k}^2}{2m}, \tag{5.71}$$

so the state of lowest energy is $\mathbf{k} = 0$. Put N identical bosons in the box and seek the many-particle ground state. The first boson goes into the zero-momentum state. The second can go into the same state, as can the third and every subsequent boson. Thus all N bosons occupy $\mathbf{k} = 0$.

This is a boson condensate: a large number of identical particles occupying the same one-particle state, here the lowest-energy state. The zero-momentum wavefunction is spatially constant and extends throughout the box. It does not localize any one boson at a particular position; each boson is correspondingly uncertain about where it is. When the occupation becomes large, the common wavefunction builds up a macroscopic field amplitude, and the field behaves increasingly like a classical field.

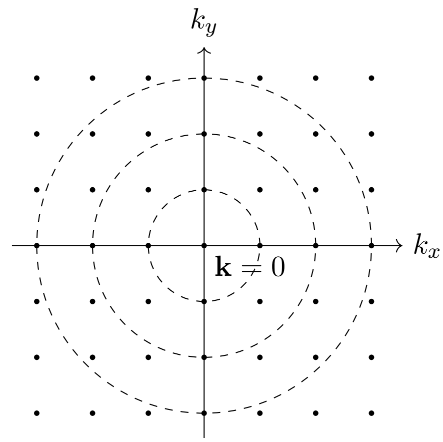

Figure 5.3: Discrete momentum states and constant-energy shells for particles in a finite periodic box.

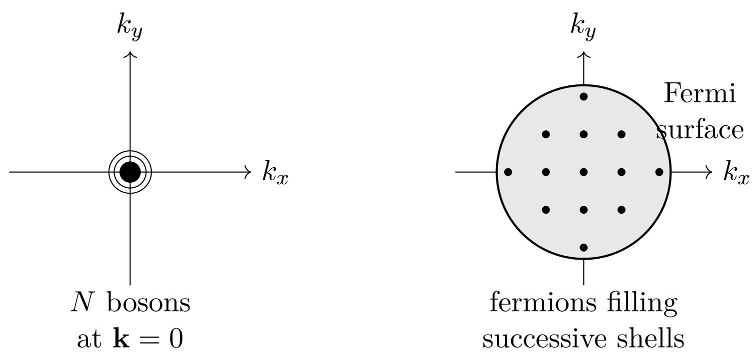

Figure 5.4: Bosons condense into the zero-momentum state, whereas fermions fill momentum states outward to a Fermi surface.

The fermionic ground state is entirely different. Ignore spin for the moment, so that each momentum point represents one available state. The first fermion goes into $\mathbf{k} = 0$, but a second identical fermion cannot join it. The next particles must occupy the next-lowest energy shell. In a two-dimensional drawing there are four nearest nonzero lattice points of equal energy; in three dimensions there are six. After those states are occupied, further particles must enter successive shells.

Because the energy is proportional to the squared distance from the origin in momentum space, minimizing the total energy means packing the occupied points as closely around $\mathbf{k} = 0$ as exclusion permits. With many fermions the occupied region approximates a sphere. Its boundary is the Fermi surface, its interior is the Fermi sea, and the complete filled region is the Fermi sphere. At finite volume the boundary is slightly jagged because the available points form a lattice, but this detail becomes negligible for a large system.

The fermionic ground state has much more energy than the corresponding bosonic ground state. All the bosons can remain at zero momentum, but most of the fermions are forced into nonzero-momentum states. As more fermions are added, the maximum occupied momentum grows. Moreover, a shell at large momentum contains more lattice points than a shell near the origin. A large Fermi sphere therefore contains many particles at substantial momentum and only a small number in the lowest-momentum states.

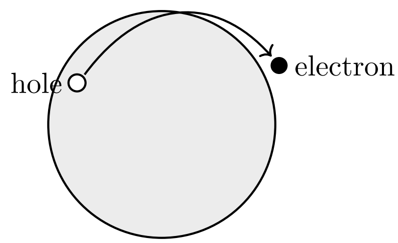

Figure 5.5: A low-energy fermionic excitation promotes an electron through the Fermi surface and leaves a hole behind.

The contrast continues in the excited states. For bosons, the first excitation is obtained by removing any one boson from the condensate and placing it in a first excited one-particle state:

$$(N, 0, 0, \dots) \longrightarrow (N-1, 1, 0, \dots). \tag{5.72}$$

It makes no difference which boson is moved; the particles are identical. Higher excitations can be made in several ways. Depending on the level spacing, one may place one boson in a higher level or place two bosons in the first excited level. Exciting the bosonic system means promoting particles out of the condensate into higher-energy states.

For fermions, no particle can be moved to another point inside the filled Fermi sphere, because every such point is already occupied. Taking a fermion from deep inside the sea and placing it above the surface is possible, but it costs a large amount of energy. The least expensive excitation takes a fermion from just inside the Fermi surface and moves it to an unoccupied state just outside. The promoted electron has only a small energy above the surface, and the vacant state left behind lies only a small energy below it.

5.5 Holes and Low-Energy Excitations

To make the charge of the hole concrete, imagine that the box contains positive charge uniformly smeared through its volume. In matter that positive charge ultimately comes from protons in nuclei, but the uniform distribution keeps the model simple. Add enough electrons to neutralize the background. In the ground state those electrons fill the Fermi sphere.

Promoting one electron from below to above the Fermi surface does not change the total charge of the box. It creates a negatively charged electron above the surface and leaves a missing negative charge below it. The missing electron can therefore be treated as a positively charged excitation: a hole.

---

[a] **Question from the audience.** Does a hole below the Fermi surface have negative energy?

Response. No. A hole is an excitation above the many-body ground state, so its excitation energy is positive. Removing an occupied electron from below the Fermi surface costs a positive amount of energy when energies are measured relative to that surface.

---

[a]Lecture timestamp: 01:31:38.

Let the removed electron have energy $\varepsilon < \varepsilon_F$, and let the promoted electron have energy $\varepsilon' > \varepsilon_F$.

The energy supplied may be separated into two positive distances from the Fermi surface.[4]

$$\Delta E = \varepsilon' - \varepsilon = (\varepsilon' - \varepsilon_F) + (\varepsilon_F - \varepsilon). \tag{5.73}$$

The first contribution is naturally assigned to the electron above the Fermi surface, while the second is assigned to the hole below it:

$$E_{\text{electron}} = \varepsilon' - \varepsilon_F > 0, \qquad E_{\text{hole}} = \varepsilon_F - \varepsilon > 0. \tag{5.74}$$

The electron carries negative charge, and the hole carries positive charge.

An electron above the Fermi surface may later drop into an available hole. Its energy then decreases, so radiation must carry the released energy away. One may describe the event as an electron dropping to an unoccupied lower state and emitting one or more photons. Equivalently, one may say that an electron and a hole annihilate into radiation. The hole is not an additional fundamental constituent of the box; it is an absence in the filled many-electron state. Nevertheless, its energy and charge make it useful to treat it as a particle-like excitation.

---

[a] **Question from the audience.** Does promoting an electron mean moving it from one lattice point to another in momentum space?

Response. It means moving the electron from one energy level to another. For free particles in the box, the momentum lattice labels those levels, but the same idea does not depend on a momentum-space lattice. In an atom the positive charge is concentrated in the nucleus. Electrons fill the lowest available atomic levels, and a photon can promote one electron to an excited level, leaving a vacancy in the level from which it came. That vacancy is the atomic counterpart of a hole.

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[a]Lecture timestamp: 01:36:52.

In the box, photon absorption likewise converts the photon energy into an electron above the Fermi surface and a hole below it. In an atom, the same language describes the excited electron together with the vacancy left among the occupied lower levels.

---

[a] **Question from the audience.** Can a low-energy photon be absorbed by an electron anywhere in the Fermi sphere?

Response. No. If the electron lies deep in the Fermi sea, every nearby state is already occupied, and a low-energy photon cannot lift it all the way to an unoccupied state above the surface. Only electrons close enough to the Fermi surface have accessible empty states within the photon's energy range. A sufficiently energetic photon can excite an electron from deep in the sea.

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[a]Lecture timestamp: 01:39:16.

The low-energy absorption cross section is therefore controlled by the surface of the Fermi sea rather than by its whole volume. Electrons deep in the sea are blocked from making small transitions; electrons just below the surface can be moved just above it.

[4]The two intervals are stated separately; the compact equality displayed here is the algebraically equivalent reconstruction.

[a] **Question from the audience.** What prevents electrons from interacting directly with one another, without an incident photon?

Response. Nothing. Electron–electron interactions are being neglected in this simple model, because including them makes the Fermi-sea picture more complicated. In this first approximation to a metal, their effective influence is treated as weak enough that it can usually be ignored.

[a]Lecture timestamp: 01:40:42.

[a] **Question from the audience.** What happens if one photon carries enough energy to excite several electrons?

Response. Any process consistent with the conservation laws is possible in principle. The photon can create more than one electron–hole pair: exciting two electrons also leaves two holes. Such a process creates more excitations and is generally less probable than using the energy to excite one electron much more strongly.

[a]Lecture timestamp: 01:41:10.

[a] **Question from the audience.** Can two photons be absorbed together when neither photon alone can produce the excitation?

Response. Yes. Absorption is probabilistic, and a two-photon process can occur when the combined photon energy reaches an available excitation even though either photon separately falls below the required energy. If the available light carries too little energy to reach any allowed excitation, it largely passes through the system.

[a]Lecture timestamp: 01:42:06.

5

[a] **Question from the audience.** Why are electron–electron interactions called weak when all electrons carry the same negative charge?

Response. The electrons are immersed in the compensating positive background. That background screens the charge of an electron and reduces the effective interaction in this simplified setting. The interaction is not absent; treating it requires the screening response of the medium.

[a]Lecture timestamp: 01:43:49.

[a] **Question from the audience.** Will screening be discussed later?

Response. Yes. Screening is important, but it is deferred until the interacting system can be treated more fully.

[a]Lecture timestamp: 01:44:41.

5.6 A One-Dimensional Fermionic Sea and Antiparticles

The hole construction points toward antiparticles. The same logical structure can be transferred from electrons in matter to a relativistic quantum field and its vacuum. The simplest version uses

[5]Editorial clarification: A brief discussion at this point connects absorbed radiation with collective electron motion, but its substantive details are not recoverable. Only the statement that such collective response is more complicated than the single particle–hole picture is secure.

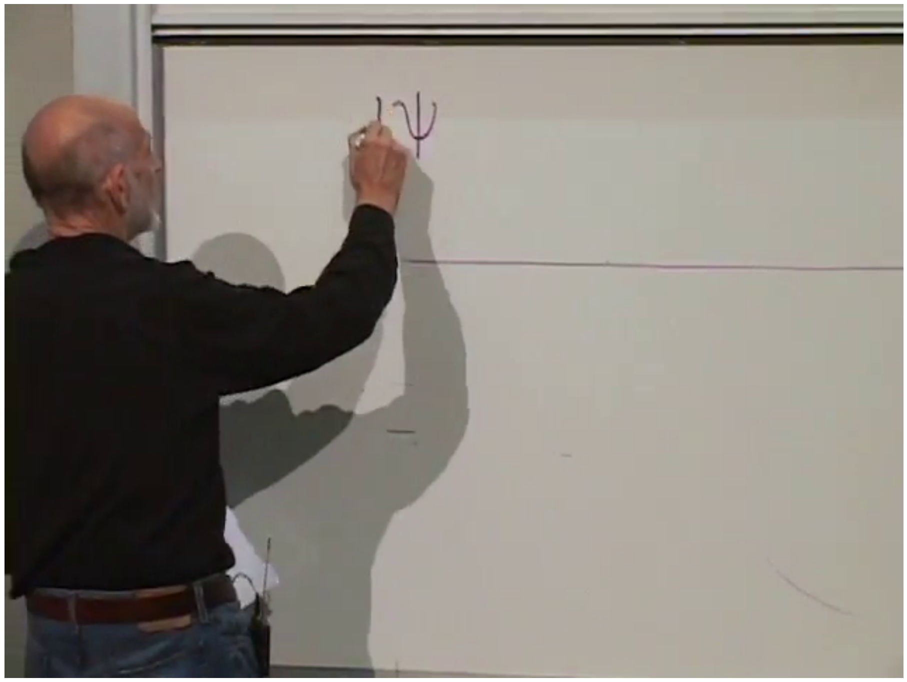

Figure 5.6: Beginning the first-order field equation whose plane waves satisfy $\omega = k$.
Lecture frame: 01:47:22.

one spatial dimension and a massless fermionic field $\psi(x,t)$. In units with the speed of light equal to one, the two possible linear dispersion relations are

$$\omega = k, \qquad \text{or} \qquad \omega = -k. \tag{5.75}$$

The first branch will be considered explicitly.
[6]

For a plane wave

$$\psi(x,t) = e^{i(kx-\omega t)}, \tag{5.76}$$

the derivatives are

$$\frac{\partial \psi}{\partial t} = -i\omega\psi, \qquad \frac{\partial \psi}{\partial x} = ik\psi. \tag{5.77}$$

Thus $\omega = k$ is equivalent to the first-order field equation

$$\frac{\partial \psi}{\partial t} = -\frac{\partial \psi}{\partial x}. \tag{5.78}$$

Any profile of the form $f(x-t)$ satisfies this equation and moves to the right. Reversing the sign gives

$$\frac{\partial \psi}{\partial t} = +\frac{\partial \psi}{\partial x}, \tag{5.79}$$

whose profiles $f(x+t)$ move to the left. Either first-order equation selects only one direction of motion. Take the field to be fermionic and, for the hole argument, suppose that its particles carry electric charge.

[6]Editorial clarification: The closing exchange alternates incompatible particle identifications. The secure model is an unspecified massless fermionic toy field in one spatial dimension, restricted to one direction of propagation.

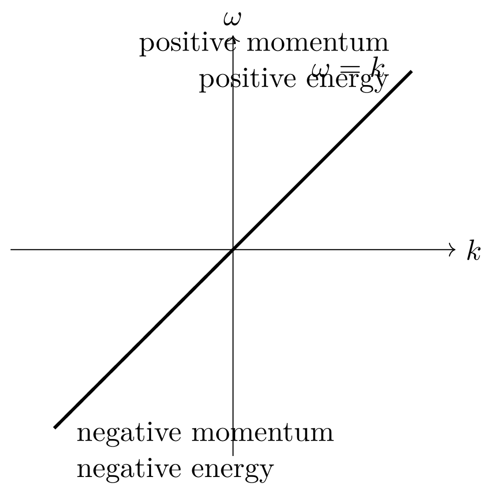

Figure 5.7: The one-branch spectrum $\omega = k$, with its negative-energy half at $k < 0$.

The branch $\omega = k$ contains an immediate difficulty. The wave number k can be negative, and therefore so can ω. Since ω is the energy in these units, the straight-line spectrum contains both positive- and negative-energy states:

$$E = \omega = k. \tag{5.80}$$

An ordinary empty vacuum cannot be the ground state of such a theory. A positive-energy fermion could radiate a photon and fall to a lower state. Nothing would force it to stop at zero: it could continue into states of negative energy, emit still more radiation, and fall without bound. The energy lost by the fermion would appear in the radiation field. A spectrum unbounded below therefore gives no stable ordinary vacuum.

Fermionic exclusion supplies a different starting point. Fill every negative-energy one-particle state, up to zero energy, with one fermion. No further fermion can fall into those states because each is already occupied. Adding another negative-energy fermion is impossible, while adding a positive-energy fermion raises the energy. The completely filled negative-energy sea can therefore be taken as the many-body ground state and called the vacuum.

The simplest excitation removes one fermion from the negative-energy sea and places it in a positive-energy state. The promoted fermion is an ordinary positive-energy particle. The vacancy left in the sea is a second excitation, a hole. If the removed fermion has negative charge, the hole has positive charge.

The hole also has positive momentum in this one-branch example. If the missing fermion occupied a state of momentum $k_- < 0$, removing it changes the total momentum by

$$p_{\text{hole}} = -k_- > 0. \tag{5.81}$$

It is a deficit of negative momentum. Thus the excitation consists of a positive-energy, negatively charged fermion and a positive-energy, positively charged hole. Both carry positive momentum and, on the $\omega = k$ branch, both move to the right.

In the full electron theory, the positively charged hole is the positron. Dirac initially considered identifying the hole with the proton. The construction requires a particle and its antiparticle to have the same mass. The hole is a distinct particle with the electron's mass and the opposite electric

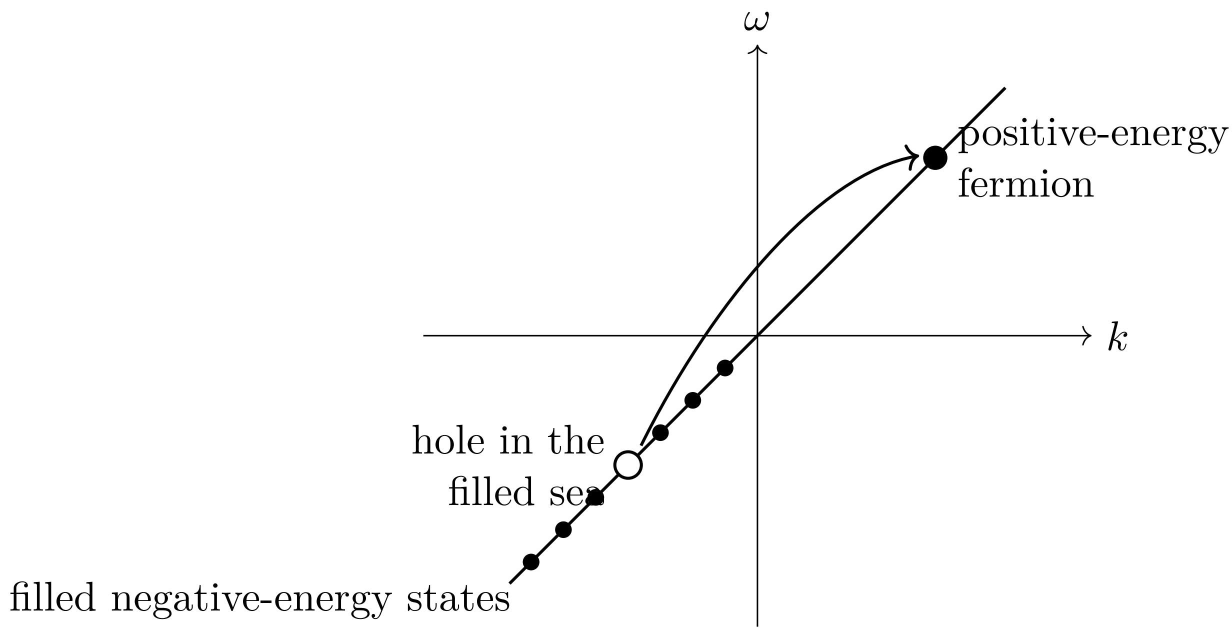

Figure 5.8: The filled negative-energy sea and an excitation formed by promoting one fermion to positive energy.

charge. This is the basic particle–antiparticle interpretation to which the negative-energy spectrum led.

The same rescue cannot be applied to bosons. A bosonic state admits arbitrarily many particles, so there is no way to fill a negative-energy level once and block further occupation. More bosons could always be added to that level, lowering the energy again and again:

$$\begin{aligned} \text{fermionic mode: } & n = 0 \text{ or } 1, \\ \text{bosonic mode: } & n = 0, 1, 2, \dots . \end{aligned} \tag{5.82}$$

The proposed bosonic system would still have no ground state. In this elementary construction, the stability of the filled sea depends precisely on fermionic exclusion.

This one-dimensional equation is only a toy version of the Dirac equation. Its field is massless, moves at the speed of light, and propagates in only one direction. A fuller Dirac equation must accommodate the missing directions of motion and the rest of the relativistic particle structure. That fuller equation, together with the treatment of antiparticles for bosonic fields, is postponed.

Chapter 6

The Dirac Sea, Fermion Mass, and Four-Component Spinors

6.1 Right-Moving Fermions and the Dirac Sea

The Dirac equation is an equation for fermions. Relativistic bosons satisfy different equations, such as the Klein–Gordon equation or Maxwell's equations. The simplest example lives in one spatial dimension and initially describes particles and antiparticles that can move only to the right.

Begin with a plane wave,

$$\psi(x,t) = \psi_0 e^{i(kx-\omega t)}. \tag{6.1}$$

Its phase velocity is

$$v_{\mathrm{ph}} = \frac{\omega}{k}. \tag{6.2}$$

The signs of k and ω matter. If they have the same sign, the phase moves to the right; if they have opposite signs, it moves to the left. When ω is proportional to k, the phase and group velocities coincide:

$$v_{\mathrm{g}} = \frac{d\omega}{dk} = \frac{\omega}{k}. \tag{6.3}$$

Consider the first-order equation

$$\frac{\partial\psi}{\partial x} + \frac{\partial\psi}{\partial t} = 0. \tag{6.4}$$

Substitution of the plane wave gives

$$ik\psi - i\omega\psi = 0, \tag{6.5}$$

and hence

$$\omega = k. \tag{6.6}$$

The wave therefore moves to the right. It is useful to label the field accordingly:

$$\frac{\partial\psi_R}{\partial t} = -\frac{\partial\psi_R}{\partial x}. \tag{6.7}$$

Restoring the speed of light gives the dimensional form

$$\frac{\partial\psi_R}{\partial x} + \frac{1}{c}\frac{\partial\psi_R}{\partial t} = 0, \tag{6.8}$$

so that

$$\omega = ck. \tag{6.9}$$

From now on $c = 1$. With $\hbar = 1$, ω is also the energy of one quantum.

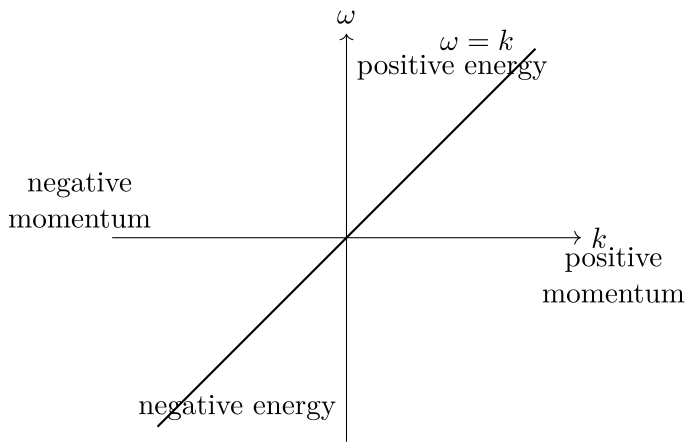

Figure 6.1: The massless right-moving dispersion $\omega = k$ contains positive- and negative-energy branches.

The straight line has a serious consequence. Positive k gives positive energy, while negative k gives negative energy. Both parts nevertheless describe right-moving waves because $\omega/k = 1$ on the whole line.

An electron placed on the positive-energy part could fall to a lower state, emit a positive-energy photon, fall again, and continue down the spectrum: down one stair, then another, and another. No electron would be stable. More seriously, the world would lower its energy by producing more and more negative-energy electrons.

Fermionic exclusion supplies the cure. Fill every negative-energy electron state. No second electron can enter an already occupied state, so the completely filled negative-energy sector blocks an unlimited fall. This filled sector is the Dirac sea, and its completely filled state is the ground state, or vacuum.

---

[a] **Question from the audience.** Are there infinitely many lower negative-energy states?

Response. Yes. With discrete momenta the spectrum is like an infinite staircase; with continuous momenta it is like a continuous slide. Filling the negative-energy sea prevents an electron from descending indefinitely, although a finite downward transition with photon emission may still be possible when an unoccupied lower state is available.

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[a] Lecture timestamp: 00:07:19.

This explains why an equation with this spectrum must be quantized as a fermionic theory. If the quanta were bosons, arbitrarily many of them could occupy the same lower state, and filling the negative-energy sector would not stabilize the vacuum.

The filled-sea picture also produces antiparticles. In the vacuum all negative-energy levels are occupied and all positive-energy levels are empty. An electron may be promoted from any occupied negative-energy level to any unoccupied positive-energy level; the initial and final levels need not be placed symmetrically.

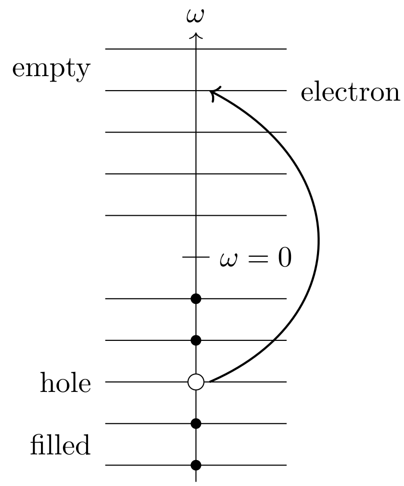

Figure 6.2: Promoting an occupied negative-energy electron produces a positive-energy electron and leaves a hole in the filled sea.

Removing an electron of negative energy raises the energy relative to the vacuum, so the hole has positive energy. On the branch $\omega = k$, the removed electron also had negative momentum. Removing negative momentum leaves a positive excess of momentum, so the hole has positive momentum. With the electron charge taken to be negative, the missing electron has positive charge. Thus the hole is a positive-energy, positive-momentum antiparticle, while the promoted electron has negative charge, positive energy, and positive momentum. Every real excitation in this one-branch theory moves to the right.

The field ψ_R is built from fermionic creation and annihilation operators. The particle and hole interpretation is therefore a statement about the quanta created and removed by the field, not only about a classical wave.

There is still an asymmetry. The right-moving sea contains negative-energy states with negative momentum. Its formal total momentum is enormous and points to the left. One may define the vacuum momentum to be zero by subtracting that constant, but the state remains structurally unbalanced: particles and antiparticles move only to the right, while the filled vacuum is made from negative-momentum states.

6.2 The Left-Moving Branch and a Balanced Vacuum

Left-right balance requires a second, independent field with its own creation and annihilation operators. Let it satisfy

$$\frac{\partial \psi_L}{\partial t} = +\frac{\partial \psi_L}{\partial x}. \tag{6.10}$$

For the same plane-wave form,

$$-i\omega\psi_L = ik\psi_L, \tag{6.11}$$

so

$$\omega = -k. \tag{6.12}$$

This second line is the mirror image of $\omega = k$. Its positive-energy branch describes left-moving particles, and its negative-energy branch must also be filled.

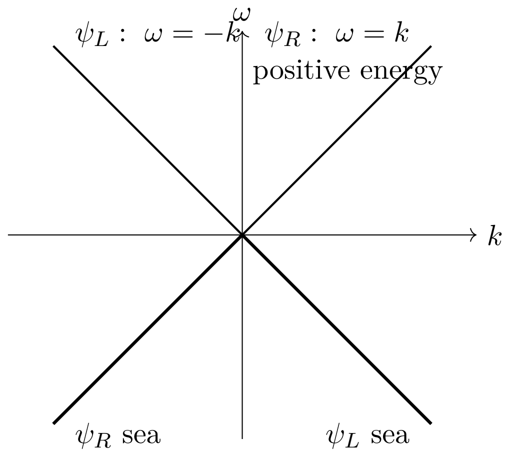

Figure 6.3: Independent right- and left-moving fields give mirrored massless branches; the two negative-energy half-branches are filled in the vacuum.

After filling both negative-energy seas, there are as many left-moving as right-moving occupied states. Their vacuum momenta balance. Particle and hole excitations can now move in either direction.

---

[a] **Question from the audience.** Can the negative-energy problem be handled without postulating filled states and holes?

Response. The same theory can be relabeled. A filled negative-energy electron state may be called an empty antiparticle state; then the physical language contains only positive-energy particles and antiparticles. The filled-sea picture remains a useful consistency check, much like a filled band in a metal, where removing an electron produces a hole while promotion produces both a particle and a hole.

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[a] Lecture timestamp: 00:15:58.

The excitations considered so far are massless. Their dispersions are linear and their velocities have magnitude 1, so they move at the speed of light. The next question is how to give them mass.

6.3 Two Components and the Mass Term

First combine the two fields into a column:

$$\Psi = \begin{pmatrix} \psi_R \\ \psi_L \end{pmatrix}. \tag{6.13}$$

This is initially only compact notation for two independent equations. Define

$$\alpha = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{6.14}$$

Then the right- and left-moving equations become

$$\frac{\partial \Psi}{\partial t} = -\alpha \frac{\partial \Psi}{\partial x}. \tag{6.15}$$

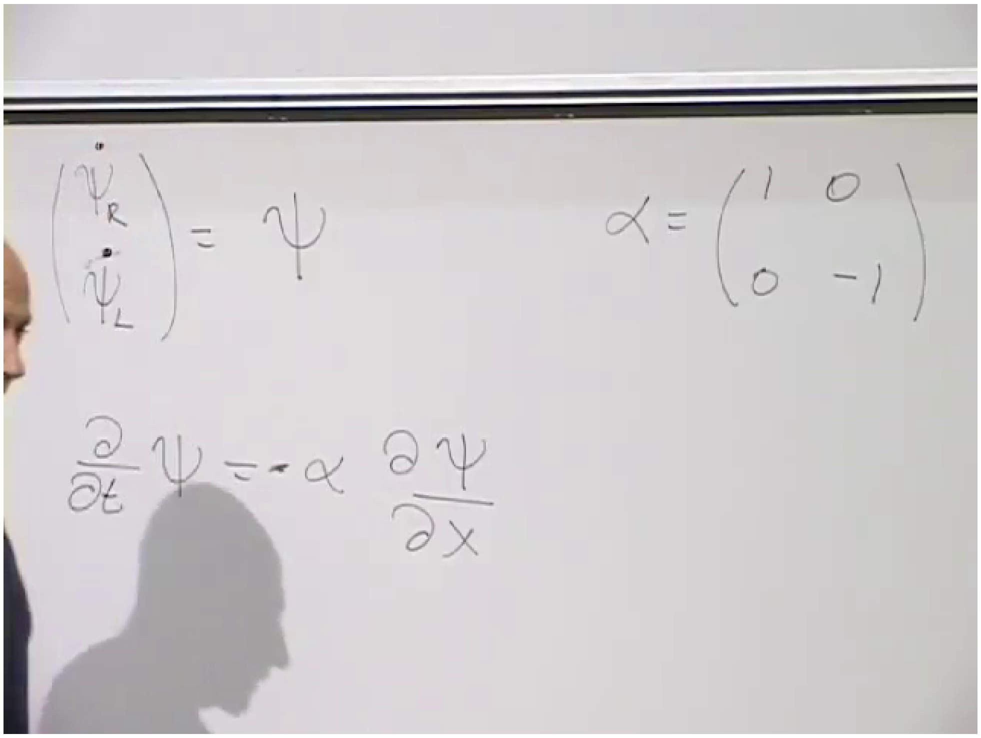

Figure 6.4: The fields ψ_R and ψ_L form a two-component spinor, with α distinguishing the two massless branches.

Lecture frame: 00:20:36.

For a plane wave, the operators

$$i\partial_t \quad \text{and} \quad -i\partial_x \tag{6.16}$$

return ω and k, respectively. The matrix equation may therefore be abbreviated as

$$\omega = \alpha k. \tag{6.17}$$

For the upper component this says $\omega = k$; for the lower component it says $\omega = -k$. The notation has compressed the two equations but has introduced no new physics.

Electrons are not massless and do not move at the speed of light. A tempting formula for their energy is the nonrelativistic kinetic energy $k^2/(2m)$, but that cannot be the starting point here: the massless limit already concerns motion at the speed of light. The relativistic relation is

$$\omega^2 = k^2 + m^2, \tag{6.18}$$

or, including both energy branches,

$$\omega = \pm\sqrt{k^2 + m^2}. \tag{6.19}$$

When $m = 0$, these reduce to $\omega = k$ and $\omega = -k$.

[a] **Question from the audience.** Are units with $c = 1$ being used, and how is the speed of a massive particle obtained?

Response. Yes. The speed of a wave packet is the group velocity, $v_g = d\omega/dk = k/\sqrt{k^2 + m^2}$ on the positive-energy branch. Thus the dispersion relation itself contains the reduction of the speed below $c = 1$ for a massive particle.

[a]Lecture timestamp: 00:25:23.

[a] **Question from the audience.** Is ω a vector or a matrix in the shorthand $\omega = \alpha k$?

Response. The shorthand stands for an operator equation acting on the two-component wavefunction. Here ω means $i\partial_t$, k means $-i\partial_x$, and α is the matrix acting on the component index. The scalar symbols ω and k become their numerical eigenvalues only on a plane wave.

[a]Lecture timestamp: 00:26:29.

Keep the equation first order and try the simplest mass-dependent extension:

$$\omega = \alpha k + \beta m. \tag{6.20}$$

The object β must be a matrix because it acts on the same two-component field as α. Its algebra follows by squaring:

$$\begin{aligned} \omega^2 &= (\alpha k + \beta m)(\alpha k + \beta m) &(6.21)\\ &= \alpha^2 k^2 + \beta^2 m^2 + \alpha\beta\, km + \beta\alpha\, mk &(6.22)\\ &= \alpha^2 k^2 + \beta^2 m^2 + (\alpha\beta + \beta\alpha) km. &(6.23) \end{aligned}$$

To obtain $k^2 + m^2$ for every k and m, require

$$\alpha^2 = I, \qquad \beta^2 = I, \qquad \alpha\beta + \beta\alpha = 0. \tag{6.24}$$

Two ordinary nonzero numbers cannot satisfy the last condition: for numbers it would give $2\alpha\beta = 0$, contrary to both squares being 1. Matrices can anticommute. In particular, β cannot equal α, since then

$$\alpha\beta + \beta\alpha = 2\alpha^2 = 2I. \tag{6.25}$$

Choose

$$\beta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{6.26}$$

Its square can be checked entry by entry:

$$\begin{aligned} \beta^2 &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0\cdot 0 + 1\cdot 1 & 0\cdot 1 + 1\cdot 0 \\ 1\cdot 0 + 0\cdot 1 & 1\cdot 1 + 0\cdot 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned} \tag{6.27}$$

The order of multiplication matters:

$$\alpha\beta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \tag{6.28}$$

$$\beta\alpha = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \tag{6.29}$$

Thus

$$\alpha\beta = -\beta\alpha, \qquad \alpha\beta + \beta\alpha = 0. \tag{6.30}$$

Other equivalent matrix choices are possible, but this one makes the component equations especially transparent.

Restoring the differential operators gives the massive equation

$$i\frac{\partial\Psi}{\partial t} = -i\alpha\frac{\partial\Psi}{\partial x} + m\beta\Psi. \tag{6.31}$$

The matrix β interchanges the two components:

$$\beta\begin{pmatrix}\psi_R \\ \psi_L\end{pmatrix} = \begin{pmatrix}\psi_L \\ \psi_R\end{pmatrix}. \tag{6.32}$$

Consequently,

$$i\dot{\psi}_R = -i\frac{\partial\psi_R}{\partial x} + m\psi_L, \tag{6.33}$$

$$i\dot{\psi}_L = +i\frac{\partial\psi_L}{\partial x} + m\psi_R. \tag{6.34}$$

The mass term couples fields that were independent in the massless theory. The equation for the upper component contains the lower component, and the equation for the lower component contains the upper one. This coupling is what a Dirac mass means. The equations are translation invariant, so they conserve momentum, and the mass of their particle excitations is m.

[a] **Question from the audience.** Is a solution automatically Lorentz invariant?

Response. A particular solution should not remain fixed under a Lorentz transformation. A state with one momentum is carried into a state with the transformed momentum, so the appropriate statement is covariance. The relation $\omega^2 = k^2 + m^2$ is one key Lorentz-invariant check, but full covariance also requires the transformation law of the spinor Ψ, which has not yet been established.

[a] Lecture timestamp: 00:38:25.

6.4 The Rest Frame and the Meaning of the Components

A massless particle cannot be brought to rest, but a massive particle can. At rest $k = 0$, so the wavefunction has no spatial variation and

$$\omega = \beta m. \tag{6.35}$$

The differential equation reduces to

$$i\frac{d\Psi}{dt} = m\beta\Psi. \tag{6.36}$$

In components,

$$i\dot{\psi}_R = m\psi_L, \tag{6.37}$$

$$i\dot{\psi}_L = m\psi_R. \tag{6.38}$$

Both off-diagonal entries of β are $+1$, so there is no extra minus sign in either of these equations.

The equations decouple after adding and subtracting the components. Define

$$\psi_+ = \psi_R + \psi_L, \qquad \psi_- = \psi_R - \psi_L. \tag{6.39}$$

Adding the two rest-frame equations gives

$$i\dot{\psi}_+ = m\psi_+, \tag{6.40}$$

while subtracting them gives

$$i\dot{\psi}_- = -m\psi_-. \tag{6.41}$$

Thus

$$\omega = +m \quad \text{for } \psi_+, \qquad \omega = -m \quad \text{for } \psi_-. \tag{6.42}$$

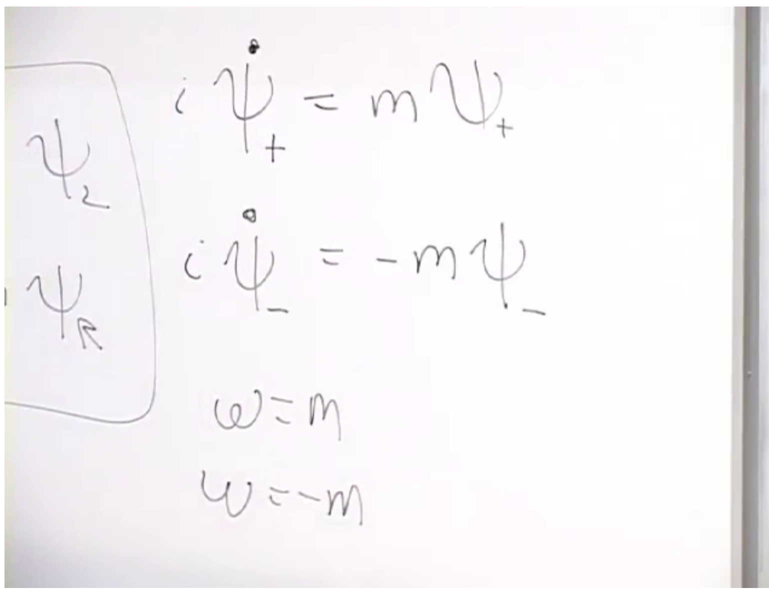

Figure 6.5: At zero momentum, $\psi_+ = \psi_R + \psi_L$ and $\psi_- = \psi_R - \psi_L$ have rest frequencies $\omega = +m$ and $\omega = -m$.

Lecture frame: 00:44:49.

The positive-energy rest field is ψ_+; the negative-energy rest field is ψ_-. The negative-energy states are filled, just as before. The rest-frame result is only the $k = 0$ slice of the full spectrum:

$$\omega_\pm(k) = \pm\sqrt{k^2 + m^2}. \tag{6.43}$$

Every momentum has a positive- and a negative-energy state. Filling every state on the negative branch leaves positive-energy particles above the sea and positive-energy holes in it. Particles and holes both have positive mass.

The essential point deserves repetition. A fermion becomes massive when its two formerly independent components are coupled. At rest, neither original component alone is an energy eigenstate. The definite-energy states are their sum and difference.

To see what the sum means, write $C_R^\dagger$ and $C_L^\dagger$ for the relevant one-particle creation pieces. Acting on the vacuum,

$$(C_R^\dagger + C_L^\dagger)|0\rangle \tag{6.44}$$

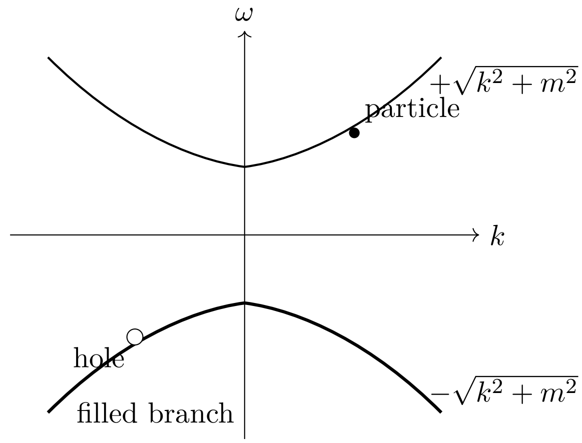

Figure 6.6: At every momentum the massive spectrum has positive- and negative-energy states. The negative branch is filled, and a missing state in it is a positive-energy hole.

creates one particle in a coherent superposition of the two components. It does not create two particles. A product,

$$C_R^\dagger C_L^\dagger|0\rangle, \tag{6.45}$$

contains two creation operators and creates a two-particle state when both modes are available. After normalization, the rest state has equal probability in the two components, but the coupled wave as a whole has zero momentum.

The labels R and L were literal motion labels only when $m = 0$. Once the mass term is present, a moving massive solution generally contains both components, and the rest solution contains them with equal magnitude. It is therefore better to regard R and L here as labels for the upper and lower spinor components.

All massive fermions in particle physics are built from left- and right-component fields: a Dirac mass couples components that are independent in the massless limit.

---

[a] **Question from the audience.** Should $\psi_R \pm \psi_L$ include a factor of $1/\sqrt{2}$?

Response. It may be included to normalize the state. It multiplies both sides of the equation and therefore does not change the equation of motion.

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[a]Lecture timestamp: 00:52:01.

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[a] **Question from the audience.** Do the relative magnitudes of the right and left components matter?

Response. For the chosen rest-frame eigenstates their magnitudes are equal. An overall normalization is arbitrary, but the relative magnitudes are fixed by the eigenvector of β.

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[a]Lecture timestamp: 00:52:12.

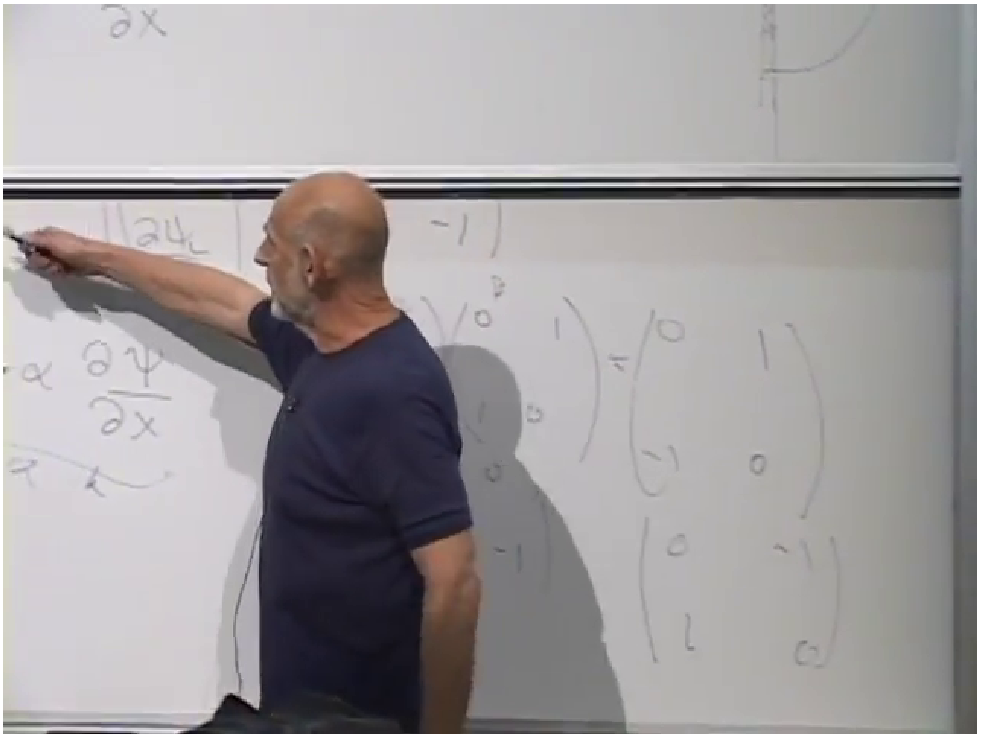

Figure 6.7: For nonzero mass, the two spinor components are coupled; their right- and left-moving interpretation applies directly only in the massless limit.

Lecture frame: 00:50:48.

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[a] **Question from the audience.** Could the two components have a relative phase?

Response. Yes. The real coefficients in $\psi_R \pm \psi_L$ follow from choosing a real β. Other matrices that square to the identity and anticommute with α are equivalent choices, and phases may then appear between the components. Once a representation is chosen, it must be used consistently.

---

[a]Lecture timestamp: 00:52:52.

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[a] **Question from the audience.** Are “left mover” and “right mover” misleading labels after mass is added?

Response. Somewhat. They describe genuine left- and right-moving waves in the massless theory. With nonzero mass, even a particle moving in one direction generally contains both components, so “upper component” and “lower component” are safer labels.

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[a]Lecture timestamp: 00:53:43.

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[a] **Question from the audience.** Does a fermion at rest already require spin angular momentum?

Response. Not in this one-dimensional model. Angular momentum is not defined in one spatial dimension, so spin has not yet entered the construction.

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[a]Lecture timestamp: 00:55:01.

[a] **Question from the audience.** Does the wavefunction contain fermionic creation and annihilation operators?

Response. Yes.

[a]Lecture timestamp: 00:55:24.

[a] **Question from the audience.** Where does the fermionic character enter apart from the choice of creation and annihilation operators?

Response. It enters through the fermionic creation–annihilation rules. An equation with this negative-energy spectrum has to be quantized with those rules; bosonic occupation rules would leave the vacuum unstable. Relativistic bosons must be described by a different wave equation.

[a]Lecture timestamp: 00:55:30.

[a] **Question from the audience.** Can a nonzero-momentum solution be Lorentz-transformed into the rest-frame solution?

Response. Yes. A boost can carry a massive solution to its rest frame, but carrying out the transformation requires knowing how Lorentz transformations act on the spinor Ψ.

[a]Lecture timestamp: 00:56:20.

6.5 Mass from a Scalar Vacuum Field

[a] **Question from the audience.** How does interaction with the Higgs field produce the Dirac mass term?

Response. Replace the constant mass coefficient by a coupling to a scalar field. If that field vanishes in the vacuum, the fermion remains massless; if it has a constant nonzero vacuum value, the coupling acts exactly like a mass.

[a]Lecture timestamp: 00:58:11.

Let $\phi(x,t)$ be a scalar field with its own equation of motion. In the Dirac equation, replace the constant m in the mass term by $g\phi(x,t)$:[1]

$$i\frac{\partial\Psi}{\partial t} = -i\alpha\frac{\partial\Psi}{\partial x} + g\phi(x,t)\,\beta\Psi. \tag{6.46}$$

The two field equations are then coupled. The fermion acts on the scalar field, and the scalar field acts back on the fermion; considered together, the system is nonlinear even though the Dirac equation is linear in Ψ for a prescribed ϕ.

Suppose first that empty space has

$$\phi(x,t) = 0. \tag{6.47}$$

Then the mixing term vanishes, the upper and lower components decouple, and the fermion is massless. Suppose instead that the lowest-energy state has a constant nonzero scalar field,

$$\phi(x,t) = \phi_0 \neq 0. \tag{6.48}$$

[1]The displayed interaction term is obtained by making this substitution in the previously established mass term $m\beta\Psi$.

The Dirac equation in that vacuum becomes

$$i\frac{\partial\Psi}{\partial t} = -i\alpha\frac{\partial\Psi}{\partial x} + g\phi_0\,\beta\Psi, \tag{6.49}$$

so that

$$m_{\text{eff}} = g\phi_0. \tag{6.50}$$

A coupling between a bosonic field and a fermionic field, together with a nonzero value of the bosonic field in empty space, therefore generates the same left-right mixing that previously appeared as a mass written directly into the equation.

---

[a] **Question from the audience.** Could the LHC provide evidence for a nonzero Higgs field in the vacuum?

Response. Absolutely. That is one of the big issues.

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[a]Lecture timestamp: 01:01:27.

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[a] **Question from the audience.** Can you say more about how the Higgs mechanism works?

Response. A scalar field has energy associated with its space and time variation, and it may also have a local potential energy $V(\phi)$. If the minima of that potential occur at nonzero values of ϕ, the vacuum contains a nonzero field. Quantized oscillations about the chosen minimum are Higgs particles.

---

[a]Lecture timestamp: 01:01:32.

A field can carry energy in two distinct ways. Some energy depends on how the field varies in space and time. The Maxwell field is the familiar comparison:

$$\mathcal{E}_{\text{Maxwell}} \propto \mathbf{E}^2 + \mathbf{B}^2, \tag{6.51}$$

and both $\mathbf{E}$ and $\mathbf{B}$ are built from derivatives of the electromagnetic potential. Thus this energy is associated with space and time variation of the underlying field.

A scalar field can also have a local potential-energy density,

$$V(\phi), \tag{6.52}$$

which depends on the value of ϕ itself rather than on its derivatives. Schematically,

$$\mathcal{E}_\phi = \mathcal{E}_{\text{space and time gradients}} + V(\phi). \tag{6.53}$$

Without a deeper principle fixing it, $V(\phi)$ could be any reasonable function compatible with the symmetries of the model.

For the simplified real-scalar illustration being used here, take the theory to be symmetric under

$$\phi \longrightarrow -\phi. \tag{6.54}$$

The potential must then be even:

$$V(\phi) = V(-\phi). \tag{6.55}$$

This symmetry alone does not say where the minimum lies. One possibility is a single minimum at $\phi = 0$. Another is a symmetric pair of minima at nonzero values, one positive and one negative.

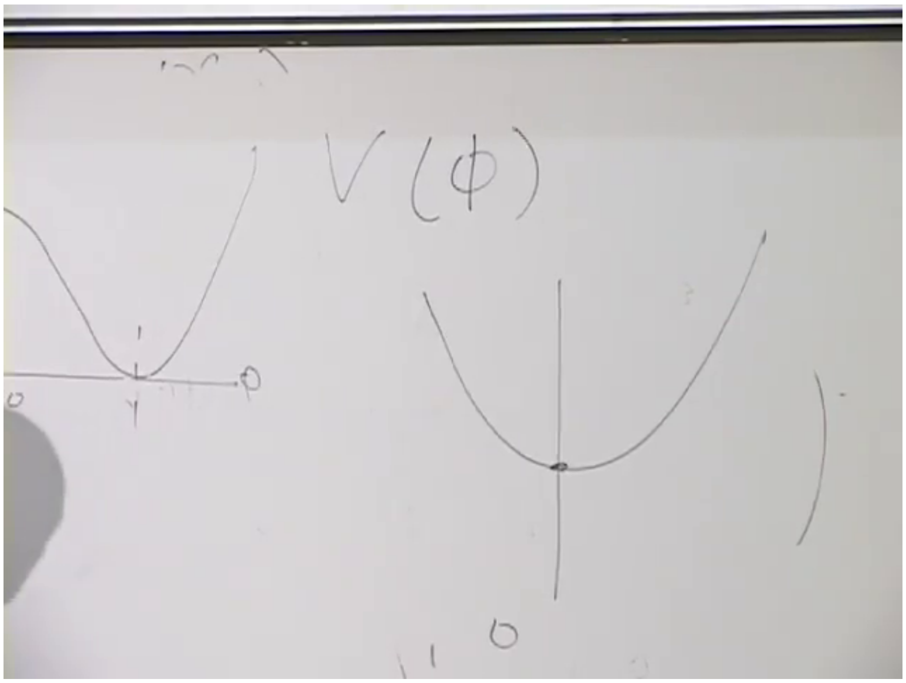

Figure 6.8: A minimum at $\phi = 0$ gives a vanishing vacuum field, whereas symmetric nonzero minima permit a vacuum value $\phi_0 \neq 0$ and hence a fermion mass $g\phi_0$.

Lecture frame: 01:04:47.

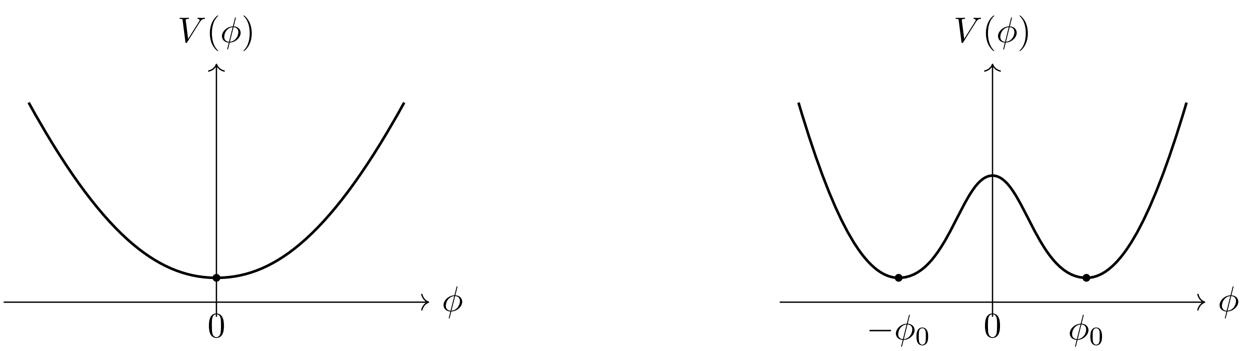

Figure 6.9: Even potentials may have a centered minimum or two symmetry-related nonzero minima.

The two possibilities can be drawn without committing to a particular polynomial form for the potential.[2]

If the minimum is at the origin, the vacuum has $\phi_0 = 0$, and this coupling gives the electron no mass. If the minima are at $\phi = \pm\phi_0$, the vacuum settles into either one. The two choices are related by the symmetry, and either produces a nonzero coefficient $g\phi_0$ in the Dirac equation. Why the potential has this qualitative shape is a deeper question, not settled by the mechanism itself.

The field need not remain exactly at its minimum. It can oscillate about the chosen value just as a particle can oscillate near the bottom of a potential well. Quantizing those oscillations gives Higgs quanta. Their characteristic frequency is related to the Higgs-particle mass: a large mass means that a large amount of energy is needed to excite even one quantum. In the vacuum the field simply sits near its minimum and supplies the fermion mass. When the field vibrates, the vibration is a Higgs particle.

The coupling entails action and reaction. Fermions respond to the value of ϕ, but sufficiently energetic fermion collisions can also excite the scalar field. A collision must supply the excitation energy; once the field is vibrating, its quanta are Higgs particles. The same field couples not only to electrons but also to quark fields and other fields whose masses it supplies.

---

[a] **Question from the audience.** If the Higgs field vibrates or is displaced, does the electron mass change?

Response. Yes. Since the effective mass is proportional to the local value of the field, displacing the field toward zero reduces the mass. Holding a macroscopic region far from the vacuum value would require enormous energy; a region large enough for such an experiment could collapse into a black hole.

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[a]Lecture timestamp: 01:08:14.

The thought experiment is direct. If a region could be held near $\phi = 0$, then within that region

$$m_{\text{eff}} = g\phi \tag{6.56}$$

would become small or vanish. An excited Higgs mode oscillates about the vacuum value, whereas producing and maintaining an appreciable displacement throughout a macroscopic region requires enormous energy. The potential-energy cost grows with the volume. Long before such a region became a practical laboratory, its energy could be large enough for gravitational collapse. The change of mass is real in principle, but arranging a large observable change is fantastically difficult.

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[a] **Question from the audience.** Why is a Higgs field needed?

Response. Because the electron has mass.

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[a]Lecture timestamp: 01:09:27.

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[a] **Question from the audience.** Does each kind of particle have its own Higgs field?

Response. No. In the Standard Model one Higgs field supplies masses to electrons, quarks, and the other particles participating in this mechanism. Supersymmetric versions require two Higgs fields rather than one separate field for every particle species.

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[a]Lecture timestamp: 01:09:41.

[2]The curves below retain only the qualitative centered and symmetry-related minima used in the argument.

The one-field picture is schematic: the full Higgs field has more structure than a single real number ϕ. Nevertheless the essential point survives. A common vacuum field can couple with different strengths to different fermions, so different masses do not require different Higgs fields for every species. The Higgs particle is an oscillation of that same field about its vacuum value.

Why not simply put the electron mass into the equation by hand and dispense with the Higgs field? The answer has to do with the weak interactions and unification, and it is deferred.

---

[a] **Question from the audience.** In a supersymmetric theory, are there two Higgs sectors?

Response. Yes. In the simplified division used here, one supplies masses to up-type quarks and the other to down-type quarks and electrons. The doubling is a technical requirement of supersymmetry, not a separate Higgs field for every particle.

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[a]Lecture timestamp: 01:11:16.

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[a] **Question from the audience.** If the Higgs particle has mass, where does that mass come from?

Response. That question is not resolved here. The sharper puzzle is why the Higgs mass is so small.

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[a]Lecture timestamp: 01:12:01.

Symmetry is inseparable from the problem of mass. In the simple Dirac model, a symmetry that keeps the left and right sectors from mixing would force the coefficient of the mixing term to vanish. If the vacuum respected that symmetry, the fermion would be massless. A nonzero Higgs vacuum value breaks the relevant vacuum symmetry and permits the left-right coupling:

$$m_{\text{eff}} = g\phi_0 \neq 0. \tag{6.57}$$

The precise symmetry transformation is not developed here; the point is its consequence for the mass term. The equations may possess the symmetry even though the particular lowest-energy state does not. It is this asymmetry of the vacuum that supplies the mass.

In the electroweak picture being sketched, electrons and quarks acquire mass in this way, as do the W and Z bosons. The photon remains massless. The Higgs particle itself is another exception: it can have mass even without the same vacuum displacement, and its mass concerns oscillations of its own field rather than left-right fermion mixing. Thus the Higgs vacuum value generates the masses of the other particles in this account, while the mass of the Higgs excitation raises a separate question.

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[a] **Question from the audience.** Does the graviton have mass?

Response. No. The graviton is massless, like the photon. Neutrinos were once also thought to be massless, but they have a very small mass; in the present language that means that their left and right components have only a very small coupling.

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[a]Lecture timestamp: 01:15:26.

Photons and, in the theory, gravitons are massless and move at the speed of light. Neutrinos do not quite belong to that class. Their tiny mass means, in this Dirac language, that the coupling which joins their left and right components is tiny. Why it is so small is another question.

6.6 The Dirac Equation in Three Spatial Dimensions

In three spatial dimensions the wave number becomes a wave vector,

$$\mathbf{k} = (k_1, k_2, k_3), \tag{6.58}$$

with components along the x, y, and z directions. The one-dimensional relation has the sign

$$\omega = \alpha k + \beta m, \tag{6.59}$$

and its simplest linear generalization is

$$\omega = \alpha_1 k_1 + \alpha_2 k_2 + \alpha_3 k_3 + \beta m. \tag{6.60}$$

Here the α_i and β are matrices. They must be chosen so that

$$\omega^2 = k_1^2 + k_2^2 + k_3^2 + m^2. \tag{6.61}$$

This invariant dispersion relation is an essential condition, although by itself it is not the whole content of relativity.

Squaring the linear expression gives

$$\begin{aligned}\omega^2 = {} & \alpha_1^2 k_1^2 + \alpha_2^2 k_2^2 + \alpha_3^2 k_3^2 + \beta^2 m^2 \\ & + (\alpha_1\alpha_2 + \alpha_2\alpha_1)k_1 k_2 + (\alpha_1\alpha_3 + \alpha_3\alpha_1)k_1 k_3 \\ & + (\alpha_2\alpha_3 + \alpha_3\alpha_2)k_2 k_3 + \sum_{i=1}^{3}(\alpha_i\beta + \beta\alpha_i)k_i m.\end{aligned} \tag{6.62}$$

The coefficients of the squared momenta and the mass require

$$\alpha_1^2 = \alpha_2^2 = \alpha_3^2 = I, \qquad \beta^2 = I. \tag{6.63}$$

The unwanted $k_i k_j$ terms disappear if distinct α matrices anticommute:

$$\alpha_i\alpha_j + \alpha_j\alpha_i = 0, \qquad i \neq j. \tag{6.64}$$

The mixed $k_i m$ terms disappear if

$$\alpha_i\beta + \beta\alpha_i = 0 \qquad \text{for each } i. \tag{6.65}$$

Thus all four matrices square to the identity, and every pair of distinct matrices anticommutes.

Three 2×2 matrices with the required mutual anticommutation properties do exist: they are the three Pauli matrices. There is no fourth 2×2 matrix that both squares to the identity and anticommutes with all three. No 3×3 realization works either. The first possible realization uses 4×4 matrices, the Dirac matrices.

The passage from two components to four raises a physical question. In one spatial dimension the two components began as right- and left-moving fields. What do four components mean? The new ingredient will be spin, but spin requires angular momentum and therefore a genuine three-dimensional discussion.

The matrices are not unique. Different choices are equivalent in much the same way that different choices of mutually orthogonal coordinate axes are equivalent: the entries change, but a transformation relates the descriptions and the physics is unchanged. One convenient representation is[3]

$$\beta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \tag{6.66}$$

together with

$$\alpha_1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad \alpha_2 = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}, \tag{6.67}$$

$$\alpha_3 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}. \tag{6.68}$$

Their structure is easier to see by dividing each matrix into 2×2 blocks:

$$\beta = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \qquad \alpha_i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}. \tag{6.69}$$

The three Pauli matrices are

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{6.70}$$

These block matrices obey the required Dirac anticommutation relations.

The wavefunction must now have four components:

$$\Psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix}. \tag{6.71}$$

The equation of motion is

$$i\frac{\partial \Psi}{\partial t} = -i\sum_{j=1}^{3} \alpha_j \frac{\partial \Psi}{\partial x_j} + \beta m \Psi. \tag{6.72}$$

This compact expression stands for four coupled linear differential equations. The off-diagonal entries of the α_j mix the four components in a more intricate way than in the one-dimensional case, but the basic structure is the same: matrices couple the components while the equation remains first order in space and time. The additional doubling of components is the clue that leads to spin.

[3]The entrywise matrices below are the expansion of the block form and Pauli matrices displayed immediately afterward; those block relations are stated explicitly in the lecture at 01:30:23–01:31:29.

[a] **Question from the audience.** Are the four components of Ψ a four-vector associated directly with the four dimensions of spacetime?

Response. No. They form a spinor, not a four-vector. The equality between the two spacetime dimensions and the two spinor components in the one-dimensional example was accidental. In higher-dimensional constructions the spinor matrix sizes occur in powers of two and do not in general equal the number of spacetime dimensions.

[a]Lecture timestamp: 01:33:53.

The rest frame again gives the simplest interpretation. At zero momentum the derivative term vanishes:

$$i\frac{\partial \Psi}{\partial t} = m\beta\Psi. \tag{6.73}$$

In the chosen representation, β has two eigenvalues $+1$ and two eigenvalues -1. There are therefore two rest states with energy $+m$ and two with energy $-m$. As before, the negative-energy states are filled. The twofold multiplicity within each energy sign is the extra structure whose physical interpretation is spin.

[a] **Question from the audience.** Should the off-diagonal blocks in α_i carry the same index, so that they are σ_i?

Response. Yes. Each matrix α_i contains the corresponding Pauli matrix σ_i in both off-diagonal blocks.

[a]Lecture timestamp: 01:35:51.

[a] **Question from the audience.** Is a factor of two missing in the compact anticommutation relation for the α_i?

Response. Yes. When $i = j$, the left-hand side is $\alpha_i^2 + \alpha_i^2 = 2I$. The correct relation is $\alpha_i\alpha_j + \alpha_j\alpha_i = 2\delta_{ij}I$.

[a]Lecture timestamp: 01:36:03.

The separate square and unequal-index conditions can therefore be combined as

$$\alpha_i\alpha_j + \alpha_j\alpha_i = 2\delta_{ij}I. \tag{6.74}$$

[a] **Question from the audience.** Was the move to four matrices a major conceptual leap in Dirac's construction?

Response. Yes. Dirac postulated the equation and followed its consequences. One consequence was the appearance of antiparticles; another was spin. Spin was already known empirically, but the equation accounted for it.

[a]Lecture timestamp: 01:36:42.

[a] **Question from the audience.** Did the empirical evidence or the theorem restricting particles to bosons and fermions come first?

Response. The empirical evidence came first and the theorem later. The boson–fermion alternative applies in three spatial dimensions. In two spatial dimensions, with one time dimension, intermediate statistics can occur; in $3 + 1$ dimensions, and in higher spatial dimensions, the alternatives are bosons and fermions.

[a]Lecture timestamp: 01:37:29.

[a] **Question from the audience.** Is there a customary index for the four components of Ψ?

Response. No essential choice is fixed here. The labels p and q may be used, running from one to four; each matrix then has one row index and one column index.

[a] Lecture timestamp: 01:39:01.

Matrix notation suppresses these component labels. If they are exposed, then

$$p, q = 1, 2, 3, 4, \tag{6.75}$$

and the equation reads

$$i\,\partial_t \Psi_p = -i \sum_{j=1}^{3} \sum_{q=1}^{4} (\alpha_j)_{pq}\, \partial_j \Psi_q + m \sum_{q=1}^{4} \beta_{pq} \Psi_q. \tag{6.76}$$

[a] **Question from the audience.** Are the explicit component indices unnecessary when matrix multiplication is used?

Response. Yes. Matrix notation suppresses those indices automatically. A more symmetric form that makes the relativistic structure more explicit can be introduced later.

[a] Lecture timestamp: 01:40:15.

The displayed equation is the Dirac equation in its simplest $3+1$-dimensional form. The next steps are to understand spin and to construct the wave equations for relativistic bosons, since the Dirac equation is not their equation and about half the particles in nature are bosons. With those ingredients in place, one can turn to quarks, leptons, weak bosons, Higgs particles, and photons, and bring some order to the particle spectrum. More symmetry is needed throughout that discussion.

Chapter 7

Angular Momentum, Spin, and Statistics

7.1 From Ordinary Rotation to Intrinsic Spin

Angular momentum is the formal subject, but spin is the real target. Spin is one of the simplest and most unintuitive properties of elementary particles, and its structure follows from the mathematics of angular momentum.

Angular momentum is a vector. Its direction lies along the axis of rotation. There are two possible directions along that axis, so a convention is needed to choose between them: curl the fingers of the right hand in the sense of rotation, and the thumb points in the direction of the angular-momentum vector. This is a definition, not a theorem.

The magnitude depends on how much mass is present, how that mass is distributed, the size of the system, and its angular velocity. The mass distribution and size are collected into the moment of inertia. A large object and a small object can therefore carry the same angular momentum while rotating at very different rates and with very different energies.

An ordinary composite object can carry two conceptually different kinds of angular momentum. A cup moving in a circle around an external axis has angular momentum because its center of mass is moving around that axis, even if the cup is not turning about any axis through itself. This is orbital angular momentum. Spin means rotation about an internal axis.

The clean definition uses the center-of-mass frame. Go to the frame in which the center of mass is at rest, so the total ordinary momentum vanishes. Any angular momentum that remains in that frame is spin angular momentum. The object need not be motionless internally; only its center of mass is at rest.

For a basketball this distinction causes no difficulty. A basketball at rest and the same basketball set rotating are plainly recognized as the same object in two different states. Quantum mechanics complicates this picture because angular momentum is discrete. There is no continuous sequence of allowed angular momenta joining a state of zero angular momentum to the first rotating state. One can therefore ask whether the first rotating state is the same object in motion or a different object altogether.

The wording is partly conventional, but the physical question is precise: how much energy is required to reach the first nonzero rotational state? If the gap is small, the excited object remains readily recognizable. If the gap is enormous, the excitation may change the object radically. An attempt to give an atom too much rotational excitation can throw off its electrons and destroy the atom rather than leave the same atom quietly rotating.

Now consider an electron, temporarily setting aside its known intrinsic spin. Can an electron be set rotating about an internal axis? A mathematical point has no visible internal structure to rotate, but experiment does not establish an electron's size by supplying a little rigid surface that can be watched turning. If the electron has an extremely small finite size, its first additional rotational excitation could simply require too much energy to reach. Because the energy rises as the size decreases, the required energy might be astronomical, perhaps even of order the Planck energy. This estimate is deliberately tentative.[1]

For an elementary particle, the magnitude of the spin is consequently treated as a fixed characteristic. Electrons do not come with a range of successively larger intrinsic spin magnitudes. If an otherwise electron-like excitation carried a different total spin, it would be classified as a different particle.

The fixed magnitude does not fix the direction. Imagine a particle called a *spintron*, identified partly by the magnitude of its angular momentum. If its spin points along one axis, rotational invariance requires that the same kind of particle can also have its spin pointed along another axis. No spatial direction is intrinsically preferred. The amount of spin remains fixed while its orientation can change. Thus the problem has two apparently conflicting ingredients: a quantized magnitude and freely rotatable directions. The angular-momentum algebra will show how they coexist.

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[a] **Question from the audience.** Why is energy required to spin a small particle?

Response. For an object of mass M and characteristic radius R, the moment of inertia is of order

$$I \sim cMR^2, \tag{7.1}$$

where the numerical factor c depends on the shape and on how the mass is distributed. If material near the outside moves with speed v, its ordinary momentum is of order Mv, and angular momentum is momentum times distance:

$$L \sim MvR. \tag{7.2}$$

Equivalently, $L = I\omega$ with $v \sim \omega R$. The rotational kinetic energy is

$$E_{\text{rot}} = \frac{L^2}{2I}. \tag{7.3}$$

At fixed M and fixed L,

$$E_{\text{rot}} \sim \frac{L^2}{2cMR^2} \propto \frac{1}{R^2}. \tag{7.4}$$

The smaller the object, the more energy a given angular momentum costs. That inverse-square dependence is classical. The additional quantum-mechanical fact is that angular momentum changes in discrete steps, so the first allowed step cannot be made arbitrarily small.

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[a] Lecture timestamp: 00:12:25.

The mathematics about to be used is abstract and initially rather unintuitive. Its interest is that experimentally testable facts emerge from a small set of algebraic rules.

[1] A short passage immediately following this estimate is not intelligible enough to support a further substantive reconstruction.

7.2 Orbital Angular Momentum

Begin with one particle moving in a circular orbit around an origin. Its angular momentum has magnitude equal to momentum times the perpendicular distance from the origin. In a plane one may call one sense of circulation positive and the reverse sense negative. The sign records the direction of the angular-momentum vector relative to the chosen axis.

This is not yet spin. The particle has angular momentum because it has ordinary momentum. Now place two particles at opposite ends of a diameter and let them move with the same sense of rotation. Their ordinary momenta cancel, so the center of mass is at rest, but their angular momenta point in the same direction and add. The system has zero total momentum and nonzero total angular momentum: a classical model of spin.

Let the position of a particle relative to the origin be

$$\vec{r} = (x, y, z) = (x_1, x_2, x_3), \tag{7.5}$$

and let its momentum be

$$\vec{p} = (p_x, p_y, p_z) = (p_1, p_2, p_3). \tag{7.6}$$

The angular momentum is the vector product

$$\vec{L} = \vec{r} \times \vec{p}. \tag{7.7}$$

Reversing the factors reverses the vector: $\vec{p} \times \vec{r} = -\vec{r} \times \vec{p}$. The standard convention is $\vec{r} \times \vec{p}$, with its direction fixed by the right-hand rule.[2]

In Cartesian components,

$$\begin{aligned} &L_x = yp_z - zp_y, \qquad L_y = zp_x - xp_z, \\ &L_z = xp_y - yp_x. \end{aligned} \tag{7.8}$$

It is enough to remember the first formula and cycle the labels

$$x \longrightarrow y \longrightarrow z \longrightarrow x.$$

Cycling both the component label and the labels on the right gives the next formula without changing the relative sign. The mnemonic is useful precisely because the sign is easy to reverse accidentally.

For a point particle, $\vec{r} \times \vec{p}$ is orbital angular momentum about the chosen origin. For a composite system, add the angular momenta of all constituents as vectors:

$$\vec{L}_{\text{tot}} = \sum_a \vec{r}_a \times \vec{p}_a. \tag{7.9}$$

When this sum is evaluated in the center-of-mass frame, it can remain nonzero even though

$$\sum_a \vec{p}_a = 0.$$

That remaining angular momentum is the spin of the composite system.

[2]Lecture video frame `lecture_07_figure_02.png`, 00:18:49.

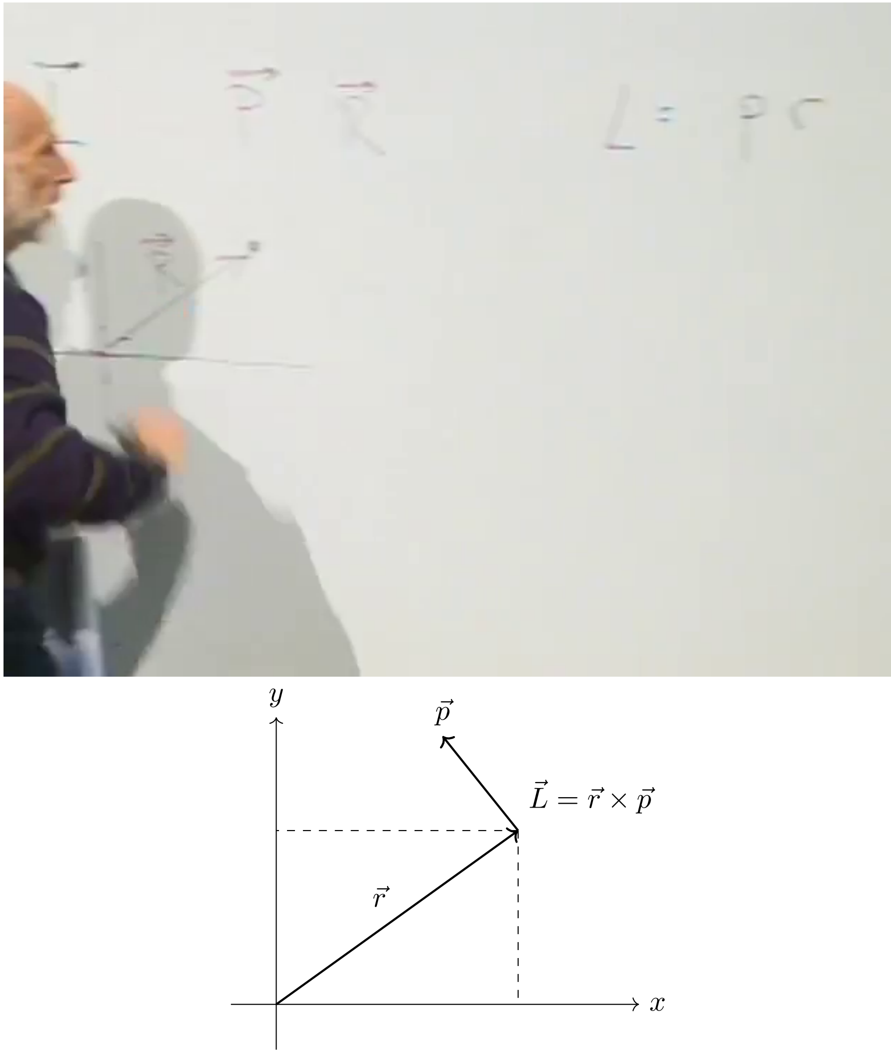

Figure 7.1: For a particle at position $\vec{r}$ with momentum $\vec{p}$, the angular momentum $\vec{L} = \vec{r} \times \vec{p}$ is perpendicular to their plane.

Lecture frame: 00:18:49.

7.3 Quantum Operators and Canonical Commutators

In quantum mechanics, observable quantities are represented by operators. Position, momentum, and each component of angular momentum are therefore operators. Their detailed realization as differential operators is not needed here. The commutation relations contain enough information to derive the angular-momentum spectrum.

For two operators A and B,

$$[A, B] = AB - BA. \tag{7.10}$$

If two observables commute, they can be simultaneously specified. If they do not commute, quantum mechanics prevents their simultaneous sharp specification.

All three position coordinates can be measured together, so the position components commute:

$$[x_i, x_j] = 0. \tag{7.11}$$

In particular, every operator commutes with itself. It would make no sense for an observable to be incompatible with another measurement of that same observable.

The momentum components likewise commute:

$$[p_i, p_j] = 0. \tag{7.12}$$

A position component commutes with a momentum component in a different direction, but not with its matching momentum component. The three canonical statements can be written together as

$$[x_i, x_j] = 0, \qquad [p_i, p_j] = 0, \qquad [x_i, p_j] = i\hbar\,\delta_{ij}. \tag{7.13}$$

Thus, for example,

$$[x, p_y] = 0, \qquad [x, p_x] = i\hbar,$$

and reversing the order reverses the sign:

$$[p_j, x_i] = -i\hbar\,\delta_{ij}. \tag{7.14}$$

Angular momentum and Planck's constant have the same dimensions. Angular momentum is distance times momentum, while the uncertainty relation has the dimensional form

$$\Delta x\,\Delta p \sim \hbar \tag{7.15}$$

up to the familiar numerical factor in the precise inequality. It is therefore natural that angular momentum will be quantized in units of $\hbar$, although the quantization has not yet been derived.

7.4 Deriving the Angular-Momentum Algebra

The first task is to compute the commutator of two angular-momentum components. From

$$L_x = yp_z - zp_y, \qquad L_y = zp_x - xp_z,$$

one has

$$\begin{aligned} [L_x, L_y] &= [yp_z - zp_y,\ zp_x - xp_z] \qquad (7.16)\\ &= [yp_z, zp_x] - [yp_z, xp_z] \\ &\quad - [zp_y, zp_x] + [zp_y, xp_z]. \qquad (7.17) \end{aligned}$$

[a] **Question from the audience.** Can the component formulas be remembered by cycling the symbols from left to right instead of cycling the three equations from top to bottom?

Response. Yes. Either cycling mnemonic is acceptable. The only purpose of the mnemonic is to generate the components in the correct cyclic order while keeping track of the sign.

[a] Lecture timestamp: 00:31:41.

A commutator can be understood by asking whether the factors in one product can be pushed through the factors in the other. In

$$[yp_z, xp_z],$$

the operator y commutes with both x and p_z, while p_z commutes with both x and itself. There is no obstruction, so this term vanishes. The same reasoning gives

$$[zp_y, zp_x] = 0.$$

Only two product pairs survive.

For the first one, the only obstruction is p_z passing through z:

$$\begin{aligned}
[yp_z, zp_x] &= yp_z zp_x - zp_x yp_z && (7.18)\\
&= y(p_z z - zp_z)p_x && (7.19)\\
&= y[p_z, z]p_x && (7.20)\\
&= -i\hbar\, yp_x. && (7.21)
\end{aligned}$$

The order of the canonical commutator matters:

$$[p_z, z] = -[z, p_z] = -i\hbar.$$

For the other surviving pair, the minus signs in the definitions of L_x and L_y multiply to a plus. The only obstruction is z passing through p_z:

$$\begin{aligned}
[zp_y, xp_z] &= zp_y xp_z - xp_z zp_y && (7.22)\\
&= x(zp_z - p_z z)p_y && (7.23)\\
&= x[z, p_z]p_y && (7.24)\\
&= i\hbar\, xp_y. && (7.25)
\end{aligned}$$

Combining the two contributions,

$$\begin{aligned}
[L_x, L_y] &= -i\hbar\, yp_x + i\hbar\, xp_y && (7.26)\\
&= i\hbar(xp_y - yp_x) && (7.27)\\
&= i\hbar L_z. && (7.28)
\end{aligned}$$

This is the important simplification. Commuting two components of angular momentum produces the third component; no position or momentum operator remains. Cyclic permutation gives

$$\begin{aligned}
&[L_x, L_y] = i\hbar L_z, \qquad [L_y, L_z] = i\hbar L_x,\\
&[L_z, L_x] = i\hbar L_y.
\end{aligned} \qquad (7.29)$$

Equivalently,

$$[L_i, L_j] = i\hbar\,\epsilon_{ijk} L_k. \qquad (7.30)$$

The algebra closes on the angular-momentum operators themselves. Once these relations have been obtained, the x_i and p_i used to derive them can be set aside. The same algebra applies when the total angular momentum is a sum over many constituents. In particular, it applies in the center-of-mass frame, where the remaining angular momentum is spin. The angular-momentum commutators can therefore be used abstractly without first constructing spin as the motion of little constituent parts.

The choice of x, y, and z axes is arbitrary. Rotate to any other mutually perpendicular set of inertial Cartesian axes, and the primed components obey the same relations:

$$[L'_x, L'_y] = i\hbar L'_z, \tag{7.31}$$

together with its cyclic partners. The algebra is rotationally invariant, so physical predictions cannot depend on which inertial axes were chosen for the calculation.

[a] **Question from the audience.** Is spin angular momentum independent of translations as well as rotations of the coordinate axes?

Response. Yes. Intrinsic spin is translation-independent. Orbital angular momentum about an origin can change when the origin is translated, but the spin remaining in the center-of-mass frame does not depend on that translation.

[a]Lecture timestamp: 00:39:09.

Abstract algebra can generate endlessly many identities, most of them physically uninteresting. Its value appears when an identity becomes a measurable fact or reveals something about the operational meaning of measurement. Up to this point the construction is still mainly algebraic. The next step extracts the discrete structure of spin.

[a] **Question from the audience.** What if the chosen x, y, and z axes rotate along with the particles?

Response. The discussion is restricted to inertial frames. A rotating coordinate system is noninertial and introduces effects such as centrifugal and Coriolis forces. Angular momentum is not invariant under a change to a co-rotating frame: a system that is visibly spinning in an inertial frame can appear stationary in a frame rotating with it. The rotational invariance used here concerns differently oriented inertial Cartesian axes, not axes that rotate in time.

[a]Lecture timestamp: 00:40:42.

7.5 Ladder Operators and the Quantization Axis

Introduce the two combinations

$$L_+ = L_x + iL_y, \qquad L_- = L_x - iL_y. \tag{7.32}$$

They are analogous to the creation and annihilation operators of the harmonic oscillator. Their role is to raise and lower one chosen component of angular momentum.

There is nothing special about the z-axis, but one axis must be chosen if a component is to be specified. Take z as the quantization axis and write

$$L_z|m\rangle = m\hbar\,|m\rangle. \tag{7.33}$$

Thus m is the measured z-component in units of $\hbar$. It is often convenient during the algebra to set $\hbar = 1$; it will be retained explicitly here.

The commutator $[L_+, L_-]$ is proportional to L_z, but it is not needed for the immediate argument. What matters first is the commutator of each ladder operator with L_z. From

$$[L_x, L_z] = -i\hbar L_y, \qquad [L_y, L_z] = i\hbar L_x,$$

one obtains

$$[L_+, L_z] = [L_x + iL_y, L_z] \tag{7.34}$$
$$= [L_x, L_z] + i[L_y, L_z] \tag{7.35}$$
$$= -i\hbar L_y + i(i\hbar L_x) \tag{7.36}$$
$$= -\hbar(L_x + iL_y) \tag{7.37}$$
$$= -\hbar L_+. \tag{7.38}$$

Changing the sign of the iL_y term gives

$$[L_-, L_z] = +\hbar L_-. \tag{7.39}$$

Equivalently, with the order of each commutator reversed,

$$[L_z, L_+] = \hbar L_+, \qquad [L_z, L_-] = -\hbar L_-. \tag{7.40}$$

Only one component of angular momentum can be sharp at a time. Any two distinct components fail to commute, so they cannot be simultaneously specified. The chosen sharp component defines the quantization axis; choosing z is a convenience, not a physical preference.

Now act with L_+ on an L_z eigenstate. The commutator can be rewritten as

$$L_z L_+ = L_+ L_z + \hbar L_+. \tag{7.41}$$

Therefore

$$L_z(L_+|m\rangle) = (L_+ L_z + \hbar L_+)|m\rangle \tag{7.42}$$
$$= m\hbar L_+|m\rangle + \hbar L_+|m\rangle \tag{7.43}$$
$$= (m+1)\hbar\, L_+|m\rangle. \tag{7.44}$$

Provided it is nonzero, $L_+|m\rangle$ is an L_z eigenstate whose eigenvalue has increased by one unit of $\hbar$. In the same way,

$$L_z(L_-|m\rangle) = (m-1)\hbar\, L_-|m\rangle. \tag{7.45}$$

Thus L_- lowers m by one.

This is already a nontrivial conclusion. Whatever the spectrum of L_z turns out to be, neighboring values are separated by integers:

$$\ldots,\ m-2,\ m-1,\ m,\ m+1,\ m+2,\ \ldots \tag{7.46}$$

The endpoints have not yet been determined.

7.6 Finite Multiplets and the Spin Label

A fixed kind of particle has a fixed angular-momentum magnitude. Its angular momentum may be reoriented, but increasing its magnitude without limit would eventually produce a different object. There must therefore be a maximum possible z-component for that particle.

How can a ladder with a raising operator have a top? The harmonic oscillator supplies the model. Its annihilation operator gives zero when it reaches the bottom state. Here the raising operator must give zero when it reaches the top state:

$$L_+|m_{\max}\rangle = 0. \tag{7.47}$$

The statement that L_+ raises an eigenvalue applies only when the vector it produces is nonzero. The highest-weight state is precisely the exceptional state at which the raising sequence terminates.

Rotational invariance fixes the lower endpoint. If the angular momentum can point maximally along $+z$, the same physical object can be rotated so that it points maximally along $-z$. Hence

$$m_{\min} = -m_{\max}. \tag{7.48}$$

The spectrum is symmetric about zero and has unit spacing. There are two possible families.

In the first family, zero occurs:

$$\begin{aligned} m = \; & -m_{\max}, -m_{\max}+1, \dots, -1, 0, \\ & 1, \dots, m_{\max}-1, m_{\max}. \end{aligned} \tag{7.49}$$

Then $m_{\max}$ is an integer, and the number of states is

$$2m_{\max} + 1. \tag{7.50}$$

The maximum value characterizes the particle and is called its spin. In conventional notation,

$$s \equiv m_{\max}, \qquad |s, m\rangle \equiv |m\rangle \tag{7.51}$$

within a fixed-spin multiplet.[3]

A spin-zero particle has only $m = 0$. A spin-one particle has

$$m = 1, 0, -1,$$

and a spin-two particle has five component values. The Higgs boson is an example of spin zero. The photon and the Z boson have spin one. A graviton, if present, has spin two. Composite objects can have spin three and much higher: nuclei can do so, and sufficiently macroscopic rotating objects such as basketballs and galaxies can carry arbitrarily large angular momentum. What is absent from the ordinary elementary-particle list is an elementary spin-three particle.

The other symmetric, unit-spaced family does not contain zero. It begins with

$$m = \frac{1}{2}, -\frac{1}{2}$$

and extends in unit steps:

$$\begin{aligned} m = \; & -m_{\max}, -m_{\max}+1, \dots, -\frac{3}{2}, -\frac{1}{2}, \\ & \frac{1}{2}, \frac{3}{2}, \dots, m_{\max}. \end{aligned} \tag{7.52}$$

[3] Lecture video frame `lecture_07_figure_03.png`, 01:01:08.

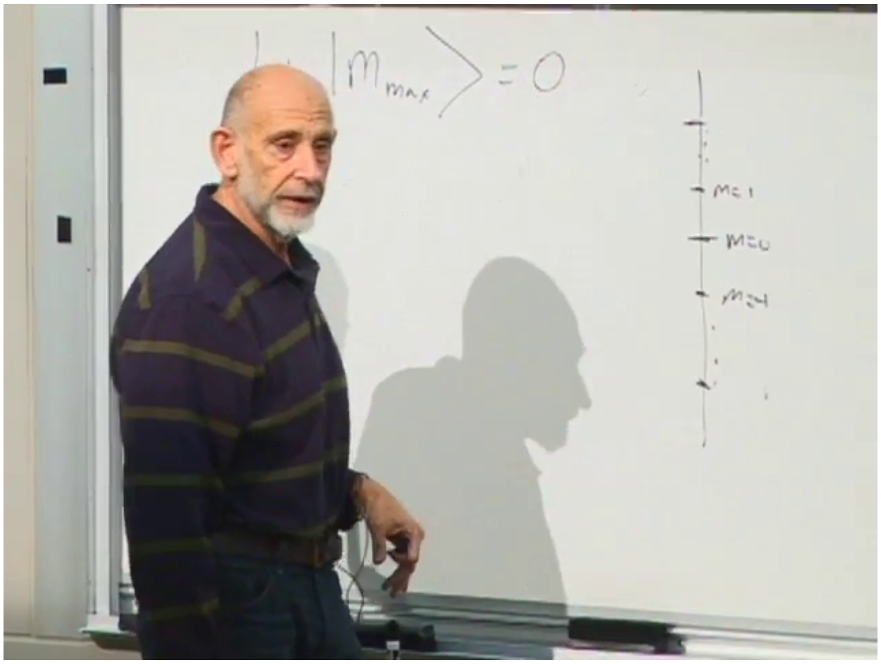

Figure 7.2: The highest-weight condition terminates the raising sequence, and the allowed m values form a finite unit-spaced ladder.

Lecture frame: 01:01:08.

Here $m_{\max}$ is a half-integer. The simplest member has spin $\frac{1}{2}$, followed by spin $\frac{3}{2}$, spin $\frac{5}{2}$, and so on. Angular-momentum multiplets therefore come in integer-spin and half-integer-spin families.

---

[a] **Question from the audience.** Does a half-integer-spin multiplet also have a maximum?

Response. Yes. The ladder again has a highest value. For a spin-$\frac{3}{2}$ particle the maximum is $m_{\max} = \frac{3}{2}$; for a spin-$\frac{5}{2}$ particle it is $m_{\max} = \frac{5}{2}$. The difference is only that the maximum and every other value in the multiplet are half-integers.

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[a]Lecture timestamp: 01:03:46.

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[a] **Question from the audience.** Are there no elementary particles of spin three or higher?

Response. Within the relativistic quantum framework under discussion, the elementary-particle spins on the usual list are $0, \frac{1}{2}, 1, \frac{3}{2}$, and 2. Composite objects are not restricted to this range and can have spin three or much larger spin.

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[a]Lecture timestamp: 01:04:04.

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[a] **Question from the audience.** Do the negative entries on the ladder represent negative spin?

Response. No. A negative m is a component value: it says that the angular momentum points oppositely relative to the chosen axis. The spin of the particle is the nonnegative maximum $s = m_{\max}$.

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[a]Lecture timestamp: 01:04:50.

[a] **Question from the audience.** If the component values are discrete, what happens to all the orientations between them?

Response. There are not merely two or three possible spatial orientations. Intermediate orientations are represented by quantum superpositions of the m-basis states. The spin-$\frac{1}{2}$ construction below makes this explicit.

[a]Lecture timestamp: 01:05:10.

[a] **Question from the audience.** What about corkscrew motion or circularly polarized waves?

Response. A circularly polarized electromagnetic wave is built from circularly polarized photons, and those photons carry angular momentum along the propagation axis. The word "corkscrew" combines rotation with forward motion, so it is not the right description for an object being considered at rest. For a moving particle, the relevant quantity is helicity: the orientation of its spin relative to its direction of motion.

[a]Lecture timestamp: 01:05:42.

[a] **Question from the audience.** Is any known elementary particle spin $\frac{3}{2}$?

Response. No spin-$\frac{3}{2}$ elementary particle has been directly confirmed. Supersymmetry conjectures a spin-$\frac{3}{2}$ partner of the graviton called the gravitino, but it has not been directly detected.

[a]Lecture timestamp: 01:07:10.

[a] **Question from the audience.** Have the angular-momentum operators stopped acting on the original wavefunction or ket space?

Response. No original spin space had yet been specified. The algebra is being used to determine that space. For fixed spin s, it has basis vectors labeled by the allowed values

$$|s, m\rangle, \qquad m = -s, -s+1, \ldots, s.$$

This intrinsic spin space is distinct from the ordinary space of spatial wavefunctions on which orbital angular momentum acts as $\mathbf{r} \times \mathbf{p}$, although both realize the same angular-momentum algebra.

[a]Lecture timestamp: 01:08:01.

7.7 The Magnitude of Angular Momentum

[a] **Question from the audience.** Is the magnitude of the angular-momentum vector observable?

Response. Yes. It is most convenient to use its square,

$$L^2 = L_x^2 + L_y^2 + L_z^2.$$

Its value on the highest-weight state is not simply $s^2\hbar^2$. Even when L_z is sharp and maximal, the noncommuting transverse components cannot both be sharply zero, so they supply an additional quantum contribution.

[a]Lecture timestamp: 01:09:32.

To calculate that contribution, multiply the ladder operators in the order for which the raising operator acts first on a ket:

$$L_-L_+ = (L_x - iL_y)(L_x + iL_y) \tag{7.53}$$
$$= L_x^2 + L_y^2 + iL_xL_y - iL_yL_x \tag{7.54}$$
$$= L_x^2 + L_y^2 + i[L_x, L_y] \tag{7.55}$$
$$= L_x^2 + L_y^2 - \hbar L_z. \tag{7.56}$$

The final step uses $[L_x, L_y] = i\hbar L_z$. Therefore

$$L_x^2 + L_y^2 = L_-L_+ + \hbar L_z, \tag{7.57}$$

and hence

$$L^2 = L_-L_+ + L_z^2 + \hbar L_z. \tag{7.58}$$

Apply this identity to the highest-weight state $|s, s\rangle = |m_{\text{max}}\rangle$:

$$L^2|s, s\rangle = \left(L_-L_+ + L_z^2 + \hbar L_z\right)|s, s\rangle \tag{7.59}$$
$$= L_-L_+|s, s\rangle + s^2\hbar^2|s, s\rangle + s\hbar^2|s, s\rangle. \tag{7.60}$$

The first term vanishes because $L_+|s, s\rangle = 0$. Thus

$$L^2|s, s\rangle = s(s + 1)\hbar^2|s, s\rangle. \tag{7.61}$$

The extra $+s$ is the quantum correction associated with the unavoidable transverse fluctuations. Classically one would have expected only $s^2\hbar^2$.

The first few values make the distinction concrete:

$$s = 0 : \qquad L^2 = 0, \tag{7.62}$$
$$s = 1 : \qquad L^2 = 2\hbar^2, \tag{7.63}$$
$$s = \frac{1}{2} : \qquad L^2 = \frac{3}{4}\hbar^2. \tag{7.64}$$

For $s = 1000$, the dimensionless factor is 1000×1001, which differs little from 1000^2 on the scale of such a large angular momentum. In general,

$$s(s + 1) = s^2\left(1 + \frac{1}{s}\right), \tag{7.65}$$

so the relative quantum correction becomes negligible for very large s. That is how the classical s^2 result emerges.

Every m-state in a fixed spin multiplet has this same eigenvalue of L^2. Changing m changes the component along the chosen axis, not the total magnitude. This is the quantum version of rotating an angular-momentum vector without changing its length.

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[a] **Question from the audience.** Can L^2 and the z-component be known simultaneously?

Response. Yes. The total square commutes with every angular-momentum component:

$$[L^2, L_i] = 0, \qquad i = x, y, z.$$

Thus L^2 and one chosen component, such as L_z, can be specified together even though two distinct components cannot. Every electron, for example, has $L^2 = \frac{3}{4}\hbar^2$, independent of its orientation.

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[a] Lecture timestamp: 01:16:43.

[a] **Question from the audience.** Is L^2 zero for spin one, or only for spin zero?
Response. Only spin zero has $L^2 = 0$. Spin one gives

$$L^2 = 1(1+1)\hbar^2 = 2\hbar^2,$$

while spin three gives

$$L^2 = 3(3+1)\hbar^2 = 12\hbar^2.$$

[a]Lecture timestamp: 01:17:41.

[a] **Question from the audience.** Why does every m-level in the multiplet have the same L^2 eigenvalue?
Response. Start from the highest-weight state and lower it. Since L^2 commutes with every component, it also commutes with L_-:

$$[L^2, L_-] = 0.$$

If

$$L^2|s,s\rangle = s(s+1)\hbar^2|s,s\rangle,$$

then

$$L^2\big(L_-|s,s\rangle\big) = L_-L^2|s,s\rangle \tag{7.66}$$
$$= s(s+1)\hbar^2\, L_-|s,s\rangle. \tag{7.67}$$

Thus the next state down has the same L^2 eigenvalue. Repeating the argument carries that eigenvalue all the way down the ladder.

[a]Lecture timestamp: 01:18:13.

The discrete m-values therefore do not mean that only a discrete set of spatial orientations exists. Rotational invariance still requires states corresponding to rotated directions. For spin $\frac{1}{2}$, those directions will appear as superpositions of the two L_z basis states.

7.8 Spin-$\frac{1}{2}$ States and Orientation

Rotational invariance requires more than the discrete L_z eigenstates. A spin may point along any axis, even though measurement of its component along a chosen axis gives only the allowed discrete values. The directions between the z-axis eigenstates are represented by superpositions.

For a spin-$\frac{1}{2}$ particle, restrict attention to the spin degrees of freedom. An electron also has orbital states, so it certainly does not have only two states altogether; its spin space alone is two-dimensional. Choose the z-axis basis

$$L_z|+\rangle = \frac{\hbar}{2}|+\rangle, \qquad L_z|-\rangle = -\frac{\hbar}{2}|-\rangle. \tag{7.68}$$

A general spin state is

$$|\psi\rangle = \alpha|+\rangle + \beta|-\rangle, \tag{7.69}$$

where α and β are complex numbers. Measurement of L_z gives $+\hbar/2$ with probability $|\alpha|^2$ and $-\hbar/2$ with probability $|\beta|^2$, so normalization requires

$$|\alpha|^2 + |\beta|^2 = 1. \tag{7.70}$$

A common phase has no physical effect:

$$(\alpha, \beta) \longrightarrow e^{i\chi}(\alpha, \beta). \tag{7.71}$$

The two complex amplitudes begin with four real parameters. Normalization removes one, and the irrelevant overall phase removes another, leaving two physical real parameters. That is exactly the number needed to specify a direction by its polar and azimuthal angles. Varying α and β rotates the spin orientation.

A measured component and its expectation value are different things. The expectation value is an ensemble average. Prepare many electrons in exactly the same state, measure L_z once on each electron, add the results, and divide by the number of experiments. Every individual result is still either $+\hbar/2$ or $-\hbar/2$, but the average may take any value between them:

$$\langle L_z \rangle = \frac{\hbar}{2}(|\alpha|^2 - |\beta|^2). \tag{7.72}$$

Equal up and down amplitudes give a spin directed along one of the x-axis orientations. Normalized representatives are[4]

$$|x;+\rangle = \frac{|+\rangle + |-\rangle}{\sqrt{2}}, \qquad |x;-\rangle = \frac{|+\rangle - |-\rangle}{\sqrt{2}}. \tag{7.73}$$

For either state, measurement along z gives the two possible outcomes with equal probability, and therefore

$$\langle L_z \rangle = \frac{1}{2}\left(\frac{\hbar}{2}\right) + \frac{1}{2}\left(-\frac{\hbar}{2}\right) = 0. \tag{7.74}$$

The plus and minus relative signs give opposite directions along the x-axis. Similarly,

$$\frac{|+\rangle + i|-\rangle}{\sqrt{2}}, \qquad \frac{|+\rangle - i|-\rangle}{\sqrt{2}} \tag{7.75}$$

give the two opposite y-axis orientations. General normalized values of α and β describe arbitrary axes. To say that a spin points along a particular axis means that measurement of the component along that axis gives a definite result. Relative to a different axis, the same state is generally a superposition.

The individual measurement outcomes remain quantized. It is their ensemble averages that vary continuously and reproduce the familiar geometry of a classical vector.

7.9 Spin, Statistics, and Composite Particles

Mass, electric charge, and spin are among the principal properties used to characterize a particle. Spin is also correlated with statistics. The spin-statistics relation is a theorem of relativistic quantum mechanics, although it will not be proved here:

$$\begin{aligned} \text{half-integer spin} &\quad\Longleftrightarrow\quad \text{fermion,} \\ \text{integer spin} &\quad\Longleftrightarrow\quad \text{boson.} \end{aligned} \tag{7.76}$$

[4]The coefficient ratios fix the directions. The factors $1/\sqrt{2}$ supply the normalization required by $|\alpha|^2 + |\beta|^2 = 1$. Relative phases $+i$ and $-i$ give opposite y-axis orientations; assigning either sign to positive y depends on the axis convention.

Two identical fermions cannot occupy the same complete quantum state. Identical bosons may occupy the same state, and large numbers of them in one state can behave collectively like a classical wave. Thus spin $\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \ldots$ particles are fermions, while spin $0, 1, 2, \ldots$ particles are bosons.

The rule extends immediately to composites. Adding two half-integer angular momenta gives an integer angular momentum. More generally, an even number of fermionic constituents makes a bosonic composite, whereas an odd number makes a fermionic composite. Adding bosonic constituents does not change this parity rule.

Ordinary hydrogen contains a spin-$\frac{1}{2}$ proton and a spin-$\frac{1}{2}$ electron. It therefore contains an even number of fermions and is bosonic as a whole. A deuteron contains a proton and a neutron, both fermions, and is also a boson. A deuterium atom adds an electron to the deuteron, giving three fermionic constituents at this level of description, so deuterium is a fermion.

Particles and antiparticles have the same mass and spin and opposite electric charge. Positronium is an electron and a positron bound to one another. It contains two fermions and is therefore bosonic.

---

[a] **Question from the audience.** Do bosons pass through things, and is superconductivity related to paired electrons behaving as bosons?

Response. The relevant property is not simply an ability to pass through matter; it is the ability of bosons to occupy the same state. Superfluid helium gives the simpler example. An ordinary helium-4 atom contains two protons, two neutrons, and two electrons, an even number of fermionic constituents, so the atom is a boson. Many helium atoms can occupy one quantum state and move collectively. A superfluid can then be pictured, approximately, as a classical wave formed by piling many atoms into the same state. That picture is not exact, but it captures the point needed here. An electron in a metal is a fermion and cannot share a complete state with another electron. Pairs of electrons, however, are bosonic composites, which is the connection to superconductivity. This raises a paradox: if the constituent electrons cannot occupy the same state, how can several composite pairs occupy one state? The resolution requires a more careful account of composite states and is deferred.

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[a] Lecture timestamp: 01:33:08.

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[a] **Question from the audience.** Since protons and neutrons are composite, why do they have half-integer spin?

Response. Each is made from an odd number of spin-$\frac{1}{2}$ quarks. Three fermionic constituents make a fermionic composite, so protons and neutrons can have half-integer spin.

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[a] Lecture timestamp: 01:35:46.

7.10 Pauli Exclusion and Atomic Orbitals

The periodic table provides the immediate motivation for the Pauli exclusion principle. The rule is not that two electrons can never have the same orbital motion. It is that two electrons cannot occupy the same complete quantum state.

This distinction matters even in the simplest atoms. Starting from hydrogen, increase the nuclear charge and add a second electron. The two electrons may occupy the same spatial orbital, but their spin states must differ. One may be spin up and the other spin down with respect to the chosen axis. When two electrons occupy one orbital, their spins combine into the spin-singlet state, so their total spin cancels. No more than two electrons may occupy a single spatial orbital because there are only two independent spin states available.

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[a] **Question from the audience.** Can the opposite spins of two electrons in one orbital be pictured as meshed gears?

Response. As a rough picture, yes: meshed gears turn in opposite senses. It is only an analogy for the opposite spin directions, not a literal mechanical model of the electrons.

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[a]Lecture timestamp: 01:37:53.

The word "orbital" can be used at two different levels. A state of orbital motion, in the broad sense, is the state obtained by solving the Schrödinger equation while ignoring spin. It includes more information than merely which shell the electron occupies.

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[a] **Question from the audience.** Does an orbital mean more than merely which shell the electron occupies?

Response. Yes. Here the orbital motion means the electron's spin-ignored Schrödinger state. It includes the principal quantum number n, which carries radial information and roughly sets the electron's distance from the nucleus, as well as the orbital-angular-momentum quantum numbers. In the spectroscopic notation used by chemists, S, P, D, and F correspond respectively to orbital angular momentum $l = 0, 1, 2, 3$.

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[a]Lecture timestamp: 01:38:09.

Thus the principal quantum number and the orbital angular momentum describe different aspects of the motion. The S, P, D, F labels refer to the value of l, while n supplies separate radial information. Pauli's rule is that no two electrons may have all of their quantum numbers the same, including the spin quantum number.

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[a] **Question from the audience.** Can the quantum numbers determine how many electrons may occupy each shell in an atom?

Response. Yes. They determine the maximum possible number, though not how many electrons a particular atom actually contains in that shell. The general shell-counting problem will not be pursued here.

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[a]Lecture timestamp: 01:40:42.

For a fixed orbital angular momentum l, the component quantum number has

$$2l + 1 \tag{7.77}$$

possible values. This is the same angular-momentum counting already used for spin. An S state has $l = 0$, so it has one orbital state. A P state has $l = 1$, so it has three orbital states. A D state has $l = 2$, so it has five.

Spin doubles each orbital count because each spatial orbital can carry one electron in each of the two opposite spin states.

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[a] **Question from the audience.** Does the S state hold two electrons, and do the three P states hold six more?

Response. Yes. The single S orbital state holds at most two electrons. The three P orbital states hold two each, giving six more. The S and P states under discussion therefore accommodate $2 + 6 = 8$ electrons in total.

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[a]Lecture timestamp: 01:41:54.

The same angular-momentum structure that organizes atomic orbitals appears again, with some differences, when quarks move in bound systems and orbit one another.

Chapter 8

Composite Bosons, Spin Representations, and Dirac Doubling

8.1 Exchange Symmetry and Composite Bosons

The opening question is a genuine paradox about composite particles. If two fermions make a boson, can two such bosons occupy the same state without violating fermionic exclusion? The answer begins with the exchange rule itself.

Suppose two identical particles are described by a wavefunction of two arguments. The symbols x and y label the two particle slots; they do not distinguish two species. For bosons the wavefunction is symmetric, while for fermions it is antisymmetric:

$$\Psi_B(x, y) = \Psi_B(y, x), \qquad \Psi_F(x, y) = -\Psi_F(y, x). \tag{8.1}$$

The symmetry says that interchanging the two indistinguishable particles does not produce a new physical state.

Two bosons can certainly occupy the same one-particle state. If that state has wavefunction φ, the two-particle product is

$$\Psi(x, y) = \varphi(x)\varphi(y). \tag{8.2}$$

Wavefunctions are ordinary complex-valued functions, not operators or matrices, so their values commute:

$$\varphi(x)\varphi(y) = \varphi(y)\varphi(x). \tag{8.3}$$

The product is therefore symmetric. The same construction cannot give a nonzero two-fermion state, because antisymmetry would require

$$\varphi(x)\varphi(y) = -\varphi(y)\varphi(x), \tag{8.4}$$

while the two sides without the minus sign are equal.

More generally, begin with any two-particle wavefunction $F(x, y)$, with no assumed exchange symmetry. Its symmetric and antisymmetric parts are, apart from normalization,

$$F_{\text{sym}}(x, y) = F(x, y) + F(y, x), \tag{8.5}$$
$$F_{\text{asym}}(x, y) = F(x, y) - F(y, x). \tag{8.6}$$

The exchange check is explicit:

$$F_{\text{sym}}(y,x) = F(y,x) + F(x,y) = F_{\text{sym}}(x,y), \tag{8.7}$$

$$F_{\text{asym}}(y,x) = F(y,x) - F(x,y) = -F_{\text{asym}}(x,y). \tag{8.8}$$

Adding the exchanged function symmetrizes it; subtracting the exchanged function antisymmetrizes it.

A pair of hydrogen atoms provides the more interesting case. There are four constituent fermions: two identical electrons and two identical protons. Let x_1, x_2 denote the electron variables and y_1, y_2 the proton variables. The four-body wavefunction must be antisymmetric separately under exchange of the electrons and under exchange of the protons:

$$\Psi(x_2, x_1; y_1, y_2) = -\Psi(x_1, x_2; y_1, y_2), \tag{8.9}$$

$$\Psi(x_1, x_2; y_2, y_1) = -\Psi(x_1, x_2; y_1, y_2). \tag{8.10}$$

To motivate two distinct one-atom wavefunctions, imagine one hydrogen atom localized near the origin, with wavefunction $\psi(x, y)$, and another localized elsewhere, with wavefunction $\phi(x, y)$. The second may simply be a translated copy of the first, obtained by translating both its electron and proton arguments, but it is still a distinct wavefunction in this setup. A naive product is

$$\psi(x_1, y_1)\phi(x_2, y_2). \tag{8.11}$$

It does not yet have either required fermionic antisymmetry.

First antisymmetrize the electron variables:

$$\Psi_x = \psi(x_1, y_1)\phi(x_2, y_2) - \psi(x_2, y_1)\phi(x_1, y_2). \tag{8.12}$$

Under $x_1 \leftrightarrow x_2$, the two terms exchange places and the whole expression changes sign. Now antisymmetrize this result in the proton variables. The four-term ordering is fixed by the two successive antisymmetrizations.[1]

$$\begin{aligned}
\Psi_{2H}(x_1, x_2; y_1, y_2) &= \Psi_x(x_1, x_2; y_1, y_2) \\
&\quad - \Psi_x(x_1, x_2; y_2, y_1) \qquad (8.13) \\
&= \psi(x_1, y_1)\phi(x_2, y_2) \\
&\quad - \psi(x_2, y_1)\phi(x_1, y_2) \\
&\quad - \psi(x_1, y_2)\phi(x_2, y_1) \\
&\quad + \psi(x_2, y_2)\phi(x_1, y_1). \qquad (8.14)
\end{aligned}$$

An electron exchange interchanges the first two terms and the last two terms, with an overall sign reversal. A proton exchange does the same. Thus the state has exactly the two separate antisymmetries required of the constituent fermions.

[1] This construction also fixes the sign pattern $+, -, -, +$ through the variable-label corrections made during the spoken calculation.

[a] **Question from the audience.** If hydrogen atoms are bosons, what prevents two of them from being placed directly on top of each other?

Response. Bosonic exchange statistics do not remove the forces between the atoms. Pauli exclusion among the constituents, together with the other interatomic forces, can make spatial overlap extremely costly. That is different from asking whether the atoms may occupy the same delocalized momentum state. A momentum eigenstate is spread throughout space, so common momentum does not mean ramming the atoms into the same point. Even hard-core objects like billiard balls could obey bosonic exchange statistics: their hard cores would prevent spatial overlap, while bosonic statistics would still allow common momentum occupation.

[a]Lecture timestamp: 00:11:11.

[a] **Question from the audience.** Would putting two hydrogen atoms in the same momentum eigenstate put their constituent fermions in the same state?

Response. No. The four-body state remains antisymmetric under exchange of the two electron variables and separately antisymmetric under exchange of the two proton variables. Only the simultaneous exchange of both constituents—the exchange of the two whole atoms—is symmetric.

[a]Lecture timestamp: 00:12:28.

Indeed, exchanging the two atoms means making both substitutions

$$x_1 \leftrightarrow x_2, \qquad y_1 \leftrightarrow y_2. \tag{8.15}$$

The first substitution contributes one fermionic minus sign and the second contributes another:

$$\begin{aligned} \Psi_{2H}(x_2, x_1; y_2, y_1) &= (-1)(-1)\Psi_{2H}(x_1, x_2; y_1, y_2) \\ &= \Psi_{2H}(x_1, x_2; y_1, y_2). \end{aligned} \tag{8.16}$$

The same result can be seen term by term in Eq. (8.14). Under the whole-atom exchange,

$$\psi(x_1, y_1)\phi(x_2, y_2) \longleftrightarrow \psi(x_2, y_2)\phi(x_1, y_1), \tag{8.17}$$

so the first and fourth terms exchange. Likewise,

$$-\psi(x_2, y_1)\phi(x_1, y_2) \longleftrightarrow -\psi(x_1, y_2)\phi(x_2, y_1), \tag{8.18}$$

so the second and third terms exchange. The complete state is symmetric under interchange of the atoms. Two constituent minus signs give the composite a plus sign.

This exchange statement is logically independent of the energy required to overlap the atoms. Strong repulsion may make such overlap inaccessible, but it does not change the symmetric exchange law.

[a] **Question from the audience.** What does the constructed four-body state represent physically, and is it specifically a momentum-space state?

Response. It is the two-atom wavefunction obtained from two one-atom wavefunctions by antisymmetrizing the electron labels and then the proton labels. It need not be written in momentum space. The variables may describe position, momentum, or another representation; the required constituent exchange symmetries are the same.

[a]Lecture timestamp: 00:14:42.

Consequently, two atoms may occupy the same delocalized momentum wavefunction even though forcing their spatial distributions into strong overlap can cost a great deal of energy.

Now set the two one-atom wavefunctions equal, $\phi = \psi$. Equation (8.14) becomes

$$\begin{aligned}\Psi_{\text{same}} = \psi(x_1,y_1)\psi(x_2,y_2) - \psi(x_2,y_1)\psi(x_1,y_2) \\ - \psi(x_1,y_2)\psi(x_2,y_1) + \psi(x_2,y_2)\psi(x_1,y_1).\end{aligned} \tag{8.19}$$

Does this four-body state vanish? It does not. The first and fourth products are equal, and the second and third products are equal. Using the commutativity of the one-atom wavefunctions,

$$\Psi_{\text{same}} = 2\left[\psi(x_1,y_1)\psi(x_2,y_2) - \psi(x_2,y_1)\psi(x_1,y_2)\right], \tag{8.20}$$

which is generally nonzero. It remains antisymmetric in the two electron variables and separately antisymmetric in the two proton variables.

There is no assignment here of one sharply located electron to each sharply located atom. An electron bound in an atom is spatially delocalized over its atomic wavefunction. Giving the two atoms the same atomic state therefore does not set $x_1 = x_2$, nor does it set $y_1 = y_2$.

The constituent exclusion is visible by testing coincidence directly. If the two modeled electron arguments coincide, $x_1 = x_2 = x$, Eq. (8.20) gives

$$\begin{aligned}\Psi_{\text{same}}(x,x;y_1,y_2) &= 2\big[\psi(x,y_1)\psi(x,y_2) \\ &\quad - \psi(x,y_1)\psi(x,y_2)\big] \qquad (8.21)\\ &= 0. \qquad (8.22)\end{aligned}$$

The corresponding expression also vanishes when $y_1 = y_2$. Thus the same-state composite wavefunction is nonzero, while its amplitude at coincident identical-fermion arguments vanishes.[2]

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[a] **Question from the audience.** Can the atoms be in the same state while their electrons and protons are not in the same constituent states?

Response. Yes. Equality of the composite atomic wavefunctions does not remove the separate electron and proton antisymmetries. The complete four-body state can describe two atoms in one atomic state while assigning zero amplitude to coincident identical-fermion arguments.

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[a]Lecture timestamp: 00:20:41.

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[a] **Question from the audience.** Did antisymmetrization project out the pieces in which the identical constituents coincide?

Response. Yes. In the variables displayed here, those coincidence components cancel exactly. The precise statement is that the antisymmetric wavefunction vanishes when the two identical-fermion arguments are equal.

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[a]Lecture timestamp: 00:21:12.

The cancellation did not depend on first making the atomic wavefunctions equal. For the general

[2]Spin and any other one-particle labels are suppressed in this model. Fermionic antisymmetry applies to the complete set of one-particle labels; the displayed cancellation concerns coincidence of the arguments retained here.

state in Eq. (8.14),

$$\begin{aligned}\Psi_{2H}(x,x;y_1,y_2) &= \psi(x,y_1)\phi(x,y_2)\\ &\quad -\psi(x,y_1)\phi(x,y_2)\\ &\quad -\psi(x,y_2)\phi(x,y_1)\\ &\quad +\psi(x,y_2)\phi(x,y_1) &(8.23)\\ &= 0. &(8.24)\end{aligned}$$

[a] **Question from the audience.** Does this cancellation still hold when the atoms are in different states and the identical constituent variables coincide?

Response. Yes. It follows from antisymmetry itself, not from setting $\phi = \psi$. The same statement holds in another representation: after transforming to momentum space, the amplitude vanishes when the two identical-fermion momentum arguments coincide, provided the other suppressed one-particle labels coincide as well.

[a]Lecture timestamp: 00:21:32.

8.2 Spin as a Finite-Dimensional Quantum System

The mathematics of spin recurs in isospin, color, and the other internal symmetries of particle physics. The analogy is formal: the same algebra appears, although the physical quantities need not have the same interpretation. The immediate task is to develop spin a little further and then return to the Dirac equation. This machinery will later be used for isospin, a concept associated historically with the 1930s.

An electron is a spin-$\frac{1}{2}$ particle, as are the proton and the muon. Its z-component of spin has two possible values. Before setting units, write

$$S_z = m\hbar. \tag{8.25}$$

From now on $\hbar = 1$. The letter L is conventionally reserved for orbital angular momentum, so intrinsic spin will be denoted by S. It obeys the same angular-momentum algebra:

$$[S_x, S_y] = iS_z, \quad [S_y, S_z] = iS_x, \quad [S_z, S_x] = iS_y, \tag{8.26}$$

and

$$S_\pm = S_x \pm iS_y. \tag{8.27}$$

The operator S_+ raises the S_z eigenvalue and S_- lowers it.

[a] **Question from the audience.** Is the discussion specifically about intrinsic spin?

Response. Intrinsic spin is the intended focus. The same angular-momentum algebra applies more generally, including to orbital angular momentum or the rotation of a macroscopic object.

[a]Lecture timestamp: 00:25:39.

For spin one-half the only S_z eigenvalues are $+\frac{1}{2}$ and $-\frac{1}{2}$. Suppress everything else about the electron for the moment: forget its position and momentum and retain only the two spin states.

Then

$$|\psi\rangle = \alpha|\uparrow_z\rangle + \beta|\downarrow_z\rangle \quad \longleftrightarrow \quad \begin{pmatrix}\alpha \\ \beta\end{pmatrix}, \tag{8.28}$$

where

$$|\alpha|^2 = P(\uparrow_z), \qquad |\beta|^2 = P(\downarrow_z), \qquad |\alpha|^2 + |\beta|^2 = 1. \tag{8.29}$$

This is a two-dimensional state space. It is not a claim that ordinary space has two dimensions.

---

[a] **Question from the audience.** Was physical space said to be two-dimensional?

Response. No. Spin state space is two-dimensional. Position and momentum have temporarily been suppressed, as if the electron were nailed to a wall and only its spin could be measured.

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[a]Lecture timestamp: 00:28:18.

Operators on this state space are represented by 2×2 matrices. It is convenient to choose the S_z basis so that S_z is diagonal, but diagonality is a basis choice rather than a physical requirement.

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[a] **Question from the audience.** Must S_z be diagonal?

Response. No. It is chosen diagonal for convenience. Other matrix representations are physically equivalent.

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[a]Lecture timestamp: 00:29:46.

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[a] **Question from the audience.** Is S_z a matrix?

Response. S_z is an observable and therefore an operator. In the chosen finite-dimensional basis, that operator is represented by a matrix acting on state vectors.

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[a]Lecture timestamp: 00:30:08.

A particularly useful representation is

$$S_i = \frac{1}{2}\sigma_i, \tag{8.30}$$

where

$$\sigma_x = \begin{pmatrix}0 & 1 \\ 1 & 0\end{pmatrix}, \quad \sigma_y = \begin{pmatrix}0 & -i \\ i & 0\end{pmatrix}, \quad \sigma_z = \begin{pmatrix}1 & 0 \\ 0 & -1\end{pmatrix}. \tag{8.31}$$

The matrices without the factor of one-half are the Pauli matrices; the spin operators themselves include the factor. Direct multiplication verifies the commutation relations. The representation is not unique: rotating the axes or changing basis produces another equivalent triplet.

This is a two-component representation of spin-one-half angular momentum.

8.2.1 Eigenstates along the coordinate axes

It is useful to solve the eigenvalue equations component by component. Since the eigenvalues of σ_i are ± 1, those of $S_i = \sigma_i/2$ are $\pm\frac{1}{2}$.

For σ_z, the two equations are

$$\begin{pmatrix}1 & 0 \\ 0 & -1\end{pmatrix}\begin{pmatrix}\alpha \\ \beta\end{pmatrix} = \begin{pmatrix}\alpha \\ \beta\end{pmatrix}, \qquad \begin{pmatrix}1 & 0 \\ 0 & -1\end{pmatrix}\begin{pmatrix}\alpha \\ \beta\end{pmatrix} = -\begin{pmatrix}\alpha \\ \beta\end{pmatrix}. \tag{8.32}$$

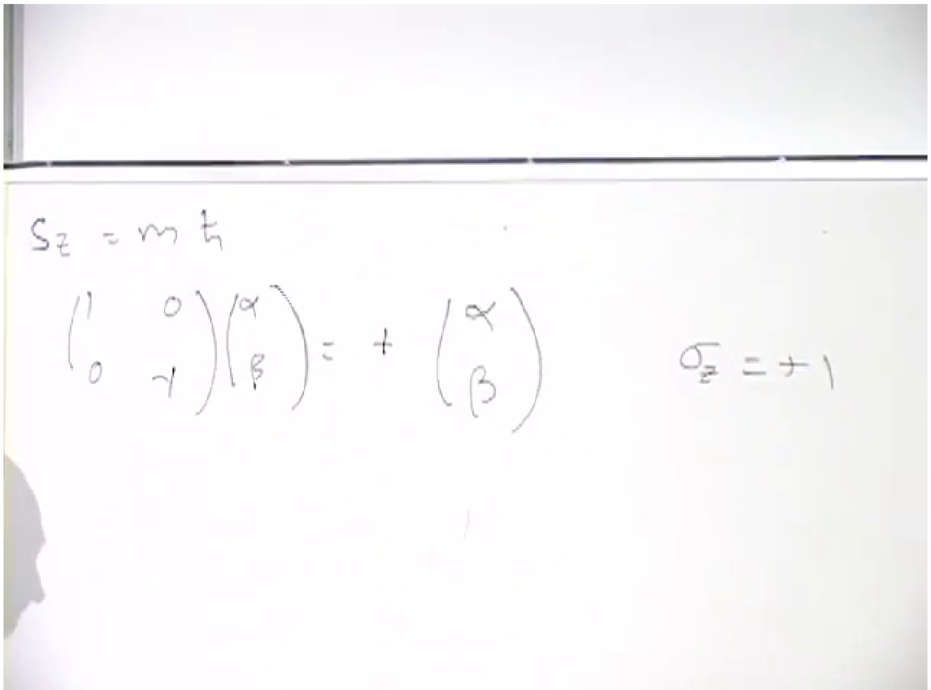

Figure 8.1: The +1 eigenvalue equation for σ_z acting on a two-component spinor.
Lecture frame: 00:34:57.

For eigenvalue +1, row-by-row multiplication gives

$$\begin{pmatrix} 1 \cdot \alpha + 0 \cdot \beta \\ 0 \cdot \alpha - \beta \end{pmatrix} = \begin{pmatrix} \alpha \\ -\beta \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \tag{8.33}$$

The upper equation is automatic, while the lower one gives $-\beta = \beta$, hence $\beta = 0$. Normalization requires $|\alpha| = 1$; choosing the irrelevant overall phase so that $\alpha = 1$,

$$|+z\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{8.34}$$

For eigenvalue -1,

$$\begin{pmatrix} \alpha \\ -\beta \end{pmatrix} = \begin{pmatrix} -\alpha \\ -\beta \end{pmatrix}. \tag{8.35}$$

Now the lower equation is automatic and the upper one gives $\alpha = -\alpha$, so $\alpha = 0$. Choosing $\beta = 1$ gives

$$|-z\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{8.36}$$

Now change the experiment and measure S_x. Since

$$\sigma_x \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 0 \cdot \alpha + \beta \\ \alpha + 0 \cdot \beta \end{pmatrix} = \begin{pmatrix} \beta \\ \alpha \end{pmatrix}, \tag{8.37}$$

the +1 equation becomes

$$\begin{pmatrix} \beta \\ \alpha \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \tag{8.38}$$

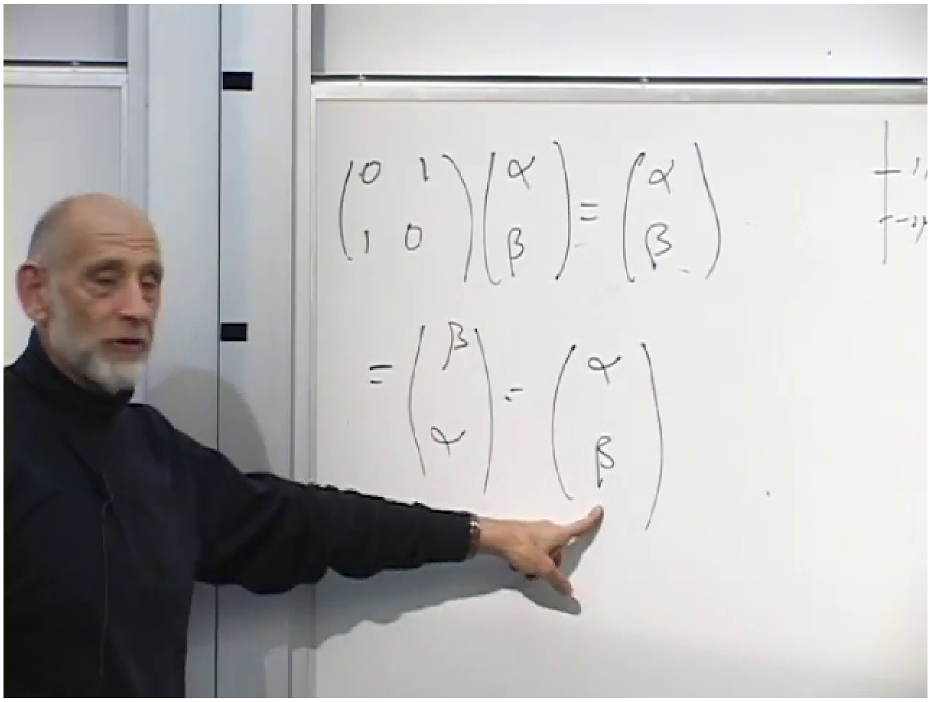

Figure 8.2: For eigenvalue $+1$, σ_x exchanges the two components, requiring $\alpha = \beta$.
Lecture frame: 00:39:42.

The two component equations, $\beta = \alpha$ and $\alpha = \beta$, are the same condition. Normalization gives

$$2|\alpha|^2 = 1, \tag{8.39}$$

so a convenient choice is

$$|+x\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}. \tag{8.40}$$

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[a] **Question from the audience.** If the factor of one-half were included, would the eigenvalue be one-half? **Response.** Yes. The Pauli matrix σ_x has eigenvalues ± 1, while $S_x = \sigma_x/2$ has eigenvalues $\pm\frac{1}{2}$.

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[a]Lecture timestamp: 00:40:13.

For eigenvalue -1,

$$\begin{pmatrix}\beta\\\alpha\end{pmatrix} = -\begin{pmatrix}\alpha\\\beta\end{pmatrix}, \tag{8.41}$$

so

$$\beta = -\alpha, \qquad \alpha = -\beta. \tag{8.42}$$

These are again the same equation. After normalization,

$$|-x\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}. \tag{8.43}$$

Thus an equal superposition of z-up and z-down is polarized along the positive x-axis, while the same superposition with a relative minus sign is polarized along the negative x-axis.

For the y-direction,

$$\sigma_y \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} -i\beta \\ i\alpha \end{pmatrix}. \tag{8.44}$$

The two normalized eigenvectors are

$$|+y\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \qquad |-y\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}, \tag{8.45}$$

up to an overall phase. Both states give probability 1/2 for z-up and 1/2 for z-down. The relative phase i, which is invisible in those two probabilities, distinguishes y-polarization from x-polarization. Linear combinations also describe spin polarized along arbitrary directions; that calculation is deferred.

[a] **Question from the audience.** Has the sign of the i component been assigned to the wrong σ_y eigenvalue?

Response. The sign may indeed have been assigned to the wrong eigenvalue. The assignment is uncertain here; it is not being fixed from memory.

[a]Lecture timestamp: 00:44:22.

The direct check is

$$\sigma_y \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -i\,i \\ i \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \tag{8.46}$$

$$\sigma_y \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} (-i)(-i) \\ i \end{pmatrix} = -\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}. \tag{8.47}$$

Thus $+i$ belongs to eigenvalue $+1$, and $-i$ belongs to eigenvalue -1.[3] Every spin-one-half state, including every state polarized along an arbitrary direction, is a linear combination of $|+z\rangle$ and $|-z\rangle$.

8.3 Spin One and Position-Dependent Spinors

For spin 1, the possible S_z values are $+1, 0, -1$. The state space is therefore three-dimensional. Spin 1 is integer spin and corresponds to bosonic statistics. We seek three 3×3 matrices satisfying the same angular-momentum commutators.

A representation symmetric among the three coordinate directions is given below.[4]

$$\begin{aligned} S_x &= i \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}, \quad S_y = i \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \\ S_z &= i \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \end{aligned} \tag{8.48}$$

[3]The spoken assignment remains unsettled after several reversals; the signs shown here are fixed by the displayed matrix multiplication.

[4]The sign of S_y follows the matrix visible near 00:48:50 and is also fixed by $[S_x, S_y] = iS_z$. Reversing that sign would reverse the required commutator.

Each matrix is i times a real antisymmetric matrix. Each contains one 1, one -1, and a vanishing row and column associated with its own coordinate: the first for S_x, the second for S_y, and the third for S_z. This gives a useful mnemonic. In this representation S_z is not diagonal.

Let

$$v = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}. \tag{8.49}$$

Then

$$S_z v = \begin{pmatrix} i\beta \\ -i\alpha \\ 0 \end{pmatrix}. \tag{8.50}$$

For eigenvalue $+1$,

$$i\beta = \alpha, \qquad -i\alpha = \beta, \qquad 0 = \gamma. \tag{8.51}$$

Thus $\beta = -i\alpha$ and $\gamma = 0$. Normalization gives $2|\alpha|^2 = 1$, so

$$|m = +1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix}. \tag{8.52}$$

For eigenvalue -1,

$$i\beta = -\alpha, \qquad -i\alpha = -\beta, \qquad 0 = -\gamma, \tag{8.53}$$

hence $\beta = i\alpha$, $\gamma = 0$, and

$$|m = -1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix}. \tag{8.54}$$

Finally, the zero-eigenvalue equation gives $\alpha = \beta = 0$, leaving γ arbitrary. Its normalized solution is

$$|m = 0\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \tag{8.55}$$

There are no further eigenvalues: these three eigenvectors exhaust the three-dimensional space. The factors $1/\sqrt{2}$ are not cosmetic. The same coefficients later enter quantitative experimental predictions when the algebra is applied to isospin.

---

[a] **Question from the audience.** How should the complex entries of these spin-one eigenvectors be read? **Response.** For $m = +1$ the components are $1/\sqrt{2}, -i/\sqrt{2}, 0$; for $m = -1$ they are $1/\sqrt{2}, i/\sqrt{2}, 0$; and for $m = 0$ they are $0, 0, 1$.

---

[a] Lecture timestamp: 00:54:10.

Other triplets of matrices satisfy the same algebra and are physically equivalent. The symmetric representation above is useful because the three directions are treated alike, and it will be used again.

8.3.1 Restoring position

Now restore the particle's motion. Spin commutes with position and with momentum, so either may be specified together with a spin component. Position and momentum do not commute, and distinct spin components do not commute with one another.

A moving spin-one-half particle is described by a position-dependent spinor,

$$\psi(x) = \begin{pmatrix} \psi_\uparrow(x) \\ \psi_\downarrow(x) \end{pmatrix}. \tag{8.56}$$

The quantities

$$\rho_{\uparrow_z}(x) = |\psi_\uparrow(x)|^2, \qquad \rho_{\downarrow_z}(x) = |\psi_\downarrow(x)|^2 \tag{8.57}$$

are joint position-and-spin probability densities. Their sum,

$$\rho(x) = \psi^\dagger(x)\psi(x) = |\psi_\uparrow(x)|^2 + |\psi_\downarrow(x)|^2, \tag{8.58}$$

is the position density. If the particle is known to have been detected at x, then the conditional probabilities are

$$P(\uparrow_z|\, x) = \frac{|\psi_\uparrow(x)|^2}{\psi^\dagger(x)\psi(x)}, \qquad P(\downarrow_z|\, x) = \frac{|\psi_\downarrow(x)|^2}{\psi^\dagger(x)\psi(x)}, \tag{8.59}$$

whenever $\rho(x) \neq 0$. Thus an electron is represented by a two-component spatial wavefunction, with the local spin amplitudes allowed to vary from place to place.

For a moving spin-one particle, write

$$\phi(x) = \begin{pmatrix} \phi_1(x) \\ \phi_2(x) \\ \phi_3(x) \end{pmatrix}. \tag{8.60}$$

At every point in space, the three components form a vector in the same spin space used above. Spin-one particles have integer spin and are bosons.

---

[a] **Question from the audience.** Is $\phi_1(x)$ itself the probability for the spin-one outcome $m = +1$?

Response. No. It is an amplitude in the chosen three-component basis. To obtain a definite spin outcome, form the bra of the corresponding normalized eigenvector, project the field onto it, and take the modulus squared.

---

[a]Lecture timestamp: 01:00:10.

For example,

$$\begin{aligned} |m = +1\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix} \\ \implies \quad \langle m = +1| &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i & 0 \end{pmatrix}. \end{aligned} \tag{8.61}$$

The projection amplitude is therefore

$$\langle m = +1|\phi(x)\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i & 0 \end{pmatrix} \begin{pmatrix} \phi_1(x) \\ \phi_2(x) \\ \phi_3(x) \end{pmatrix} \tag{8.62}$$

$$= \frac{\phi_1(x) + i\phi_2(x)}{\sqrt{2}}. \tag{8.63}$$

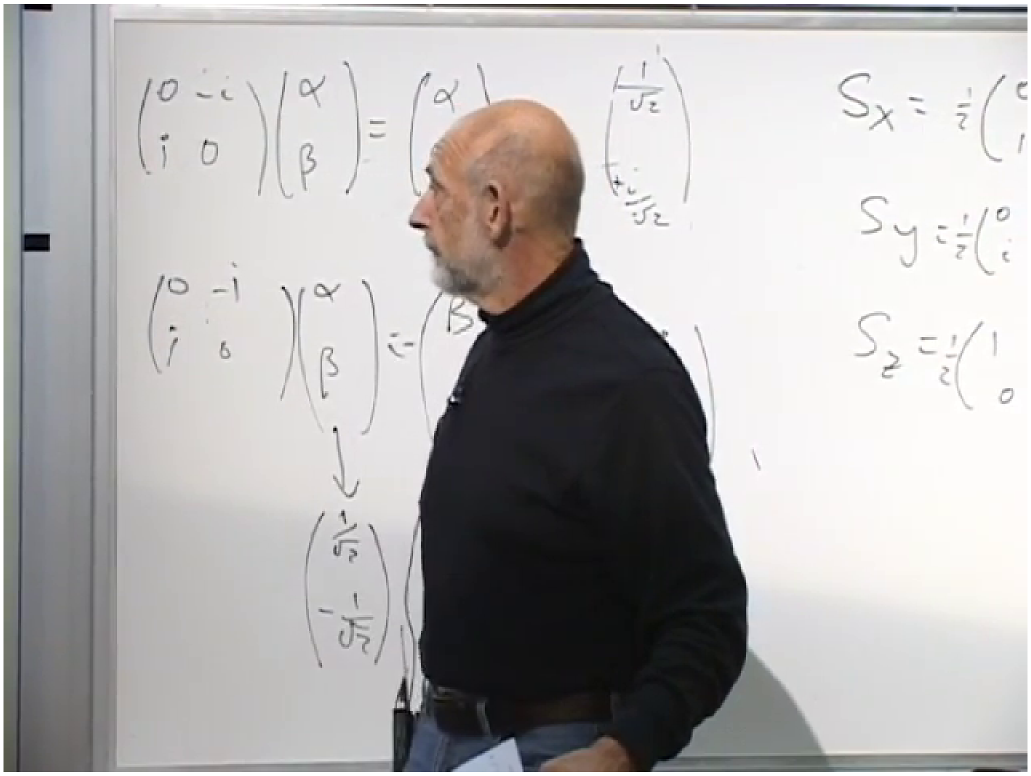

Figure 8.3: Normalized σ_y eigenvectors alongside the three spin-one-half matrices.
Lecture frame: 01:00:12.

Applying the same rule to all three eigenvectors gives the spin-resolved densities.[5]

$$\rho_{+1}(x) = \frac{1}{2}\left|\phi_1(x) + i\phi_2(x)\right|^2, \tag{8.64}$$

$$\rho_{-1}(x) = \frac{1}{2}\left|\phi_1(x) - i\phi_2(x)\right|^2, \tag{8.65}$$

$$\rho_0(x) = |\phi_3(x)|^2. \tag{8.66}$$

They satisfy

$$\begin{aligned}\rho_{+1}(x) + \rho_{-1}(x) + \rho_0(x) &= |\phi_1(x)|^2 + |\phi_2(x)|^2 \\ &\quad + |\phi_3(x)|^2 \\ &= \phi^\dagger(x)\phi(x).\end{aligned} \tag{8.67}$$

If the particle is conditioned to have been detected at x, then

$$P(m \mid x) = \frac{\rho_m(x)}{\phi^\dagger(x)\phi(x)}, \tag{8.68}$$

provided $\phi^\dagger(x)\phi(x) \neq 0$. The general measurement rule is always the same: form the bra of the desired normalized eigenvector, project the actual state onto it, and take the modulus squared. The resulting density retains its position dependence.

[5]These formulas complete the projection algebra after the explicit correction that ϕ_1 and ϕ_2 need not be real. Treating them as real would incorrectly erase the interference that distinguishes the $m = +1$ and $m = -1$ densities.

Higher-spin matrices are more complicated, but the principle does not change.

---

[a] **Question from the audience.** How large is the state space for spin three-halves?

Response. It has four states, with $m = \frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}$. Spin two has five states.

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[a]Lecture timestamp: 01:07:31.

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[a] **Question from the audience.** Why has the discussion involved x, y, z but no time direction? Is this treatment nonrelativistic?

Response. So far it is nonrelativistic. Spin, like mass, is most simply characterized in the particle's rest frame; its description in another inertial frame is obtained by the appropriate Lorentz transformation. That relativistic step is deferred.

---

[a]Lecture timestamp: 01:07:55.

The Dirac equation supplies the relativistic setting and explains why its four-component wavefunction contains a spin doublet.

8.4 Dirac Matrices and the Four-Component Spinor

The four-component Dirac wavefunction satisfies

$$i\frac{\partial \Psi}{\partial t} = -i\alpha_i \frac{\partial \Psi}{\partial x_i} + m\beta\Psi, \qquad i = 1, 2, 3, \tag{8.69}$$

where a repeated spatial index is summed. Thus the derivative term means

$$\alpha_1 \frac{\partial \Psi}{\partial x} + \alpha_2 \frac{\partial \Psi}{\partial y} + \alpha_3 \frac{\partial \Psi}{\partial z}. \tag{8.70}$$

For a plane wave this becomes the algebraic equation

$$\omega\Psi = (\alpha_i k_i + m\beta)\Psi. \tag{8.71}$$

---

[a] **Question from the audience.** Do the two occurrences of m denote the same thing?

Response. No. One was being used as a spatial index and the other is the mass of the particle. Rename the spatial index i; when i occurs twice it is summed from 1 to 3.

---

[a]Lecture timestamp: 01:11:32.

Squaring the frequency operator must reproduce the relativistic relation

$$\omega^2 = \mathbf{k}^2 + m^2. \tag{8.72}$$

The square terms therefore require, for each i,

$$\alpha_i^2 = I_4, \qquad \beta^2 = I_4. \tag{8.73}$$

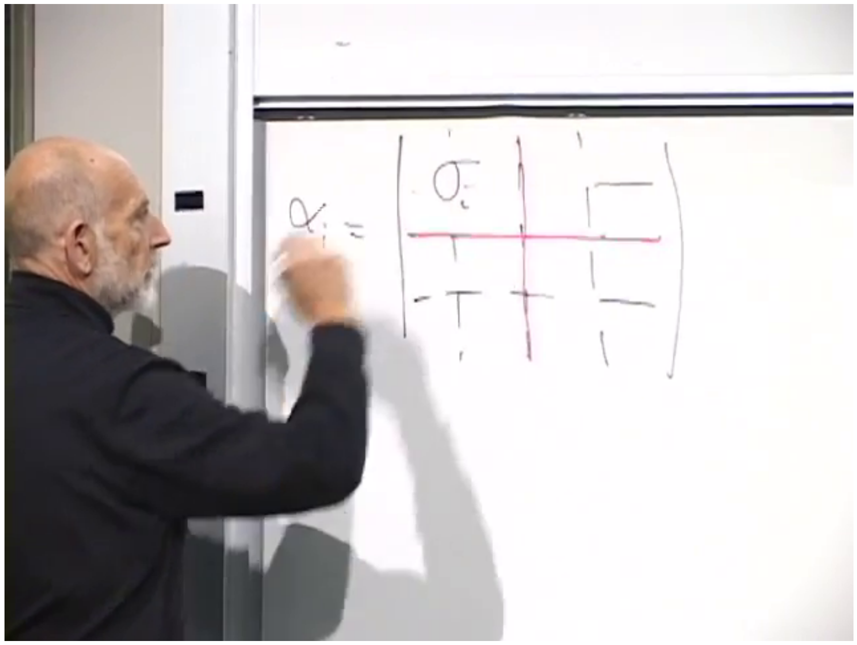

Figure 8.4: The Dirac matrix α_i written in 2×2 blocks, with σ_i and $-\sigma_i$ on the diagonal.

Lecture frame: 01:17:23.

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[a] **Question from the audience.** Is β different for each of the four components of Ψ?
Response. No. There is one matrix β, acting on the whole four-component spinor, and it satisfies $\beta^2 = I_4$.

---

[a]Lecture timestamp: 01:13:01.

The mixed terms must vanish. For different spatial directions, and also between every α_i and β, this gives

$$\alpha_i\alpha_j + \alpha_j\alpha_i = 0 \quad (i \neq j), \qquad \alpha_i\beta + \beta\alpha_i = 0. \tag{8.74}$$

Equivalently,

$$\{\alpha_i, \alpha_j\} = 2\delta_{ij} I_4, \quad \{\alpha_i, \beta\} = 0, \quad \{\beta, \beta\} = 2I_4. \tag{8.75}$$

Each matrix anticommutes with every different matrix, while its anticommutator with itself is twice the identity.

Without β, the three Pauli matrices would already provide three 2×2 matrices that square to the identity and anticommute pairwise. The mass term demands a fourth such matrix. There is no fourth 2×2 matrix with the required properties, and 3×3 matrices do not solve the algebra either. The smallest representation is therefore 4×4.

A 4×4 matrix may be divided into four 2×2 blocks. In one convenient representation,

$$\alpha_i = \begin{pmatrix} \sigma_i & 0 \\ 0 & -\sigma_i \end{pmatrix}, \qquad i = 1, 2, 3. \tag{8.76}$$

[a] **Question from the audience.** Were the σ_i blocks arranged the other way in the earlier representation?

Response. Yes. They were placed off the diagonal in another representation. The two arrangements are equivalent and are related by a similarity transformation.

[a]Lecture timestamp: 01:17:41.

In the present representation the remaining matrix is

$$\beta = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \qquad I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{8.77}$$

Block multiplication reduces checks of the 4×4 algebra to ordinary multiplication of 2×2 matrices.

[a] **Question from the audience.** Did the earlier representation use a different sign or a diagonal form for β?

Response. It used a different but physically equivalent basis. A basis change mixes the four components of Ψ and changes the appearance of the matrices without changing the physics.

[a]Lecture timestamp: 01:19:17.

The four entries of Ψ are not the four coordinates of spacetime. They are coordinates in the smallest vector space on which this anticommutation algebra can act. There are two distinct twofold structures: the division into 2×2 blocks and the two-component Pauli structure within each block. Anticipating the interpretation established below, the four possibilities may be organized as

$$\begin{array}{c|cc} & S_z = +\frac{1}{2} & S_z = -\frac{1}{2} \\ \hline \omega > 0 & (\omega > 0, +\frac{1}{2}) & (\omega > 0, -\frac{1}{2}) \\ \omega < 0 & (\omega < 0, +\frac{1}{2}) & (\omega < 0, -\frac{1}{2}) \end{array}. \tag{8.78}$$

The lower row consists of negative-frequency electron states. It is not a row of positrons; a positron will be interpreted as a hole in the filled set of those states.

8.5 Rest-Frame Energy Doubling and the Positron

The twofold energy structure is easiest to expose at zero momentum. When $\mathbf{k} = 0$, the plane-wave state is spatially constant and the spatial-derivative term disappears. The rest-frame equation[6] is

$$i\frac{\partial \Psi}{\partial t} = m\beta\Psi. \tag{8.79}$$

Divide the four-component column into an upper and a lower two-component object:

$$\Psi = \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}. \tag{8.80}$$

[a] **Question from the audience.** Do the labels ψ_+ and ψ_- already mean positive and negative energy?

Response. No. At this stage they are only names for the upper and lower two-component blocks. Their energy combinations have not yet been found.

[a]Lecture timestamp: 01:24:47.

[6]The frame at 01:23:38 retains a transient extra factor of i on the mass term; the spoken correction at 01:24:05 removes it.

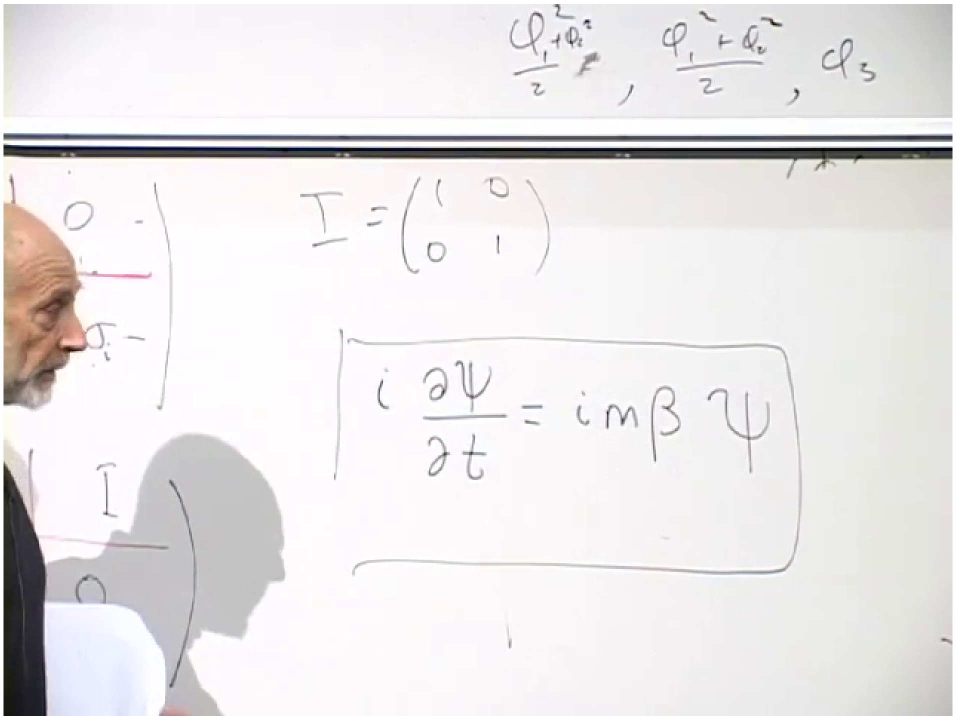

Figure 8.5: At zero momentum the spatial derivatives vanish, leaving the corrected rest-frame equation $i\partial_t \Psi = m\beta\Psi$.

Lecture frame: 01:23:38.

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[a] **Question from the audience.** What does the divided four-component column represent?

Response. It is the same four-component Dirac spinor, organized to match the block form of the matrices. Each half is a two-component object, and two such objects make the original four-component column.

---

[a] Lecture timestamp: 01:25:10.

The matrix β swaps the two blocks:

$$\beta \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix} = \begin{pmatrix} \psi_- \\ \psi_+ \end{pmatrix}. \tag{8.81}$$

The rest-frame equation therefore becomes

$$i\frac{\partial \psi_+}{\partial t} = m\psi_-, \qquad i\frac{\partial \psi_-}{\partial t} = m\psi_+. \tag{8.82}$$

There are two natural suggestions. Taking another time derivative would decouple the system, but adding and subtracting the two equations diagonalizes them immediately. Indeed,

$$i\frac{\partial}{\partial t}(\psi_+ + \psi_-) = m(\psi_+ + \psi_-), \tag{8.83}$$

$$i\frac{\partial}{\partial t}(\psi_+ - \psi_-) = -m(\psi_+ - \psi_-). \tag{8.84}$$

Defining

$$\chi_+ = \psi_+ + \psi_-, \qquad \chi_- = \psi_+ - \psi_-, \tag{8.85}$$

gives two independent equations,

$$i\partial_t\chi_+ = m\chi_+, \qquad i\partial_t\chi_- = -m\chi_-. \tag{8.86}$$

For definite-frequency dependence $e^{-i\omega t}$, the sum has

$$\omega = +m, \tag{8.87}$$

while the difference has

$$\omega = -m. \tag{8.88}$$

This is exactly the positive- and negative-energy doubling found in the earlier one-dimensional example: the relativistic equation again has positive-energy and negative-energy electron solutions. Thus the block doubling separates into positive- and negative-energy branches. Pure rest-frame states may be written

$$\Psi_{\omega=+m} = \begin{pmatrix}\xi\\ \xi\end{pmatrix}, \qquad \Psi_{\omega=-m} = \begin{pmatrix}\xi\\ -\xi\end{pmatrix}, \tag{8.89}$$

where ξ is still an arbitrary two-component object.

The sum and difference have disposed of one twofold structure, but the two components of ξ remain. The Pauli matrices act on this residual pair. This is the pair associated with spin, subject to the rotation and conservation-law check discussed below.

[a] **Question from the audience.** Can spin be treated as the outer twofold structure and energy sign as the inner one?

Response. Yes. The two factor orderings are equivalent. One may interchange which index is written first; the change is only a reshuffling of the four components.

[a]Lecture timestamp: 01:32:23.

The negative-frequency branch leads to the Dirac-sea interpretation. A negative frequency is assigned negative electron energy. Filling every negative-energy electron state lowers the energy as far as the exclusion principle permits, since only one electron can occupy each complete fermionic state. The completely filled negative-energy sea is then called the vacuum.

Promote one electron from this filled sea into a positive-energy state. The result contains a positive-energy electron and an unoccupied negative-energy state. Removing a negative energy raises the energy, so the hole has positive excitation energy; because an electron is missing, it also carries the opposite electric charge. The hole is a positron.

[a] **Question from the audience.** Can the positron interpretation be explained once more?

Response. The sum of the two blocks has positive frequency and the difference has negative frequency. The negative-frequency solutions are negative-energy electron states, and the vacuum is obtained by filling all of them. Promoting one filled electron to positive energy leaves both a positive-energy electron and a hole. That positive-energy, oppositely charged hole is the positron.

[a]Lecture timestamp: 01:32:57.

The algebra has therefore produced two consequences at once: positive- and negative-energy branches, which become the particle–antiparticle structure, and a residual two-component structure associated with spin one-half. The striking point is that both emerged from the smallest matrices capable of satisfying the relativistic anticommutation relations. Spin had only recently appeared as an empirical property when this mathematical construction produced particles with spin together with an antiparticle branch; in that historical setting the result was dramatic.

---

[a] **Question from the audience.** Are the negative-energy electrons themselves positrons?

Response. No. The negative-energy electron states are filled in the vacuum. A positron is the absence of one electron from that filled set: a hole, not a negative-energy electron.

---

[a]Lecture timestamp: 01:35:14.

A hole may participate in transitions. A positron has positive energy because it is the absence of a negative-energy electron. If a positive-energy electron fills the hole, the electron and positron annihilate, but their energy must be carried away, ordinarily by photons. In sea language, an electron has filled an unoccupied negative-energy state. The same release of energy can be pictured as an excited electron in an atom dropping to a lower level: the downward transition emits photons.

---

[a] **Question from the audience.** Why do higher negative-energy electrons not fall into the hole?

Response. That question mixes two cases. A positive-energy electron filling the hole is electron–positron annihilation, with photons carrying away the energy. A transition by a negative-energy sea electron is a different process, taken up below.

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[a]Lecture timestamp: 01:36:39.

The zero-momentum calculation by itself has established a twofold degeneracy after energy sign is separated. A degeneracy of two states is not, by itself, a proof that those states carry angular momentum.

---

[a] **Question from the audience.** How does the zero-momentum calculation prove that the remaining pair of states is spin?

Response. The pair may be called up and down, and the Pauli matrices appear in the full equation, but that alone is not yet the proof. One must show either how the pair transforms under spatial rotations or that assigning it spin makes total angular momentum conserved in interactions and collisions. That demonstration is deferred.

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[a]Lecture timestamp: 01:37:44.

For a positive-energy rest state the upper and lower blocks are equal,

$$\Psi_{\omega=+m} = \begin{pmatrix} \xi \\ \xi \end{pmatrix}. \tag{8.90}$$

Establishing that ξ is an angular-momentum doublet requires the rotation or conservation argument just described; it does not follow merely from counting two components.

---

[a] **Question from the audience.** Can a different electron annihilate with the positron even if its energy is different?

Response. Yes. Electrons retain no individual identity. Either available electron may fill the hole, and differences in the initial energy and momentum are reflected in the number and energies of the emitted photons.

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[a]Lecture timestamp: 01:40:46.

The hole picture also explains a transition wholly within the negative-energy sea. Suppose an occupied negative-energy state moves into the hole. The original hole is filled, but the state from which that electron came is now empty. The hole has moved. Equivalently, the positron has changed its energy and has emitted radiation.

---

[a] **Question from the audience.** Why cannot neighboring negative-energy levels successively fall into the hole?

Response. They can move the hole from one level to another. An electron dropping into the old hole leaves a new hole at its former energy, so the initial positron becomes a positron of different energy. The energy difference is carried by photons.

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[a]Lecture timestamp: 01:42:12.

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[a] **Question from the audience.** Why does this not produce an endless downward radiative cascade?

Response. Because energy and momentum must be conserved simultaneously. Those two conservation laws do not permit the unrestricted process. The same issue arises for an ordinary free electron: it cannot simply keep lowering its energy by emitting photons in empty space.

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[a]Lecture timestamp: 01:44:39.

Chapter 9

Lagrangians, Fields, and Local Particle Processes

Boson fields, fermion fields, and the electromagnetic field all obey wave-like equations. Those equations involve the fields themselves together with their space and time derivatives. One could list a separate equation of motion for every field, but a Lagrangian collects the entire system into one compact expression. Applying the Euler–Lagrange rule to that expression produces the equations for all the fields at once.

This construction is closely related to stationary action. Here the immediate task is more concrete: identify what a field Lagrangian depends on, use it to obtain a few classical equations, and then reinterpret the same field expressions quantum mechanically. In the quantum theory the Lagrangian governs particle motion and interaction, and ultimately the amplitudes and cross sections for collisions.

9.1 A Compact Code for Field Equations

For one scalar field, the Lagrangian density depends on the field and on its first derivatives,

$$\mathcal{L} = \mathcal{L}(\phi, \partial_t\phi, \partial_x\phi, \ldots) = \mathcal{L}(\phi, \partial_\mu\phi)\,. \tag{9.1}$$

The explicit list of time and spatial derivatives is often more useful while calculating; the notation $\partial_\mu\phi$ packages the same list in relativistic form.

The field Euler–Lagrange equation is

$$\frac{\partial}{\partial t}\frac{\partial\mathcal{L}}{\partial\dot{\phi}} + \frac{\partial}{\partial x}\frac{\partial\mathcal{L}}{\partial\phi_x} + \frac{\partial}{\partial y}\frac{\partial\mathcal{L}}{\partial\phi_y} + \frac{\partial}{\partial z}\frac{\partial\mathcal{L}}{\partial\phi_z} = \frac{\partial\mathcal{L}}{\partial\phi}, \tag{9.2}$$

where

$$\dot{\phi} = \frac{\partial\phi}{\partial t}, \qquad \phi_x = \frac{\partial\phi}{\partial x}, \qquad \phi_y = \frac{\partial\phi}{\partial y}, \qquad \phi_z = \frac{\partial\phi}{\partial z}. \tag{9.3}$$

If $\mathcal{L}$ contains several fields, this rule is applied once for each field. The result is the system of equations of motion.

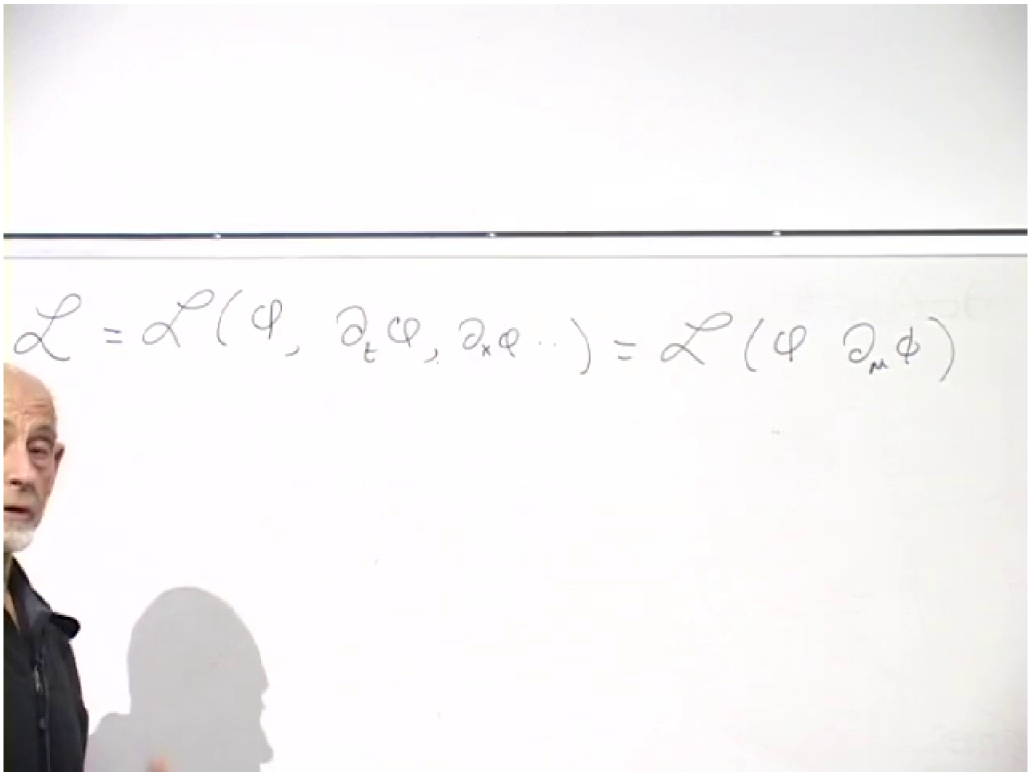

Figure 9.1: Dependence of a scalar-field Lagrangian on the field and its spacetime derivatives.
Lecture frame: 00:03:56.

The simplest scalar example is

$$\begin{aligned} \mathcal{L} &= \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}\phi_x^2 - \frac{1}{2}\phi_y^2 - \frac{1}{2}\phi_z^2 - V(\phi) \\ &= \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}(\boldsymbol{\nabla}\phi)^2 - V(\phi). \end{aligned} \tag{9.4}$$

The spatial derivative terms carry the opposite sign from the time-derivative term. Differentiating term by term gives

$$\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi}, \qquad \frac{\partial \mathcal{L}}{\partial \phi_i} = -\phi_i, \qquad \frac{\partial \mathcal{L}}{\partial \phi} = -V'(\phi), \tag{9.5}$$

and therefore

$$\ddot{\phi} - \nabla^2\phi = -V'(\phi). \tag{9.6}$$

This is the equation generated by the scalar Lagrangian.[1]

Here $V(\phi)$ is the field's potential-energy density. A constant contribution to V has no effect on this equation because its derivative vanishes. A linear potential is algebraically possible, but it has no lower bound: the field energy can run toward negative infinity. The first stable example is therefore

$$V(\phi) = \frac{1}{2}m^2\phi^2, \qquad V'(\phi) = m^2\phi, \tag{9.7}$$

[1]Editorial clarification: The retained frame is cropped at the left edge; the displayed completion is fixed by the immediately surrounding spoken derivation.

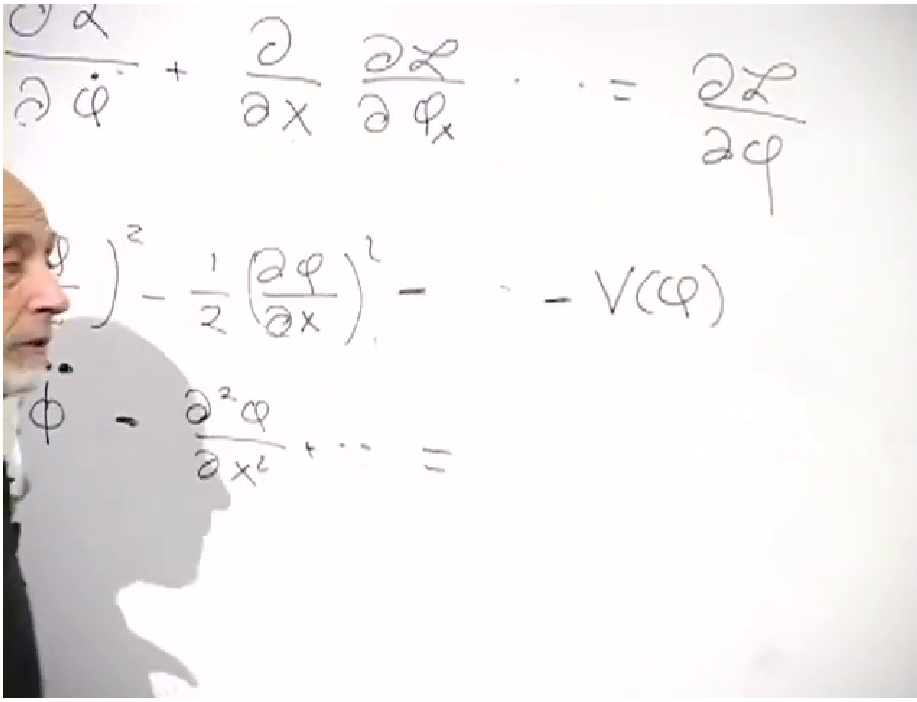

Figure 9.2: Euler–Lagrange equation, scalar-field Lagrangian, and the field equation generated by $V(\phi)$.

Lecture frame: 00:08:23.

which yields

$$\ddot{\phi} - \nabla^2\phi = -m^2\phi. \tag{9.8}$$

To expose the meaning of m, insert a plane wave,

$$\phi(\mathbf{x}, t) = \phi_0 e^{i\omega t - i\mathbf{k}\cdot\mathbf{x}}. \tag{9.9}$$

Then

$$\ddot{\phi} = -\omega^2\phi, \qquad \nabla^2\phi = -\mathbf{k}^2\phi, \tag{9.10}$$

so the wave equation becomes

$$-\omega^2\phi + \mathbf{k}^2\phi = -m^2\phi. \tag{9.11}$$

After canceling ϕ,

$$\omega^2 = \mathbf{k}^2 + m^2. \tag{9.12}$$

In units with $\hbar = c = 1$, frequency is energy and wave number is momentum. Thus

$$E^2 = \mathbf{p}^2 + m^2, \tag{9.13}$$

or, with powers of c restored, $E^2 = \mathbf{p}^2c^2 + m^2c^4$. The parameter m is the mass of one quantum of the field.

Relativistic invariance can be imposed at the level of the compact expression. If $\mathcal{L}$ is a Lorentz scalar, its equations of motion are Lorentz invariant. The relative sign between temporal and spatial derivatives is the familiar spacetime sign of special relativity.

9.2 Nonlinearity and Coupled Classical Fields

Higher powers of a field introduce a new kind of behavior. If

$$V(\phi) = \frac{1}{2}m^2\phi^2 + g\phi^3, \tag{9.14}$$

then

$$\ddot{\phi} - \nabla^2\phi = -m^2\phi - 3g\phi^2. \tag{9.15}$$

Terms quadratic in the Lagrangian give equations linear in the field. Cubic and higher terms give nonlinear equations.

The distinction is visible in the motion of wave packets. For a linear equation, a right-moving packet and a left-moving packet may be added to form another solution. They overlap and interfere for a while, then pass through one another and emerge with their original shapes. With nonlinear terms, the packets influence one another. They can scatter, deform, or break into more complicated waves. The temporary interference of linear waves is not an interaction; a lasting change in the packets signals nonlinearity.

The same pattern appears with two scalar fields. Let

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) + \frac{1}{2}(\partial_\mu\rho)(\partial^\mu\rho) - V(\phi,\rho), \tag{9.16}$$

$$V(\phi,\rho) = \frac{1}{2}m^2\phi^2 + \frac{1}{2}M^2\rho^2 + \rho\phi^2. \tag{9.17}$$

Without the last term, the two equations are independent,

$$\ddot{\phi} - \nabla^2\phi = -m^2\phi, \qquad \ddot{\rho} - \nabla^2\rho = -M^2\rho. \tag{9.18}$$

They describe quanta of two species, with masses m and M. With the mixed term included,

$$\ddot{\phi} - \nabla^2\phi = -m^2\phi - 2\rho\phi, \tag{9.19}$$

$$\ddot{\rho} - \nabla^2\rho = -M^2\rho - \phi^2. \tag{9.20}$$

Each equation now contains the other field. The equations are coupled and nonlinear: a ϕ packet scatters a ρ packet, and conversely. The derivative terms govern the energy–momentum relation, the quadratic undifferentiated terms set the masses, and the mixed higher-order term encodes interaction. This is the basic classical pattern.

9.3 Dirac Fields and a Scalar Coupling

For a Dirac field the equation is first order in time. In one-dimensional shorthand, with the extension to three spatial directions understood, its term structure is

$$i\dot{\Psi} \simeq i\alpha\frac{\partial\Psi}{\partial x} + m\beta\Psi, \qquad i\boldsymbol{\alpha}\cdot\boldsymbol{\nabla}\Psi = i\sum_i \alpha_i\frac{\partial\Psi}{\partial x_i}. \tag{9.21}$$

The matrices α_i and β act on the components of Ψ. A corresponding Lagrangian contains a term linear in $\partial_t\Psi$, a spatial-derivative bilinear involving α_i, and the mass bilinear $m\Psi^\dagger\beta\Psi$. Only that structural fact is needed here.[2]

[2]Editorial clarification: The signs, parentheses, and placement of i in the attempted Dirac Lagrangian are explicitly left unsettled in the source. No exact convention-dependent formula is inferred from it.

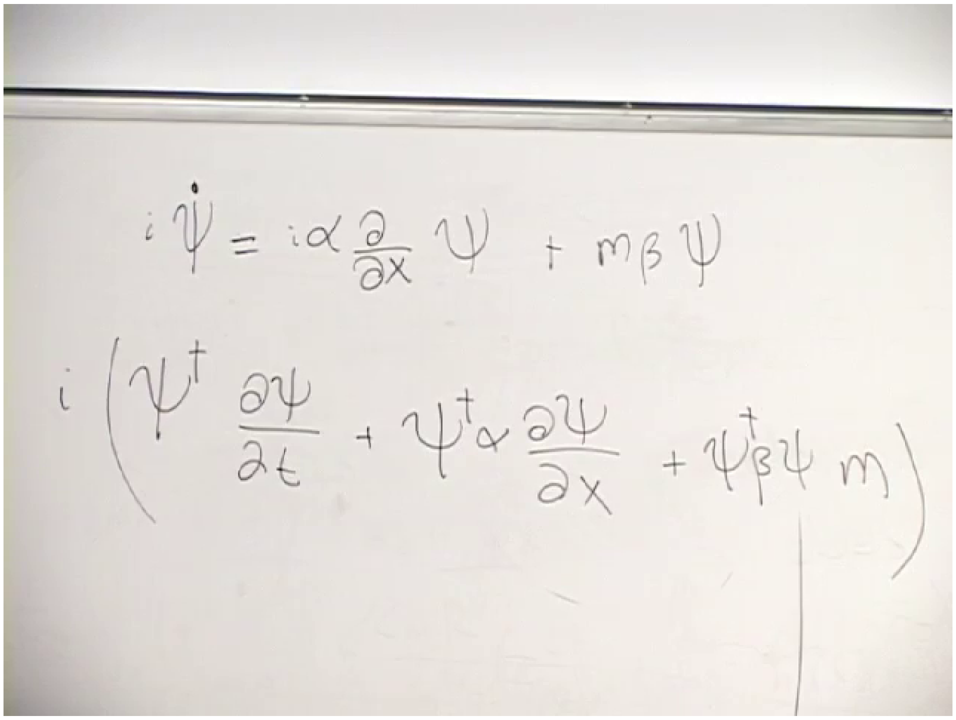

Figure 9.3: Time-derivative, spatial-derivative, and mass terms in the Dirac-field discussion.
Lecture frame: 00:25:37.

[a] **Question from the audience.** Is the parenthesis in the attempted Dirac Lagrangian correctly placed, with i multiplying the whole expression?

Response. The placement is not being fixed with confidence here, and it is not needed for the structural point: a Lagrangian exists that generates the Dirac equation.

[a]Lecture timestamp: 00:26:37.

Now add a scalar field ϕ. The naive product $\Psi^\dagger\Psi\,\phi$ is not a Lorentz scalar. Inserting the same Dirac matrix that occurs in the mass term gives

$$\mathcal{L}_{\text{int}} \propto \Psi^\dagger\beta\Psi\,\phi. \tag{9.22}$$

Equivalently, $\Psi^\dagger\beta\Psi$ is the scalar Dirac bilinear. Varying the full Lagrangian adds a term proportional to ϕ to the Dirac equation and a source proportional to $\Psi^\dagger\beta\Psi$ to the scalar equation. The two fields scatter one another.

[a] **Question from the audience.** Is the β inserted in the scalar–fermion interaction the same β that appears in the Dirac mass term?

Response. Yes. It is the same Dirac matrix.

[a]Lecture timestamp: 00:28:03.

The comparison

$$m\Psi^\dagger\beta\Psi \qquad \text{and} \qquad \Psi^\dagger\beta\Psi\,\phi \tag{9.23}$$

makes the coupling look like a mass parameter replaced by a scalar field. This observation will return when a field with a nonzero vacuum value is considered near the end.

So far the interpretation has been classical: wave packets scatter other wave packets. The same algebra has a sharper particle meaning when the fields become operators.

9.4 What Powers of Quantum Fields Do

A real scalar quantum field contains both creation and annihilation operators. Schematically,

$$\phi(x) \sim A^+(x) + A^-(x), \tag{9.24}$$

where A^+ creates a quantum and A^- annihilates one. Consequently, $\phi^3(x)$ contains four kinds of local process:

$$\phi^3 \sim (A^+)^3 + (A^+)^2 A^- + A^+(A^-)^2 + (A^-)^3, \tag{9.25}$$

where ordering details and numerical coefficients are suppressed in this schematic reading. The four particle-number changes are

$$3 \longrightarrow 0, \qquad 0 \longrightarrow 3, \qquad 2 \longrightarrow 1, \qquad 1 \longrightarrow 2. \tag{9.26}$$

The three field factors all act at the same point x. Thus a single cubic term includes three-particle annihilation, three-particle creation, fusion of two quanta into one, and splitting of one quantum into two. In a physical amplitude some of these local readings may later be removed by energy–momentum conservation, but they are all present in the operator product itself.

The quadratic free terms have a different role. First suppress the normalization of the difference between neighboring points x and x':

$$\partial_x \phi \;\propto\; \phi(x) - \phi(x'). \tag{9.27}$$

Squaring and expanding gives the term structure

$$\frac{1}{2}(\partial_x \phi)^2 \;\propto\; \frac{1}{2}\big[\phi(x) - \phi(x')\big]^2 \tag{9.28}$$

$$= \frac{1}{2}\phi(x)^2 + \frac{1}{2}\phi(x')^2 - \phi(x)\phi(x'). \tag{9.29}$$

Each same-point square has several operator readings: it may create two quanta, annihilate two, or annihilate and re-create one at that same point. Those terms do not transport a particle. The cross-term can annihilate a quantum at x' and create it at x, or the reverse. It therefore acts as a nearest-neighbor hopping term. Repeated factors carry a particle from x to x', then to x'', and onward. Quadratic derivative terms govern free propagation; cubic and higher terms scatter, create, and annihilate particles.

---

[a] **Question from the audience.** Should the finite-difference representation of the derivative include the separation $x - x'$?

Response. Yes. Calling the separation L supplies the factor L^{-2} after the difference is squared. That numerical factor changes the coefficient but not the hopping interpretation.

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[a]Lecture timestamp: 00:36:07.

Restoring the separation gives

$$\frac{1}{2}(\partial_x \phi)^2 \longrightarrow \frac{1}{2L^2}\big[\phi(x) - \phi(x')\big]^2 \tag{9.30}$$

$$= \frac{1}{2L^2}\phi(x)^2 + \frac{1}{2L^2}\phi(x')^2 - \frac{1}{L^2}\phi(x)\phi(x'). \tag{9.31}$$

The mass term is also quadratic but contains no derivative:

$$\frac{1}{2}m^2\phi(x)^2. \tag{9.32}$$

Its one-particle reading is an incoming quantum absorbed and re-emitted at the same point. It does not move the quantum; it weights, or in that sense counts, its presence there. The same operator square also contains its two-creation and two-annihilation readings. The motion-relevant distinction is therefore between a cross-term joining neighboring points and a mass insertion acting within one point.

9.5 The Local Vertex of Quantum Electrodynamics

These elementary particle-process sketches are the beginnings of Feynman diagrams. The central example is quantum electrodynamics, the theory of electrons, positrons, and photons.

The Dirac equation has positive- and negative-energy modes. In the older language, Ψ contains annihilation operators for both kinds of mode, while $\Psi^\dagger$ contains the corresponding creation operators. Removing a negative-energy electron is reinterpreted as creating a positive-energy positron. After that relabeling, the useful schematic decomposition is

$$\Psi \sim C^-_{e^-} + C^+_{e^+}, \tag{9.33}$$
$$\Psi^\dagger \sim C^+_{e^-} + C^-_{e^+}. \tag{9.34}$$

Thus Ψ annihilates an electron or creates a positron, while $\Psi^\dagger$ creates an electron or annihilates a positron. No normalization, spin label, or momentum integral is meant in this abbreviation.

The QED interaction couples the Dirac current to the electromagnetic vector potential. In the component notation used here, its structure is

$$\mathcal{L}_{\text{QED,int}} \sim e\left(\Psi^\dagger\Psi\, A_0 + \Psi^\dagger\boldsymbol{\alpha}\Psi\cdot\mathbf{A}\right), \tag{9.35}$$

where

$$J^0 = \Psi^\dagger\Psi, \qquad \mathbf{J} = \Psi^\dagger\boldsymbol{\alpha}\Psi. \tag{9.36}$$

The symbol $\sim$ is important: the relative sign between the temporal and spatial terms depends on the metric and vector-potential conventions. The invariant statement is that the four-current is contracted with the four-potential.[3]

The photon field A_μ itself contains photon creation and annihilation operators. Its vector character also carries the photon's polarization—a small directional "flag" in addition to its momentum.

[a] **Question from the audience.** How can the interaction be a scalar when A_μ is a vector and the fermion factor first appears to be a scalar?

Response. The fermions supply a four-vector too: $\Psi^\dagger\Psi$ is its time component and $\Psi^\dagger\boldsymbol{\alpha}\Psi$ its spatial component. Contracting that current with A_μ produces a scalar.

[a] Lecture timestamp: 00:44:28.

Multiplying the operator contents of $\Psi^\dagger$, Ψ, and A_μ gives four familiar readings of one local vertex:

[3] Editorial clarification: The source fixes the temporal and spatial components of the current but does not settle a metric convention for the component contraction.

1. an electron emits a photon;
2. an electron absorbs a photon;
3. an electron and a positron annihilate into a photon;
4. a photon creates an electron–positron pair.

For the first reading, Ψ annihilates the incoming electron, $\Psi^\dagger$ creates the outgoing electron, and the photon field creates the photon. Choosing the photon-annihilation part gives absorption. Choosing the positron pieces gives pair annihilation or pair creation. One local term contains the whole family.

In a fermion-flow diagram, the arrow on a positron line points opposite to the positron's direction of propagation. With that convention the fermion arrow passes continuously through every allowed QED vertex. The four readings differ by moving external legs between the incoming and outgoing sides, not by breaking the fermion-flow line.

Every one of these readings carries the same coefficient e, the electric charge in the microscopic convention being used. This coefficient is the electric charge e, not the fine-structure constant, although the two are closely related. The vertex amplitude is proportional to e; after all kinematic and normalization factors are included, the corresponding probability scale is proportional to e^2.

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[a] **Question from the audience.** Does the photon-to-electron–positron reading have the same coefficient as the other readings of the vertex?

Response. Yes. It carries the same electric-charge coefficient e.

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[a]Lecture timestamp: 00:49:44.

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[a] **Question from the audience.** Must there be a scaling factor, since a probability has no unit?

Response. In the natural microscopic convention being used, the charge e is dimensionless. It is the amplitude scale, while its square supplies the probability scale.

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[a]Lecture timestamp: 00:49:59.

A concrete picture is an electron striking a wall and stopping abruptly. There is some probability that energy and momentum leave in the form of a photon. The elementary electromagnetic factor in that radiation amplitude is e, so its probability scale contains e^2. This statement isolates the coupling factor; the actual probability also depends on the electron's state and on the kinematics.

---

[a] **Question from the audience.** Does electric charge have units?

Response. Not in the natural microscopic convention used here. Macroscopic charge is often reported in coulombs, but particle theory measures charge relative to a chosen elementary unit. In Planck or other natural units the quantities are expressed as pure numbers.

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[a]Lecture timestamp: 00:51:15.

The contrast is one of conventions and scales. Coulombs are convenient for current in wires; a dimensionless coupling is convenient for microscopic amplitudes.[4]

[4]Editorial clarification: A brief question from 00:52:37–00:53:15 about "grouping" charge-bearing terms is substantially damaged and elicited no definite construction.

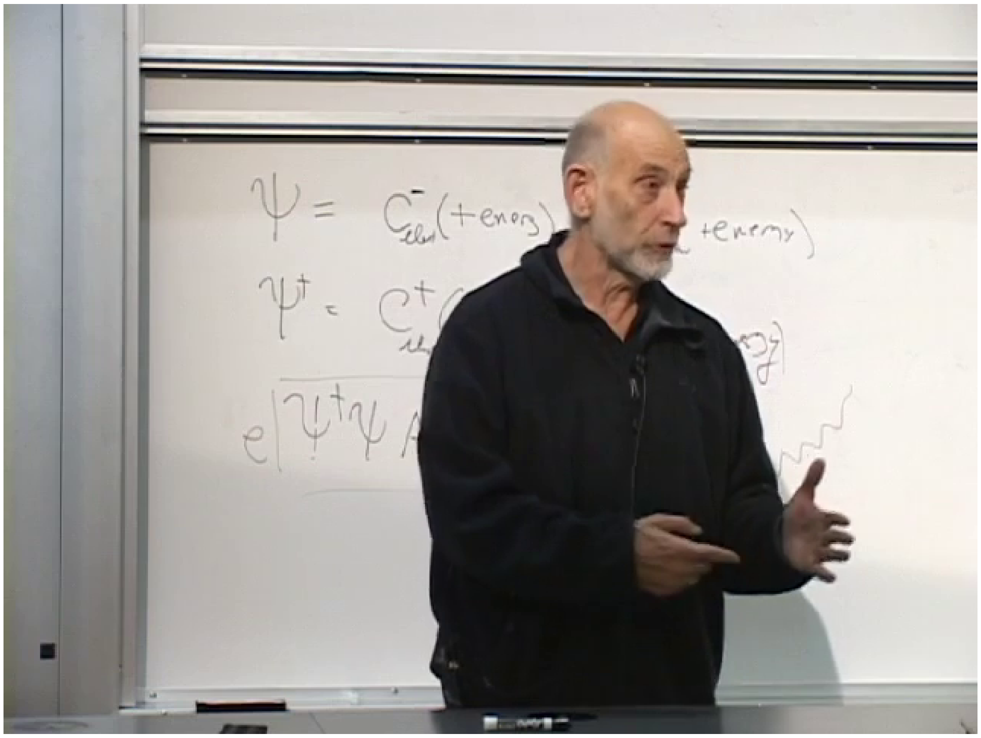

Figure 9.4: Electron and positron operator content of Ψ and $\Psi^\dagger$, with an electron–photon interaction sketch.

Lecture frame: 00:50:39.

All three fields in the elementary QED interaction are evaluated at one spacetime point. No internal time delay is assigned to the vertex itself.

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[a] **Question from the audience.** Are the absorption and re-emission of the electron and the emission of the photon simultaneous, or does the equation build in a delay?

Response. At the elementary vertex all the operators are evaluated at the same spacetime point. The basic process therefore contains no delay between those acts.

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[a]Lecture timestamp: 00:53:19.

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[a] **Question from the audience.** Must each conserved quantity be conserved separately?

Response. Yes. Electric charge is one of the quantities that must be conserved separately.

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[a]Lecture timestamp: 00:53:57.

Charge conservation is visible both in the diagram and in the algebra. A process that turned an incoming electron into an outgoing positron would reverse the charge and is not an allowed QED vertex. With time drawn upward, an electron arrow follows time upward and a positron arrow points downward; the valid vertex always has a through-going fermion arrow. Algebraically, the interaction contains one Ψ and one $\Psi^\dagger$. The first raises charge by one unit when it annihilates an electron or creates a positron, and the second lowers it by one unit. Their effects cancel.

[a] **Question from the audience.** Is the boxed electron–photon expression the whole QED Lagrangian?

Response. No. It is the interaction term. The quadratic free-photon terms and the quadratic Dirac terms, which govern propagation, have not been written.

[a]Lecture timestamp: 00:56:05.

[a] **Question from the audience.** Is a classical Lagrangian deduced while a quantum Lagrangian is guessed and tested, and does the kinetic-minus-potential structure remain in quantum mechanics?

Response. The kinetic-minus-potential pattern remains: time derivatives provide the kinetic form, spatial gradients generalize it, and undifferentiated field terms provide the potential form. The quantum Lagrangian is the more fundamental description. In a state containing very many quanta, the same Lagrangian produces the classical wave equations. Candidate Lagrangians are proposed using physical principles and deeper theory, then established or rejected through collision experiments.

[a]Lecture timestamp: 00:56:35.

The large-occupation limit explains why the same expression can govern a quantum particle theory and its classical waves. This agreement is not a separate miracle imposed afterward; the classical field is the collective behavior of many quanta. Conversely, experimental collisions expose the primitive vertices directly enough to determine the QED and Standard Model interactions with great precision.

9.6 Crossing and Spacetime Conservation Laws

The legs of a local vertex may be crossed between incoming and outgoing positions. An outgoing particle moved to the incoming side is reinterpreted as its antiparticle, and conversely. In this sense the same electron–positron–photon operator has emission, absorption, pair-annihilation, pair-creation, and even an all-incoming local reading.

[a] **Question from the audience.** If a basic vertex diagram is rotated, does it remain a valid local reading?

Response. Yes. Its external legs may be moved between incoming and outgoing positions, with the corresponding particle–antiparticle reinterpretation.

[a]Lecture timestamp: 00:59:34.

The local operator reading is not yet a complete physical process. Its amplitude must be integrated over every spacetime point at which the vertex could occur. The time integral enforces energy conservation and the space integral enforces momentum conservation. A picture in which an electron, a positron, and a photon all disappear into the vacuum is present as a formal local reading, but the integrated amplitude vanishes because the process cannot conserve energy and momentum.

[a] **Question from the audience.** Does the all-incoming picture mean that an electron, a positron, and a photon can disappear into nothing?

Response. It is a reading of the local operator, but the physical amplitude includes integration over the vertex position. Those integrals impose energy and momentum conservation, so this kinematically forbidden process has zero amplitude.

[a]Lecture timestamp: 01:00:30.

[a] **Question from the audience.** Does the converse vacuum-to-electron–positron–photon reading exist, and do excluded processes disappear after integration?

Response. The converse local reading is also present. Integration over space and time removes any process incompatible with the conservation laws.

[a]Lecture timestamp: 01:01:57.

Derivatives in the Lagrangian fit the operator expansion without differentiating the creation and annihilation symbols themselves. A field has the schematic form

$$\Psi(x) \sim \int d^3k \, [C(\mathbf{k})e^{i\mathbf{k}\cdot\mathbf{x}-i\omega t} + \cdots]. \tag{9.37}$$

The operators $C(\mathbf{k})$ are coefficients in this spacetime expansion. Derivatives act on the plane waves, pulling down $\mathbf{k}$ or ω.

[a] **Question from the audience.** When differentiating a field expansion, is the creation operator itself differentiated?

Response. No. The creation and annihilation operators are independent of the spacetime coordinate in this expansion. The derivatives act on the exponential factors and bring down momentum and frequency.

[a]Lecture timestamp: 01:02:27.

9.7 Nucleons and Pions as an Effective Theory

The same language can describe protons, neutrons, and pions. Protons and neutrons are fermions represented by fields Ψ_p and Ψ_n. For this interaction-counting discussion the pion fields are treated as scalar fields in three charge varieties,

$$\pi^+, \qquad \pi^-, \qquad \pi^0. \tag{9.38}$$

The proton and π^+ carry one positive unit of electric charge, the π^- carries one negative unit, and the neutron and π^0 are neutral. At a deeper level pions are quark–antiquark systems, but that internal description can initially be left unresolved.

A schematic charge-conserving list of nucleon–pion field products is

$$\mathcal{L}_{N\pi} \sim \left\{ \Psi_p^\dagger \Psi_n \, \pi^+, \; \Psi_n^\dagger \Psi_p \, \pi^-, \; \Psi_p^\dagger \Psi_p \, \pi^0, \; \Psi_n^\dagger \Psi_n \, \pi^0 \right\}. \tag{9.39}$$

Only the field content and charge assignments are being displayed; relative coefficients, signs, and the spinor-matrix structure are not fixed here. The charge on the first pion field is fixed by testing a concrete reading: a neutron together with a positive pion may produce a proton, whereas using a negative pion there would turn negative incoming charge into positive outgoing charge. The conjugate term gives the opposite charged process.[5]

Each term again has several crossed readings. Moving an outgoing neutron to the incoming side changes it into an incoming antineutron. The charged vertex can therefore describe a proton and an antineutron combining into a π^+, as well as processes such as $p \to n + \pi^+$ when the kinematics

[5]Editorial clarification: The pion sign is corrected several times in the source before the charge-conserving pair shown here is reached; the displayed ordering records that final schematic structure.

permit. The neutral terms let a proton or neutron emit or absorb a π^0. Charge conservation selects the allowed family, while energy, momentum, and any further symmetries determine which member can occur in a particular experiment.

This was a useful effective description before quark substructure was known. It remains useful whenever a probe cannot resolve that structure. A proton has a finite distribution of constituents, rather as an atom has internal structure, although on a much smaller scale. If an experiment transfers too little momentum to break up or resolve the proton, the proton may be treated as a point particle with its own field. A sufficiently hard collision exposes the quarks, and the point-particle proton field is then no longer adequate.

Because the proton is charged, it also has a local proton–photon interaction, schematically $J_p^\mu A_\mu$, where J_p^μ is the proton current. The electron has its corresponding electron–photon interaction. A proton cannot instead turn into a neutron by emitting a photon: $p \to n + \gamma$ fails charge conservation.

[a] **Question from the audience.** What happens when a proton and an electron interact?

Response. Each charged particle has a photon vertex. The proton can emit a photon while remaining a proton, and the electron can absorb the photon while remaining an electron. Proton–electron scattering is assembled from those two local vertices and propagation of the photon between them.

[a]Lecture timestamp: 01:10:28.

9.8 Locality and Repeated Elementary Processes

There is no special nonlocal term for the complete proton–electron exchange. One local vertex emits the photon at the proton, quadratic photon terms carry it from neighboring point to neighboring point, and another local vertex absorbs it at the electron. The extended process is a product of those elementary pieces.

$$\begin{aligned}[J_p^\mu A_\mu](x_p) &\longrightarrow \underbrace{x_p \to x_1 \to x_2 \to \cdots \to x_e}_{\text{quadratic photon propagation}} \\ &\longrightarrow [J_e^\nu A_\nu](x_e).\end{aligned} \tag{9.40}$$

The arrows in this schematic chain denote successive neighboring-point insertions, not a new nonlocal interaction.

[a] **Question from the audience.** How would one write the Lagrangian for the whole proton–electron exchange?

Response. One does not add a term for the whole extended event. It is built from a local proton–photon vertex, repeated quadratic propagation terms, and a local electron–photon vertex.

[a]Lecture timestamp: 01:12:06.

The local structure has two forms. An undifferentiated interaction term acts at one spacetime point. A derivative term, expressed as a finite difference, connects that point only with a neighboring point. Neither kind transfers a particle directly across a finite distance. Long-range propagation is assembled from many infinitesimal moves; there is no elementary action at a distance.

[a] **Question from the audience.** Does the Lagrangian contain terms only at one point?

Response. Interaction terms act at one point. Derivative terms connect one point with an infinitesimally neighboring point. Emission, many neighboring hops, and absorption together produce a finite exchange.

[a]Lecture timestamp: 01:14:36.

[a] **Question from the audience.** Are all the pieces of the extended process present, and does compounding them mean multiplication?

Response. Yes. Each elementary emission, hop, or absorption is encoded in the Lagrangian. Products of repeated insertions compound those pieces into the larger process.

[a]Lecture timestamp: 01:15:36.

The relativistic quantum field theory under discussion presumes the spacetime setting on which its fields are defined. It should not be confused with a complete quantum theory of a dynamical spacetime geometry.

[a] **Question from the audience.** Does this compounding remain valid in a strongly relativistic setting such as near a black-hole horizon?

Response. The black-hole extension is outside the scope of this construction. The quantum field theory being used here does not by itself provide a complete description of black-hole dynamics, where gravity is essential.

[a]Lecture timestamp: 01:16:28.

A field is an operator for an elementary local act. One field factor can absorb a particle or emit one. A product of two factors can absorb one particle and emit another at the same point; higher products can change the particle content further. Products of local operators join these acts into histories. In spacetime, motion is represented by absorption in one small region, re-emission into a neighboring region, another absorption, and another re-emission. Taking the step size toward zero while the number of steps grows gives the continuum trajectory.

[a] **Question from the audience.** How should fields, distance, and time be understood in this local-step picture?

Response. Fields are operators for local absorption and emission. Products join those elementary acts, and motion through spacetime is built from infinitesimal neighboring steps before the continuum limit is taken.

[a]Lecture timestamp: 01:17:30.

A regulated starting point is therefore a division of spacetime into small cells, with the field operators attached to the cells.[6] Derivatives become differences. Quadratic cross-terms connect neighboring cells, while mass and interaction terms act inside one cell. The cells are a device for defining the limiting theory, not additional particles or a new physical mass scale.

Special-relativistic effects need not be added separately at every step. When the Lagrangian is a Lorentz scalar, the amplitudes assembled from it are Lorentz invariant, and effects such as time dilation follow from that structure.

[6]Editorial clarification: The audience wording at 01:17:30–01:20:40 is partly obscured; the paraphrased question preserves only its recoverable subject, while the response about local operators and successive spacetime steps is clear.

[a] **Question from the audience.** Do high-energy motion and time dilation create extra problems in a multistep process?

Response. No separate correction is required. If the Lagrangian is a Lorentz scalar, special-relativistic effects such as time dilation are already built into the result. The precise concern behind the question is otherwise not fully clear.

[a] Lecture timestamp: 01:21:10.

Creation and annihilation operators may be represented in a position basis or in a momentum basis. A position-space creator is a Fourier superposition of momentum-space creators, and conversely. There is no operator that creates a state with both exactly specified position and exactly specified momentum. Localizing the particle sharply requires a wide band of momenta; localizing it sharply in momentum spreads it over position. The Fourier relation between the two bases is the uncertainty principle in this language.

[a] **Question from the audience.** How are creation and annihilation operators related to position, momentum, and the uncertainty principle?

Response. They can be represented as functions of position or as functions of momentum. The two representations are Fourier transforms of one another. A sharply localized position state requires many momenta, while a sharply localized momentum state is spread over position; there is no creator for simultaneously sharp x and p.

[a] Lecture timestamp: 01:22:48.

The cell construction can now be stated completely. Label the cells by n, replace a derivative by a finite difference such as

$$\partial_\mu \phi(x) \longrightarrow \frac{\phi_{n+\hat{\mu}} - \phi_n}{a}, \tag{9.41}$$

and place every field species in every cell. A quadratic cross-term removes a quantum from one cell and creates it in a neighbor. The continuum theory is obtained by taking the spacing a toward zero after the regulated expressions have been defined.

$$\underbrace{\phi_n \phi_{n+\hat{\mu}}}_{\text{hop } n \leftrightarrow n+\hat{\mu}}, \qquad \underbrace{m^2 \phi_n^2}_{\text{same-cell mass insertion}} . \tag{9.42}$$

[a] **Question from the audience.** Can the subdivision of spacetime into cells be completed?

Response. Yes. The fields are indexed by cells, derivatives become finite differences, and quadratic terms hop particles between neighboring cells. The continuum is recovered as the cell size tends to zero.

[a] Lecture timestamp: 01:25:16.

Electrons hop between cells through their quadratic terms, and photons do the same. When an electron and photon meet in one cell, the QED interaction may absorb and re-emit the electron while creating or annihilating the photon. The scalar mass term has a same-cell reading: a particle enters and leaves that cell with a coefficient proportional to m^2. It does not supply a hop.

[a] **Question from the audience.** Is the cell size related to the mass coefficient?

Response. No. The cell size is a regulator spacing that is ultimately taken to zero; the mass coefficient is an independent parameter in the Lagrangian.

[a]Lecture timestamp: 01:28:18.

[a] **Question from the audience.** Is multiplying the local factors a joint-amplitude construction over spacetime rather than merely a sequence in time?

Response. Yes. The cells cover spacetime, and their local contributions combine across that spacetime grid. The product supplies a joint amplitude for the required local events and propagation steps.

[a]Lecture timestamp: 01:28:43.

9.9 Sources, Detectors, and the Dynamics Between Them

A complete process is a history in spacetime. Its initial condition is fixed by what the experimenter prepares, and its final condition by what the detectors register. A simple source is a hot filament that emits an electron. A field operator at the filament creates that incoming electron; field operators at detector positions describe its eventual absorption.

[a] **Question from the audience.** What is the initial stage of such a spacetime process?

Response. It is prepared at the sources. An electron emitted by a hot filament is a simple example: a field operator at the filament creates the electron, while field operators at the detectors specify the final absorption events.

[a]Lecture timestamp: 01:29:30.

The same construction can be stated from the vacuum. Creation operators at the sources build the incoming state on the vacuum. Annihilation operators associated with the detectors select the outgoing state. These operators specify boundary conditions; they do not replace the dynamics throughout the region between source and detector.

[a] **Question from the audience.** Would it be wrong to place the dynamics only at the emitters rather than throughout the system?

Response. Yes. Start with the vacuum, use source operators to prepare the incoming particles, and use detector operators to select the final particles. Between them, repeated Lagrangian insertions act throughout spacetime, moving particles, creating them, and annihilating them.

[a]Lecture timestamp: 01:31:11.

With enough insertions, several particles can move simultaneously through many neighboring cells; particles can also be created or annihilated along the way. The initial operators say what enters, the final operators say what is registered, and the repeated Lagrangian operations supply everything between them.

[a] **Question from the audience.** If arbitrarily many Lagrangian insertions can occur, are their positions being integrated over?

Response. Yes. Their spacetime positions are integrated over. The exact Feynman rules are not being derived here; the immediate point is that simple local field expressions encode the primitive interactions from which a full process is assembled.

[a]Lecture timestamp: 01:33:24.

9.10 Phase Symmetry, Charge, and Amplitudes

Writing the primitive processes in a Lagrangian also makes their symmetries visible. Consider a global phase change of the electron field,

$$\Psi \longrightarrow e^{i\theta}\Psi, \qquad \Psi^\dagger \longrightarrow e^{-i\theta}\Psi^\dagger, \qquad A_\mu \longrightarrow A_\mu. \tag{9.43}$$

A term with equal numbers of Ψ and $\Psi^\dagger$ is unchanged because the phases cancel. A term with unequal numbers would acquire a phase. The equality of those numbers is the algebraic form of charge conservation.

This is a central advantage of the Lagrangian description. Its field factors display the elementary processes, their combinations display the symmetries, and the symmetries display the conservation laws. Repeated insertions evolve the prepared initial state into the selected final state. The result is a quantum amplitude. Exact evaluation requires the Feynman rules, but the structural content is already visible: local operations repeat in every order allowed by the conservation laws.

9.11 Field Measurements and Commutators

Field operators retain the usual quantum-mechanical meaning of operators. Their eigenvalues are possible results of field measurements, and their eigenvectors describe states associated with those values. An electric field can be probed by placing a charged particle in it and observing how the particle accelerates.

[a] **Question from the audience.** Do the eigenvalues and eigenvectors of field operators play their usual quantum-mechanical role?

Response. Yes. Eigenvalues are possible measured field values, and fields can be probed through their effects on particles. Commutators determine the restrictions on simultaneous measurements.

[a]Lecture timestamp: 01:37:06.

The trivial identity

$$[\phi(x), \phi(x)] = 0 \tag{9.44}$$

does not settle the measurement question. One must specify two observables and their spacetime relation. A field and its conjugate momentum form a canonical noncommuting pair, and field operators at timelike-related points need not commute; local bosonic fields at spacelike separation do commute, as locality requires. The familiar connection between commutators and uncertainty therefore carries over, but only after the operator pair has been identified correctly.[7]

[7]Editorial clarification: The source states noncommutation at distinct points too broadly. The qualification here preserves its measurement point while enforcing the microcausality of spacelike-separated local bosonic fields.

[a] **Question from the audience.** Is the electric or electromagnetic field measured in volts, or in volts per meter?

Response. The electric field is measured in volts per meter. Volts and meters are not the most useful language for microscopic field processes, which are more naturally organized in terms of photons. Conversely, quantum field theory is not the useful language for deciding how to water a house.

[a]Lecture timestamp: 01:39:57.

The units natural to one description need not be the illuminating units in another. Volts and volts per meter fit macroscopic apparatus; photon creation and annihilation fit microscopic processes.

[a] **Question from the audience.** Is the central idea that Lagrangian terms are building blocks combined into larger processes?

Response. Yes. Elementary processes repeat from one point to the next. Each basic element acts at one spacetime point, or connects neighboring points through a derivative term. Apparently nonlocal propagation is built from these local steps; there is no elementary action at a distance.

[a]Lecture timestamp: 01:41:10.

9.12 Products of Local Processes and Sums over Routes

There is a compact heuristic way to display the repetition of local processes. Divide spacetime into cells and write

$$\prod_x \big(1 + \mathcal{L}_I(x)\big), \tag{9.45}$$

where x labels the cells and $\mathcal{L}_I(x)$ is the local interaction Lagrangian. This is not the exact continuum formula. It is a combinatorial stand-in that exposes how products generate every allowed selection of local events.

[a] **Question from the audience.** Can many creation, propagation, and destruction steps be compacted into one expression?

Response. Heuristically, multiply one factor $1 + \mathcal{L}_I(x)$ for every spacetime cell. Expanding the product generates no event, one event at any selected point, or events at several points.

[a]Lecture timestamp: 01:42:18.

For two cells,

$$\begin{aligned}\big(1 + \mathcal{L}_I(x)\big)\big(1 + \mathcal{L}_I(x')\big) &= 1 + \mathcal{L}_I(x) + \mathcal{L}_I(x') \\ &\quad + \mathcal{L}_I(x)\mathcal{L}_I(x').\end{aligned} \tag{9.46}$$

The first term means that neither interaction occurs. The next two place one elementary interaction at x or at x'. The last term places interactions at both points. With many factors there are terms involving one point, two points, three points, and so on. A term with several vertices can describe, for example, two incoming particles producing three outgoing particles.

Quadratic propagation pieces enter the same construction. A finite displacement is represented by a chain of neighboring insertions. Somewhere in the expanded product are factors at one cell, its neighbor, the next neighbor, and onward, carrying a particle between the endpoints.

[a] **Question from the audience.** If finite propagation requires many local motion terms, how is the required motion recovered?

Response. The large product contains chains of neighboring insertions that carry the particle from its initial point to its final point. There are generally many such chains along different routes. Each route contributes an amplitude, and the total amplitude is their sum.

[a]Lecture timestamp: 01:46:29.

If r labels a route, then schematically

$$\mathcal{A}_{\text{total}} = \sum_r \mathcal{A}_r, \qquad P = |\mathcal{A}_{\text{total}}|^2 . \tag{9.47}$$

Ordinary probabilities are not assigned to the routes and then added. The amplitudes are added first:

$$\left|\sum_r \mathcal{A}_r\right|^2 = \sum_r |\mathcal{A}_r|^2 + \sum_{r \neq s} \mathcal{A}_r \mathcal{A}_s^*. \tag{9.48}$$

The cross-terms are interference.

[a] **Question from the audience.** Is this consistent with the double-slit experiment?

Response. Exactly. The alternatives through the two slits contribute amplitudes. Squaring their sum produces interference terms, just as the many spacetime routes in the local-step construction interfere.

[a]Lecture timestamp: 01:48:27.

[a] **Question from the audience.** Where does the 1 in each factor $1 + \mathcal{L}_I$ come from?

Response. It is a combinatorial device that lets a cell contribute no interaction as well as an interaction. It is related to path-integral methods, but the displayed product is only a heuristic approximation to the technical expression.

[a]Lecture timestamp: 01:49:00.

The exact formalism supplies the proper spacetime integrations, ordering, normalization, and phases. The product is useful because it reveals the combinations without pretending to be the full formalism.

[a] **Question from the audience.** Is this the principle of stationary action?

Response. No. Classical stationary action selects a distinguished classical trajectory. Quantum mechanics instead sums amplitudes over trajectories, with their weights governed by the action. Even an extremely indirect route—going through Paris on the way from an office to a nearby house—belongs to the quantum sum.

[a]Lecture timestamp: 01:49:57.

The contrast is simple even though the relation is subtle. Classical mechanics selects a stationary path. Quantum mechanics sums paths; in the appropriate limit, phases reinforce around the classical trajectory.

9.13 Inferring the Lagrangian and Choosing a Sector

Once a Lagrangian and a specified process are given, the rules for computing its probability are definite. A calculation may still demand ingenuity, as difficult mathematics often does, but the deeper uncertainty lies in discovering which fields and couplings nature uses.

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[a] **Question from the audience.** Is the art in expressing the Lagrangian or in using it?

Response. The main art is inferring the Lagrangian from experimental data. Once it is known, a process such as two particles entering and four leaving has a definite amplitude to be calculated by precise rules. Carrying out the mathematics can require cleverness, but that is not an ambiguity in the theory.

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[a]Lecture timestamp: 01:50:43.

Particle collisions expose the elementary vertices. Terms are proposed, often under the guidance of symmetry and deeper principles, and the resulting amplitudes are tested against the observed outcomes. Success fixes both the allowed combinations of fields and their numerical coefficients.

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[a] **Question from the audience.** Is there a separate Lagrangian for every interaction—a large book of Lagrangians?

Response. The working picture is instead one large Lagrangian containing all the fields and interactions, although that picture may ultimately be false. Recognizable sectors occur as pieces of the sum. A new particle adds a field; a new primitive process adds an interaction term.

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[a]Lecture timestamp: 01:52:24.

The complete expression can be untidy without being book-length. As a deliberately rough scale, one may imagine of order a hundred objects called elementary particles and several hundred terms. Quantum electrodynamics, the strong and weak interactions, and a gravitational sector could still occupy only a few densely written pages. The exact count is not the point. The working picture is one dynamical expression assembled from many recognizable pieces, and it is explicitly a picture that may eventually prove incomplete.

Whether the fields determine a more fundamental spacetime geometry lies beyond the standard quantum-field framework being used here.[8]

---

[a] **Question from the audience.** Are the experimentally determined numbers the coupling constants?

Response. Coupling constants are coefficients of terms in the Lagrangian. The coefficient m^2 of a quadratic mass term is one example; coefficients of ϕ^4, ϕ^6, or still higher powers are others.

---

[a]Lecture timestamp: 01:55:22.

One coefficient controls a family of related amplitudes because each field product has several creation and annihilation readings. Knowing the Lagrangian therefore determines far more than one isolated reaction.

[8]Editorial clarification: The question at 01:55:00–01:55:20 is too damaged to identify the manifold or geometric construction reliably; only this boundary-of-scope response is secure.

[a] **Question from the audience.** Must the entire Lagrangian be known in order to calculate quantum electrodynamics?

Response. Infinite precision would require every sector through which electrons and photons can interact. At atomic energies, however, the electron–photon sector can normally be isolated. There is not enough energy to resolve tiny quark structure, so other sectors can be neglected to the required accuracy.

[a]Lecture timestamp: 01:56:05.

Practical calculations use the sector relevant to the experimental scale. This resembles hydrodynamics: a complicated microscopic system can admit a precise coarse-grained description without displaying all its deeper constituents. Quantum field theory can therefore be phenomenological and exact at once—phenomenological in the fields and coefficients inferred from experiment, exact in the rules applied after those ingredients are specified.

[a] **Question from the audience.** Does the Standard Model include gravity, and do these relativistic Lagrangians include general relativity?

Response. The theory officially called the Standard Model does not include gravity, although gravity may be appended when discussing a broader theory. The field theories considered here combine quantum mechanics with special relativity. They do not by themselves include general relativity.

[a]Lecture timestamp: 01:58:09.

9.14 A Vacuum Value as a Mass Coefficient

The mass of a scalar field appears as the coefficient of ϕ^2. An interaction with another field can therefore imitate a mass term when that other field has a nonzero vacuum value.

[a] **Question from the audience.** If particle masses arise through Higgs interactions, can a Higgs coupling function as a mass term?

Response. Yes. A term $H\phi^2$ is cubic and ordinarily represents an interaction with two ϕ legs and one H leg. If the potential $V(H)$ selects a vacuum with nonzero H, that vacuum value is a number multiplying ϕ^2, so it contributes an effective quadratic mass term.

[a]Lecture timestamp: 01:59:16.

Consider schematically

$$\mathcal{L} \supset H\phi^2 - V(H). \tag{9.49}$$

As an operator term, $H\phi^2$ can absorb a ϕ, emit a ϕ, and emit or absorb an H quantum. Suppose the potential has two minima away from $H = 0$, and the vacuum selects one of them:

$$\langle H \rangle = v \neq 0. \tag{9.50}$$

Write the field as its vacuum value plus a fluctuation,

$$H = v + h. \tag{9.51}$$

Then

$$H\phi^2 = v\phi^2 + h\phi^2. \tag{9.52}$$

The first term is quadratic in ϕ and contributes to its mass coefficient; the second remains an interaction with the fluctuating field h.[9]

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[a] **Question from the audience.** Does the Standard Model include virtual-particle creation and annihilation?

Response. Absolutely. Virtual creation and annihilation are intrinsic to the repeated quantum processes encoded by the Standard Model Lagrangian.

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[a]Lecture timestamp: 02:01:29.

[9]Editorial clarification: The source ends the sentence while stating that the nonzero vacuum value becomes a mass coefficient. The displayed substitution is the immediate algebraic completion; its normalization and sign depend on the full Lagrangian conventions.

Chapter 10

From Path Integrals to Propagators and Vertices

Before turning to the inventory of particles and their interactions, it is worth making the central construction clear. A Lagrangian does more than encode classical field equations. Through the path integral it also supplies amplitudes for particles to propagate, split, join, and scatter. This is the most direct route from quantum field theory to Feynman diagrams.

10.1 Least Action for Particles and Fields

Feynman's path integral is the quantum counterpart of the principle of least action. Begin with an ordinary Newtonian particle, whose Lagrangian depends on its coordinates and velocities,

$$\mathcal{L} = \mathcal{L}(x, \dot{x}), \tag{10.1}$$

and the action assigned to a trajectory is

$$S[x] = \int \mathcal{L}(x, \dot{x})\, dt. \tag{10.2}$$

Every candidate trajectory from one spacetime point to another has an action, whether or not it satisfies Newton's equations. The classical trajectory is distinguished by stationary action. In the traditional "least-action" language it minimizes the action, although stationary is the more general statement. The stationarity condition gives the classical differential equations of motion.

The same structure applies to a classical field. Write all the fields generically as ϕ. A local field Lagrangian density may depend on the fields and on both their spatial and temporal derivatives:

$$\mathcal{L} = \mathcal{L}\left(\phi, \frac{\partial \phi}{\partial x^\mu}\right) = \mathcal{L}(\phi, \partial_\mu \phi). \tag{10.3}$$

Relativity requires $\mathcal{L}$ to transform as a Lorentz scalar. Subject to that requirement, scalar combinations built from fields and their derivatives furnish the possible terms in the Lagrangian.

For a field, the action is integrated over a region of spacetime:

$$S[\phi] = \int_\Omega d^4x\, \mathcal{L}(\phi, \partial_\mu \phi), \qquad d^4x = dt\, dx\, dy\, dz. \tag{10.4}$$

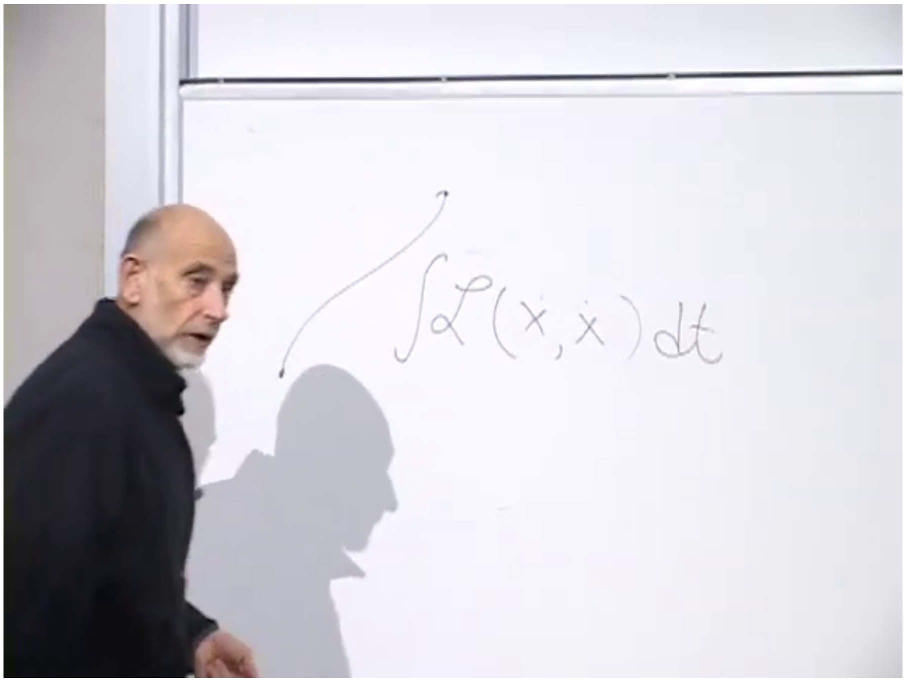

Figure 10.1: Action integral for a particle trajectory.
Lecture frame: 00:02:14.

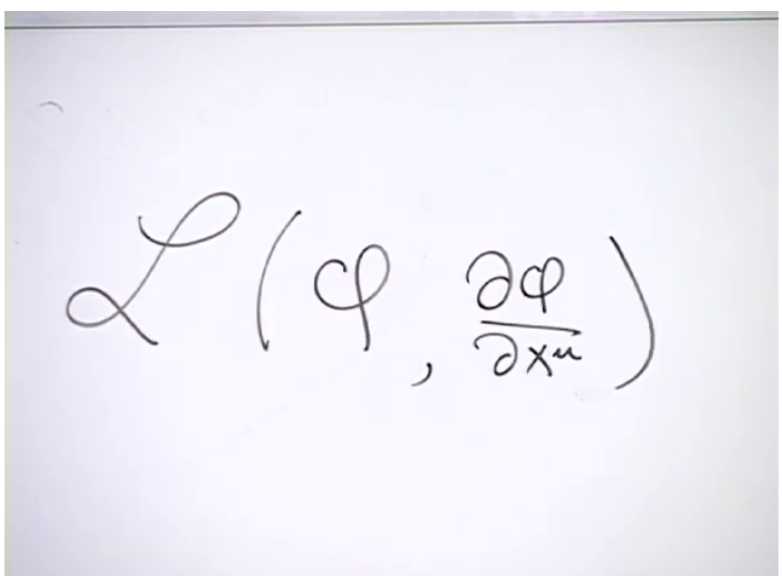

Figure 10.2: Dependence of the field Lagrangian on the field and its spacetime derivatives.
Lecture frame: 00:04:09.

The particle action is an integral over time along a path. The field action is the integrated Lagrangian density throughout a spacetime region.

The analog of fixing the endpoints of a particle path is to prescribe the complete field configuration on an initial time slice and on a final time slice. If the initial field is $\phi_{\mathrm{i}}(\mathbf{x})$ and the final field is $\phi_{\mathrm{f}}(\mathbf{x})$, an interpolating field fills the whole slab between them. When the boundary data admit a classical interpolating solution, that solution is a stationary point of the action with the boundary configurations held fixed. [1]

The geometry is the field-theory counterpart of a particle path.[2]

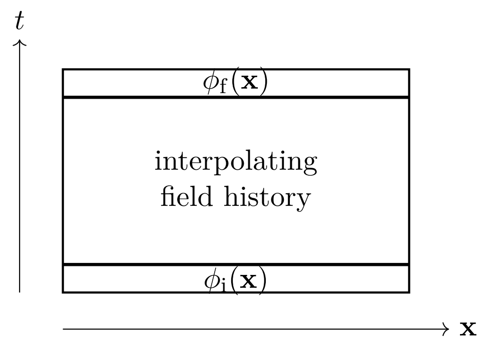

Figure 10.3: Initial and final field configurations bounding a spacetime region.

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[a] **Question from the audience.** Can a field return to the same value?

Response. Yes. There is no obstruction to a field returning to an earlier value, just as there is no obstruction to a particle coordinate returning to an earlier position.

[a]Lecture timestamp: 00:07:34.

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[a] **Question from the audience.** Why speak of spacetime events or a trajectory when the boundary data are field values?

Response. The particle trajectory is replaced by a field history. Each endpoint in field space is the whole field on a chosen time slice, and a history specifies the field at every point in the spacetime region between the two slices.

[a]Lecture timestamp: 00:07:46.

Requiring stationary action for the interpolating field produces partial differential equations. Depending on the fields and the Lagrangian, these include equations such as Maxwell's equations and, for a gravitational action, Einstein's equations. The detailed variational derivation is not needed here; the important sequence is

$$\mathcal{L} \longrightarrow S[\phi] \longrightarrow \text{classical field equations.} \tag{10.5}$$

[1]Editorial clarification: Arbitrary complete field configurations on two time slices need not admit a classical solution. The stationary-action statement is therefore restricted to boundary data for which an interpolating solution exists.

[2]The spacetime slab is reconstructed from the diagram at approximately 00:04:29.

10.2 Feynman's Sum over Particle Paths

Quantum mechanics replaces the question of which trajectory occurs by a question about amplitudes. Imagine injecting a particle at (x,t), closing one's eyes for a while, and then turning on a detector at (x',t'). The amplitude

$$\mathcal{A}(x,t;x',t') \tag{10.6}$$

is a complex number. Its magnitude squared is the probability:

$$\begin{aligned} P(x,t;x',t') &= |\mathcal{A}(x,t;x',t')|^2 \\ &= \mathcal{A}(x,t;x',t')\mathcal{A}(x,t;x',t')^*. \end{aligned} \tag{10.7}$$

Feynman's rule assigns an action to every candidate path and associates with it the phase

$$\exp\left(-\frac{i}{\hbar}S[x]\right) = \exp\left[-\frac{i}{\hbar}\int \mathcal{L}(x,\dot{x})\,dt\right]. \tag{10.8}$$

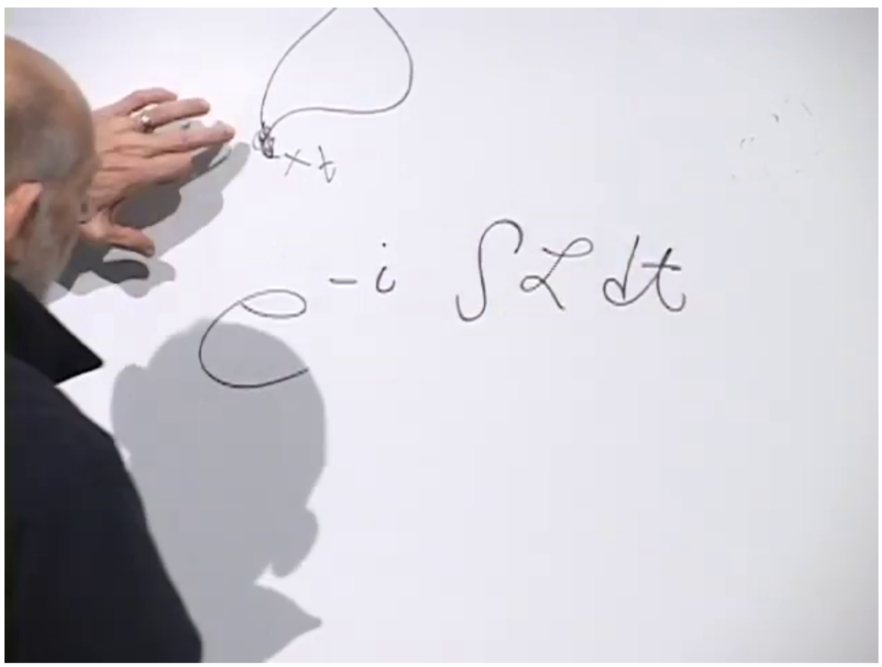

Figure 10.4: Complex phase associated with a particle path.

Lecture frame: 00:11:52.

The factor of $\hbar$ is required to make the exponent dimensionless.[3] The transition amplitude is the sum of these phases over every possible path connecting the endpoints:

$$\mathcal{A}(x,t;x',t') = \sum_{\substack{\text{paths } x(\tau) \\ (x,t)\to(x',t')}} \exp\left(-\frac{i}{\hbar}S[x]\right). \tag{10.9}$$

Here "possible path" does not mean a solution of Newton's equations. It means any path included in the sum. In nonrelativistic quantum mechanics the paths progress forward in time rather than looping backward. Planck's constant has the dimensions of action, so $S/\hbar$ is dimensionless.

[3] At 00:11:52 the board reads naturally as $e^{-i\int \mathcal{L}\,dt}$; the factor of $\hbar$ is supplied in speech at 00:12:02.

The exponential is natural because amplitudes for successive small steps multiply. If a path is divided into consecutive pieces, the action adds,

$$S = S_1 + S_2 + \cdots , \tag{10.10}$$

while the phases multiply:

$$e^{-iS/\hbar} = e^{-iS_1/\hbar} e^{-iS_2/\hbar} \cdots . \tag{10.11}$$

The factor i is not a bookkeeping device; it is part of the quantum-mechanical rule.

[a] **Question from the audience.** Is the exponential connected with multiplying the amplitudes for successive small steps?

Response. Yes. Successive step amplitudes multiply, while the corresponding pieces of the action add.

[a]Lecture timestamp: 00:13:27.

[a] **Question from the audience.** Is the sum over paths itself the probability?

Response. No. The path sum is a generally complex probability amplitude. Multiplying it by its complex conjugate gives the probability.

[a]Lecture timestamp: 00:13:49.

10.3 The Sum over Field Histories

The same formulation extends to quantum field theory, although the construction here will remain deliberately nonrigorous. Begin with an initial field configuration $\phi(\mathbf{x})$ and a separately specified final configuration $\phi'(\mathbf{x})$. These are complete functions of space, not values at two isolated points.

One may imagine an idealized apparatus that prepares the field at every spatial point, lets the system evolve, and later measures the field at every point. Such an apparatus is not realistic, but it defines the conceptual question cleanly: what is the amplitude to pass from the initial complete field configuration to the final one?

Every imaginable interpolating field history contributes. For each history, evaluate

$$S[\phi] = \int d^4x \, \mathcal{L}(\phi, \partial_\mu \phi) \tag{10.12}$$

and attach the corresponding phase. Schematically,

$$\mathcal{A}[\phi(\mathbf{x}) \to \phi'(\mathbf{x})] = \sum_{\substack{\text{field histories} \\ \phi_{\mathrm{i}}=\phi,\ \phi_{\mathrm{f}}=\phi'}} \exp\left(-\frac{i}{\hbar} S[\phi]\right) . \tag{10.13}$$

A history need not solve the classical field equations. It is simply one complete assignment of field values throughout the intervening spacetime region.

[a] **Question from the audience.** There are infinitely many histories, and each phase has magnitude one. How can their sum converge? Are the histories being minimized?

Response. The histories are summed, not minimized. The terms are phases distributed around the unit circle, so they can cancel one another. This differs from adding infinitely many positive terms bounded away from zero. Oscillatory integrals can converge; the Fourier representation of a delta function is a familiar example. If the phases were distributed uniformly around the circle, their vector sum would vanish. Most histories cancel strongly. Histories near a stationary-action trajectory cancel least because the phase changes least rapidly there, and this is how the classical limit emerges.

[a]Lecture timestamp: 00:17:39.

4

The cancellation can be pictured as a vector sum in the complex plane.[5]

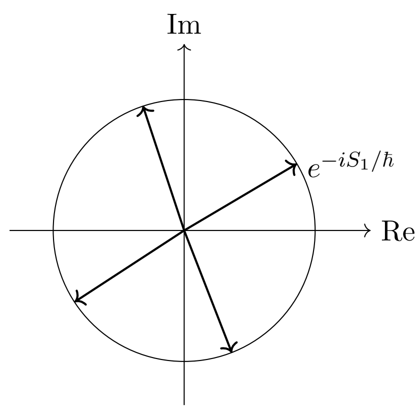

Figure 10.5: Unit-magnitude phases can cancel when their directions differ.

Indeed,

$$e^{-iS/\hbar} = \cos\left(\frac{S}{\hbar}\right) - i\sin\left(\frac{S}{\hbar}\right). \tag{10.14}$$

Each contribution is a unit vector. Contributions pointing in different directions subtract as well as add. Near a stationary point of S, nearby histories have nearly the same phase, so their contributions reinforce one another more effectively. Away from stationary action, the phase varies rapidly and destructive interference dominates.

This path-integral structure underlies essentially all modern quantum field theory. The next practical step will be stated rather than derived. It is an oversimplified route into the calculation, but it makes visible how the same object can be used in two languages.

To calculate a process is to calculate an initial-to-final transition probability. One description specifies an initial field configuration and a final field configuration. The other specifies incoming and outgoing particles, the quanta of one or several fields. These questions look different, but both are calculated from the same field-theory path integral.

[4]Editorial clarification: A real-time path integral ordinarily requires a convergence prescription. The unit-circle discussion gives the oscillatory-cancellation argument used here, not a general proof of convergence.

[5]The unit-circle construction is reconstructed from the board discussion at 00:18:32–00:20:21.

10.4 A Discrete Spacetime Form

To give the field-history sum a concrete meaning, divide spacetime into small cells and assign a field value to each cell. The continuum limit is approached by shrinking the cell size, but for now the cells remain finite. This converts the formal sum over functions into a sum over assignments of numbers to lattice cells.

An earlier heuristic uses a product of factors resembling $1 + \mathcal{L}_i$, one for each cell, and then identifies terms describing particle propagation and collision. The construction is central enough to repeat, but the factor must be corrected. The small quantity is not the Lagrangian itself. It is the action contained in a cell.

If a cell has spacetime volume

$$a^4 = \Delta t \, \Delta x \, \Delta y \, \Delta z, \tag{10.15}$$

then its action is

$$A_i = \mathcal{L}_i a^4. \tag{10.16}$$

Since the cell is small, A_i is small. Temporarily suppressing $\hbar$, the corrected elementary factor is

$$1 - iA_i, \tag{10.17}$$

rather than $1 + \mathcal{L}_i$.

For any small ϵ,

$$1 + \epsilon \simeq e^{\epsilon}, \tag{10.18}$$

because

$$e^{\epsilon} = 1 + \epsilon + \frac{\epsilon^2}{2!} + \frac{\epsilon^3}{3!} + \cdots . \tag{10.19}$$

Thus, in these temporary units, $1 - iA_i$ approximates e^{-iA_i} for a sufficiently small cell. It is cleaner to use the exact exponential factor from the beginning.

---

[a] **Question from the audience.** Does the exponential require a factor of $\hbar$?

Response. Yes. The dimensionless cell factor is $1 - iA_i/\hbar$ to first order, and the exact exponential is $\exp(-iA_i/\hbar)$. One may later adopt units in which $\hbar = 1$.

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[a] Lecture timestamp: 00:27:57.

Restoring $\hbar$, the corrected intermediate factor and its exponential form are

$$1 - \frac{i}{\hbar} A_i \simeq \exp\left(-\frac{i}{\hbar} A_i\right). \tag{10.20}$$

For a fixed history, replace the spacetime integral by a sum of cell actions:

$$S[\phi] = \int d^4x \, \mathcal{L} \longrightarrow \sum_i A_i. \tag{10.21}$$

The phase for that history becomes

$$\exp\left(-\frac{i}{\hbar} \sum_i A_i\right). \tag{10.22}$$

An exponential of a sum is a product of exponentials:

$$\exp\left(-\frac{i}{\hbar}\sum_i A_i\right) = \prod_i \exp\left(-\frac{i}{\hbar}A_i\right). \tag{10.23}$$

The order of operations matters. First choose one complete history and multiply its cell weights. Only then sum over the different histories.

The lattice also makes the boundary data explicit.[6]

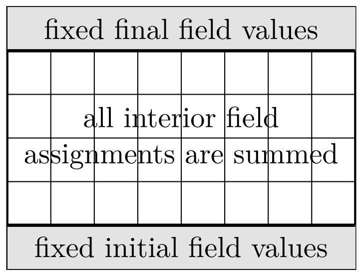

Figure 10.6: Discrete field histories with fixed initial and final lattice rows.

The first row specifies the initial field configuration, and the last row specifies the final configuration. Every interior cell is allowed every field value included in the discretization. Thus the transition amplitude takes the schematic form

$$\mathcal{A}[\phi_\mathrm{i} \to \phi_\mathrm{f}] = \sum_{\{\phi\}_\mathrm{interior}} \prod_i \exp\left(-\frac{i}{\hbar}A_i[\phi]\right), \tag{10.24}$$

with the first and last rows held fixed. Equivalently, for each interior assignment one computes the exponential of the total action and then adds that contribution to the contributions from all other assignments.

10.5 From Field Configurations to Particle Configurations

The same transition amplitude may be described in terms of fields or in terms of particles. Instead of fixing the field on the first and last time slices, one may specify an incoming collection of field quanta and an outgoing collection of field quanta, then ask for the amplitude connecting them. The object being evaluated is the same. Only the boundary description has changed.

Quantum fields contain creation and annihilation operators. Products of fields therefore act on particle states by creating, destroying, or moving quanta. This operator reading is what turns the lattice path integral into a set of particle rules.

[6] The checkerboard and its fixed first and last rows are reconstructed from the diagrams at approximately 00:23:54 and 00:30:49.

[a] **Question from the audience.** Are field quanta necessarily identifiable as particles?

Response. Not always in the conventional sense of separately observable particles. Quarks are quanta of quark fields, for example, but confinement prevents an isolated quark from escaping and being detected by itself. More generally there need not be a one-to-one correspondence between the fields named in a theory and the particle species that appear as conventional observable states. Some field theories do not admit that simple correspondence. If "particle" is used broadly to mean an indivisible quantum carrying energy and other quantum numbers, then the distinction can partly become one of terminology.

[a]Lecture timestamp: 00:34:11.

[7]

The lattice sum itself still ranges over field values. It is not restricted to configurations that resemble smooth classical solutions.

[a] **Question from the audience.** Must the field histories be continuous, or at least approximately continuous?

Response. No. On a discrete lattice, neighboring cells simply carry field values, and every assignment is included in the sum. There is no restriction to histories that are approximately continuous. Quantum fields fluctuate strongly; typical histories are not smooth classical fields.

[a]Lecture timestamp: 00:36:44.

The variables assigned independently are the fields at the lattice sites or cells. Their derivatives are obtained from differences between neighboring values.

[a] **Question from the audience.** Are the fields and their derivatives assigned independent values?

Response. No. Field values are assigned at the sites. A derivative is then reconstructed from the difference between neighboring field values. Different finite-difference conventions are possible, but the derivatives are not additional independent variables.

[a]Lecture timestamp: 00:37:56.

Thus, for lattice spacing a, one may write schematically

$$\partial_\mu \phi(x) \longrightarrow \frac{\phi(x + a\hat{\mu}) - \phi(x)}{a}. \tag{10.25}$$

The field-value axis may itself be divided into small intervals, leaving a countable set of possible values at each site.

[a] **Question from the audience.** What changes when there is more than one field?

Response. A value of each field is assigned at every site, and the sum runs over all such assignments.

[a]Lecture timestamp: 00:38:53.

A complex-valued field should not be confused with an analytic function of a complex variable. Analyticity is an exceptionally strong smoothness condition. The histories in a field path integral need not obey it and, in general, are not even approximately smooth.

[7]Editorial clarification: The source's weak-coupling clause at 00:35:45–00:36:08 is grammatically ambiguous and is left unresolved here. The secure claim is that field labels and conventionally observable particle species need not correspond one-to-one.

[a] **Question from the audience.** If the fields are complex-valued, must their values satisfy the smoothness conditions obeyed by analytic functions?

Response. No. Complex-valued does not mean analytic. The theory of analytic functions concerns an extremely smooth and tightly constrained class of functions. The field histories summed here need not be analytic functions of position; typical histories are highly irregular and may be regarded as highly discontinuous in the continuum limit.

[a]Lecture timestamp: 00:39:07.

The resulting computation looks formidable. Even after spacetime and field values have been discretized, there can be an enormous number of assignments to sample.

[a] **Question from the audience.** How does a computer know how many lattice assignments to sample, or when the calculation has converged?

Response. That question belongs to lattice gauge theory, or more generally lattice quantum field theory. By now it is an effective computational subject. I was involved at the very beginning in setting up the rules for lattice gauge theory, not in the later computer technology. Setting up the rules took about three weeks; developing the technology to compute these things took something like thirty years. Much of the practical wisdom came from statistical mechanics, where partition functions are also obtained by summing over configurations of a lattice—for example, over whether a site in a crystal is occupied. Determining whether enough configurations have been sampled is a technical matter. The experts know—or think they know—when the calculation is under control, and the resulting calculations agree with experiment.

[a]Lecture timestamp: 00:40:04.

The conceptual rules for lattice gauge theory came quickly; the technology required to compute with them took much longer. The checkerboard is not merely an analogy. It is also the starting point of a practical computational method.

10.6 Quadratic Terms and Particle Propagation

Return now to the operator interpretation of the fields. For two neighboring sites x and x', the product

$$\phi(x)\phi(x') \tag{10.26}$$

can supply, in an appropriate one-particle matrix element, the annihilation of a particle at one site and its creation at the other. In that particle sector it may therefore be pictured as a one-link hop. This is one reading of the operator product, not its unique meaning.

The terms responsible for such links are the derivative terms. Strictly speaking, they are associated not with one cell but with a neighboring pair. A temporal derivative, for example, becomes

$$\partial_t\phi(t,\mathbf{x}) \longrightarrow \frac{\phi(t+a,\mathbf{x})-\phi(t,\mathbf{x})}{a}, \tag{10.27}$$

and a spatial derivative is treated in the same way along a spatial link.

[a] **Question from the audience.** Do the Lagrangians under discussion contain second derivatives?

Response. The standard theories being considered here are written with first derivatives. Their lattice versions therefore use first differences between neighboring sites.

[a]Lecture timestamp: 00:44:39.

Squaring a first difference produces both same-site terms and a neighboring cross term:

$$\frac{1}{2}\big[\phi(x)-\phi(x')\big]^2 = \frac{1}{2}\phi(x)^2 + \frac{1}{2}\phi(x')^2 - \phi(x)\phi(x'). \tag{10.28}$$

The cross term can supply motion from one site to the other. The two squares are same-site terms. The quadratic derivative terms are the kinetic terms, and it is their neighboring cross terms that will be used first to build propagation.

To isolate the combinatorics, collect the neighboring products into

$$K = \sum_{\langle x,x'\rangle} \phi(x)\phi(x'). \tag{10.29}$$

The corresponding factor in the path-integral weight is written schematically as

$$e^{-iK}. \tag{10.30}$$

[8] Its expansion is

$$e^{-iK} = 1 - iK - \frac{1}{2!}K^2 + \frac{i}{3!}K^3 + \cdots, \tag{10.31}$$

or, with the neighboring sums exposed,

$$\begin{aligned} e^{-iK} = 1 - i &\sum_{\langle x,x'\rangle} \phi(x)\phi(x') \\ &- \frac{1}{2!}\sum_{\langle x,x'\rangle}\sum_{\langle y,y'\rangle} \phi(x)\phi(x')\phi(y)\phi(y') + \cdots. \end{aligned} \tag{10.32}$$

Each factor independently selects a neighboring pair.

The particle diagrams obey a no-dangling-end rule. An exposed endpoint is allowed only where the boundary condition explicitly inserts a particle on the initial slice or removes one on the final slice. An isolated link in the interior has unmatched ends and does not represent a complete contribution to the prescribed transition. Internal link ends must join.

Consider first a lattice only two time layers high. If the initial state inserts a particle at x and the final state removes it at the neighboring point x', the first-order term can connect the two boundary insertions:

$$x \longrightarrow x', \qquad \mathcal{A}_{\text{one link}} \sim -i. \tag{10.33}$$

The coefficient $-i$ is only the coefficient in this schematic expansion; the suppressed lattice and normalization factors remain understood.

---

[a] **Question from the audience.** Does the first-order one-link term contribute only when the relevant initial and final sites are adjacent?

Response. Yes. That term selects one neighboring pair, so it can connect those boundary sites only when they are neighbors. More distant propagation comes from higher-order products.

---

[a]Lecture timestamp: 00:52:16.

[8]Editorial clarification: In this schematic lattice factor, powers of the lattice spacing, $\hbar$, metric signs, normalization constants, and the coefficients of the links are suppressed.

At second order the two factors are independently summed over neighboring pairs:

$$K^2 = \sum_{\langle x,x'\rangle} \sum_{\langle y,y'\rangle} \phi(x)\phi(x')\phi(y)\phi(y'). \tag{10.34}$$

---

[a] **Question from the audience.** Must the second neighboring pair adjoin the pair selected by the first factor?

Response. No. Each factor independently selects a neighboring pair. The two pairs may share a site, they may be completely separate, or they may even be the same pair. The sum includes every possibility.

---

[a]Lecture timestamp: 00:52:40.

For one particle propagating across three time layers, the useful second-order selection has a shared intermediate site:

$$\big[\phi(x_0)\phi(x_1)\big]\big[\phi(x_1)\phi(x_2)\big], \qquad x_0 \longrightarrow x_1 \longrightarrow x_2. \tag{10.35}$$

The end of the first link is joined to the beginning of the second. Only the initial endpoint at x_0 and the final endpoint at x_2 remain exposed, so the term contributes to the required one-particle transition. In the schematic exponential its coefficient is

$$-\frac{1}{2!}. \tag{10.36}$$

The selections may instead form separate hops.[9]

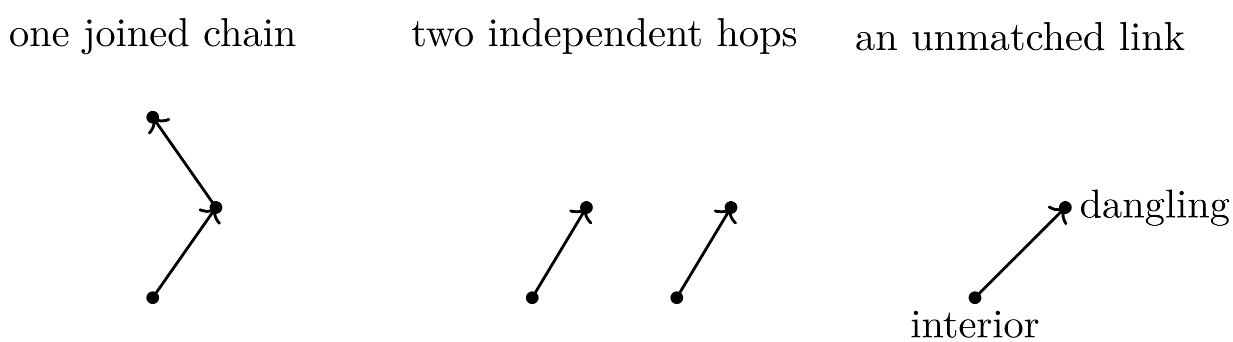

Figure 10.7: Joined, independent, and unmatched selections of neighboring links.

There can already be more than one route at this order. For a displacement by one temporal and one spatial lattice step, the two orders

$$x \longrightarrow x + \hat{t} \longrightarrow x + \hat{t} + \hat{s}, \tag{10.37}$$

$$x \longrightarrow x + \hat{s} \longrightarrow x + \hat{s} + \hat{t} \tag{10.38}$$

reach the same endpoint.

---

[a] **Question from the audience.** Is there another two-step path to the same diagonal endpoint?

Response. Yes. One may take the temporal step first and the spatial step second, or take them in the opposite order. Both routes contribute to the same transition amplitude.

---

[a]Lecture timestamp: 00:55:53.

[9]The link diagrams are reconstructed from the sequence at 00:49:18–00:55:12.

There is no reason to stop at two links. The third-order term supplies chains of three neighboring links. A route may advance directly, make a detour, or move away and return before reaching the final point. Still higher orders include longer detours and repeated backtracking. Expanding to arbitrary order therefore produces a contribution for every route assembled from local links. The point-to-point amplitude is the sum of all of them, with coefficients containing the appropriate powers of i and factorials:

$$\mathcal{A}(x_{\text{in}} \to x_{\text{out}}) = \sum_{\text{joined lattice routes } r} c_r, \qquad c_r \text{ read from the exponential expansion.} \tag{10.39}$$

The coefficients are generally complex, so the resulting amplitude is complex.

Relativistic propagation adds another possibility. A route may run both upward and downward in the time direction, and it may traverse the same link repeatedly.

---

[a] **Question from the audience.** Can a path pass through the same two cells over and over again?

Response. Yes. Repeated traversals are included, and in the relativistic expansion the route may also run backward in time.

---

[a]Lecture timestamp: 00:58:47.

A backward-time segment admits two equivalent graphical readings. It may be treated as a particle line running backward in time. Alternatively, follow time only forward: a particle–antiparticle pair is created, the particle member continues forward, and the antiparticle later meets and annihilates the original particle. The same connected line encodes both descriptions.[10]

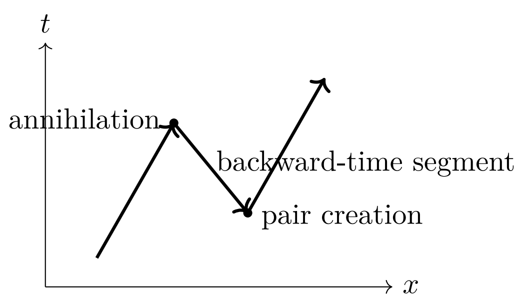

Figure 10.8: A backward-time segment read forward as pair creation and annihilation.

The factorials appearing at high orders control the exponential power series, but that is not the same as controlling the full sum over field configurations.

---

[a] **Question from the audience.** Do the factors of $1/n!$ guarantee convergence?

Response. They guarantee convergence of the exponential series for a fixed argument. They do not by themselves guarantee convergence of the separate sum over all field configurations. Those are distinct questions. On a finite lattice with a finite discretized set of field values the configuration count is finite, but the continuum and infinite-volume limits require further control.

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[a]Lecture timestamp: 01:00:00.

[10]The two readings follow the diagram and explanation at 00:59:00–00:59:51.

10.7 One Link Product, Several Particle Readings

The second-order product contains more than one-particle propagation. When its two neighboring pairs share an endpoint, it can move one particle through a two-link chain:

$$\big[\phi(x_0)\phi(x_1)\big]\big[\phi(x_1)\phi(x_2)\big]. \tag{10.40}$$

When the selected pairs are separate, the same second-order term can move two particles by one link each.[11]

$$\big[\phi(x_0)\phi(x_1)\big]\big[\phi(y_0)\phi(y_1)\big], \qquad x_0 \neq y_0. \tag{10.41}$$

The first factor transports one particle and the second transports another. Thus the quadratic lattice expansion contains propagation amplitudes in every multiparticle sector. The same algebraic order may describe one longer connected route or several shorter disconnected routes.

This is a great deal of information, but it is not yet a genuine interaction vertex. With only quadratic propagation terms, connected particle lines do not split into a different number of lines. Particle-number-changing processes require additional interactions, even though local operator products may admit several creation-and-annihilation readings.

[a] **Question from the audience.** Could an electron and a positron annihilate and leave no particles at all?

Response. A contribution with nonzero incoming energy and no energy-carrying final state violates energy conservation. When its possible spacetime positions are summed, the amplitudes cancel. Electron–positron annihilation can occur when another field carries away the energy—for example, when electromagnetic quanta are produced.

[a]Lecture timestamp: 01:03:30.

12

The derivative terms therefore do much, but not everything. Further terms in the Lagrangian are fixed by the experimentally observed properties and interactions of the fields.

10.8 The Mass Term

For a scalar field, the derivative structure under discussion has the schematic Lorentzian sign pattern

$$\mathcal{L}_{\text{kin}} \sim \frac{1}{2}\left[(\partial_t\phi)^2 - (\partial_x\phi)^2 - (\partial_y\phi)^2 - (\partial_z\phi)^2\right]. \tag{10.42}$$

This display fixes the temporal-minus-spatial structure relevant to the discussion; overall signs and normalization conventions are left implicit.

The Lagrangian may also contain powers of the undifferentiated field. The quadratic example has the form

$$\text{mass term} \;\propto\; \frac{1}{2}m^2\phi^2, \tag{10.43}$$

[11]The comparison between one-particle propagation and two independent one-link hops is reconstructed from the board discussion at 01:00:46–01:02:34.

[12]Editorial clarification: The abbreviated exchange moves directly from quadratic hopping to electron–positron annihilation. The secure distinction is that quadratic terms supply free propagation, while physical annihilation requires an interaction with another field; the no-output amplitude cancels by energy conservation.

with m^2 identified as the mass coefficient.[13] In the schematic particle reading, this term annihilates a particle and recreates it at the same lattice site. It does not displace the particle. It inserts a same-site factor into a path:

$$\text{particle at } x \longrightarrow \boxed{m^2\phi(x)^2} \longrightarrow \text{particle still at } x. \tag{10.44}$$

Every time such an insertion acts, the path acquires another factor proportional to m^2.

If only the derivative terms are retained, the resulting propagation is the massless theory in this schematic expansion. For a massive field, a route may include any number of same-site mass insertions between its hopping terms. The mass term changes the relative weights of the paths so that the resulting propagator carries the particle's mass.

10.9 Cubic Terms and Split–Rejoin Histories

The mass term changes the weight of a path without moving the particle. More interesting behavior begins with terms of degree higher than two. Take a cubic interaction,

$$\mathcal{L}_{\text{int}} = g_3\phi^3. \tag{10.45}$$

At one lattice cell, the three field factors have four particle readings:

$$1 \longrightarrow 2, \qquad 2 \longrightarrow 1, \qquad 0 \longrightarrow 3, \qquad 3 \longrightarrow 0. \tag{10.46}$$

Thus a cubic factor may annihilate one particle and create two, annihilate two and create one, create three, or annihilate three.

The quadratic neighboring-pair factors still supply propagation. A sequence of them can carry a particle to a cell, a cubic factor at that cell can absorb it and emit two particles, and further quadratic factors can carry those two particles away. This contributes to an amplitude with one incoming particle and two outgoing particles.[14]

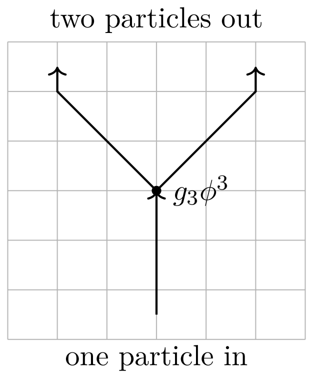

Figure 10.9: Quadratic factors propagate a particle into and away from a cubic splitting vertex.

Even an amplitude with one particle both initially and finally is no longer merely a sum of single uninterrupted routes. One contribution is direct propagation by a chain of quadratic hops. Another

[13] Editorial clarification: The source does not fix the overall sign convention for the complete scalar Lagrangian in this passage.

[14] The lattice geometry is reconstructed from the diagram at 01:08:48–01:09:42.

contribution contains two cubic factors: the particle splits into two and the two later rejoin. The intermediate state may therefore contain more particles than the external state.[15]

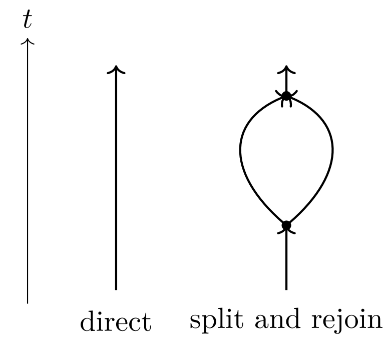

Figure 10.10: Direct propagation and a split–rejoin contribution to the same one-particle amplitude.

In electrodynamics the corresponding example is an electron that emits a photon and later reabsorbs it. That emission-and-reabsorption history is part of the amplitude for the electron to propagate from one point to another. Higher orders contain several emissions and absorptions. A photon may convert into an electron–positron pair, and the electron and positron in that intermediate pair may themselves exchange a photon. Nothing requires the intermediate particle content to look simple merely because the initial and final particle content is simple.[16]

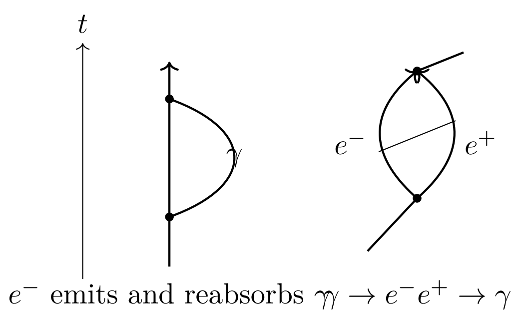

Figure 10.11: An electron self-emission contribution and a photon history containing an electron–positron pair.

Each such history comes from a definite product of terms in the expansion of the exponential. Its coefficient carries the appropriate power of i, a factorial from the exponential series, and the couplings attached to its vertices. The coefficient supplies the contribution of that history, but the amplitude for the specified initial and final states is the sum of all histories with those same external states.

[a] **Question from the audience.** Can the same overall process occur in several different histories, so that all of them contribute to its amplitude?

Response. Yes. The amplitude for the process is obtained from all the different histories that join the same initial state to the same final state.

[a]Lecture timestamp: 01:13:18.

[15]The two alternatives are reconstructed from the diagram at 01:09:44–01:11:36.
[16]The examples in the following reconstruction occur at 01:11:22–01:12:34.

There are therefore two equivalent ways to organize quantum field theory. In the field description, one specifies an initial field configuration and a final field configuration and sums over every intervening field history. In the particle description, the field is resolved into quanta: one specifies the incoming particle content and the outgoing particle content and sums all the particle histories connecting them. The same Lagrangian governs both descriptions.

10.10 Quartic Terms, Conservation Laws, and Couplings

A quartic interaction,

$$\mathcal{L}_{\text{int}} = g_4 \phi^4, \tag{10.47}$$

has four field factors at one point. Its local particle readings include

$$\begin{gathered} 2 \longrightarrow 2, \qquad 1 \longrightarrow 3, \qquad 3 \longrightarrow 1, \\ 0 \longrightarrow 4, \qquad 4 \longrightarrow 0. \end{gathered} \tag{10.48}$$

A single four-leg vertex can therefore describe two-particle scattering, one particle splitting into three, three joining into one, or all four legs oriented outward or inward.[17]

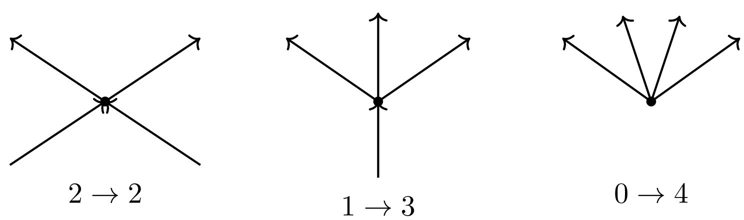

Figure 10.12: Three local orientations of a four-leg interaction vertex.

The local reading $0 \to 4$ does not by itself say that four particles can appear from the vacuum as a physical process. The vertex must still be summed over every possible spacetime position. When the proposed external state cannot conserve energy, the phases from those positions cancel.

[a] **Question from the audience.** Does the four-out orientation describe the spontaneous arrival of four particles in the final state?

Response. It is one local reading of the four-field operator, but a process with no incoming energy violates energy conservation. When the amplitudes from all possible spacetime positions of the vertex are added, they cancel to zero.

[a]Lecture timestamp: 01:15:18.

[a] **Question from the audience.** Do conservation of energy and conservation of momentum emerge from the calculation itself?

Response. Yes. Integrating the vertex over all time enforces energy conservation, and integrating it over all space enforces momentum conservation. The sum over vertex positions vanishes unless both are conserved.

[a]Lecture timestamp: 01:15:58.

[17]The orientations are reconstructed from the diagram at 01:14:28–01:15:18.

Subject to those conservation laws, every graph allowed by the Lagrangian contributes some amplitude. The size of that contribution depends in part on the coefficient of each interaction term. Write the cubic and quartic coefficients as G_3 and G_4. Every three-leg vertex contributes one factor of G_3, and every four-leg vertex contributes one factor of G_4. A graph with V_3 cubic vertices and V_4 quartic vertices therefore has the schematic coupling dependence

$$\mathcal{A}_{\text{graph}} \propto G_3^{V_3} G_4^{V_4}. \tag{10.49}$$

If these couplings are small, a graph with few vertices generally contributes more than a graph with many vertices. Each additional vertex brings another small factor. This gives the perturbative sum a chance to be useful: begin with the simplest graphs, then add successively smaller corrections.

Whether the complete series actually converges is a difficult question. Small couplings make the ordering plausible, but they do not by themselves settle every convergence problem. At large coupling the naive series may fail altogether. Such failure does not show that the underlying theory is wrong; it may show only that expansion in powers of the coupling was a poor way to calculate it. The methods under discussion are most directly useful when the couplings are small enough that a graph with four vertices is much smaller than one with two.

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[a] **Question from the audience.** Can color theory be treated in the same way?

Response. Yes. Color theory can be formulated in the same way and will be discussed later.

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[a]Lecture timestamp: 01:19:24.

10.11 Propagators, Vertices, and Complementary Descriptions

The next task is to list the particles and encode their dynamics either in a Lagrangian or in Feynman rules. These are two forms of the same information. Quadratic terms describe propagation from one point to another. A line representing that propagation is called a propagator. Terms involving three or more fields describe places where lines meet. Those meeting points are vertices. For example,

$$\begin{aligned} \text{quadratic term} &\longleftrightarrow \text{propagator}, \\ G_3\phi^3 &\longleftrightarrow \text{three-leg vertex}. \end{aligned} \tag{10.50}$$

One can specify a theory by giving a quantum field for every particle and writing one Lagrangian containing all their masses and interactions. Equivalently, one can list every propagator, every allowed vertex, and the coefficient attached to each vertex. Either list tells us what processes are possible and how to assign amplitudes to their graphs.

For example, two electrons can scatter by exchanging a photon. Each electron line contains an electron–electron–photon vertex, and the photon propagator joins the two vertices.[18]

The Lagrangian determines the amplitude for this graph and contains the symmetries and conservation laws that constrain it. In a time-oriented drawing, particles placed at the bottom form the initial configuration and particles at the top form the final configuration.

[18]The diagram is reconstructed from the example at 01:21:38.

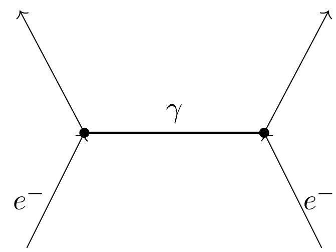


Figure 10.13: Two-electron scattering through the exchange of a photon.

[a] **Question from the audience.** In a Feynman diagram, are the particles at the bottom the initial configuration and those at the top the final configuration?

Response. Yes. The bottom is the initial particle configuration, and the top is the final particle configuration.

[a]Lecture timestamp: 01:22:06.

The particle description and the field description are complementary, much as momentum and position are complementary descriptions in ordinary quantum mechanics. They describe the same system in different bases. There is also a corresponding uncertainty: if particle number is specified precisely, the field value is uncertain; if the field value is specified precisely, particle number is uncertain. Neither description replaces the other, and the Lagrangian is useful in both.

[a] **Question from the audience.** Are the coupling constants predicted by theory, or are they determined experimentally?

Response. Symmetries can sometimes relate one coupling constant to another, but their numerical values generally come from experiment.

[a]Lecture timestamp: 01:23:26.

10.12 Measured Inputs and Broad Predictions

Quantum field theory itself has elegance and beauty. But when one starts listing the facts—what particles there are, what their masses are, and what their coupling constants are—the list is a mess. It is a very ugly mess, with very little coherence except the coherence supplied by symmetries. Symmetries relate different kinds of particles and their processes, but only to a limited extent. Much of the inventory remains a large collection of comparatively random facts about a somewhat unmotivated collection of different kinds of particles.

[a] **Question from the audience.** Why should the measured coupling constants be trusted so confidently?

Response. Once a coupling has been measured, changing it gives incorrect predictions. Its value is tested by the many calculations that use it.

[a]Lecture timestamp: 01:24:58.

There are too many particles, which means too many fields—too many for any aesthetic sense. There are roughly a hundred or so. There are many coupling constants, more or less numbers pulled from experiment, and many masses, which range all over the map. There are a few relations among them, but only a few.

But once you know them, you can start to calculate. Once you know them, that is it: one can calculate with great precision anything about those particles. Anything about quarks, electrons, photons, and the rest reaches into atomic physics, particle physics, nuclear physics, chemistry, and perhaps biology. The amount of input may be larger than one would like, but the amount of output is huge.

10.13 Quantum Electrodynamics as a Perturbative Example

Quantum electrodynamics gives a concrete example of the coupling expansion.

[a] **Question from the audience.** For electrons and photons, how many interaction couplings of the G_n type are there?

Response. There is one elementary interaction coupling: the electric charge e. It is a G_3-type coefficient attached to a vertex with an electron entering, an electron leaving, and a photon emitted or absorbed.

[a]Lecture timestamp: 01:26:12.

Suppressing the spinor matrices and indices, the interaction has the schematic field content

$$\mathcal{L}_{\mathrm{QED,int}} \sim e\,(\text{electron bilinear})\,A, \tag{10.51}$$

where A denotes the photon field. The detailed Dirac-matrix structure is not needed for the counting argument. The elementary QED rule is the emission or absorption of a photon by an electron; all the more complicated QED graphs are assembled from repeated uses of that one vertex.[19]

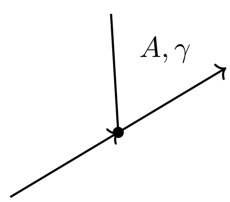


Figure 10.14: The elementary electron–electron–photon vertex of quantum electrodynamics.

[a] **Question from the audience.** What about electron–proton scattering?

Response. In the elementary atomic treatment, the proton can be idealized as a fixed, infinitely heavy point source—a heavy particle nailed down in place. Its own quantum-field dynamics enter only when the internal theory of protons and neutrons is included.

[a]Lecture timestamp: 01:27:38.

[a] **Question from the audience.** Can the electron move nearly at the speed of light relative to that proton?

Response. Yes.

[a]Lecture timestamp: 01:28:30.

[19]The schematic vertex is reconstructed from 01:26:42; the detailed matrix structure is deliberately suppressed in the source.

Consider photon scattering by an electron. At lowest order the graph has two electron–electron–photon vertices. The amplitude is therefore proportional to e^2:

$$\mathcal{A}_{\text{lowest}} \propto e^2. \tag{10.52}$$

The probability, or equivalently the coupling dependence of the cross section, comes from the absolute square and is proportional to e^4:

$$P_{\text{lowest}} \text{ or } \sigma_{\text{lowest}} \propto e^4. \tag{10.53}$$

With the conventional dimensionless normalization, the electric coupling is small, so genuine scattering is much less likely than the photon simply passing without scattering.

There are already two graphs at this order. In one ordering the incoming photon is absorbed and the outgoing photon is emitted later. In the other, the outgoing photon is emitted before the incoming photon is absorbed. Both amplitudes contribute.[20]

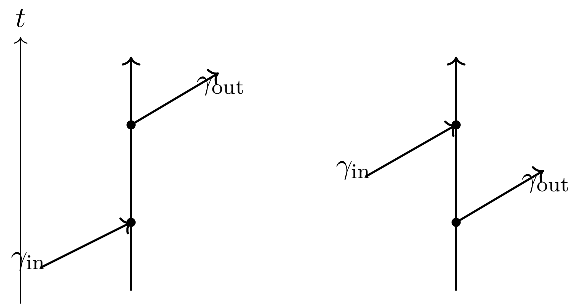


Figure 10.15: The two lowest-order time orderings for electron–photon scattering.

The lowest-order amplitude is consequently not one graph but the sum of the two:

$$\begin{aligned} \mathcal{A}^{(2)} &= \mathcal{A}_{\text{absorb then emit}} + \mathcal{A}_{\text{emit then absorb}}, \\ \mathcal{A}^{(2)} &\propto e^2. \end{aligned} \tag{10.54}$$

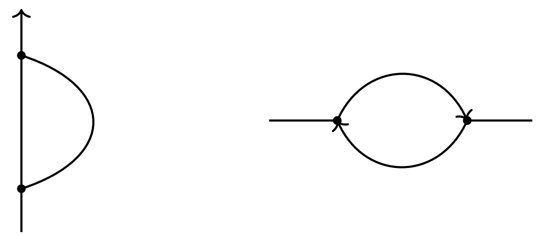
Higher-order corrections add more elementary vertices. An additional internal photon attached twice to the electron line adds two factors of e, giving a contribution proportional to e^4 in the amplitude and e^8 in its squared magnitude. A photon line may also contain an electron–positron insertion, again adding elementary vertices and additional powers of e. These are examples, not an exhaustive list; the extra photon can be attached in many positions.[21]

Thus the QED expansion proceeds in successive powers of e^2. Each comparable new complication adds two powers of the electric charge to the amplitude. In the normalization being used, that makes such a correction roughly a hundred or a couple of hundred times smaller in amplitude than the preceding order. The more complicated graphs are progressively weaker, so the perturbative sum has a chance to be useful.

The probability is not obtained by squaring each graph separately and then adding those squares. First add every amplitude contributing to the same external process:

$$\mathcal{A}_{\text{total}} = \sum_r \mathcal{A}_r. \tag{10.55}$$

[20] The two time orderings are reconstructed from the diagrams at 01:28:42 and 01:30:18.
[21] The higher-order structures are reconstructed from 01:30:38–01:31:16.



additional internal photon electron–positron insertion

Figure 10.16: Two higher-order QED structures that add elementary vertices and further powers of e.

Only then form

$$P = |\mathcal{A}_{\text{total}}|^2 = \left|\sum_r \mathcal{A}_r\right|^2. \tag{10.56}$$

The cross terms between different graphs are interference terms. The rule is simple and essential: add first, then square.

10.14 Toward the Particle Inventory and the Standard Model

With propagators, vertices, coupling counting, and interference in place, the particle-physics description can begin in earnest. One can name the particles, specify their masses, write the Lagrangians that govern them partly from theory and partly from experiment, and study their symmetries directly in those Lagrangians. The same framework then leads to the Standard Model and its properties.

[a] **Question from the audience.** Could the constant mass coefficient m^2 be replaced by the Higgs field?

Response. Yes. That is the right idea, and the mechanism will be discussed later.

[a]Lecture timestamp: 01:33:22.

[a] **Question from the audience.** Can the neighboring lattice points x and x' lie at two different times?

Response. Yes. They may be neighbors in the time direction; that is how vertical propagation on the lattice is represented.

[a]Lecture timestamp: 01:33:42.

For neighboring points, the finite-difference square contains both same-site and cross-link terms:

$$\left[\phi(x) - \phi(x')\right]^2 = \phi(x)^2 + \phi(x')^2 - 2\phi(x)\phi(x'). \tag{10.57}$$

[a] **Question from the audience.** Which term represents a particle simply standing still?

Response. The same-site terms $\phi(x)^2$ and $\phi(x')^2$ in the finite-difference square represent standing still, while the cross term $\phi(x)\phi(x')$ supplies the moving interpretation. The explicit mass term also contributes same-site, standing-still factors.

[a]Lecture timestamp: 01:33:56.